



Artificial Misinformation

Exploring Human-Algorithm
Interaction Online

Donghee Shin

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Artificial Misinformation

“This book discusses how misinformation is wielded to manipulate the public and deny facts and truth, why humans are susceptible to fake news, and how the spread of misinformation can be controlled using technologies such as AI. This interdisciplinary discussion suggests how people can be better supported to combat misinformation through human judgment and AI.”

—Mohammed Ibahrine, *Northwestern University, Evanston, IL, USA*

“This book takes a multidisciplinary approach to contribute to the ongoing development of human–misinformation interaction, with a particular focus on the “human” dimension, and provides insights to improve the design of AI that could be genuinely beneficial and effectively used in society.”

—Frank Biocca, *New Jersey Institute of Technology, Newark, New Jersey, USA*

“Bringing the psychology of misinformation to AI, the book is the definitive guide to navigating the misinformation age. This book tells us that we are all vulnerable to believing misinformation. Informed by years of research, the book provides insightful analytics on the misinformation dynamics that lie at the intersection of human minds and the double-edged sword of AI.”

—John Pavlik, *Rutgers, the State University of New Jersey, New Brunswick, NJ, USA*

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ABOUT THE EDITOR

Donghee Shin is Professor of Digital Media and Professional Communication at the College of Media and Communication at Texas Tech University. Over the last 25 years, he has worked at various universities in the U.S., South Korea, and UAE. Broadly, his research areas include human-algorithm interaction, social computing, and media analytics. His research explores the impact of algorithmic platforms in terms of ethical considerations, algorithms, human-computer interaction, and media studies. In his recent research, he has examined various mechanisms to investigate users' behavior around opaque algorithmic systems, redesign these systems to communicate opaque algorithmic processes to users, and provide them with a more informed, satisfying, and engaging interaction.

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PART I

The Cognitive Science of
Misinformation: Why We Are
Vulnerable, and How Misinformation
Beliefs Are Formed/Maintained



Introduction: The Epistemology of Misinformation—How Do We Know What We Know

DIFFUSION OF MISINFORMATION

In the time of AI, Everett Rogers' theory of *Diffusion of Innovation* (1962) recently received a great deal of revisit with the prevalence of misinformation—diffusion of misinformation. The diffusion of fake news and disinformation is a growing problem with a serious and negative social impact. The diffusion of misinformation becomes even more problematic when it addresses issues related to health, as it can affect people at both the individual and societal levels. Not only does misinformation about health facts contribute to feeding anxieties, but it also has harmful social, political, and economic consequences. The diffusion of falsehood is like the spread of viral contagion, requiring a new paradigm for countering misinformation. Researchers identify the drivers of the misinformation diffusions that characterize the decreasing trust in our contemporary society: obscuring the line between fact and opinion; increasing the relative volume, and resulting influence, of opinion and personal experience over fact; mounting dissonance in the judgment of factors and interpretations of facts and data; and deteriorating trust in formerly respected sources of factual information. These diffusion trends, to the extent that they continue, imply that misinformation will continue to find highly vulnerable users in our society. There is a pressing need for us to create a healthier information ecosystem and safeguard against misinformation and infodemics.

MISINFORMATION ON MISINFORMATION: *THE MISINFORMATION PARADOX*

People understand the harmful nature of misinformation but continue to engage with misinformation and continue to accept and share misinformation. This phenomenon is called the misinformation paradox (Munyaka et al., 2022), the discrepancy between users' attitudes toward misinformation and how they actually behave in online misinformation. As one of the misconceptions about misinformation, the relationship between individuals' intentions to debunk misinformation and their actual personal misinformation-accepting behaviors has shown to be very different. People may assert openly that they criticize misinformation but behave in ways that enjoy consuming, reproducing, and sharing misinformation. While people might express doubt and disbelief toward information sources, they engage with misinformation content in the same ways they engage with content they trust. People express high awareness and alertness of misinformation, yet in reality, they continue to interact with misinformation in harmful ways. Research has shown this misinformation paradox in people that a complete distrust of misinformation does not result in disengagement from that misinformation (Kim et al., 2023; Munyaka et al., 2022; Shin, 2023). People understand cognitively the negative sides of misinformation. Nevertheless, they still engage with it online: the more they know about the adversarial effects of misinformation, the more they consume and share it. As AI technologies advance, the divergence between users' misinformation concerns and behavior becomes clearer.

Normal People Like Us Fall for Misinformation

Why does misinformation persist and spread so quickly, why are falsehoods so contagious, and why do people believe disinformation so easily? There are divergent views on the reasons for misinformation spreading. Some researchers view social platforms' structure of rewarding users for habitually sharing misinformation as a source of misinformation problems (Ceylan et al., 2022). Other researchers consider that the human mind is a strong force that drives the consumption and spread of misinformation (Shin, 2023). Whether it is a function of the structure of the social media platforms or the users' habitual behavior of spreading misinformation, the

underlying factor is the human mind behind misinformation. Research has shown that human cognition plays a bigger role, and social media platforms take advantage of weak human factors when it comes to misinformation. It is the humans to share online, and it is the humans to transmit networks. Misinformation is analogous to a virus that can infect people and spread within and between the human nexus of networks. The more it is shared, the more transmittable it becomes. Once some human is exposed to misinformation, it can bolt onto our cognition and embed deep into human unconsciousness, making it very hard to delete.

We live in a world of conspiracy theories, inflammatory memes, political disinformation, and fake news headlines. Misinformation is not new and has been around along with information. Likewise, disinformation is an old story fueled by rising AI. The platforms driven by algorithms have become fertile ground for artificially intelligent misinformation. With the instant and rapid distribution of user-created content on social media, there are no barriers or intervening forces to entry. Everyone can publish, and anyone can distort or misrepresent the truth online. AI has spawned a deluge of misinformation and spreads fake news faster, farther, further, deeper, and broader than the truth. Misinformation has the added edge of being novel as compared to true news, and novel information is more likely to be retweeted and shared. Misinformation spreads more pervasively than the true information.

The boundary between misinformation and truth is blurring and intentionally being obscured with the rise of AI, which has fundamentally impacted the way misinformation is created, consumed, and transmitted. The gray area of misinformation is hard to debunk both by human and automatic fact-checkers due to the variety of twists and tweaks in place, which proliferate across social media and that cannot be reduced to a binary problem of true versus false information. Humans are cognitive misers that we think in less effortful and simpler ways rather than in more reasonable and conscientious modes. By default, people trust anything they see or read as their minds often seek to avoid spending cognitive effort and rely on heuristics and attributional biases. They are limited in their capacity to process information, and they take shortcuts whenever they can. This cognitive misery is what makes our cognitions so vulnerable to misinformation. No one is entirely immune to misinformation, in part

because of how our cognition is structured and how misinformation manipulates it. Human cognitive tendencies can make us susceptible to misinformation if we are not careful.

Sticky Misinformation and Self-enforcing Beliefs

Misinformation sticks even when humans realize it is untrue. The continued influence effect of misinformation persists even after the misinformation has been retracted. Once we know disinformation, it is difficult to discard it from our memories, partly because it contains some truth about the real issues, which makes it difficult to discern truth from lies. Often, it can be a single bogus paragraph inserted into an otherwise genuine document. Disinformation makes people build a cognitive model of information, and once a model has been built, any corrective efforts to identify a critical element of the model as false should fail since the deletion of that element would undermine the whole cognitive model and mental schema. The more people realize the information is false, the more they nevertheless enhance their wrong bias. Misinformation takes advantage of these cognitive biases, often cited as confirmation bias; people pursue information that confirms what they already believe. With confirmation bias, true information, at times, can be reframed and converted by biases and malicious data to come to a collective fabricated idea. In *What the Fact?*, the author Seema Yasmin (2022) writes of confirmation bias: “*Our brains seek more dopamine, more oxytocin, more information that backs up what we’ve come to believe, while conveniently ignoring evidence that contradicts our beliefs.*” Inside the human brain, there is a status quo bias that avoids cognitive dissonance and supports humans’ existing attitudes and views. Information is thus inevitably reconstructed at the level of cognition and selectively retransmitted at the level of emission. With this confirmation bias in place in the age of AI, a key question raised is: What role does human decision-making play, and how can AI enable humans to make better decisions? Misinformation has consequences not only for our civil society and democracy but also for human mental and physical health. Misinformation has already fanned the flames of distrust toward social media around the world. Misinformation can latch onto our cognition and destroy our fundamental brain function.

How to Counter Misinformation and Fight Against Infodemics

Can we unstick misinformation? Researchers have agreed that the spread of misinformation can be prevented in two main ways: (1) by correcting the perceptions of those who believe the misinformation by disseminating corrective information and (2) by detecting its spread. In this light of preventative methods, the book has two parts of discussion: (1) what are the people's cognition behind misinformation and how to correct their perception; and (2) how to detect its spread and how to increase people's literacy on misinformation. The book systematically analyzes the different dimensions of misinformation through cognate disciplinary perspectives, taking into account the related contexts of communication, cognitive science, and psychology.

This book admits that misinformation is sticky and difficult to dislodge. But we know misinformation can be prevented or at least the harm reduced by alerting humans to how they might be misled. That is why the book discusses why our cognitions are so vulnerable to misinformation, how misinformation spreads online, and what we can do to protect ourselves and others. Rather than looking at misinformation through a single lens, the book maps the various kinds of misinformation through several different disciplinary perspectives, taking into account the overlapping contexts of psychology, technology, and journalism.

The book focuses on four main building blocks:

- Individuals' and societies' vulnerability to misinformation
- The ways people interact with misinformation (how people view and process misinformation)
- Factors and interventions that can increase individuals' (and societies') resistance to misinformation
- Misinformation and AI

Chapter 2, susceptibility to misinformation, focuses on factors that affect the endorsement and persistence of misinformation, particularly from a bias point of view. What happens if the data fed to AI are biased? What happens if the response of a chatbot spreads misinformation? Unlike many people hope, AI is as biased as humans are. Bias can originate from various venues, including but not limited to the design and unintended or unanticipated use of the algorithm or algorithmic decisions about the way data are coded, framed, filtered, or analyzed to train machine learning.

Algorithmic bias has been widely seen in advertising, content recommendations, and search engine results. Algorithmic prejudice has been found in cases ranging from political campaign outcomes to the proliferation of fake news and misinformation. It has also surfaced in health care, education, and public service, aggravating existing societal, socioeconomic, and political biases. These algorithm-induced biases can exert negative impacts on a range of social interactions, ranging from unintended privacy infringements to solidifying societal biases of gender, race, ethnicity, and culture. The significance of the data used in training algorithms should not be underestimated. Humans should play a part in the datafication of algorithms, as preventing the spread of misinformation is difficult by technology alone, especially considering the rate at which information can spread online.

Chapter 3 examines the role that social media algorithms play in recommending extreme content by discussing how misinformation is related to belief polarization and proposing the radicalization process model. TikTok has ushered in a novel era of misinformation, exposing its user base to extreme information regularly. Misinformation can be a direct cause of radicalization due to its tendency to trigger strong emotions. Aggressive messages that arouse anxiety can be highly persuasive—messages that point to a threat, particularly one that is sensitive and socially hot, create a cognitive drive for more content about that threat and generate support for responsive action. TikTok’s role in fostering radicalized content was examined by tracing how users become radicalized on TikTok and how its recommendation algorithms drive this radicalization. The results revealed that the pathways by which users access far-right content are manifold and that a large part of this can be ascribed to platform recommendations through a positive feedback loop. The results are consistent with the proposition that the generation and adoption of extreme content on TikTok largely reflect the user’s input and interaction with a platform. It is argued that some features of misinformation are likely to promote radicalization among users. It concludes how trends in artificial intelligence (AI)-based content systems are forged by an intricate combination of user interactions, platform intentions, and the interplay dynamics of a broader AI ecosystem.

Chapter 4 proposes the misinformation paradox and discusses the nudge maneuver to mitigate the paradox. It examines the effects of accuracy nudges on judging misinformation and how user trust moderates this effect. Applying the nudge principle to misinformation and sharing

intention, we empirically test (1) whether accuracy nudges (accuracy alerts/warning messages) trigger accuracy judgments and thus deter the sharing of news based on falsehoods on social media and (2) whether the effect is moderated by news sources and whether this moderation depends on users' trust in algorithms. The results from a 2 (nudge: accuracy nudge vs. no nudge) $\times$ 2 (news source: algorithmic news vs. nonalgorithm media) ($N = 400$) experiment showed significant main and interaction effects, indicating that algorithmic source effects are present in the process of nudge acceptance. Misinformation sharing intention was largely lower for nonalgorithmic news than for algorithm-based news, but there was a greater decrease in algorithmic news when nudging was employed. Moderation from algorithmic trust was found, and users' trust in algorithmic media amplified the nudge effect only for news from algorithmic media and not nonalgorithmic online media sources. The results suggest that there is a need for an efficient mechanism combining AI and cognitive nudges that can support humans in making judgments regarding the information spreading online. A cognitive AI framework can augment humans' capability in judging the veracity of the information online and reinforce positive information-sharing behavior in individuals thereby reducing the spread of misinformation.

Chapter 5 examines the psychological, cognitive, and social factors involved in the processing of misinformation people receive through algorithms and artificial intelligence. Modeling cognitive processes has long been of interest for understanding user reasoning, and many theories from different fields have been formalized into cognitive models. Drawing on theoretical insights from information processing theory with the concept of diagnosticity, it examines how perceived normative values influence a user's perceived diagnosticity and likelihood of sharing information and whether explainability further moderates this relationship. The findings showed that users with a high heuristic processing of normative values and positive diagnostic perception were more likely to proactively discern misinformation. Users with a high cognitive ability to understand information were more likely to discern it correctly and less likely to share misinformation online. When exposed to misinformation through algorithmic recommendations, users' perceived diagnosticity of misinformation can be predicted accurately from their understanding of normative values. This perceived diagnosticity would then positively influence the accuracy and credibility of the misinformation. With this focus on misinformation processing, this chapter provides theoretical insights and relevant

recommendations for firms to be more resilient in protecting themselves from the detrimental impact of misinformation.

Chapter 6 introduces the principle of diversity-aware AI and discusses the need to develop recommendation models to embed AI with diversity awareness to mitigate misinformation.

Nudge principles have been applied to algorithms so that they maneuver search results through news recommendations, target messages, steer recommendations, and mix commercials with information in social media feeds. Algorithmic personalization through nudges is a cause of increasing concern for the sustainable development of algorithmically curated news platforms. Algorithmic nudging in news recommender systems (NRSs) has become important in ensuring users' right to view diverse news and viewpoints. This chapter proposes a conceptual framework for personalized recommendation nudges that can promote diverse news consumption on online platforms. It empirically tests the effects of algorithmic nudges by examining how users make sense of algorithmic nudges and how nudges influence users' views on personalization and attitudes toward news diversity. The findings show that algorithmic nudges play a key role in understanding normative values in NRS, which then influence the user's intention to consume diverse news. The findings imply the personalization paradox that personalized news recommendations can enhance and decrease user engagement with the systems. This paradox provides conceptual and operational bases for diversity-aware NRS design, enhancing the diversity and personalization of news recommendations. It proposes a conceptual framework of algorithmic nudges and news diversity, and from there, we develop theoretically grounded paths for facilitating diversity and inclusion in NRSs.

Chapter 7 discusses the design of nudging interventions in the context of misinformation, including a systematic review of the use of nudging in human-AI interaction that has led to a design framework. By using algorithms that work invisibly, nudges can be maneuvered in misinformation to individuals, and their effectiveness can be traced and attuned as the algorithm improves from user feedback based on a user's behavior. It seeks to explore the potential of nudging in decreasing the chances of consuming and spreading misinformation. The key is how to ensure that algorithmic nudges are used in an effective way and whether the nudge could also help to achieve a sustainable way of life. This chapter discusses the principles and dimensions of the nudging effects of AI systems on user behavior in response to misinformation.

Chapter 8 proposes the inoculation idea that it might be possible to deliver a cognitive vaccine against misinformation. Can we inoculate people against the misinformation epidemic by cultivating scientific habits of cognition? Based on inoculation theory and a heuristic-systematic model, this chapter discusses the cognitive mechanisms of inoculation effects on using AI chatbots by addressing questions on how users construe inoculation messages and how the messages influence users' resistance against misinformation. How inoculation confers resistance to users provides important implications for theory and practice. The chapter found that inoculation messages alleviate the negative effects of misinformation from AI chatbots on user interaction. A more involved variant of inoculation not only provides an overt caution of the impending threat of misinformation but it furthermore refutes an anticipated argument that exposes the imminent fallacy. It renders a critical perspective of how the theory can be conceptually extended to misinformation and how the theoretical frame can be used practically.

Chapter 9 is motivated by the rapidly improving capabilities and accessibility of generative AI and rapidly increasing misinformation problems. This chapter discusses the misinformation effect by examining how users process and respond to misinformation in generative artificial intelligence (GenAI) contexts. Misinformation is by no means a new phenomenon, yet its trend is highlighted by the emergence of AI. It might be useful to see misinformation in the context of a new and rapidly evolving AI landscape, which has facilitated the spread of unparalleled volumes of information at lightning speeds. When exposed to misinformation from GenAI, users' construed diagnosticity of misinformation can be accurately predicted from their understanding of ethical values. With this focus on misinformation processing, this chapter provides theoretical insights and relevant recommendations for firms to be more resilient in protecting users from the detrimental impact of misinformation.

The conclusion chapter ends with a review of deepfakes and the related discussion of how algorithms generate and manipulate misinformation. The advent of AI and machine learning has crucially changed the way misinformation is created, shared, spread, diffused, and consumed. The rapid advancement of AI is a driving force behind the proliferation and growing impact of misinformation. The growing prominence of deepfakes has triggered an ongoing discussion of authenticity online and the line between fact and fiction. The future online environment should reflect how a healthy society naturally acts rather than an algorithm to manipulate

our attention to boost corporate profit. AI systems should bear transparency, provide fair results, establish accountability, and operate under a clearly defined data governance policy. This concluding chapter gives insights into designing responsible AI to curb misinformation.

The book offers an integrated analysis of the logic and social implications of misinformation processes. Reporting on years of empirical scientific studies, the results of such integrated analyses are useful and constructive for understanding the relationships between humans and misinformation. Most of the empirical data in this book were collected from international contexts, analyzed by global perspectives, and discussed by non-U.S. scholars. This broad aspect is important and heuristic because disinformation is a global problem, extending beyond the political sphere to all aspects of human lives. To date, however, many of the empirical studies on misinformation, its conceptualization, and theorization have stemmed from North America, where the global AI firms are located and operated.

This study presents an imperative debate about universal users' engagements in the spread of misinformation, methods for countering misinformation, and what is at stake while industry and government deal with misinformation in everyday business. By examining the immense repercussions that misinformation will have on people and society, the book brings together various perspectives on algorithms into an integrated conceptual framework. It provides a broad sociotechnical analysis, addressing the critical and ethical issues of combating misinformation. Illustrating through models and descriptions how that works, both theoretically and statistically, is helpful in parsing out how misinformation takes advantage of human epistemic vulnerabilities.

Cutting across all the chapters raises the need for an urgent cross-sectoral interdisciplinary effort to investigate, protect against, and mitigate the risks of misinformation. This book proposes a new framework, human-misinformation interaction (Karduni, 2019). As an extension of human-computer interaction, it is an understanding of the interdisciplinary approach needed to address and combat misinformation. Solutions to the problem of the spread of misinformation have come from a variety of disciplines. Misinformation is a product of human interaction processes (engagement, interpretation, and representation) where it is formed, spread to others, judged, and consumed, all against a backdrop of social, technological, and cultural dynamics. The prevalence of AI and machine learning has amplified the interaction bandwidth and styles and further the

effects of misinformation on our societies. Countering misinformation requires systematic multi-stakeholder coordination and lasting investment in building societal resilience and media and information literacy. In this light, we need a new disciplinary field that focuses on designing AI systems to curb misinformation and evaluate misinformation that people interact with. The research can study the source, content, technology, and humans as the main components involved in the process of misinformation. The multidisciplinary field is concerned with understanding and improving the interaction and relation between humans and misinformation by utilizing AI as a tool to control misinformation. The research within the field will consider how to develop and deploy AI systems that effectively detect and monitor misinformation, how to nudge users to have literacy against misinformation, and how misinformation covertly modifies and influences human behavior. The main areas of human-misinformation interaction research include:

1. Misinformation detection and fact-checking using content.
2. Identifying untrustworthy sources and malicious AI.
3. Human literacy and cognitive immune tools and methods aim to make humans resilient to misinformation.

REFERENCES

- Ceylan, G., Aderson, I., & Wood, W. (2022). Sharing of misinformation is habitual, not just lazy or biased. *Proceedings of the National Academy of Sciences*, 120(4), e2216614120. <https://doi.org/10.1073/pnas.2216614120>
- Karduni, A. (2019). Human-misinformation interaction: Understanding the interdisciplinary approach needed to computationally combat false information. [arXiv:1903.07136](https://arxiv.org/abs/1903.07136). <https://doi.org/10.48550/arXiv.1903.07136>
- Kim, J., Lee, J., & Dai, Y. (2023). Misinformation and the paradox of trust during the COVID-19 pandemic in the U.S.: Pathways to risk perception and compliance behaviors. *Journal of Risk Research*, 26(5), 469–484. <https://doi.org/10.1080/13669877.2023.2176910>
- Munyaka, I., Hargittai, E., & Redmiles, E. (2022). The misinformation paradox: Older adults are cynical about news media, but engage with it anyway. *Journal of Online Trust and Safety*, 1(4) <https://doi.org/10.54501/jots.v1i4.62>
- Rogers, E. M. (1962). *Diffusion of innovations*. Free Press of Glencoe.
- Shin, D. (2023). *Algorithms, humans, and interactions: How do algorithms interact with people? Designing meaningful AI experiences*. Routledge. <https://doi.org/10.1201/b23083>



Misinformation and Algorithmic Bias

WHY DO PEOPLE FALL FOR MISINFORMATION?

People like us fall for misinformation (Young, 2023). Everyone is susceptible to falsehood because misinformation plays with whatever existing biases we might have. Humans make decisions based on true and accurate information. However, human psychological predispositions and biases make people vulnerable to misinformation. Hence, misinformation is more likely to be accepted, trusted, and amplified even after it turns out to be false. It has been said that no people are completely immune to misinformation, mostly because of the way the human cognitive system is set up and the tactics with which misinformation breaks into the cognitive system (Hameleers & Brosius, 2022). People easily take mental shortcuts in making judgments to simplify problems and avoid cognitive overload (Sundar, 2008). However, this heuristic process can make humans susceptible to misinformation if they are not careful and conscious of their actions. In general, humans will trust anything they view or hear unless it is a downright falsehood. In people's everyday lives, most information that they are exposed to is truth, partially true, or at least contains an element of truth. However, the more people see some information repeatedly, the more likely they are to accept it as a truth. This default truth effect (by default, humans tend to trust information they are exposed to) emerges since people tend to rely on ease of understanding and familiarity when they seek true information. Normally, people think given information is

likely to be true or have beliefs consistent with it. They are likely to have previous familiarity with the information. The more information is recurrent, the more familiar and less controversial it feels whether it is false or true. When people see something repeatedly, it starts to turn into a fact, and people believe in their mind that it is true. This phenomenon is the so-called popularity cue—people tend to believe popular news as true. This popularity cue is similar to a bandwagon effect—Many people like something making us more likely to support it. In terms of misinformation, a popularity cue occurs when we see how many times something has been shared or tweeted over the content itself. How many excellent reviews get, or what percentage of users rated a service positively, affects trust (Walther et al., 2022). The popularity of information allows us to bypass assessing information. If thousands of other people have shared a piece of news, surely someone else has verified it. The main function of AI is to make certain information look repeatedly read and suggested, which can also inflate the apparent popularity of misinformation (Metzger et al., 2021). Popularity cues can affect not only human attention but also human behavior.

In many cases, however, these mental heuristics do not function well all the time but work only within certain contexts and specific rhetorical situations. In our current social media where AI dominates and information can be forged and amplified, heuristics at times lead to poor decision-making based on a false dataset and can easily fall for misinformation. In the current social media environment, even a single exposure to fake news can lead to misinformation fallacy. Propaganda and political campaign often repeat falsehood as they know the power of the default truth effect, which leads to “illusory truth,” a perception of familiarity that human cognitions begin to interpret as the feeling of truth, despite the obvious falsehood of the claims and the simple repetitions of a falsehood. Humans by nature are more susceptible to misinformation that fits into our existing perspectives or societal values, and people can easily fall into *confirmation bias*, a psychological tendency to cherry-pick information that supports their preexisting beliefs while dismissing contradictory information. Exaggerated stories and flashy headlines are easier to accept and more lucrative than factual objective information.

Per confirmation bias, humans see only what they want to see or expect, without examining information thoroughly: the internal propensity of people to filter out information that confirms and corroborates what they already believe, instead of examining critically all the aspects of

information. Whenever a presented piece of news includes information that conflicts with the currently held cognitive beliefs of people, it immediately induces cognitive dissonance (Badke, 2018). People are driven to reduce this dissonance, hence they circumvent or even distort fact that contrasts their personal views. This tendency is related to the question of why misinformation resists correction. When people have been exposed to and hooked by misinformation, it is difficult to root out even when people try to seek the truth. Relevant research has confirmed that misinformation can still impact human cognition despite people receiving a correction and accepting it as a fact (e.g., Ecker & Antonio, 2021). This continued influence effect (or sleeper effect) is a primary reason for resisting correction, and numerous follow-up studies have found its strong and persistent effect. Walter and Tukachinsky (2020), for example, found that correcting misinformation decreases but does not completely remove the effect of misinformation. Many other studies confirm that hearing the truth after misinformation does not eliminate misinformation from human cognition once it accepts misinformation. In fact, accepted misinformation and people's efforts to correct it exist side-by-side and constantly clash and compete. A study by Lewandowsky and van der Linden (2021) found that human brains encode both the original piece of misinformation and its correction, but it is difficult to correct falsehood once it is established within the cognitive scheme as a truth. Canceling misinformation engages a complete extra cognitive effort of labeling it as false in the human brain. This uprooting process is not easy since it is already encoded in human memory. After a while human effort and will to correct or conduct fact-checking may fade, leaving them vulnerable to more extreme misinformation.

Correcting misinformation is even more difficult if it is embedded into human identity or the system of belief. It is almost impossible to remove human bias without the whole thing reformatted because people form cognitive models of the world to understand unfolding situations. Misinformation is deeply embedded in human belief systems, and it is cognitively impossible to root it out with a simple corrective message.

WHY AND HOW MISINFORMATION SPREAD ONLINE

Misinformation is running rampant and spreading rapidly on multiple platforms—social media, chatbots, AI, and even word of mouth. Falsehoods are often mixed with elements of truth such as a virus

online. Misinformation evolves over time, transforming into more contagious false tales, fanning across social media platforms through constant change of routes and swift jumping from one platform to the other.

The way that misinformation spreads online depends largely on the features of a network and its users. Research shows that the spread of misinformation is a function of the structure of social media (Giansiracusa, 2021). Algorithm-based systems on social media form users' habits of sharing information that draws other people's attention. Once habits are established, information-sharing becomes automatically activated by machine learning on the platforms without users considering critical response consequences. Thus, without considering its long-term effects, twitting, sharing, exaggerating, and posting on social media have become habitual practices.

A group of researchers argue that much of the spread of disinformation can thus be attributed to human action. Pennycook and Rand (2022) argue that most misinformation results from humans, and the most powerful suppliers and amplifiers of misinformation are humans. A study of Instagram during the COVID pandemic found that in a sample of more than 20,000 users, 5% of those who shared health information also shared misinformation. However, 90% of the misinformation was generated from only 0.2% of users. Falsehood is amplified even more when power bloggers, such as superspreaders, online opinion leaders, and media celebrities, have access to trillions of users on legacy media and AI-based social media.

Once misinformation has been generated by its originators, the actions of individual social media users spread it through wider channels. Some users may disseminate the information to their own people of networks through sharing tools such as likes, links, and comments, which trigger the algorithms of social media platforms to show it to other users. This spreading mechanism can lead to misinformation spreading exponentially and often in distorted ways. Extreme information is more propelled and is spread quickly and widely through social media due to such human motivation.

Misinformation can continue to influence reasoning after correction; this is known as the continued influence effect of misinformation (Swire & Ecker, 2018). The effect occurs when there is initial, persuasive information, followed by a new piece of true information (e.g., falsehood warning cue) that discounts the initial information. Over time, the discount label and the initial information become separated, such that the original information grows in persuasiveness in a person's cognition as its connection to

the reason to doubt it wanes (Grady et al., 2021). The failure of two cognitive processes to be causal, namely memory updating and suppression of misinformation reliance (Pluviano et al., 2022). Once we have heard misinformation, it is hard to uproot even when we want to know the truth. Misinformation can still influence our thinking even if we receive a correction and believe it to be true, a phenomenon known as the sticky nature of misinformation. Efforts are often made to correct misinformation by identifying it as false. Yet research on the continued influence effect of misinformation finds that such corrections are mostly unsuccessful at eliminating belief in the misinformation and/or its use in subsequent inferential reasoning.

The motivations of social media users are a more substantial driver of misinformation spread than the system attributes of social media. Some people do not process information correctly or critically, and others shape opinions based on certain social/political prejudices, which also influences their capability to judge falsehood online. A study by Shin (2022) examined more than 2400 Facebook users and found that people's social media habits were enhanced and fortified: the number of shared fake news items tripled. Habitual users shared five times more hoaxes than normal or new users. Social media users look for information that confirms their existing opinions and beliefs. Habits were more powerful in sharing misinformation than other reasons, such as lack of literacy and biased beliefs. This kind of action has been incentivized by algorithms that prioritize addictive engagement when choosing which posts users view in their recommended news and by the design and structure of social media (Mosleh et al., 2020).

Understanding the mechanisms of misinformation spread is critical considering its societal, political, health, and economic consequences. Addressing misinformation propensity would require reframing the online ecosystems that enhance and support its sharing. Social media platforms should take a more proactive measure than simply monitoring what information is posted and instead pursue structural changes in their reward structure to discourage the spread of misinformation.

WHY IS AI VULNERABLE TO BIAS?

There is no such thing as neutral data. While machine learning can learn human biases, such as in health care, they can also reinforce them or introduce new ones in their code. For example, an AI hiring program may favor

males for a position in computing as it has been learned most computer positions are male.

In 2023, the World Health Organization called for caution in relying AI on public health care since data used by AI to reach decisions could be biased or misused. The way AI collects data and the way they are used to train AI can be biased and generate inaccurate or misleading information, and machine learning models may be misused to generate misinformation (Park, 2023a). The potential of AI but had concerns over how it will be used to improve access to critical information as a decision-support tool for humans. This limitation comes from algorithms embedded in AI. Algorithms are human artifacts in that they are made, designed, trained, and applied by humans. Contrary to popular beliefs, AI is neither objective nor fair (Benjamins, 2021). An algorithm's performance largely hinges on the people who designed them, the code they used, the data they analyzed for the machine learning models, and the way they trained the models (Park, 2023b). As long as algorithms are human-made, they are naturally at risk of an inherent bias. Humans are under a cognitive bias, a pattern of deviation from rationality in judgment. This bias often leads to misrepresentation, wrong judgment, or mistaken interpretation. Humans create their own individual social reality more from their perceived world of reality and less from the objective input from it. Humans' cognitive bias often transpires into the algorithms they design, and thus, AI produces and amplifies the bias that humans enter. For AI to be aligned with human preferences, it must first learn those preferences. AI systems that are trained on user behavior can mistake human biases as human values and then optimize these biases. Learning human values by machine learning carries inherent risks. The underlying cause of AI bias lies in the conscious or unconscious human bigotry embedded in it throughout the development of algorithms. Hence, biases are quietly encoded during the process of algorithm creation.

Algorithms are programmed by people who—even with good will—can be prejudiced and discriminate within an unfair social world; thus, algorithms reflect and amplify the larger prejudices of reality (Shin et al., 2022). This type of amplification, which is called algorithmic amplification, is common in the platforms with which we interact every day because platforms such as Google, TikTok, Instagram, and YouTube are designed toward organized data gathering, automated processing, distribution, and maximizing monetization of customer data. These platforms use vast amounts of data for algorithmic systems and have far expanded the

capacities that they can do to drive people to decisions and behaviors that maximize monetization. The history of people's likes, clicks, comments, and retweets are the data that power the algorithmic amplification. By purposely ranking certain information higher, algorithms can amplify specific information while suppressing the representation of the rest. Some social communities benefit more from algorithmic amplification than others. Algorithmic amplification is furthered, and some online content becomes popular at the expense of other viewpoints. This is a reality on most of the platforms we currently use. The history of our likes, shares, and comments are the data driving algorithmic amplification.

Herbert Simon (1957) presented the idea of bounded rationality, which is limited by imperfect information, cognitive ability, and time constraints. Bounded rationality plays a key part in the way humans design algorithms and in the way automation bias comes from. YouTube's recommendations amplify sensational content to increase the number of people on the screen and the duration for which they are on the screen. Platforms recommend videos concerning political information, inappropriate content, and hate speech as ways to maximize revenue. News recommender systems suggest either negative or incorrect information—likely to provoke rage—and fake news headlines are designed to be exaggerated news rather than real facts (Shin, 2021). Programmers code the algorithms, markets choose the data used by the algorithms, and the algorithm designers decide how to apply the results of the algorithms. If the data analyzed by algorithms or used to train machine learning do not reflect the various parameters of users correctly, the results would be biased. Bias can enter algorithms and machine learning because of preestablished social, cultural, and political inequalities in society and thus people, which can impact decisions regarding how data are gathered, filtered, coded, or selectively analyzed to frame machine learning. It can be easy for people to let biases enter, which AI then algorithmizes and automates. That is why AI often makes decisions that are systematically unfair to certain groups of people. Algorithmic bias is a set of systematic faults in algorithmic systems that generate unfair discriminations, such as favoring a certain group of users over others (Fiske, 2022). A bias can be either intended or unintended, and it can emerge from a misunderstanding or a misinterpretation, that is, intentional design of certain algorithms or unexpected decisions associated with the way data are gathered, analyzed, or included to train machine learning. Relevant inquiries have found that these biases can potentially cause significant harm to the public (Sundar et al., 2020). For example, facial recognition

algorithms misjudge certain races more often than other people of color. A report published by the U.S. Department of Justice reported that facial recognition AI misjudges the races of colored people more often than noncolored people. If used by law enforcement, facial recognition could run the risk of discriminatorily arresting people of color. Another study by the University of California, Berkeley, showed that some mortgage automations have charged colored people higher interest rates. Another study confirmed the finding that some bank loan algorithms have levied people of color to higher interest rates than white people. A study at Harvard University revealed that AI-driven speech recognition systems show significant racial disparities, with voice recognition misunderstandings in 40% of words from minority users and only 11% of those from white users (Zarocostas, 2020). Additionally, the study found that virtual assistants with submissive Asian female voices reinforce stereotyped gender role bias. In 2021, it was revealed that Naver's recruitment system was biased against female candidates because the machine learning models were configured on datasets for weighing applications that reflected the male majority in the technology field. Algorithmic bias has commonly been found in social media platforms and search engine results. This bias can have serious effects on intensifying social stereotyping, biases, and prejudice. Algorithmic bias is prevalent in every aspect of our lives, as biased algorithms are embedded throughout health care, criminal justice, and employment systems, influencing critical decisions, operational work processes, and working rules. Machine learning helps technologies understand human rhetoric, bias, and discourse and has been found to reflect gender, racial, and class inequalities. Then, human biases, such as stereotyped sentiments attached to certain races, high-salary professions linked to a specific gender and race, and negative imaging of certain sexual orientations, become popularized in a wide variety of services. Indeed, algorithms can amplify human biases and societal stereotypes.

How do such biases enter into a set of algorithms? Humans write the codes in algorithms, select the data used by algorithms, and decide how to present the results of the algorithms. As humans develop an algorithmic structure, human biases are inevitably written into the algorithms. Biases are implanted through algorithmic data, machine learning embeds these biases, and AI reflects these biases in its performance (Huszar et al., 2021). Additionally, algorithms themselves contribute to biases. AI systems do not process and generate results based only on user data. They can also operate self-learning and self-programming algorithms based on

secondary data, nonobservational data, and situational data such as synthesized data, simulations, bootstrapped data, or a combination of generalized assumptions or rules. Machine learning processes such data and learns from the data. Human rights may be negatively influenced by the use of such self-programming algorithms. People whose data are not processed or who have not otherwise been taken into consideration may also be directly involved and negatively impacted, particularly when algorithmic systems are used to inform critical decision-making. As Sartori and Theodorou (2022) state, “(Algorithms) are embedded within larger social systems and processes, inscribed with the rules, values and interests of a typically dominant group.” Algorithms reveal glimpses on the existing structure of bias and inequalities that are embedded within our social, economic, and political systems. Without conscientious and rigorous mental investigations, it is easy for humans to intentionally and unintentionally input human biases into algorithms. Then, the biases are amplified, regenerated, and propagated. It is urgent and critical for AI researchers and industry leaders who develop AI technologies to test their algorithms to identify problems and potential biases.

Algorithmic bias can be seen across various platforms. Social media platforms are deliberately opaque, and their algorithms bolster existing societal biases (Abul-Fottouh et al., 2020). Social media platforms that contain biased algorithms exacerbate misinformation, fake news, and disinformation. However, how these platforms distribute misinformation is not open to the public. The 2020 U.S. presidential election result was partially influenced by algorithmic bias because platforms such as Facebook and Instagram acted as major biased news providers, as they failed to filter out fake news. Additionally, fake news concerning COVID-19 has triggered panic, fear, and hatred in many places. There has been fake news accusing racial groups, illegal immigrants, and even governments of the diffusion of COVID-19. Certain political groups propagate fake news for the sake of political gains. Social media algorithms have come under severe scrutiny for facilitating the spread of fake news, radicalized content, and political polarization (Abul-Fottouh et al., 2020). Misinformation and disinformation about political campaigns harm democracy because people lose trust in the political system (Fiske, 2022). The real threat of algorithms is that they can amplify and magnify biases that already exist in the world. Machine learning models can encode and propagate at any scale the biases programmed intentionally or unintentionally. Realizing the seriousness of the bias issue, recent efforts have been made in legal

frameworks such as the Artificial Intelligence Act (2021) and the European Union's General Data Protection Regulation (2018). Most firms have started to run programs to fight against bias and societal inequalities. Amazon operates thorough antidiscrimination policies, recruits diverse races, trains to recognize potential employee bias, and promotes diversity. Regulators are pushing hard and tough drives. Legislation is underway in the state of California to require firms using AI in their business to examine AI for possible bias and report the audit results to authorities. However, all these efforts may be in vain if the AI models continue to operate in routine operations and the delivery of results remains inadvertently discriminating. Removing or mitigating algorithmic bias involves efforts beyond technical fixes. The tools and methods used to remove bias and reduce variance tend to cause another bias. Removing algorithmic bias should involve not only changing the algorithms or the systems but also changing cultural biases and social structures. Bias can perpetuate algorithmic inner systems as a result of preestablished social and cultural values. Society should continuously request that critical decisions be transparent, fair, and accountable, even as they become increasingly algorithmized. If the self-regulatory model or ethical algorithm design does not work well properly, more systematic, ongoing, and legal measures auditing algorithms may be necessary.

TYPES OF ALGORITHMIC BIAS

There is no such thing as neural data. Data are not objective, and neither are decision algorithms. For example, an algorithm-based hiring system may favor males for a computing-related position since it has been learned that most computing professionals are male. Algorithm is inherently biased. The term "algorithmic bias" refers to systematic and repeatable faults in AI systems with unfair consequences (Fiske, 2022). When algorithmic systems are designed by private sectors, their foremost aim is to maximize profit for users. Because machine learning in AI is capable of learning how users behave, it can maneuver users toward specific behaviors that are lucrative for firms. Potential threats from AI in terms of manipulating and maneuvering human behavior have been extensively researched. Researchers have identified several categories of bias in AI: measurement bias, selection bias, framing bias, confounding bias, and confirmation bias. These are the most common types of AI bias that creep into the algorithms. The forms of algorithmic bias are dependent upon

where bias is coming from, the source of bias, and whether it is from humans, algorithms, and data (Zarocostas, 2020). Thus far, it has been indicated that the majority of biases result from humans because human prejudice causes and offsets other sources of bias. The types of algorithmic bias range from measurement bias to exclusion bias.

Measurement bias occurs because of underlying problems with the precision of the training data and how it is quantified and gauged. A model with flawed data collection methods or fallacious measurements leads to measurement bias and unfair results. For example, if an automatic survey system asks interviewees about preloaded questions, the responses will be inaccurate and biased due to systematic errors in the survey measurements. If an algorithm-driven recruiting system is programmed to select predetermined qualifications or schools, the system is biased toward specific criteria. A health care risk predictive algorithm is racially biased because of its reliance on a faulty metric (systematically biased) for determining need.

Selection bias occurs when the data utilized in machine learning are not representative of the overall population and produce a distorted version of the real trend. This bias occurs in a database that has more cases of a particular type than others. If a database in algorithms contains overrecognized groups and underrecognized groups, which commonly occurs because algorithms are trained to collect information online, the results would be biased. Chatbot technology trained only with the audio language data generated by males will have distorted voice recognition results when females interact with it. A face-recognition AI may train more faces of light-colored races than dark-colored races, thus producing biased performance in identifying darker-colored races. Biased sampling can lead to biased or invalid conclusions. This part of the sampling bias is a part of the overall selection bias.

The framing effect is a cognitive bias that influences human decision-making in different ways. People tend to believe in previously trustworthy cases and accept how the problem is framed and the information is presented; thus, the results obtained can vary and probably be biased. ProPublica, a nonprofit organization, found Correctional Offender Management Profiling for Alternative Sanctions (COMPAS) biased in predicting the possibility of recidivism. ProPublica alleged that the model used was biased against colored offenders, as the group was connected to a higher false-positive rate. ProPublica cited proof that the fairness metrics used in the COMPAS system breached the fairness criteria of equalized likelihoods and equality of opportunity. Most state and local governments

now use algorithms to inform bail and parole decisions, and we should demand that such decisions be transparent, fair, and accountable.

Confounding bias can result from the AI model if the model acquires incorrect associations by missing some information in the data or if it does not consider exhaustive information. This is a systematic misrepresentation in the measure of the relationship between exposure and the event caused by confounding the effect of that exposure of primary interest with irrelevant factors. This kind of bias stems from incorrect causalities that influence both inputs and outputs. For example, algorithms collect data on ice cream purchases and sunburns. Machine learning learns that higher ice cream buying is related to a higher chance of sunburn. Ice cream buying does not cause sunburn, and incorrect causality between inputs and outputs is established, thus producing algorithmic bias.

Confirmation bias arises when the observer of the data experiment includes their envisaged results in the data. This bias is similar to ideological bias, which is a reflection of underlying implicit values and assumptions about particular groups and political positions (Hameleers & Brosius, 2022). Bias can occur when humans begin with data research with biased thinking about their study, consciously or unconsciously. This bias commonly occurs when people favor information that confirms previously existing thoughts or customs. There has been a misconception that left-handed people are more athletically effective or artistically creative than right-handed people. This would result in inaccurate data because this has never been proven, and algorithms can take this confirmation and turn into bias.

Other forms of algorithmic bias include exclusion bias, recall bias, outliers, overfitting, underfitting, and reporting bias. Understanding potential types of biases in AI is the first step toward eradicating them. With that knowledge, it is critical to check the data analytic processes carefully to ensure that the datasets are as error free as possible before they are put into the training stage.

NEGATIVE FEEDBACK LOOP AND BIAS

A feedback loop is the part of a system in which some parts of the system's output are used as input for future operations (Shin et al., 2022). In AI, a feedback loop refers to the process of using the output of an AI system and corresponding user actions to train and reinforce models over time. The predictions and recommendations that are generated by AI are compared

against the output, and feedback is provided to the model, making it learn from its errors. Feedback loops help AI systems learn what they did right or wrong, feeding them data that enable them to adjust their parameters to perform better in the future. This is a form of positive feedback loop that is sustained and supported by trust between humans and algorithms. A positive feedback loop is normally considered to have those components of a system that jointly increase each other's values when a stimulus occurs in one component. User profiles and recommendations shape a feedback loop. The users and the system are in a process of reciprocal confluence, where user profiles are continuously maintained and updated over time through interaction with the algorithm systems, and the quality of the algorithmic system is also improved by the user profiles. Using this feedback loop, social media platforms have reward systems that incentivize users to stay on their platforms and keep posting and sharing. Users who post and share catchy and sensational misinformation are likely to draw traction. Due to the reward-based feedback loop on social media, people form habits of sharing information that gets recognition from others. Once habits are developed, misinformation sharing is repeatedly activated by cues on the platform without users considering critical response consequences.

Most AI applications in real-world systems follow this loop: data collection, data analysis, annotation and labeling, modeling, training, deployment, operationalization, use, feedback into the system, and use again. At every single point of this loop, bias can be added. Any model can spread biases in the training data. Humans should define their data sources and populations, their sampling methods for data collection, and the rules in the annotation and labeling task. The training model can proliferate and amplify the biases arising from prior analyses. The training model can enhance the feedback loop, which can contribute to rising bias. A relevant example of feedback loops producing bias can be found in Dressel and Farid's (2018) COMPAS study. It predicts the probability of an inmate recommitting a crime in the future. By analyzing the COMPAS algorithms, it was revealed that the algorithms are not accurate and are biased toward colored races. Nonwhite subjects were falsely labeled as having a high risk of committing a crime. This can be inferred by the fact that the recidivism model can produce different results with different parameters, such as gender, location, probability of family members of convicts, and crime history. With these parameters, the models are biased toward specific ethnicities and races. This finding is consistent with ProPublica's

investigative reporting that COMPAS is biased against colored defendants. Currently, COMPAS has been commonly criticized for propagating systematic racial bias.

Generally, certain popular items are suggested selectively, while the majority of other items are overlooked. These recommendations are then perceived by the users, and their reactions are logged and written into the system. This is called a negative feedback loop, which leads to generating and amplifying bias and misinformation. With negative feedback loops in place, bias causes further bias. Once a bias is included in the system, it can not only be compounded to generate discriminatory results that generate and amplify existing biased notions, but also lead to algorithms developing their own ways to form biases. Because of the autonomous feature of negative feedback loops, algorithms can deviate from their intended goals, which can create significant obstacles when this bias mechanism goes unnoticed. Thus, designers and providers of algorithms should constantly monitor potential negative feedback loops that can lead to algorithms that generate biased decisions or even perpetuate gender and racial stereotypes.

MISINFORMATION AND AI: INHERENT BIASES

The advent of AI has significantly transformed the way people access, consume, and share information. The AI information ecosystem has forged new participatory ways of news generation and consumption, but it has also worsened misinformation problems by eroding trust and further reducing people's appetite for information (Shin et al., 2022). The algorithms embedded in social media work as the basis for echo chamber phenomena, the cause of the rise of misinformation, particularly in health communication and political discourse.

Misinformation is defined as false information that is spread, regardless of whether there is intent to mislead (Pennycook & Rand, 2022). While this definition is widely accepted, it is a broad term to explain complex phenomena, for which there is no universally agreed-upon definition. Extensive literature on misinformation implies that misinformation is the most commonly used label for studies investigating the impacts and proliferation of false information. Part of this has to do with the fact that misinformation has become something of a catch-all term for related concepts such as disinformation, malinformation, fake news, deep fake, and

conspiracy theories. Its status as a catch-all term has sometimes resulted in broad use of the concept and imprecise definitions.

Misinformation poses a considerable challenge to public debate during political campaigns, and disinformation about Ukraine's war with Russia has led to confusion and anxiety. For the last several years, the volume and dissemination of fake news and misinformation have grown considerably (Zarocostas, 2020). There are several reasons for this exponential growth, including a deepening of the digital divide, economic disparities, a decline in information literacy, the existence of partisan media, and a decrease in public trust in media. Usually, while the wide use of social media is the main reason for the upsurge of misinformation, conventional media coverage can also contribute to the problem.

One of the significant problems with social media platforms is their tendency to spread misinformation. This can occur when algorithms prioritize controversial or sensational content, regardless of its relevance and accuracy, with the goal of keeping people involved and staying on the platform longer. This can lead to the spread of misleading, false, and inflammatory information, which can have serious consequences for individual wellbeing, public health, and political democracy. These concerns result from the design and implementation of social media algorithms. The vulnerability of social media algorithms comes from two sources: data and humans. The output is molded by the data they are trained on, reflecting both the broad spectrum of human knowledge and the inherent biases within the data. The data fed to train these algorithms are largely collected from users in a predisposed manner, which can extend existing biases and bolster power structures. The programmers and designers of these algorithms may maintain their own personal prejudices, which can be represented in the algorithms they design. This can result in unfair consequences and exacerbate societal inequalities. Social media algorithms can be used to manipulate public opinion and undermine the diversity and benefits of the marketplace of ideas.

As the effort to counter misinformation intensifies, the industry will continue to work to find effective ways to sort out facts from fiction and to improve algorithmic tools that can help reduce disinformation. Many experts have argued that measures of misinformation should be contextualized within a broad understanding of digital ecology, taking into account societal and political factors (Lewandowsky et al., 2017). This implies the need for a human-centered approach to fake news and misinformation. As much as fake news is related to human involvement in adoption,

circulation, and dissemination, countering fake news should also be a human-centered approach to identifying fake news dissemination behavior, for example, by designing an explainable fake news spreader classifier based on the cognitive and psychological cues of users. The human-centered approach to fake news demands an understanding of the human aspects of fake news. Social and psychological user motivations play a key role in fake news generation and diffusion. Fake news-spreading is underpinned by these motivations as well as by interactions among users and social contexts that enable them to write fake news. We cannot expect AI to solve the problems of misinformation and fake news because it is more than detecting fake news; it is a problem of trust and a lack of critical literacy. Beyond technological solutions, increasing people's media literacy through fact-checking and displaying the results can be a more effective solution. Educating people and increasing their literacy to critically and objectively assess algorithmic content is important (Jones-Jang et al., 2021). Combatting fake news, misinformation, and even deep fakes will require both AI and human efforts. While algorithmic tools are good at automatizing high quantities of information at rapid speeds, they lack the nuanced examination that human journalists can provide.

Although large social media platforms have recently initiated self-regulatory approaches by continuously monitoring their sites, governments and regulatory agencies have a role in holding social media firms responsible for the algorithms they design and their repercussions on society. This could involve establishing policies, enforcing regulations and rules to proactively thwart the spread of misinformation and incendiary content, and holding firms accountable for their algorithms' biases (Shin et al., 2022).

A group of researchers have proposed possible solutions for the problems of social media algorithms (e.g., Huszar et al., 2021). First, increasing transparency and accountability seems to be the most important step for the problems. Social media firms can be more transparent about their data practices and internal algorithms, and they should accept accountability for the impact of their algorithms on society. This can involve public audits and reporting on algorithmic biases and their impact on society. Firms can release the datasets used to train the algorithms and procedural information on how the data were gathered and analyzed, what the data are, and any possible concerns or predisposed biases they may include. Governments can ensure that social media algorithms are implemented and operated according to democratic values and principles. Second, it is

suggested to have diverse ownership of social media firms. Promoting a more diverse and competitive landscape of social media firms can decrease the dominance of power in the hands of limited dominant players in the market and encourage innovation and diversity in the marketplace of ideas (Molina et al., 2021). Third, for the user side, algorithm users should be required to increase their misinformation literacy to be empowered to make informed decisions about their acceptance and use of social media. Users should be encouraged to produce high-quality and desirable content by prioritizing relevant information, and social media firms incentivize this creative activity by rewarding creators who produce high-quality content. Most importantly, collaboration between the civil community, government, and industry is needed, as there is not a single solution to misinformation. Addressing the challenges posed by social media algorithms will require close cooperation among social media firms, governments, and civil society organizations. This collaboration involves establishing forums for discussion and collaboration, providing documentation and support for programmers, and engaging with the civic community to address their concerns and suggestions. To ensure that social media algorithms are equitable, fair, and unbiased, social media firms should be transparent about the data they collect and use to train their algorithms. The sharing of data and best practices, the development of common standards and regulations, and the implementation of public education and awareness programs require a multilayered and multi-stakeholder partnership.

Why Is the Human Role the Most Important in Misinformation?

Constant exposure to the same misinformation makes it likely that someone will accept it. If AI detects misinformation and cuts the rate of its circulation, this can stop the cycle of reinforced information consumption and dissemination patterns. Machine learning algorithms can identify misinformation based on text nuance and the way stories are shared and distributed. However, AI detection of misinformation remains technically and operationally inaccurate. The contemporary algorithmic detection method is based on the evaluation of textual content. Although the method can determine the origin of the sources and the dissemination mechanism of misinformation, the essential limitation lies in how algorithms substantiate the actual nature of the information. Here are the real problems of fake news and misinformation: fake news and misinformation

are more about philosophical questions of how people deal with the truth than technical or mathematical questions of algorithms and AI. Many organizations now have fact-checker software tools for identifying fake news for reporters and journalists. Most fact-checking software performs basic functions, such as content-independent detection, by using tools that target the form of the content and by using deep fake identifying tools to check any manipulated content, image, and video. However, algorithms cannot check whether the content itself provides false information. This task should be performed by human fact-checkers by searching for social media posts or online information with similar queries and information. Most current fact-checking approaches examine content by analyzing the semantic features of fake news. This approach may work at a basic level but faces larger problems; for example, platforms such as WeChat have language barriers, and fact-checkers cannot access the content of WhatsApp because it is an encrypted message. In addition, misinformation in many cases is an image that is difficult to investigate using textual techniques. If the amount of training data is sufficient, the AI-powered detection model should be able to detect fake news and misinformation. However, the reality is that detecting such news demands prior social, cultural, and political knowledge, which AI algorithms still lack. We can prevent someone from making fake news and from spreading and promoting misinformation by applying AI-powered analytics that use anomaly detection, but eventually, AI cannot catch all problems related to fake news. Most current fake news detection systems require humans to work with AI to check the accuracy of information.

HOW TO STOP THE SPREAD OF MISINFORMATION

Researchers agree that countering misinformation is difficult for several reasons. Cognitive factors, including biases and emotions, may render efforts to respond to misinformation by providing corrective measures unsuccessful. This is why interventions, such as debunking or fact-checking, have been shown to be ineffective. Additionally, the volume and a wide range of social media easily promote the creation and maintenance of echo chambers by making it easy for individuals to self-curate their news and pursue similar news through automated algorithms. These capabilities decrease the possibility that users who are part of a community in which fake news is circulating will be vulnerable to information that differentiates the prevalent opinion within their community (Bhadani et al., 2022).

A group of studies on the psychology of misinformation supports several methods of countering misinformation. Debunking incorrect information after it has spread has been the most common method. Recently, prebunking—inoculating people against future misinformation before they are experiencing—has been found to be more effective than debunking. Debunking tactics such as fact-checking and postverification services can leave a falsehood still festering. Similar to medical immunization, we can expose people to a weak version of misinformation and train them on how they might be lied and misled. Many studies (Pennycook & Rand, 2022) have confirmed that prebunking can alleviate misinformation on health care, climate change, and other political issues. Alerting people who a specific piece of information is false and explaining why a source might lie or be misinformed about it before they encounter the information. Roozenbeek and Van der Linden (2020) developed and tested this technique using “*Bad News*,” a gamified intervention that simulates a social media feed to train participants in how to discern real and fake news headlines on politicized topics. Tests of the game reveal that playing it once can boost participants’ capability to judge misinformation but that the inoculation effect decays after approximately two months.

Another way to counter misinformation is to encourage people to reflect on the accuracy of claims they see. Pennycook et al. (2020) conducted research on testing COVID-19 fake news and found that a single accuracy nudge improved users’ literacy in distinguishing between real and fake news. Respondents viewed a series of headlines—a combination of true and false—and evaluated whether they would share each item. Those in the experimental condition, who were also asked to rate the veracity of each headline, shared more accurate news content compared with participants in the control group.

Increasing user literacy has been used widely for countering misinformation. Media literacy organizations such as First Draft and Citizen News Literacy are applying literacy strategies in an effort to remove misinformation and disinformation online. First, Draft’s virtual meeting room offers a range of lessons on topics such as misinformation, conspiracy theories, and deep fakes, drawing on cognitive theories on confirmation bias, motivated reasoning, and cognitive dissonance. The Poynter Institute’s MediaWise for Senior program and AARP’s Fact Checking initiatives have developed media literacy resources geared toward adults, who are just as capable of spotting hoaxes but have been disproportionately targeted by disinformation sources. All efforts are geared to help people realize the exploitative nature of misinformation.

RESPONSIBLE AI

With a general definition of algorithmic bias, it is difficult to define it as being systematically unfair. Current AI systems (e.g., ChatGPT) are unable to suitably fulfill editorial roles since they are often unreliable due to the unbiasedness and factual accuracy of information. Thus far, there has been no widely accepted definition of the term systematic unfairness. Each firm has its own standard of fairness and its own measure of bias. Firms such as Microsoft, Google, and TikTok have their own policies. One problem with this is that the notion of responsible AI may be practiced unevenly across sectors (Mikalef et al., 2022). AI researchers and industry have proposed over 80 metrics, each of which defines bias by identifying how an algorithm considers different groups represented in a dataset differently. These groups are defined through a combination of sensitive characteristics (such as race, gender, age, sexual orientation, ethnicity, or religion). Deciding which bias metric is most effective requires a contextual interpretation of a use case. One place to start is with two general kinds of fairness: by representation or by error.

Despite active initiatives, addressing AI bias is challenging. As it is impossible to completely eradicate algorithmic bias, an alternative is considered: responsible AI. Now, society realizes the need for responsible AI that is trustworthy, ethical, reliable, robust, well-governed, and understandable.

Responsible AI is a rising field of AI governance, and it is a blanket term that includes both legality and ethics. Responsible AI is the practice of designing, developing, and implementing AI with good intention to empower people, and it fairly impacts users and society—allowing firms to generate trust and scale AI with confidence (Bastian, 2021). It serves as a reference framework that details how a particular organization addresses the risks around AI from both legal and ethical perspectives.

Responsible AI is the right thing to do, as it ensures that any AI system will be effective, follow regulations and norms, operate based on ethical values, and avert potential social and financial harm along the way (Mikalef et al., 2022). An increasing number of government regulations are increasing their supervision of AI services and proposing legal frameworks that include guidelines and ethical standards. AI effect assessments and algorithmic auditing procedural frameworks are introduced to ensure that AI works responsibly and legitimately. An algorithmic audit is a key method in responsible AI that evaluates and rigorously frames fairness and

systematically investigates how algorithms can be responsible by looking at the role they play in our society. Algorithmic audits operate on the assumption that algorithms are critically understood, framed, and regulated as they become fixtures of human life and social activities.

The European Media Freedom Act has called for news organizations to draft standards for the responsible use of AI in journalism. In the field of journalism, responsible AI is the design and implementation of algorithmic systems without “infringing on core values and human rights” (Dignum, 2019, p. 119). For example, Benjamins Field (Benjamins, 2021) argued that news organizations should make generic technical choices regarding privacy, security, and safety that could lead to more responsible use of AI in the media. Currently, in most countries in the world, including the United States, there is no regulation of responsible AI. Nevertheless, leading industries, such as Microsoft, Meta, and Apple, have requested effective data and AI regulation.

Microsoft established its responsible AI principles in 2021. The guidelines propose six key principles in responsible AI: fairness (AI systems should treat all entities equitably), reliability and safety (AI systems should operate consistently and securely), privacy and security (AI systems should be secure and respect privacy), inclusiveness (AI systems should empower users and engage stakeholders), transparency (AI systems should be interpretable), and accountability (developers should be accountable for AI systems). Eventually, a widely accepted governance framework of responsible AI best practices will make it easier for firms in different sectors to ensure that their AI coding is human-centered, explainable, and fair. Shin et al. (2022) argues that responsible AI should include the principles of explainability, justification, monitorability, unbiasedness, reproducibility, human-centeredness, and security.

Samsung has established responsible AI guidelines to assist its workers and customers in understanding how machine learning models of the firm use routine work and what the AI’s drawbacks are. AI researchers at Samsung test the entire lifecycle of their algorithm models, keeping a constant check on their fairness and errors. These researchers share an explainable AI method with staff and clients for bias detection. Amazon has its own ethical task force committee dedicated to the issues surrounding AI, which supports the establishment of ethical and responsible AI throughout the firm. Their guidelines contain AI trust and transparency, everyday ethics for AI, open-source community resources, and research into trusted AI. Prudential insurance firms have launched AI into their claim-handling

process along with a responsible AI frame. The insurance firm has incorporated a governance system that designates accountability for AI and allows for making transparent and more informed decisions while improving customer experiences. The firms' responsible AI model, called the dynamic property assessment model, helps predict a total financial loss with a high level of confidence while enabling transparency regarding the tool of processing insurance claims.

A responsible AI framework incorporates details on the kind of data that can be gathered and analyzed, the way models can be assessed, and the best way to implement and supervise models. The framework can also define who is responsible for any harmful AI results. The goal of responsible AI is to design understandable, transparent, and socially acceptable AI systems. The most important component of the framework is explainability. When we interpret models, we discover an explanation of how they generate personalization. An AI system can diagnose patients with cancer, and an algorithmic model can deny loan applications. Users would require an explanation regardless of the correctness of the decisions. Responsible AI can describe how we can design understandable models or when it is appropriate to use one that is less understandable.

The term fairness is related to explainability. While there is no consensus on a single definition, it is known that fairness in AI should treat all people fairly. AI systems can easily make decisions that victimize underserved groups of people, which results from bias in the data used to train machine learning. The more explainable the model, the easier it is to warrant fairness and detect possible biases. There is a need for a responsible AI framework to define how people perceive fairness and what to do when algorithms are found to create biased predictions.

Eventually, trust is the most significant factor in responsible AI. If users trust AI, they will continue to use the services. People do not trust AI systems that use data that we are unsure about sharing (such as sensitive profile data of race, sexual orientation, or political beliefs), considering that such systems produce biased results. People do not trust AI if they believe it may harm them. Explanations for decisions and accountability for those decisions go a long way toward building this trust. This need for trust drives self-regulation among companies that use AI. In 2022, the European Commission announced ethics guidelines for trustworthy AI—the core principles that AI should have to be deemed responsible and faithful. These guidelines would help firms ensure that their AI systems meet the same standards. However, a more important question is: Can we trust that firms can regulate themselves?

FAIRNESS AND TRANSPARENCY IN CORPORATE DATA RESPONSIBILITY

Fairness and transparency are becoming important considerations in the use of algorithms for the recommendation and delivery of digital content (Ananny & Crawford, 2018). Automated data gathering and sharing may involve processes that are unfair, flawed, opaque, or unaccountable (Crain, 2018). Over-the-top platform content recommendations embody these issues in highly visible applications. Fairness and transparency are vital prerequisites in the design and development of algorithm-supported platforms (Kitchin, 2017), which are purportedly designed to offer accurate and reliable recommendations for users (Lepri et al., 2018). It remains unresolved whether such recommendations match user interest, how the analytic processes are done, and whether the outcomes are legally responsible (Diakopoulos & Koliska, 2016). Thus, fairness and transparency emerge as fundamental requirements in the use of algorithms on media platforms (Montal & Reich, 2017). When transparent, open, and fair services are provided, users are more likely to consider that the recommendations are of high quality (Shin & Park, 2019). Highly transparent platforms can grant users a sense of personalization, as responsible and fair recommendations afford users a sense of trust that promotes satisfaction and a willingness to continue using them and subscribe to them (Soffer, 2019). Open visibility and clear transparency of relevant recommendations boost the user interpretability of the system and search performance (Shin, 2021).

Algorithms are widespread and accepted in practice, but their popularity comes at the expense of limited transparency, systematic prejudice, and nebulous responsibility (Wölker & Powell, 2021). Algorithmic filtering procedures may lead to more impartial, and thus possibly fairer, processes than those processed by humans (Möller et al., 2018). However, algorithmic recommendation processes have been criticized for their bias to intensify/reproduce prejudice, information asymmetry, distortion of facts, and the black-box process of decision-making (Sandvig et al., 2016). Algorithmic bias may increase algorithmic inequality in that machine learning automates and propagates unjust and discriminatory patterns (Hoffmann, 2019). Recent over-the-top (OTT) platforms (Hulu, Disney, or Netflix) have faced similar problems (Shin, 2021). While these platforms offer personalized and relevant content in innovative and interactive manners, ethical and privacy issues remain intertwined with algorithmic personalization in a complicated way (Araujo, 2018). Hence, the issues

regarding how to safeguard the goals, values, and personalizing processes of platforms, to what extent users need to share personal information with algorithms and how to balance privacy and algorithmic personalization remain controversial (Hoffmann et al., 2022). Underlying these questions are concerns about how to mitigate bias and discrimination in data and the urgent need to understand how to design algorithmic systems that are transparent and fair (Crain, 2018). As ethical concerns have peaked recently with the rise of OTT algorithms, the opacity of black-box algorithm processes has led to calls for investigation on transparency and fairness (Sandvig et al., 2016). Recent research (e.g., Shin & Park, 2019; Dörr & Hollnbuchner, 2017) has highlighted the normative implications and problems of these algorithms, which are summarized by fairness, accountability, and transparency (FAccT). In particular, the transparency and fairness of algorithmic systems while processing user-sensitive data emerge as key attributes for those algorithmic systems to be trustworthy (Shin et al., 2022). This issue will be even more important when OTT increasingly relies on algorithms, and people rely more on algorithms than social influence when making judgments. AI has become pervasive across all media industries and service functions.

Given the growing importance of FAccT, it becomes necessary for businesses to develop policies and practices that address user data on how their data are being collected, filtered, used, and protected and to be forthright and empathetic in creating and sharing those policies and practices. By taking the right approach to data—becoming more transparent and giving consumers more control—businesses can build consumer trust and solidify access to this critical resource. Corporate Data Responsibility, a novel extension of Corporate Social Responsibility, is based on the identical values of ethics, trust, and algorithmic features of FAccT. It can be said that Corporate Data Responsibility is a result of the combination of Corporate Social Responsibility and General Data Protection Regulation (GDPR). Consumers and users are becoming increasingly worried about how firms safeguard their personal data against misinformation and misuse (Lee, 2018). Firms using AI, in particular, are urged to take proactive steps to keep pace with FAccT expectations.

Many AI ethics studies (Kieslich et al., 2022) have investigated how people process algorithmic judgments instead of human judgments and how this process impacts algorithm adoptions. Consumer perceptions of

corporate data responsibility (fairness, transparency, and accountability) are primarily influenced by moral principles that individuals use to weigh the benefits and drawbacks of AI (Malmio, 2023). Although individuals' sociotechnical ethics are initially formed through institutionalized education, narratives in the media can substantially influence their perspectives toward the algorithmic recommendation systems (Shin et al., 2022). For example, Facebook's data breaches and TikTok's data privacy settlement have generated concerns over corporate data practices in the AI era. As criticism of the opacity and bias of increasingly sophisticated AI-based personalized recommendation systems grows, companies are starting to improve data governance (Burrell, 2016). Data governance entails determining who can make decisions about data usage and for what purposes, as well as how data are collected, stored, processed, used, and destroyed (Shah & Murthi, 2021). Corporate data responsibility can be proposed as a normative and conceptual standard for corporate data practices. While data governance establishes the ethical principles and standards for data use within an organization, corporate data responsibility represents a company's voluntary commitment to responsible data practices toward external stakeholders. Thus, consumer perceptions of corporate data responsibility are crucial in assessing corporate ethics in data use practices.

Ethically sound recommendation systems should be seen as transparent about how a firm collects, uses, stores, and processes customer information. Recommendation systems must also be perceived as fair and unbiased toward specific genders, ethnic groups, or marginalized members of society. There is currently no standardized set of dimensions defining consumer perceptions of responsible and ethical data use, as different consumers may have different expectations and priorities regarding companies' ethical data use.

Corporates, particularly AI firms, are suggested to provide clear, timely updates on data governance, limit the use and sale of user data, and allow more user control over their data. Fairness, transparency, and data accountability remain essential functions of firms' responsibilities. Users want to see clear and transparent communication from firms to ensure their data are safeguarded and ethically processed. Developing a series of data responsibility frameworks becomes necessary to improve user trust.

WHAT DOES FUTURE WORK ON MISINFORMATION NEED TO CONSIDER?

Future efforts need to address three critical areas. First, it should be further revealed how and to what extent misinformation in a specific context of interest influences people's beliefs and behaviors. This issue may be key since social media and AI platforms are replete with fake news, yet accurate pictures of its effect and preventative measures are rare (e.g., Adams et al., 2023). Hence, an exact assessment of the media ecology to estimate the extent of the trend is a critical procedure to understand its growing impact on people's opinions and social discourse. Even a small amount of misinformation can seriously influence an individual's attitudes and beliefs. Given the detrimental effects of small amounts of misinformation on psychological/societal outcomes, it should be clarified to what extent misinformation influences the public's beliefs on a large scale.

Second, it should be clearly researched whether misinformation constitutes a major factor in individuals' actions and how misinformation plays a role in forming and strengthening socially harmful behavior (Pennycook et al., 2021). We have taken for granted the direct causal link between the pervasiveness and penetration of misinformation and ensuring damaging outcomes. A key question for policymaking is the characteristics and magnitude of the link between misinformation and action (Osman et al., 2022). Often, a simple causal relation of misinformation—opinion—behavior is assumed primarily because of (1) complicated relations between beliefs and human action, (2) unestablished conceptual bases of how beliefs are generated, and (3) inconsistent empirical findings. Thus far, the relationship has not been clarified clearly, and future work should focus on this topic.

Third, while research has been conducted on how misinformation leads to harmful behavior and results in negative outcomes, a key question remains as to how to combat and what measures and strategies should be given for misinformation issues. This dilemma is related to broader debates concerning behavior modification tools and strategies (Pluviano et al., 2022) as well as in AI environments (Shin et al., 2022) and the limited power of these tools in forming meaningful motivational attitudinal change (Soutter et al., 2020).

HUMAN-MISINFORMATION INTERACTION

The prevalence of social media and rapidly developing AI has amplified the effects of misinformation on our societies. Similar to the human-computer interaction (HCI) area, which has made great strides with the emergence of computing, it may be necessary to create computational tools to mitigate their effects effectively, such as human-misinformation interaction (Karduni, 2019). With this framework in place, more research is needed on how to combat misinformation. A new approach will focus on misinformation that pays attention to the interaction, source, and users as the three main elements involved in the process of misinformation. Such an approach should be inherently interdisciplinary, involving computer science, communication, psychology, cognitive science, and cognate human discourses.

The vast majority of works in computer science and information technology are concerned with the crucial tasks of detection and verification of content and sources of misinformation. Computational research focusing on humans of misinformation in HCI and related fields is very sparse and often does not address the subtleties of this process. The majority of existing interfaces and systems are less concerned with the interface and usability of the tools rather than the robustness and accuracy of the detection methods. It is imperative to call for an interdisciplinary approach toward human-misinformation interaction that focuses on building tools and methods that robustly deal with such complex cognitive/psychological/societal phenomena.

CAN AI STOP MISINFORMATION?

Can AI Stop Fake News?

By using advanced algorithms to discover and reach the demographic that may easily absorb a message, legitimate companies utilize AI to locate and target the most likely consumers of a message or point of view. Naver, for example, has already implemented its AI Recommender System (AiRS) to refine its capabilities to recognize accurate results and identify fake news at the same time.

Detecting misinformation can be done by a combination of algorithms, machine learning models, and humans. To distinguish computer-generated material from human-produced articles, AI-based technologies can

perform linguistic analysis on textual content and find clues such as syntactic structure, meaning patterns, and readability. These algorithms can analyze any text to find instances of incendiary content by examining word vectors, word placement, and connotation. Among all AI tools, a key question is who is responsible for controlling, if not stopping, the spread of misinformation once it is detected. Only social media companies are truly in the position to exercise control over the spread of information through their networks.

Fake news sources often come from bogus sources before the information proliferates. Snopes, Politifact, and FactChecking use human editors to perform the main analysis necessary to check the authenticity of a text or an image. Once misinformation has been detected, AI systems search the web for similar information that can cause controversy. Applications can then assign a reputation score to a website article if it is determined that the material is genuine. Several AI engines currently use sentiment analysis, opinion analysis, propaganda analysis, and data mining to give a clear picture of how trustworthy an article is and what we're up against.

Generative AI applications (e.g., GPT-3 and Bard) can already generate texts, novels, and images based on a one-line cue. AI is close to perfecting the production of material that resembles a person. AI has made it so easy to manipulate all sorts of information that open-source programs may make inexperienced new users potentially the epicenter of social unrest. These issues are made worse since these semantic analysis algorithms cannot decipher the nature of misinformation pictures that have not been changed but are instead spread with harmful or malicious context.

Additionally, detecting misinformation by AI still has a long way to go to be reliable. Present detection is based on the assessment of text and its social network to determine its credibility. Despite determining the origin of the sources and the dissemination pattern of fake news, the fundamental problem lies within how AI substantiates the actual nature of the content.

The AI-supported classification model can detect whether information includes falsehoods if a large amount of training data is available. However, the reality is that such detection requires prior cultural, rhetorical, social, and political knowledge or common sense, which generative algorithms still lack.

Once deceitful information has been found, eliminating it is more challenging than it would seem. Organizations may be criticized for censorship and for attempting to change information that one group or another

believes to be untrue. Finding an optimum point between the battle against false information and the right to free expression is challenging. AI also normally lacks the ability to recognize parody and satire. Thus, it may classify falsehood or disinformation as malevolent misinformation if it is utilized in an entertaining context or for a joking purpose.

Misinformation is a complex problem and is not a one-dimensional problem that algorithms alone can correct (Giansiracusa, 2021). There should be a paradigm change in how we approach information processing and acquisition. One feasible suggestion would be to let humans work alongside AI to secure the authenticity of information. For instance, while the crowdsourcing of collaborative knowledge among expert groups is essential to evaluating raw data, communities of knowledgeable users may also support ethical monitoring operations. A lack of collaborative actions involving all parties can worsen the problems of misinformation.

REFERENCES

- Abul-Fottouh, D., Song, Y., & Gruz, A. (2020). Examining algorithmic biases in YouTube's recommendations of vaccine videos. *International Journal of Medical Informatics*, 140, 104175. <https://doi.org/10.1016/j.ijmedinf.2020.104175>
- Adams, Z., Osman, M., Bechliyanidis, C., & Meder, B. (2023). (Why) Is misinformation a problem? *Perspectives on Psychological Science*. <https://doi.org/10.1177/17456916221141344>
- Ananny, M., & Crawford, K. (2018). Seeing without knowing. *New Media & Society*, 20(3), 973–989. <https://doi.org/10.1177/1461444816676645>
- Anna, L. H. (2019). Where fairness fails: Data, algorithms, and the limits of antidiscrimination discourse. *Information, Communication & Society*, 22(7), 900–915. <https://doi.org/10.1080/1369118X.2019.1573912>
- Araujo, T. (2018). Living up to the chatbot hype: The influence of anthropomorphic design cues and communicative agency framing on conversational agent and company perceptions. *Computers in Human Behavior*, 85, 183–189.
- Badke, W. (2018). Fake news, confirmation bias, the search for truth, and the theology student. *Theological Librarianship*, 11(2), 4–7. <https://doi.org/10.31046/tl.v11i2.519>
- Bastian, J. A. (2021). Mine, yours, ours: Archival custody from transaction to narrative. *Archival Science*, 21, 25–42.
- Benjamins, R. (2021). A choices framework for the responsible use of AI. *AI and Ethics*, 1(1), 49–53. <https://doi.org/10.1007/s43681-020-00012-5>

- Bhadani, S., Yamaya, S., Flammini, A., et al. (2022). Political audience diversity and news reliability in algorithmic ranking. *Nature Human Behavior*, 6, 495–505. <https://doi.org/10.1038/s41562-021-01276-5>
- Burrell, J. (2016). How the machine ‘thinks’: Understanding opacity in machine learning algorithms. *Big Data & Society*, 3(1), 2053951715622512.
- Crain, M. (2018). The limits of transparency. *New Media & Society*, 20(1), 88–104. <https://doi.org/10.1177/1461444816657096>
- Diakopoulos, N., & Koliska, M. (2016). Algorithmic transparency in the news media. *Digital Journalism*, 5(7), 809–828. <https://doi.org/10.1080/21670811.2016.1208053>
- Dignum, V. (2019). *Responsible artificial intelligence: How to develop and use AI in a responsible way*. Springer Nature.
- Dörr, K., & Hollnbuchner, K. (2017). Ethical challenges of algorithmic journalism. *Digital Journalism*, 5(4), 404–419. <https://doi.org/10.1080/21670811.2016.1167612>
- Dressel, J., & Farid, H. (2018). The accuracy, fairness, and limits of predicting recidivism. *Science Advances*, 4(1). <https://doi.org/10.1126/sciadv.aao5580>
- Ecker, U. K. H., & Antonio, L. M. (2021). Can you believe it? An investigation into the impact of retraction source credibility on the continued influence effect. *Memory & Cognition*, 49, 631–644. <https://doi.org/10.3758/s13421-020-01129-y>
- Fiske, S. (2022). Twitter manipulates your feed: Ethical considerations. *Proceedings of the National Academy of Sciences*, 119(1). <https://doi.org/10.1073/pnas.2119924119>
- Giansiracusa, N. (2021). *How algorithms create and prevent fake news: Exploring the impacts of social media, deepfakes, GPT-3, and more*. Springer.
- Grady, R., Ditto, P., & Loftus, E. (2021). Nevertheless, partisanship persisted: Fake news warnings help briefly, but bias returns with time. *Cognitive Research*, 6, 52. <https://doi.org/10.1186/s41235-021-00315-z>
- Hameleers, M., & Brosius, A. (2022). You are wrong because I am right: The perceived causes and ideological biases of misinformation beliefs. *International Journal of Public Opinion Research*, 34(1), edab028. <https://doi.org/10.1093/ijpor/edab028>
- Hoffmann, H., Vogt, V., Hauer, M., & Zweig, K. (2022). Fairness by awareness? On the inclusion of protected features in algorithmic decisions. *Computer Law & Security Review*, 44, 105658.
- Huszar, F., et al. (2021). Algorithmic amplification of politics on Twitter. *Proceedings of the National Academy of Sciences*, 119(1). <https://doi.org/10.1073/pnas.2119924119>
- Jones-Jang, S. M., Mortensen, T., & Liu, J. (2021). Does media literacy help identification of fake news? Information literacy helps, but other literacies don't. *American Behavioral Scientist*, 65(2), 371–388.

- Karduni, A. (2019). Human-misinformation interaction: Understanding the interdisciplinary approach needed to computationally combat false information. arXiv:1903.07136v1
- Kieslich, K., Keller, B., & Starke, C. (2022). Artificial intelligence ethics by design. Evaluating public perception on the importance of ethical design principles of artificial intelligence. *Big Data & Society*, 9(1).
- Kitchin, R. (2017). Thinking critically about and researching algorithms. *Information, Communication & Society*, 20(1), 14–29. <https://doi.org/10.1080/1369118X.2016.1154087>
- Lee, M. K. (2018). Understanding perception of algorithmic decisions: Fairness, trust, and emotion in response to algorithmic management. *Big Data & Society*, 5(1).
- Lepri, B., et al. (2018). Fair, transparent, and accountable algorithmic decision making processes. *Philosophy and Technology*, 31(4), 611–627.
- Lewandowsky, S., Ecker, U. K., & Cook, J. (2017). Beyond misinformation: Understanding and coping with the posttruth era. *Journal of Applied Research in Memory and Cognition*, 6(4), 353–369. <https://doi.org/10.1016/j.jarmac.2017.07.008>
- Lewandowsky, S., & van der Linden, S. (2021). Countering misinformation and fake news through inoculation and prebunking. *European Review of Social Psychology*, 32(2), 348–384. <https://doi.org/10.1080/10463283.2021.1876983>
- Malmio, I. (2023). Ethics as an enabler and a constraint—Narratives on technology development and artificial intelligence in military affairs through the case of Project Maven. *Technology in Society*, 102193.
- Metzger, M., Flanagin, A., Mena, P., Jiang, S., & Wilson, C. (2021). From dark to light: The many shades of sharing misinformation online. *Media and Communication*, 9(1). <https://doi.org/10.17645/mac.v9i1.3409>
- Mikalef, P., Conboy, K., Lundstrom, J., & Popovič, A. (2022). Thinking responsibly about responsible AI and the dark side of AI. *European Journal of Information Systems*. <https://doi.org/10.1080/0960085X.2022.2026621>
- Molina, M., Sundar, S., Le, T., & Lee, D. (2021). Fake news is not simply false information. *American Behavioral Scientist*, 65(2), 180–212. <https://doi.org/10.1177/0002764219878224>
- Möller, J., Trilling, D., Helberger, N., & Es, B. van. (2018). Do not blame it on the algorithm: An empirical assessment of multiple recommender systems and their impact on content diversity. *Information, Communication & Society*, 21(7), 959–977. <https://doi.org/10.1080/1369118X.2018.1444076>
- Montal, T., & Reich, Z. (2017). I, robot. you, journalist. who is the author? *Digital Journalism*, 5(7), 829–849. <https://doi.org/10.1080/21670811.2016.1209083>

- Mosleh, M., Pennycook, G., & Rand, D. G. (2020). Self-reported willingness to share political news articles in online surveys correlates with actual sharing on Twitter. *PLoS One*, 15, e0228882.
- Osman, M., Adams, Z., Meder, B., Bechlivanidis, C., Verduga, O., & Strong, C. (2022). People's understanding of the concept of misinformation. *Journal of Risk Research*, 25(10), 1239–1258.
- Park, Y. J. (2023a). Let me tell you, ChatGPT-like AI will not change our world. *Internet Policy Review*. <https://policyreview.info/articles/news/let-metell-you-chatgpt-ai-will-not-change-our-world/1694>
- Park, Y. J. (2023b). How we can create the global agreement on generative AI bias: Lessons from climate justice. *AI & Society*.
- Pennycook, G., Epstein, Z., Mosleh, M., Arechar, A., Eckles, D., & Rand, D. (2021). Shifting attention to accuracy can reduce misinformation online. *Nature*, 592(7855), 590–592.
- Pennycook, G., McPhetres, J., Zhang, Y., Lu, J. G., & Rand, D. G. (2020). Fighting COVID-19 misinformation on social media: Experimental evidence for a scalable accuracy-nudge intervention. *Psychological Science*, 31(7), 770–780. <https://doi.org/10.1177/0956797620939054>
- Pennycook, G., & Rand, D. G. (2022). Accuracy prompts are a replicable and generalizable approach for reducing the spread of misinformation. *Nature Communications*, 13, 2333. <https://doi.org/10.1038/s41467-022-30073-5>
- Pluviano, S., Watt, C., Pompéia, S., Ekuni, R., & Della, S. (2022). Forming and updating vaccination beliefs: Does the continued effect of misinformation depend on what we think we know? *Cognitive Processing*, 23, 367–378.
- Roozenbeek, J., & van der Linden, S. (2020). Breaking harmony square: A game that inoculates against political misinformation. *Harvard Kennedy School Misinformation Review*, 1(8). <https://doi.org/10.37016/mr-2020-47>
- Sandvig, C., Hamilton, K., Karahalios, K., & Langbort, C. (2016). When the algorithm itself is a racist. *International Journal of Communication*, 10, 4972–4990.
- Sartori, L., & Theodorou, A. (2022). A sociotechnical perspective for the future of AI: Narratives, inequalities, and human control. *Ethics and Information Technology*, 24(4), 1–11. <https://doi.org/10.1007/s10676-022-09624-3>
- Shah, D., & Murthi, B. P. S. (2021). Marketing in a data-driven digital world: Implications for the role and scope of marketing. *Journal of Business Research*, 125, 772–779.
- Shin, D. (2021). The perception of humanness in conversational journalism. *New Media & Society*. <https://doi.org/10.1177/1461444821993801>
- Shin, D. (2022). How do people judge the credibility of algorithmic sources? *AI and Society*, 37, 81–96. <https://doi.org/10.1007/s00146-021-01158-4>

- Shin, D., & Park, Y. (2019). Role of fairness, accountability, and transparency in algorithmic affordance. *Computers in Human Behavior*, 98, 277–284. <https://doi.org/10.1016/j.chb.2019.04.019>
- Shin, D., Zaid, B., Biocca, F., & Rasul, A. (2022). In platforms we trust? Unlocking the black-box of news algorithms through interpretable AI. *Journal of Broadcasting & Electronic Media*. <https://doi.org/10.1080/08838151.2022.2057984>
- Simon, H. A. (1957). Background of decision making. *Naval War College Review*, 10(9), 2.
- Soffer, O. (2019). Algorithmic personalization and the two-step flow of communication. *Communication Theory*. <https://doi.org/10.1093/ct/qtz008>
- Soutter, A. R. B., Bates, T. C., & Möttus, R. (2020). Big Five and HEXACO personality traits, proenvironmental attitudes, and behaviors: A meta-analysis. *Perspectives on Psychological Science*, 15(4), 913–941.
- Sundar, S., Kim, J., Beth-Oliver, M., & Molina, M. (2020). Online privacy heuristics that predict information disclosure. *CHI '20*, April 25–30, 2020. <https://doi.org/10.1145/3313831.3376854>
- Sundar, S. S. (2008). The MAIN model: A heuristic approach to understanding technology effects on credibility. In M. J. Metzger & A. J. Flanagin (Eds.), *Digital media, youth, and credibility* (pp. 72–100). The MIT Press.
- Swire, B., & Ecker, U. K. (2018). Misinformation and its correction: Cognitive mechanisms and recommendations for mass communication. In B. Southwell, E. A. Thorson, & L. Sheble (Eds.), *Misinformation and mass audiences* (pp. 195–211). University of Texas Press.
- Walter, N., & Tukachinsky, R. (2020). A meta-analytic examination of the continued influence of misinformation in the face of correction. *Communication Research*, 47(2), 155–177. <https://doi.org/10.1177/0093650219854600>
- Walther, J., Lew, Z., Quick, J., & Edwards, A. (2022). *The effect of social approval on attitudes toward the focus of fake news in social media*. Presented at the Annual Meeting of the International Communication Association, Paris, May 2022.
- Wölker, A., & Powell, T. (2021). Algorithms in the newsroom? *Journalism*, 22(1), 86–103. <https://doi.org/10.1177/1464884918757072>
- Young, D. G. (2023). *Wrong: How media, politics, and identity drive our appetite for misinformation*. Johns Hopkins University Press.
- Zarocostas, J. (2020). How to fight an infodemic. *The Lancet*, 395(10225), 676.



Misinformation, Extremism, and Conspiracies: Amplification and Polarization by Algorithms

INTRODUCTION: WHY PEOPLE FALL FOR RADICALIZATION

Eric Schmidt, a Google technologist, has stated, “The Internet is the first thing that humanity has built that humanity doesn’t understand, the largest experiment in anarchy that we have ever had.” The rapid advancement in AI technologies has posed serious problems to society, firms, and researchers in its catalyzation and diffusion of disinformation, radical content, and hate speech. Those issues have become even more complicated factoring in the progress of social media platforms and search engines’ algorithms, which curate and diffuse content based on the preference of users, amplifying already established beliefs and thus instigating societal polarization (Vicario et al., 2019).

Social media platforms using algorithms to personalize user content raise significant concerns for society because this phenomenon has the potential to intensify “reinforcing spirals” (Slater, 2007, p. 206) that can lead users toward radicalism and extremism. Social media platforms’ design structures that are embedded with algorithms are exploited to stimulate extremism and promote radicalism (Ledwich & Zaitsev, 2020). Social media users are algorithmically categorized into similar peers and groups, driving them into steadily more insular online environments (what is commonly known as filter bubbles).

As recommendation algorithms are embedded in the manner in which platforms operate, social media platforms progressively draw people into conspiracy theorist groups or those geared toward illicit online activity (Kaiser & Rauchfleisch, 2020). People who join online communities are funneled toward more extreme, insular communities and ultimately into invite-only online communities that require a filtering process to make users more parochial and insular (Wolfowicz et al., 2023). Weimann and Masri (2020) demonstrated that algorithmic personalization is higher for populations with extreme ideological beliefs; thus, there is an urgent need to identify the factors that make one vulnerable to radicalization online.

This argument is in line with the Facebook whistleblower case of Frances Haugen, which reveals that the algorithms of dominant social media platforms are abused to further radicalization and extremism. The more provocative the content, the more it keeps users involved, and the more it is reinforced by an algorithm. Haugen revealed that Facebook employees were required to upload more incendiary content online, thus promoting more polarized political views to get messages accessed and constantly forwarded under the platform's scheme of algorithmic amplification (Huszar et al., 2022). When the platform users' comments or messages became noninflammatory, they did not receive as intense attention as when they attacked each other. The mechanisms of social media platforms are neither neutral nor fair, and core product mechanics, including recommendations, engagement optimization, and virality, are key to why bias and hate proliferate on social media platforms (Cervi et al., 2021). The body of research has consistently confirmed that this is also prevalent on other platforms (see, e.g., Ledwich & Zaitsev, 2020). For example, TikTok's user algorithm similarly stimulates content from QAnon, Three Percenters, and Oath Keepers (Weimann & Masri, 2020; Zulli & Zulli, 2022).

The existence and operation of a radicalization pipeline on platforms in the wake of increasing misinformation and deteriorating trust online has been criticized (Abul-Fottouh et al., 2020). Currently, people are skeptical of conventional media outlets and increasingly consume content and news on social media and other online platforms (Shin et al., 2022). YouTube, for example, is becoming increasingly popular, particularly among young populations, and if this streaming platform is radicalizing its users, this could push peripheral beliefs, such as those of white supremacists, further into the mainstream (Yesilada & Lewandowsky, 2022).

YouTube routinely suggests further intake of content that a user had recently visited, commented on, and reviewed.

In response to mounting concerns regarding radicalization, theories have been developed to analyze the diffusion and adoption of algorithmic radicalization online (see, e.g., Ribeiro et al., 2020). Although there are manifold driving forces behind radicalization, relevant research indicates that it is deeply rooted in people's cognitive and psychological mechanisms (Pennycook & Rand, 2022). This view is based on the proposition that algorithmic radicalization is a multifaceted problem that requires understanding how individuals encounter and experience it to derive a constructive and sustainable solution (Huszar et al., 2022).

Knowledge of people's radicalization processes can provide insight into how radical content is produced, legitimized, shared, and amplified (Tully et al., 2020). The concerning impact of radicalization warrants robust empirical research to examine the process of such an effect. Particularly for social platforms that embed artificial intelligence (AI), the gap between the impact of radicalization and the practice of social norms necessitates a better understanding of how users become engaged in radicalization via AI, which is crucial as industries and governments strive to develop effective strategies to combat radicalization.

The processes and mechanisms of radicalization (despite their importance), particularly in algorithmic contexts, have remained underresearched, as the exact nature of how users become engaged in radicalization and with what effect, how users respond to misinformation, and the impact of misinformation on user behavior remain inconclusive. Although there is burgeoning research in the algorithmic radicalization literature on the impacts and broad effects of radical practices on users' behaviors (Jahng, 2021) and technical tools, such as automated fact-checking (Epstein et al., 2021), there is a lack of research that focuses on users' engagement with misinformation and radicalization and the long-term process of extremist behaviors.

This gap in the literature relates to a need for more knowledge about the mechanisms that explain the process of radicalization in human cognition (Walter & Tukachinsky, 2020). We attempted to narrow this research gap by investigating users' radicalization processes in algorithmic contexts where algorithms are embedded in platforms as a design feature. The roles these social platforms and their behind-the-scenes recommendation algorithms play in driving online radicalization have been posited by both academics and industry alike (Roberts-Ingleson & McCann, 2023).

Understanding users' radicalization processes on algorithmic platforms could advance theoretical development in algorithmic radicalization and help pave the way for professionals to incorporate appropriate strategies to combat radicalization via AI.

LITERATURE REVIEW: TIKTOK'S ALGORITHM FOR RADICALIZATION

Algorithmic radicalization is the idea that social media platforms use algorithms that frame someone's personal online activities and channel him or her down "rabbit holes" of ideologies (Fernandez & Alani, 2021). These algorithms control what people see and when they see it, and they learn from their past activities; people gradually and subconsciously begin to think like the rabbit hole down which they have been pushed. Social media platforms drive people toward progressively more radical content over time, leading them to become radicalized by extreme sociopolitical perspectives (Ledwich & Zaitsev, 2020).

TikTok and For You Page: The Dark Side of the FYP

TikTok has been the fastest growing online platform around the globe (Cervi et al., 2021; Zulli & Zulli, 2022). This popularity is attributed to its unique algorithm, which is different from those of its counterparts (Montag et al., 2021). TikTok uses machine learning to show you a personalized feed known as For You Page (FYP), which is different for every user. This algorithm is designed on an interest graph (similar to a social graph on other platforms), which works based on users' interests as well as the videos with which they interact, capturing people's likes and dislikes and relating users with other user groups who share those interests (Basch et al., 2022).

The FYP makes every user's page different by recommending a curated feed of videos from creators they might follow based on their interests and past interactions (Montag et al., 2021). TikTok's FYP is built on the sharing of content with users, based on what the TikTok algorithm has learned to be the most pertinent and of the highest relevance to them. FYP feeds are customized to a user's interests, preferences, and beliefs by tracing entertaining short-form videos. With the FYP loop, TikTok can be seen as affording solid feedback loops of sentiments and specific topics for its users

(Weimann & Masri, 2020). TikTok's FYP has received popularity because it is able to identify the kind of content that users wish to view and keep them watching through recommended videos. However, this feature has also led to criticism that the largely unexplained algorithm could also nudge people into radicalizing kinds of videos. How FYPs collect and feed content has been criticized in terms of privacy and data protection, the spread of hate speech, and the promotion of extremism on the platform (Montag et al., 2021). A group of expert groups condemn the amount of data that is collected by TikTok as well as its possible data leak to third parties. Relevant studies have shown that the FYP has contributed to showing more incendiary and radical videos over time based on users' previous interactions with the FYP (Ledwich & Zaitsev, 2020).

The Growth of Radicalization Loops: Why People Fall into Extreme News

Radicalization is defined as a "change in beliefs, attitudes, and behaviors in directions that increasingly justify intergroup violence and demand sacrifice in defense of the ingroup" (McCauley & Moskaleiko, 2008, p. 416). From a procedural perspective, Harrington (2013) defined radicalization as a process by which people develop violent extremism. Using a longitudinal approach, Ribeiro et al. (2020) found that YouTube has played a role in radicalizing people via exposure to far-right ideologies. The study revealed evidence that users who engaged with a middle ground of extreme right-wing content migrated to commenting on the most fringe, far-right content. This finding is in line with other studies that criticize YouTube as a radicalization engine (e.g., Tufekci, 2018). These studies warned of the increasing trend of far-right radicalization on this platform. For example, Baugut and Neumann (2020) identified a loosely connected network of YouTubers, ranging from mainstream libertarians and conservatives to overt neo-Nazis and white racists, who were spreading their bigoted ideas to other viewers. Tethered to mainstream media, they often collaborated and strengthened each other's content. In the process, they made it simple for viewers to gradually shift to more extremist content.

Shin (2023) argues that social media platforms intentionally and unintentionally help far-right political groups by assisting them in radical social divisions. Platforms often demonize and further marginalize minority communities by describing them as dangerous and harmful in order for the public to generate hatred and fear against them. Many other studies

have echoed this point by showing how exposure to derogatory rhetoric against minorities could pave the way for social radicalization and involvement in hate crimes and violence (Soral et al., 2022).

O'Connor (2021) reported that TikTok's user-recommender algorithm pushes users toward accounts with far-right viewpoints that are supposedly banned from the platform. The report found that the platform promotes content from far-right sites, such as the Patriot Party, QAnon, Three Percenters, and Oath Keepers on the platform's FYP, which recommend content they believe each specific user would like. FYPs display algorithmically recommended content to users based on inferred interests. The accuracy of the FYP is cited as a primary factor in TikTok's popularity (Bhandari & Bimo, 2022). FYP feeds are based on numerous parameters, such as video information (captions, sounds, and hashtags), user experience (profiles, liked and shared videos, accounts followed, created content, and shared content), and account settings (device type, language preference, and country setting) (Little & Richards, 2021). With the FYP, TikTok's algorithms increasingly present users with far-right TikTok accounts to follow. This methodology can send TikTok users spiraling down a rabbit hole of progressively more aggressive, hateful, and violent content (Cotter et al., 2022).

These findings are consistent with Shin's argument (2023) that far-right content is highly prevalent in TikTok's algorithmic services and that many far-right influencer communities are growing there. This argument indicates that the current dominant social platforms have recommendation algorithms, and similar concerns have been raised regarding them. These platforms have commonly struggled with extremist and far-right content. In 2021, TikTok took several measures to address aggressive falsehoods concerning the COVID-19 pandemic. Facebook has made attempts to address the growing extremism on its platform. YouTube banned content related to far-right conspiracy theory, QAnon (Shin, 2023).

The Loop Effect and TikTok

Recommendation algorithms are generally regarded as platforms' secret mechanisms for providing their users with hypertargeted recommendations (Cervi et al., 2021). These algorithms are exceptionally good at identifying what would keep users viewing content on the service, although users do not know what they themselves need (Yesilada & Lewandowsky, 2022). Algorithms written by data-rich platforms, such as TikTok and

YouTube, can actively create a loop that conveys designed content in pursuit of user engagement.

The loop effect posits that algorithmic systems will continue suggesting similarly prejudiced and potentially more violent content on their homepages and sidebars (Haroon et al., 2022). This is similar to what Ribeiro et al. (2020) referred to as a “radicalization pipeline” or “radicalization pathways.” Social media platforms are helping spread divisive messages online at an unprecedented rate and potentially fuel offline violence (see Fig. 3.1). Recent studies have highlighted the role of recommender systems, which may steer users toward radicalized content, eventually establishing radicalization pathways in which users can be further directed toward radicalized content, such as a “rabbit hole” or a “loop effect” (Kaiser & Rauchfleisch, 2019; Ledwich & Zaitsev, 2020). However, the issue concerns whether these platforms are an *essential cause* of radicalization (without which these events could not have happened) or merely reflect real-world tensions.

There have been many claims that online platforms are the source of algorithmic amplification, where certain online content becomes popular at the expense of other perspectives on a certain topic (Haroon et al., 2022). The history of people’s clicks, reviews, likes, shares, and comments are the data sources of algorithmic personalization. Researchers argue that algorithms simply stir divisive behaviors already seen in the

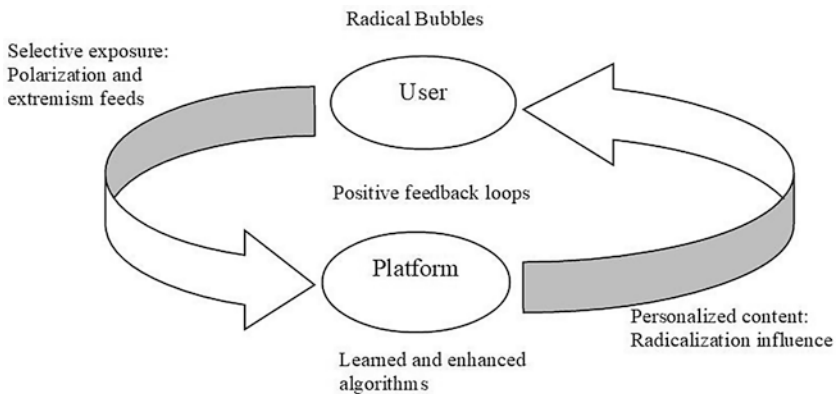


Fig. 3.1 The loop effect on platforms and users. (Source: modified from Haroon et al., 2022)

real world (Shin, 2023). A report by Facebook established that user self-selection is to blame for the type of content seen in a news feed. Users' psychological drive for more incendiary content that satisfies their subsequent action is compounded by the tendency of extreme individuals to seek out extreme information. Content that elicits anger may also lead users to seek out existing belief-confirming information. Options can become increasingly narrow, and user choices can be limited to drastically radical content. This phenomenon is known as algorithmic confounding (Chaney et al., 2018). AI platforms are often fed or trained with data from users who have already been exposed to algorithmic recommendations, which creates a pernicious feedback loop wherein users become more dependent on algorithmic recommendations. This loop uses data confounded in many ways and homogenizes the user's behavior without increasing utility.

TikTok's algorithms and architecture present several reasons for concern (Zulli & Zulli, 2022). TikTok is an open platform that depends on user-created shared content and, hence, allows those with hateful or extremist views to compete directly with other media and social media sources. TikTok's algorithms are designed on an interest graph (like a social graph on other platforms), which works based on users' interests as well as the videos they interact with, capturing people's likes and dislikes and relating users with other user groups who share those interests. TikTok's FYP is built on the sharing of content with users based on what the TikTok algorithm has learned to be the most pertinent and of the highest relevance to them. FYP feeds are customized to a user's interests, preferences, and beliefs by tracing entertaining short-form videos. With the FYP loop, TikTok can be seen as affording solid feedback loops of sentiments and specific topics for its users (Weimann & Masri, 2020).

To better understand how cognitive bonding is established during the consumption of extreme content and how radicalization is shaped and sustained, the loop effect framework in the radicalization mechanism in platforms can be adopted.

Misinformation Spirals and Insular Social Groups

Numerous studies have investigated the nexus between recommendation algorithms and radical content, with a predominance of TikTok (see, e.g., Weimann & Masri, 2020) and YouTube (see, e.g., Schmitt et al., 2018). Whittaker et al. (2021) analyzed far-right content, finding that more

extreme content can be presented to users and that they may find themselves in an ideological chamber that can be devoid of content that does not fit their belief systems. Similarly, Ribeiro et al. (2020) found that YouTube's recommendation algorithm often recommends far-right content, and once in these communities, people receive increasingly more far-right content. A number of studies support the notion that YouTube cultivates a radicalization loop (Schmitt et al., 2018), suggesting that radical content may be amplified via its recommendation algorithms.

In contrast, some studies have demonstrated that YouTube recommendation algorithms effectively deter users from viewing extreme content online, which they claim refutes popular radicalization claims (Hosseinmardi et al., 2021). They found no evidence of migration to extreme-right channels; their findings indicate that recommendation algorithms curb traffic toward these categories and that users are instead nudged toward mainstream channels. Gaudette (2020) researched Reddit's upvoting and downvoting algorithms, arguing that the upvoted sample contained extreme discourse that facilitated "othering" toward two outgroups: the Islamic and left-wing political ideology. Their findings revealed that the upvoting algorithm played a key role in facilitating an extreme collective identity on the subreddit. Based on interviews with extreme Islamists, Baugut and Neumann (2020) found that users began with only a basic interest in ideology but followed platforms' recommendations when they were shown radical propaganda, which pushed them to engage in violence. Waters and Postings (2018) found that Facebook and Twitter recommend radical jihadist accounts for users to follow after an individual initially engages with extreme content, arguing that the platforms unintentionally forge a network that supports connecting extremists.

Considering the aforementioned research, platform algorithms indeed shape and promote radicalization by displaying extreme content. However, this research tended to focus on analyzing interactions between users and the content of recommendations from YouTube's open application programming interfaces (APIs). Actual interactions based on personalization remain unexplored. Despite increasing concerns about radicalization, the evidence base for this concern is limited partly because most research tends to underplay the effect of extreme content or ascribe it to users' decision-making on extreme content.

Hypothesis and Research Questions

The related literature indicates that radicalization is an interactive incremental process in which users' involvement can contribute to radicalization. It has been debated whether the existence of a radicalization pipeline on platforms systematically leads users to progress toward more extreme content on the platform (Gaudette, 2020). Platforms have been considered a key potential venue for extremist and harmful content online (Ledwich & Zaitsev, 2020). TikTok has been criticized for spreading and popularizing virulent and hateful content by the way its FYP algorithm directs user traffic. However, an empirical investigation of the platform's role in radicalization is lacking. Having established the notion of the loop effect, we hypothesize the following:

Hypothesis The more people watch extreme content via platforms, the likelier the platforms are to provide similar content, creating a loop between people and platforms (i.e., the loop effect).

Although relevant research and industry reports have revealed the negative side of TikTok, a significant gap has remained in the knowledge of how platform design specifications are utilized at scale to direct hatred at others. To narrow this gap, we propose the following questions based on Ledwich and Zaitsev's (2020) radicalization criteria:

RQ1 Do algorithmic recommendations nudge users toward more extreme content (i.e., radicalization influence)?

RQ2 How do automated platforms radicalize users, and how do recommendations foster polarization (i.e., polarization and extremism)?

RQ3 To what extent do users systematically gravitate toward more extreme content (i.e., radical bubbles)?

Given the increasing concern over algorithmic radicalization, we aim to advance our understanding of a potential remedy for the radicalization that we currently face. Insights from the research questions can contribute to the management of radicalization on platforms to nudge users away from radicalization and misinformation.

METHODS: FINDING THE LINK BETWEEN MISINFORMATION AND RADICALIZATION

To answer these questions, we used mixed multiple methods by combining (1) longitudinal methods (a t test and in-depth interviews) and (2) thematic analysis. We conducted a longitudinal study to test the radicalization hypothesis, utilizing a panel selection of 50 users and their everyday consumption behavior of the platform over three months. We conducted in-depth interviews with the panel (asking about their experiences) and performed a matched t test on their pre- and postexperiment extremism scores. We also performed a thematic analysis of TikTok videos in accordance with the categories of hateful and extremist content developed by O'Connor (2021).

Longitudinal Methods

Our data collection process included (i) obtaining a large pool of relevant channels from TikTok, (ii) collecting comments and metadata for each TikTok video, (iii) annotating channels belonging to several different interest communities, and (iv) tracking TikTok videos and channel recommendations. Our data were extracted from Qualtrics's nationally representative online panel in the United States, spanning February through November 2022 (see the Appendix), which traces respondents' viewing of specific content. The researchers crowdsourced TikTok to which participants were exposed via their FYPs, similar to the methods used by Cotter et al. (2022).

We used a population of panelists who had at least one TikTok-recorded page view. The recruited participants were first interviewed about their attitudes, political views, social beliefs, and other values. We measured their views (ranging from 1 conservative to 7 liberal) with scales we developed and were validated by expert groups. The scale was composed of seven questions on contemporary issues of politics, society, and the economy, asking their views on each issue or on an event. The issues include stories regarding abortion, immigration, feminism, LGBTQ+, race, welfare, and government regulation. The scale measurements were reviewed by a group of experts specializing in misinformation and fake news. We used the categories of hateful and extremist content developed by O'Connor (2021). The categories were composed of 12 content schemes: (1) anti-Asian, (2) anti-colored (Black/Hispanic), (3) anti-LGBTQ+, (4)

anti-immigrant/refugee, (5) anti-Muslim, (6) antisemitic, (7) features extremist symbols embedded in normal media, (8) glorifies an extremist individual/group/viewpoint, (9) misogynist, (10) promotes white supremacy, (11) terrorism footage, and (12) COVID-19 conspiracies/misinformation threaten or attack a person or group.

With these categories in place, the panel respondents were required to view TikTok for a designated amount of time during the span of the study. The panel respondents were required to use TikTok services regularly for their search information and browsing. Specific instructions were given on how to view, the time, and what comments to leave after using TikTok. The respondents were asked to routinely consume TikTok content by scrolling through its FYP and were requested to record their daily consumption (see Table 3.1). The respondents took notes and described every video consumed. After the experiment, in-depth interviews were

Table 3.1 Experiment log

<i>Participant number</i>	<i>Time spent daily (hours)</i>	<i>Daily videos viewed</i>
1	4	534
2	1	98
3	2.5	204
4	2	504
5	1	98
6	2.5	204
7	0.5	152
8	3	87
9	2	104
10	1.5	204
11	2	68
12	1	104
13	3	204
14	5.5	504
15	1	171
16	2.5	202
17	0.5	151
18	3	99
19	2	114
20	1.5	244
21	2	234
22	3	404

(continued)

Table 3.1 (continued)

<i>Participant number</i>	<i>Time spent daily (hours)</i>	<i>Daily videos viewed</i>
23	2	298
24	2.5	204
25	0.5	152
26	3	94
27	2	104
28	1.5	204
29	0.5	68
30	2	104
31	1	204
32	2.5	381
33	0.5	108
34	3	304
35	2	252
36	1.5	89
37	2	104
38	2	209
39	1	224
40	2.5	99
41	0.5	203
42	3	152
43	2	183
44	1.5	114
45	2	224
46	1	69
47	2.1	124
48	2	104
49	1	214
50	2	94
Total average	1.922	189.78

conducted with the panels. The interviews were designed to test (1) any changes in their views on certain issues and (2) how their views were changed by TikTok or others. The radicalization scores were compared to those recorded three months prior. We adopted the repeated t test method to analyze the data, which involved matched-pair measurements with the same respondents.

Thematic Analysis

We collected TikTok videos in dual phases. In the first phase, we employed a multipronged strategy to systematically collect a diverse sample of

videos. The centrality of algorithmic curation on TikTok and its opaque nature make it difficult to collect data by searching the platform itself (Bhandari & Bimo, 2022). Since each user’s FYP is distinctive and changes temporally, it is not possible to build a typical or stereotypical dataset of videos. As such, we relied on three sources of videos that eventually formed our dataset: (1) an online survey, (2) researchers’ curated TikTok, and (3) TikTok collected through relevant keyword searches and videos that include a popular song circulated on the platform.

Members of the research team added 87 additional videos to the primary corpus, as identified in their FYPs, via routine TikTok use. Finally, the videos were collected using search terms via TikTok’s search function (see Table 3.2). These terms were selected based on Cotter et al. (2022) and other similar phrases/sounds emerging in earlier steps. Fifteen videos were randomly selected from each of the searches (random numbers between 1 and 100 were generated, and the corresponding video in the search results was randomly assigned a number and selected). This added 89 videos to the primary corpus after eliminating one duplicate. Across the three data sources, 196 videos were collected to form the primary corpus of data.

Table 3.2 Thematic coding analysis

<i>No</i>	<i>Genre</i>	<i>No. of videos</i>
1	Anti-colored (Black/Hispanic)	39
2	Anti-Asian	86
3	Anti-LGBTQ+	91
4	Anti-immigrant/refugee	75
5	Anti-Muslim	29
6	Antisemitic	39
7	Features extremist symbols embedded in normal media	43
8	Glorifies an extremist individual/group/viewpoint	67
9	Misogynist	59
10	Promoting white supremacy	69
11	Terrorism footage	58
12	Uses COVID conspiracies/misinformation to threaten or attack a person or group	52

How Algorithmic Recommendations Promote Radicalizations

Multiple-method analyses revealed some signals of radicalization. Our analyses imply that exposure to content from hateful or extremist channels on TikTok remains routine. The results are consistent with previous findings (see, e.g., Weimann & Masri, 2020; Ribeiro et al., 2020), implying that when users view extreme feeds, they are more likely to continue to follow recommendations for similar content on platforms. Although there is no direct empirical evidence of causality, our findings indicate that recommendation algorithms can play a facilitating role in societal polarization and extremism. When the algorithm configures that users view numerous biased problematic videos, it can trap them in a loop in which the system will continuously suggest similarly prejudiced and potentially more excessive content on their main page and side menu. This can lead to polarization and further division of opinions among various social groups, such as left-wing and right-wing users (Hosseinmardi et al., 2021). Recommendation systems perform as intended, but they have issues that are external to the system. Unless users are determined to cut that loop, all the recommendations on that system will funnel into a single particular niche interest in which they have been ditched. This can lead to partisanship and increase the societal divide among diverse social groups. Our findings can be broken into several themes.

FYP and Radicalization

On TikTok, the FYP feed echoes preferences personalized to each individual. The FYP suggests videos to users based on the items they have viewed and preferred in the past by classifying videos based on a combination of parameters, ranging from the interest/preference users express as new users to adjusting for content they indicate they are not interested in to form a personalized FYP feed. This feed can be considered a source of radicalization (Bhandari & Bimo, 2022).

One respondent who used and liked the FYP on TikTok mostly for transphobic content and far-right issues stated that the algorithm gradually began to populate their page with far-right content and called for extreme violence (“*I feel like TikTok has become a strong venue for my viewing formation*”). Another respondent who used the FYP function said that the algorithm influenced users of mainstream and center-left channels by nudging extreme right-wing videos, content that targeted vilifying

left-wing or centrist narratives (*“I think that TikTok’s recommendations are increasingly extreme in the narrative, themes, and tone”*).

Other respondents claimed that TikTok created a sort of radical bubble in that recommendations affect users of radical videos to view more related videos than they would otherwise, making it less likely that alternative viewpoints would be rendered (*“By the end of the experiment, I feel the gradual development of an echo chamber of an idea”*). The respondent noted, *“the platform is much more assertively curating than other platforms like YouTube or Facebook.”* A similar response came up: *“TikTok secretly fuels the right-wing outrage”* Many respondents ascribed bubbles to TikTok’s Interest Graph, which captures users’ preferences and associates them with other user communities of similar interests. TikTok’s FYP is based on this Interest Graph, in which users watch only a few videos, and the algorithm quickly curates what users would view next. One respondent stated, *“TikTok is like more an editor rather than a forum on social media.”*

TikTok’s emphasis on an interest graph instead of a social group perhaps made users feel more loop effects. The interest graph, which embeds AI, captures people’s changing interests in real time without users having to do much work. The algorithm identifies what is interesting to users while they passively watch and scroll. This removes the burden of the social aspect entirely, as many people use TikTok solely as an entertainment platform just for discovery. This inference is consistent with other respondents who indicated this as being addictive. Given that TikTok is personalized specifically for individual users, the algorithm works as a speedy matchmaker, discovering content for users who they do not even know they want to see. This claim is consistent with Ledwich and Zaitsev’s findings (2020). The watch-next algorithm on YouTube recommends that a video be seen subsequently when autoplay is enabled. This function enables users to create radical bubbles by recommending further and stronger content. In this light, one respondent noted, *“TikTok’s algorithm keeps users glued to their sites.”* TikTok creates a strong loop not only by connecting them with other users but also by thoroughly tracing users’ behavior on the service. Another respondent explained how TikTok is different from other platforms: *“On Twitter and Facebook, what people follow largely decides what message you view. On TikTok, we don’t have to follow anyone.”* TikTok’s algorithm quickly learns how users interact with videos

to decide what videos to recommend. This is why TikTok's establishing loop is powerful; by users merely viewing and reacting to videos, the algorithm quickly learns their preferences. One respondent commented, "*I feel like passive personalization.*"

Addictive Engagement

To differing extents, respondents commonly noted that misinformation, divisive video, and salacious content drive significant, or at times addictive, engagement. Although users may acknowledge the false nature of videos and flag them, this often occurs only after they have already consumed them. By then, they had given a clear sign to the algorithm that they were in a content rabbit hole. Extreme content has been favored, preferred, and viralized, which encourages creators to upload more of it. Driven by algorithms incentivized to bolster attributes that are positive for engagement, more of this kind of extreme content filters into recommendation systems. One respondent echoed this idea by saying, "*The recommendation from FYP appears to lead us down far-right rabbit holes.*" Similarly, "*I felt like got sucked in by FYP when viewing controversial content.*" When platform algorithms learn (by machine learning) how they hook up one user, they can reproduce the same mechanism on millions of users.

Numerous respondents noted that the platform encourages hatred targeting Asians, the Black race, Muslims, refugees, females, and LGBTQ+. Many commonly stated that they saw content eulogizing the killing of people of these minority groups, sarcastic victims of racism, and both explicit and implicit displays of hate against these communities. One respondent noted, "*I had the experience of looking at some hate sites and then seeing similar violent content for it all over TikTok feed.*" Other respondents also echoed that features of the platform, such as video filters, sounds, and profiles, contribute to enabling targeted extremism and hate to proliferate on the platform. Products endorsed by TikTok to encourage involvement, virality, and creativity are being used to help spread extremism and radicalization. Sensational content reaches significant numbers of viewers. Hate speech-related videos have reached thousands of tractions in some instances and have generated millions of followers (Soral et al., 2022). The algorithmic systems underlying TikTok's service obviously serve to stimulate and promote this video to users who may not otherwise have identified it.

Polarization Scores

The findings from the qualitative interviews are consistent with the polarization scores the respondents had recorded after viewing for three months (see Table 3.3). Generally, the respondents' scores increased from the initial records before the experiment. These increased scores are common over all the items from politics, religion, gender, immigration, and LGBTQ+.

The Loop Effect

The triangulated results from the qualitative and quantitative methods support the existence and exertion of the loop effect of consuming TikTok content. The *t* test for longitudinal data revealed that the average scores differed significantly in most of the far-right categories ($p < 0.001$). There was a significant difference between the average scores in the two tests, with a *p* value of 0.000 at a 0.001 level of significance. The *p* values were

Table 3.3 Pre- and post-comparison (paired *t* test, *n* = 50)

<i>Category</i>	<i>Pretest</i>		<i>Posttest</i>		<i>Significance</i>	<i>t score</i>	<i>p</i>
	<i>Mean</i>	<i>SD</i>	<i>Mean</i>	<i>SD</i>			
Anti-Asian	3.45	1.032	4.98	1.242	0.000*	-7.117	<0.001
Anti-colored (Black/ Hispanic)	3.52	1.001	4.53	1.023	0.000*	-5.249	<0.001
Anti-LGBTQ+	3.56	1.052	5.35	0.023	0.000*	-5.377	<0.001
Anti-immigrant/refugee	3.66	2.592	5.64	2.043	0.000*	-5.080	<0.001
Anti-Muslim	3.63	0.252	5.72	0.145	0.000*	-6.241	<0.001
Antisemitic	3.91	1.402	4.23	1.302	0.345	-12.230	0.080
Features extremist symbols embedded in normal media	3.12	2.025	3.98	2.145	0.000*	-5.435	<0.001
Glorifies an extremist person/group/viewpoint	3.55	1.523	3.88	0.023	0.000*	6.310	<0.001
Misogynist	3.68	1.521	5.23	1.234	0.534	-3.255	0.061
Promoting white supremacy	3.12	0.052	4.66	0.302	0.000*	-5.030	<0.001
Terrorism footage	2.91	0.234	5.33	2.042	0.000*	-13.235	<0.001
Uses COVID conspiracies	3.19	1.023	4.62	0.145	0.000*	-19.030	<0.001
Misinformation to attack or threaten a person or group	3.56	0.294	4.11	1.249	0.000*	-9.412	<0.001

insignificant only for two categories (misogynist and antisemitic). This suggests that there was a significant change overall in the perception of the far-right after using TikTok for a certain time. The test indicated that far-right scores increased significantly throughout the experiment.

Positive Feedback Loop

Overall, we identified a plausible link in which respondents with polarized views on issues such as race frequently viewed videos from extremist channels on TikTok. Consumption of this detrimental content was concentrated among people who already had a significant racial bias, the group that was seemingly most vulnerable to its influence. Moreover, despite recent modifications to the TikTok algorithm, the platform still regularly suggests content from alternative or extremist channels when people watch videos from those channels. Consequently, various racially resentful groups are not only watching a large amount of content from such channels but are also shown recommendations for more such content when they do so, further intensifying exposure to potentially polarized and radical content.

The findings support the loop effect hypothesis of the possibility of a negative feedback loop wherein users become more dependent on radical content as they rely on platforms for content. The loop effects are created and sustained through several mechanisms: (1) users spend more time on TikTok, thus having a great impact on the algorithms; (2) the content they engage with garners more likes and views; (3) content creators are aware of this trend and create more of these extreme videos; (4) users spend even more time watching those extreme videos; and (5) users become stabilized into a radicalized loop. This loop is firmly supported by the incentive structure underpinning TikTok's recommendation engine. Recommendation algorithms need users to engage with their platform as often and as much as possible since it is in their operational interest. This business interest of the platform also incentivizes users to stay on the platform as much as possible because the time and likes they receive can be turned into a monetary reward. Eventually, the machine learning algorithms of the platform turn into social profiling and racial tension cues that create and deepen radicalization.

The regulations TikTok has on the development of radicalization or extremism are uncertain. Each of the interviews and insights from the respondents raises concerns over the ease with which implicit or explicit

endorsements of extremism and hate speech are permitted through TikTok profiles. The findings allude to the platform’s relaxed guidelines, profile functions, and FYP algorithms, which have the potential to further a user’s radicalization process. Together, the respondents’ interviews suggest that through the loosely defined enforcement of community guidelines and the clever activities of content creators, TikTok’s FYP algorithms have the potential to act as facilitators in bridging certain users to increasingly hateful and conspirational narratives and echo chambers. These analyses suggest that TikTok may act against content that is not obvious in its promotion of radicalization, hate speech, or extremism, particularly content that expresses support via symbols and emojis (Table 3.4 and Fig. 3.2).

Table 3.4 Summary of findings

	<i>Findings</i>	<i>Data</i>
Hypothesis: Loop effect	The more users view extreme content, the more likely TikTok provides similar content, which creates a positive feedback loop.	Overall supported
RQ1: Radicalization influence	TikTok’s FYP frames users by nudging them to more extreme content than they would otherwise seek out.	Partially supported
RQ2: Polarization/extremism	The <i>t</i> test results indicate that TikTok’s automated algorithms radicalize users and foster polarization.	Partially supported
RQ3: Radical bubbles	TikTok’s FYP influences users of radical content to view more similar content than they would otherwise.	Partially supported

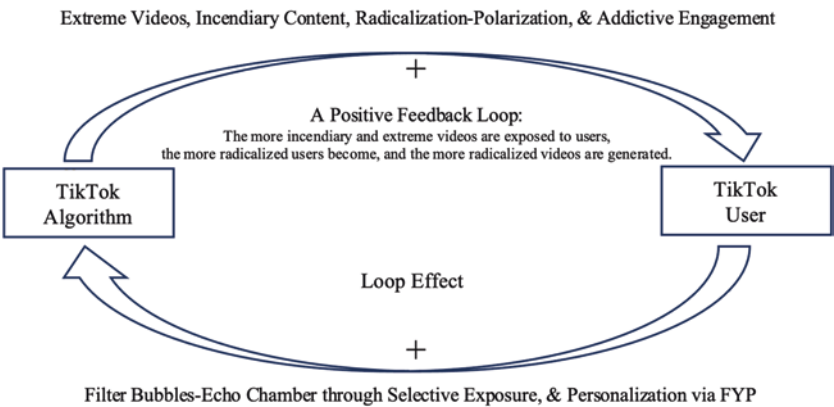


Fig. 3.2 Illustration of the loop effect

DISCUSSION: THE ROLE OF MISINFORMATION IN FUELING EXTREMISM ONLINE

We examined various categories of radical content on TikTok and analyzed extremist ideologies steering an individual's viewership. Our findings showed that the platform has an inherent content manipulation issue: extremist and even hate crime-related videos are easily accessible on the platform. TikTok facilitates radicalization through algorithms, as well as by legitimizing ideological biases and amplifying social polarization through filter bubbles. Despite the contravention of the platform's policies and terms of service, such extreme content is easily discoverable by users because the platform is trained using data from users who have already been subjected to algorithmic recommendations, which generates a feedback loop. Respondents indicated that content eulogizing the extreme actions of terrorists and extremists was reflected in their searches, illustrating how the platform functions as a new avenue for radicalization-promoting extreme ideologies. These results resonate with the argument that the feedback loop amplifies the radicalization of users and the effects of the FYP on the spread of video consumption.

Based on these findings, suggestions to limit algorithmic radicalization can be made. First, greater transparency and more individualized control of personal data should be ensured. Transparency should be publicized, and users should be involved in the transparency-promoting process. For example, applying the bipartisan Honest Ads Act to TikTok would ensure the same level of transparency that is required for broadcasting election advertising. Thus, it is suggested that TikTok release regular audits of its algorithms or that they open source its algorithms so that third parties can audit them for ideological bias. Furthermore, platforms should be required to have robust systems for archiving ads and information on how those ads were made searchable and sortable through an API.

Second, regulatory audits can be effective in curbing and moderating algorithmic amplification. Algorithms on their own are not accountable for radicalization, yet they produce a large array of recommendations that guide and influence users' online and offline behaviors. Left unregulated, algorithms have the potential to guide casual or coincidental users toward extreme content to which they may not otherwise be exposed. Thus, the government could use its existing user protection authorities, which safeguard against unfair or deceptive service practices, to probe platforms' potential material failures to honor commitments made in their own terms

of service or biased actions in failing to safeguard against harm to users. For example, Facebook allegedly exempted six million VIP users from content moderation standards, and they systematically allowed content that infringed on terms of use to go viral before taking it down.

Third, platforms can develop a self-regulatory model, such as best practices guidance or a code of conduct, to moderate algorithmic functions that promote extreme content or illegal activity. Social media platforms, including TikTok, need to appropriately handle users who find ways to dodge the present systems that spot and respond to offenses to terms of service on the platform. A well-established framework would work best alongside legislation to establish baseline transparency and accountability. As such, governments should work with platforms to develop transparent and voluntary codes of conduct to handle radicalization and content manipulation processes. There is no one-size-fits-all approach to addressing algorithmic radicalization, but it is important that this task be addressed to create a set of measures that would make a difference at the margins while continuing to honor First Amendment freedoms and values.

WHAT SOCIAL MEDIA PLATFORMS SHOULD DO ABOUT RADICALIZATION

The legality of legislating content moderation leads to issues in the First Amendment and free speech. In a democratic society, can the government legally regulate the spread of radical content? Private firms are duly protected by First Amendment freedom of speech and can limit speech under their control (Otero, 2021).

Social media such as Google, Facebook, and Twitter have openly quoted this principle of free speech as a virtue to which they are committed as a reason for not restricting speech on their platforms. Such platforms do not block the sharing of news, even if they have been identified as false by fact-checkers since they do not want to infringe on the free speech rights of their users. Instead of downright elimination, platforms may share the fact-checking information or other sources to view, leaving it up to users to judge their own biases and make a reasoned assessment of the likely truth of information. Platforms are not legally obligated to do so under the free speech clause of the First Amendment. Yet these statements of dedication to free speech often give wrong messages to the public, which often misreads their commitment to free speech as an ethic as a legal obligation not to suppress their speech. Consequently, some users wrongly

interpret the removal of social media posts as an infringement on their fundamental legal rights.

Regulating polarization and misinformation may involve legislating social media content moderation itself, which is a problem at two legal and practical levels. Instagram, Twitter, and YouTube algorithms perform highly sophisticated operations. Nowadays, they more proactively present media content to users to drive higher engagement. The platforms foster, rank, recommend, selectively target, discriminatory show, and eclectically conceal content. They also trail data related to personalizing content and product recommendations. Thus, people who engage with uncertain, ambiguous, problematic, or deceptive content receive more of it; people who engage with confirmation bias, radicalizing, polarizing, and radical videos acquire more of that too. These engagement optimization mechanisms escalate the time users spend on these platforms in support of the platforms' profits. Still, the result is that platforms efficiently prioritize the most radical content and hateful messages.

The recent report on TikTok's recommendation effects provides adverse details about these algorithms and what the firm was cognizant about their operation and the repercussions. Such details have caused new levels of public criticism and calls for action. New legislation should reflect the technological realities of how much power social media platforms have to magnify misinformation and aggravate radicalization.

In addition to regulations, practical steps can be suggested for social media platforms that could operate independently.

- *Categorizing radical content and misinformation:* Taxonomies can be developed to categorize news and news-like content for bias. Developing a transparent and fair framework for doing so is as critical as any platform's regulatory compliance framework. Useful taxonomies and transparent measurements will allow platforms to label content internally and address key questions informing new policies.
- *Offer relevant cues on content in-platform:* The relevant information should allow for judging whether delivered content is or is not trustworthy or biased. Instagram has provided information about news sources that could be more helpful in making such assessments. For example, they could provide content from vetted fact-checkers. Also, they could tag content publicly as they tag it internally. This tagging could require an 'opt-in' from users since some users do not want such tags.

- *Provide users choice architecture about algorithmic curation of content.* Users should understand how algorithms present particular types of content over time and give them choices to control it. Such options should be noticeable, easy to navigate, operate, and choose. Most platforms nowadays have limited explanations and options for configuring one's feed, but they are about as valuable for most users. Social media platforms should clarify salient conditions to their users by periodically providing usage statistics to users about the content they use.

CONCLUSION: MISINFORMATION-ALGORITHMIC RADICALIZATION NEXUS

This study validated the proposition that feedback loops in a platform's recommendation systems can give rise to loop effects, which can solidify users' biases and ultimately radicalize their views. The design features of TikTok are exploited for nefarious purposes, including boosting the extremism that FYP algorithms allow through inadequate enforcement of community guidelines and the shrewd activities of content creators, which direct users to increasingly extreme and conspirational narratives. This practice implies the existence and operation of a radicalization pathway that affects the recommendation output of watch-next algorithms. Finding ways to deradicalize platform content is an urgent issue for governments and social media platforms. In this regard, we present a series of recommendations to minimize and/or neutralize radicalizations.

First, we recommend that illegal hateful materials be removed from platforms. TikTok's user guidelines assert that users are banned from sharing, streaming, uploading, or posting a video that promotes "dangerous individuals or organizations" or their extreme ideas. TikTok classifies terrorist individuals and organizations in this category, which is described as "nonstate actors" that intimidate or use violence or "serious crimes (such as crimes against humanity) against civilian groups in search of ethnic, religious, political or ideological goals." Banning the posting, sharing, or praising of terrorists of any kind should be a normative requirement and is a prime basic requirement for content manipulation on all platforms. Our study established that a large amount of such content is still permitted to

be posted without constraints. TikTok can prioritize the accurate, robust, and comprehensive enforcement of these practices to control the proliferation of such content across its ecosystem.

Second, extremist individuals and groups online need to be clearly defined and controlled. Platform user guidelines include a set of requirements aimed at targeting violent extremism, which contains specific definitions of hate, organized hate, and terrorism, referring to individuals or groups that attack people based on protected attributes. On TikTok, attacks are defined as “actions that incite violence or hatred, dehumanize individuals or groups, or embrace a hateful ideology.” In principle, this definition implies a category of illicit content that goes beyond a narrowly described concept of extremism defined in terms of terrorism and violence. Although TikTok considers other platforms and information from offline sources to catch extremist and violent users, it is unsure how extremism is defined by each platform and what exactly hateful ideology is. Do platforms clamp down on political figures that incite and practice extreme ideologies, who might not explicitly call for violence? Our findings imply key loopholes in how such challenges are conceptualized or approached. Platforms are required to offer a transparent and operational definition of radicalization in their terms of service. This is a precondition for effectively limiting the promotion and support of destructive individuals and groups on the platform, along with associated promotional extremism and misinformation (Shin, 2023).

Third, the enforcement of regulations around extreme content and the promotion of protected attributes are recommended. Veiled hateful references, slurs, and racist terms that attack people based on their protected attributes are common among hateful creators and extremists on TikTok. Platforms should work to systematically update and enforce their policies around extremism and broaden their understanding of how creators take advantage of user profile features or nontext and nonvisual cues in TikTok videos to increase hatred against minority people based on their protected demographics. TikTok should address affected groups to better incorporate their experiences of abuse and hatred on the platform into its decision-making, content moderation, and policy and strategy formulation. Users should be able to understand which AI algorithms are working for them and which are working against them.

LIMITATIONS AND FUTURE RESEARCH

This study illustrates how extremism on TikTok is growing and how certain groups take advantage of features of their profile to stimulate hatred, advocate extremists, or publicize their support for extremist views to like-minded users on this platform. However, as with any other study, the results should be treated with caution due to weaknesses that need to be addressed in future work.

First, we did not consider the confounding effect (Chaney et al., 2018) of recommendation algorithms and consumption patterns (algorithmic compounding). One confounding factor is that both recommendation algorithms and usage patterns on TikTok may change over time. Numerous studies have shown the confounding effect (see, e.g., Munger & Phillips, 2022) that radicalization could compound the effects of exposure to potentially hateful content. For example, Hussein et al. (2020) found that viewing conspiracy theory content amplifies recommendations for more related videos. Personalization on TikTok enhances recommendations for pseudoscientific videos. Although we argue for relevant evidence for the existence of radicalization loop effects on TikTok, our study provides little insight into why these loops occur. Investigating the reasons for radicalization is the appropriate direction to better capture user radicalization and the effects of platforms on social media viewership.

Second, we mainly used qualitative methods, relying on users' perceptions, experiences, and self-reported reflections on their uses. Although we used measures to bring objectivity into the analysis and incorporate pre- and postpolarization scores, our analysis of the results is mainly subjective, and a common method bias may exist. Furthermore, the interpretative data are insightful and heuristic; however, the analyses could be missing clear quantitative evidence. As recommendation algorithms are constantly changing, people's perceptions and interpretive heuristics may be limited in understanding the essence of algorithms, as they are designed to work secretly. Additionally, the findings have limited generalizability due to the small amount of panel data. Future studies may use robust empirical methods to identify the causality between users' actions and radicalization. Future studies could pursue topics such as radicalization modeling or the model of misinformation spread.

Third, our method focused only on a single platform for a case study. There may be wide variations in how content recommendations are created, how algorithms are structured and operated, and the models that

users engage with on different platforms. TikTok may be different from other platforms in terms of algorithms, user profiling methods, interfaces, usability, and recommendation parameters. As this study has limitations due to being a single case study, future research could conduct a comparative cross-platform study, particularly in cross-cultural contexts, to examine the contextual influence on radicalization.

Despite some drawbacks, our study contributes to the radicalization literature by examining the trends of a major social media platform. The longitudinal study framework offers a useful framework that researchers interested in the user behavior of radicalization in an AI environment should find useful (Faddoul et al., 2020). This insight can foster consideration and provoke excitement among researchers as we attempt to test and advance theory and solve radicalization on algorithmic platforms. Future studies can further address the underlying processes of radicalization or inflammatory behavior in various contexts, with consideration of the effects of personality traits on the processes of radicalization.

REFERENCES

- Abul-Fottouh, D., Song, M. Y., & Gruz, A. (2020). Examining algorithmic biases in YouTube, recommendations of vaccine videos. *International Journal of Medical Informatics*, 140. <https://doi.org/10.1016/j.ijmedinf.2020.104175>
- Basch, C. H., Donelle, L., Fera, J., & Jaime, C. (2022). Deconstructing TikTok videos on mental health: Cross-sectional, descriptive content analysis. *Journal of Medical Internet Research*, 6(5), e38340. <https://doi.org/10.2196/38340>
- Baugut, P., & Neumann, K. (2020). Online propaganda use during Islamist radicalization. *Information, Communication & Society*, 23(11), 1570–1592. <https://doi.org/10.1080/1369118x.2019.1594333>
- Bhandari, A., & Bimo, S. (2022). Why's everyone on TikTok now? The algorithmized self and the future of self-making on social media. *Social Media + Society*, 8(1). <https://doi.org/10.1177/20563051221086241>
- Cervi, L., Tejedor, S., & Marín Lladó, C. (2021). TikTok and the new language of political communication. *Culture, Language and Representation*, 26, 267–287. <https://doi.org/10.6035/clr.5817>
- Chaney, A., Stewart, B., & Engelhardt, B. (2018). How algorithmic confounding in recommendation systems increases homogeneity and decreases utility. *RecSys '18: Proceedings of the 12th ACM Conference on Recommender Systems*. September 2018. Pages 224–232. <https://doi.org/10.1145/3240323.3240370>

- Cotter, K., DeCook, J., Kanthawala, S., & Foyle, K. (2022). In FYP we trust: The divine force of algorithmic conspirituality. *International Journal of Communication*, 16, 1–23. <https://ijoc.org/index.php/ijoc/article/view/19289>
- Epstein, Z., Berinsky, A., Cole, R., Gully, A., Pennycook, G., & Rand, D. (2021). Developing an accuracy-prompt toolkit to reduce COVID-19 misinformation online. *Harvard Kennedy School Misinformation Review*, 2(3), 1–12. <https://doi.org/10.37016/mr-2020-71>
- Faddoul, M., Chaslot, G., & Faird, H. (2020). A longitudinal analysis of YouTube's promotion of conspiracy videos. *CoRR* abs/2003.03318.
- Fernandez, M., & Alani, H. (2021). Artificial intelligence and online extremism. In J. McDaniel & K. Pease (Eds.), *Predictive policing and artificial intelligence* (Routledge frontiers of criminal justice) (pp. 132–162). Routledge. <https://doi.org/10.4324/9780429265365-7>
- Gaudette, T. (2020). Upvoting extremism: Collective identity formation and the extreme right on Reddit. *New Media & Society*, 23(12), 3491–3508. <https://doi.org/10.1177/1461444820958123>
- Haroon, M., Chhabra, A., Liu, X., Mohapatra, P., Shafiq, Z., & Wojcieszak, M. (2022). YouTube, the great radicalizer? Auditing and mitigating ideological biases in YouTube recommendations. arXiv:2203.10666v1
- Harrington, N. (2013). Irrational beliefs and sociopolitical extremism. *Journal of Rational-Emotional Cognitive-Behavior Therapy*, 31, 167–178. <https://doi.org/10.1007/s10942-013-0168-x>
- Hosseinmardi, H., Ghasemian, A., Clauset, A., Mobis, M., Rothschild, D., & Watts, D. (2021). Examining the consumption of radical content on YouTube. *Proceedings of the National Academy of Sciences*, 118(32), e2101967118. <https://doi.org/10.1073/pnas.2101967118>
- Hussein, E., Juneja, P., & Mitra, T. (2020). Measuring misinformation in video search platforms: An audit study on YouTube. *Proceedings of the ACM on Human-Computer Interaction*, 4. <https://doi.org/10.1145/3392854>
- Huszar, F., et al. (2022). Algorithmic amplification of politics on Twitter. *Proceedings of the National Academy of Sciences*, 119(1), e2025334119. <https://doi.org/10.1073/pnas.2025334119>
- Jahng, M. (2021). Is fake news the new social media crisis? *International Journal of Strategic Communication*, 15(1), 18–36. <https://doi.org/10.1080/1553118X.2020.1848842>
- Kaiser, J., & Rauchfleisch, A. (2019). The implications of venturing down the rabbit hole. *Internet Policy Review*, 8(2), 1–22.
- Kaiser, J., & Rauchfleisch, A. (2020). Birds of a feather get recommended together. *Social Media + Society*, 6(4). <https://doi.org/10.1177/2056305120969914>

- Ledwich, M., & Zaitsev, A. (2020). Algorithmic extremism: Examining YouTube's rabbit hole of radicalization. *First Monday*, 25(3). <https://doi.org/10.5210/fm.v25i3.10419>
- Little, O., & Richards, A. (2021). TikTok's algorithm leads users from transphobic videos to far-right rabbit holes. *Media Matters for America*. <https://www.mediamatters.org/tiktok/tiktoks-algorithm-leads-users-transphobicvideos-far-right-rabbit-holes>
- McCauley, C., & Moskalenko, S. (2008). Mechanisms of political radicalization. *Terrorism and Political Violence*, 20(3), 4153–4333. <https://doi.org/10.1080/09546550802073367>
- Montag, C., Yang, H., & Elha, D. (2021). On the psychology of TikTok use: A first glimpse from empirical findings. *Frontiers in Public Health*, 9, 1–20. <https://doi.org/10.3389/fpubh.2021.641673>
- Munger, K., & Phillips, J. (2022). Right-Wing YouTube: A supply and demand perspective. *The International Journal of Press/Politics*, 27(1), 186–219. <https://doi.org/10.1177/1940161220964767>
- O'Connor, C. (2021). Hatescape: An in-depth analysis of extremism and hate speech on TikTok. A research report by ISD. August 2021. <https://www.isd-global.org/isd-publications>.
- Otero, V. (2021). What social media platforms should do about misinformation and polarization. *Cornell Policy Review*. <http://www.cornellpolicyreview.com>
- Pennycook, G., & Rand, D. (2022). Accuracy prompts are a replicable and generalizable approach for reducing the spread of misinformation. *Nature Communications*, 13, 2333. <https://doi.org/10.1038/s41467-022-30073-5>
- Ribeiro, M., Ottoni, R., West, R., Almeida, V., & Meira, W. (2020). Auditing radicalization pathways on YouTube. *Conference on Fairness, Accountability, and Transparency*, January 27–30, 2020, Barcelona, Spain. ACM. <https://doi.org/10.1145/3351095.3372879>
- Roberts-Ingleson, E., & McCann, W. (2023). The link between misinformation and radicalization. *Perspectives on Terrorism*, 17(1), 36–49. <https://www.jstor.org/stable/10.2307/27209215>
- Schmitt, J., Rieger, D., Rutkowski, O., & Ernst, J. (2018). Countermessages as prevention or promotion of extremism. *Journal of Communication*, 68(4), 780–808. <https://doi.org/10.1093/joc/jqy029>
- Shin, D. (2023). *Algorithms, humans, and interactions: How do algorithms interact with people?* Routledge, Taylor & Francis. <https://doi.org/10.1201/b23083>
- Shin, D., Lim, J., Ahmad, N., & Ibarahim, M. (2022). Understanding user sense-making in fairness and transparency in algorithms. *AI & Society*. <https://doi.org/10.1007/s00146-022-01525-9>

- Slater, M. D. (2007). Reinforcing spirals: The mutual influence of media selectivity and media effects and their impact on individual behavior and social identity. *Communication Theory*, 17(3), 281–303. <https://doi.org/10.1111/j.1468-2885.2007.00296.x>
- Soral, W., Malinowska, K., & Bilewicz, M. (2022). The role of empathy in reducing hate speech proliferation. Two contact-based interventions in online and off-line settings. *Journal of Peace Psychology*, 2(3), 361–371. <https://doi.org/10.1037/pac0000602>
- Tufekci, Z. (2018, August 14). *How social media took us from Tahrir Square to Donald Trump*. MIT Technology Review.
- Tully, M., Bode, L., & Vraga, E. (2020). Mobilizing users? *Social Media + Society*, 6(4). <https://doi.org/10.1177/2056305120978377>
- Vicario, M. D., Quattrociochi, W., Scala, A., & Zollo, F. (2019). Polarization and fake news: Early warning of potential misinformation targets. *ACM Transaction Web*, 2, 174. <https://doi.org/10.48550/arXiv.1802.01400>
- Walter, N., & Tukachinsky, R. (2020). A meta-analytic examination of the continued influence of misinformation in the face of correction. *Communication Research*, 47, 155–177. <https://doi.org/10.1177/0093650219854600>
- Waters, G., & Postings, R. (2018). *Spiders of the caliphate: Mapping the Islamic State's global support network on Facebook* [Report]. Counter Extremism Project. <https://www.counterextremism.com/sites/default/files/Spiders%20of%20the%20Caliphate%20%28May%202018%29.pdf>
- Weimann, G., & Masri, N. (2020). Spreading hate on TikTok. *Studies in Conflict & Terrorism*, 14(5), 752–765. <https://doi.org/10.1080/1057610X.2020.1780027>
- Whittaker, J., Looney, S., Reed, A., & Votta, F. (2021). Recommender systems and the amplification of extremist content. *Internet Policy Review*, 10(2). <https://doi.org/10.14763/2021.2.1565>
- Wolfowicz, M., Weisburd, D., & Hasisi, B. (2023). Examining the interactive effects of the filter bubble and the echo chamber on radicalization. *Journal of Experimental Criminology*, 19(1), 119–141. <https://doi.org/10.1007/s11292-021-09471-0>
- Yesilada, M., & Lewandowsky, S. (2022). Systematic review: YouTube recommendations and problematic content. *Internet Policy Review*, 11(1). <https://doi.org/10.14763/2022.1.1652>
- Zulli, D., & Zulli, D. J. (2022). Extending the Internet meme: Conceptualizing technological mimesis and imitation publics on the TikTok platform. *New Media & Society*. <https://doi.org/10.1177/1461444820983603>

PART II

How People View and Process
Misinformation: How People
Respond to Corrections of
Misinformation



Misinformation, Paradox, and Heuristics: An Algorithmic Nudge to Counter Misinformation

INTRODUCTION: UNDERSTAND THE HUMAN PSYCHE BEHIND MISINFORMATION

The nudge principle is widely accepted and used in numerous contexts in which positive reinforcement and indirect influences guide the behavior and decision-making of individuals or groups (Sunstein & Thaler, 2014). Nudge is the idea that influences people's behavior into behaving in a certain way that will improve their user choices and experience (Lazer et al., 2018). Nudging with information engages in providing cues or information to raise awareness. For instance, in the context of misinformation, this may include giving a label or alert about the authenticity or the political leaning of a source.

Because of its effectiveness, nudging has been widely used in user analytics, system design, and public policy arenas (Mertens et al., 2021). In the digital media literature, nudges have been researched and used as a concept of affordance (e.g., media affordance), allowing users greater choice to control what, when, and how they view content online (Vermeulen, 2022). Nudging in the digital context has been used as an indirect form of using design, interactive, and psychometric components to influence user behavior in online environments without limiting users' freedom of choice.

Recent research has considered nudging for media applications, such as the evaluation of media credibility in digital media (Bhuiyan et al., 2018)

or the affordance of social interactions (Shin, 2022). Meanwhile, nudging has become a key research topic as a means of addressing misinformation and fake news (Andi & Akesson, 2021). In the context of social media use, nudging can potentially be used to address online misinformation in the form of fact-check alerts and labeling. Nudging provides a label of some kind to raise users' alertness about the trustworthiness and reliability of a source, and content posted on news feeds can be accompanied by alerting flagging that source as potentially factually wrong or inaccurate (Bryanov & Vziatysheva, 2021). An accuracy nudge is a kind of priming or brief reminder intended to shift people's attention toward the accuracy of the news content they come across online (Pennycook et al., 2021). Accuracy nudges come in the form of flagging features (e.g., tags, embedded tweets, and pop-up banners) applied to break news-sharing momentum by encouraging readers to quickly check misinformation in online news and content (Roozenbeek et al., 2021).

Against the backdrop of the rising problem of fake news, calls for evidence-based interventions aimed at mitigating the spread of misinformation have been increasing (Nekmat, 2020). Relevant studies have addressed how simple interventions that nudge or prime people to ponder the accuracy of online information can decrease the spread of misinformation (Fazio, 2020). Pennycook et al. (2021) revealed that pausing to consider why a headline is true or false contributes to a reduced intention to share false information. A number of methods for combating misinformation have been examined, and studies have investigated why people share false or misinformation on social media (Bryanov & Vziatysheva, 2021). As there are diverse reasons for the use of misinformation, it would be useful to examine what influences users' intention to share misinformation on social media and how intervention or nudging can influence such behavior. As a form of indirect intervention, accuracy alerts have been examined and used to nudge people to discern truthfulness, which can improve their discernment of what to share on social media (Vermeulen, 2022). Fact-checkers such as Full Fact, PolitiFact, FeedReflect, and Associated Press warn readers about information that has been the subject of debate. While the effectiveness of such nudges has not been confirmed (Choi et al., 2022), some of them, such as fact-checkers, have been growing in popularity, becoming a major trend in social media and algorithmic platforms (Roozenbeek et al., 2021).

Despite their importance, the effects and mechanisms of accuracy nudges on user behaviors, particularly in algorithmic contexts, have been

underresearched, and the exact nature of how nudges work and with what effect, how users respond to them, and what impact they have on user behavior relating to misinformation and fake news remain inconclusive. While there is extensive research in the fact-checking literature on the utility and effectiveness of fact-checking practices, technologies, and policies, there is a dearth of research attention on users' perception of nudges and the ensuing effect on content sharing. Little is known about the potential of an accuracy nudge to prevent users from sharing questioned information with others on social media. Consequently, there have been calls (e.g., Shin et al., 2022) for research on a theoretical and practical framework for the effectiveness of accuracy nudges in decreasing or stemming the proliferation of misinformation on social media.

We respond to this call for research to narrow the research gap by investigating (1) whether the presence of accuracy nudges flagging false news will lead users to refrain from sharing questioned information on social media and (2) how the nudge effect is moderated by sources and trust (algorithmic source effect) in the Korean context. A majority of users in Korea (89% of the population) conduct news searches and consumption from online sources, including algorithmic media platforms (Shin, 2021). It is critical to understand the proliferation of misinformation on social media and the factors contributing to and moderating it. The kinds of misinformation that proliferate online have ramifications for the actions of individuals as well as larger-scale societal and political processes and decisions. The effectiveness of accuracy nudges and the potential for information sources to moderate the nudge effect conditional upon users' trust in algorithmic media are also examined. Furthermore, the article expands on the theoretical foundations of nudging using source effects and trust evaluation perspectives. Our attempts will also offer practical guidelines on the implementation and use of accuracy nudges in the prevalent algorithmic news environment on social media.

LITERATURE REVIEW: HOW EFFECTIVE ARE ACCURACY NUDGE INTERVENTIONS FOR MISINFORMATION?

Online misinformation has become an increasingly serious issue on social media and throughout society. The relevant literature has examined the use of nudges in helping steer users toward fact-checking the content they read online (Epstein et al., 2021). Recent empirical research has found

that shifting user attention to accuracy improves the relevance of the news they subsequently share on social media (Pennycook & Rand, 2022). We develop this finding of news accuracy discernment in algorithmic contexts by testing the moderating roles of sources and trust in news sharing. As previous studies are based on a U.S. sample (Pennycook & Rand, 2022; Pennycook et al., 2020), we extend the relationships to the Korean context. We derive our research questions by addressing how to effectively discourage users from sharing misinformation.

News Recommender Systems and Misinformation

Mis(dis)information, such as fake news, are current trends in media and society. They are defined as deliberately or accidentally sharing false information presented in a manner that is intentionally or unintentionally designed to mislead people (Lazer et al., 2018). The terms misinformation and fake news are both referenced to describe the nature of the information as being untrue. While misinformation refers to false or misleading information in general, the motivation to spread misinformation is either unknown or not necessarily done with the intent to manipulate (Lewandowsky & van der Linden, 2021). Relatedly, the term fake news is a form of misinformation and disinformation, typically with fabricated, false information presented in ways that are similar to news media content (Lazer et al., 2018). Given the ambiguity of intent to manipulate in the context of this study, the term misinformation is mainly used to inclusively infer the related trends.

Although the phenomenon of misinformation is not new, the prevalent use of online and social media has increased its detrimental impact. Not only is misinformation more easily transmitted by sharing features on social media, but the algorithm that favors engagement (e.g., likes, retweets, clicks) could also lead to “popular” content being mistaken as credible information without prudent judgment regarding its accuracy (Shin, 2021). In the context of social media, disinformation with manipulative intent could later be shared when perceived to be true, thereby becoming misinformation. For example, in the Coca-Cola Dasani water parasite case, false information was first disseminated on a website in the form of fake news and later disseminated widely on social media. Such misinformation could put firms and brands in the spotlight and cause longer-term harm to corporate reputation and consumer trust if no appropriate response is provided (Jahng, 2021). To effectively combat

misinformation and mitigate its detrimental effects, communication professionals need to understand the mechanism through which it could impact consumer responses.

Alerts that remind users of the value of accurate information may counteract the effects of platform engagement strategies. Pennycook et al. (2021) demonstrated that priming or arousing users' attention toward accuracy effectively decreased the likelihood of sharing COVID-19-related misinformation. In a field experiment conducted on a sample of social media users, Pennycook et al. (2020) demonstrated the effectiveness of the intervention by proving that referring Twitter users to an accuracy nudge significantly reduced the extent to which they shared inaccurate information on the platform.

Interestingly, these effects have not been tested in algorithmic contexts. Given the widespread use of AI in social media and news consumption, it is worthwhile to understand how nudge effects are differentially applied to algorithm-driven systems. In the era of algorithmically mediated environments, misinformation can be particularly powerful, especially through self-managed sharing on social media (Lewandowsky & van der Linden, 2021). Thus, individual decisions to accept and spread misinformation have reached new levels of importance, as they can be a microlevel driver of the scaled spread of false information and can collectively put democratic decision-making at risk (Vosoughi et al., 2018). The algorithmic effect on nudge priming can be particularly relevant because recommender systems have been considered one of the major sources of misinformation spreading in the online sphere (Shin et al., 2022). For example, news recommender systems (NRSs) have recently been the subject of intense criticism for contributing echo chambers and information cocoons, which can influence the dynamics of social news sharing and the ways in which misinformation originates, becomes legitimized, and propagates (Cardenal et al., 2019). News networks, such as *CNN*, *the BBC*, *USA Today*, and *The New York Times*, provide ubiquitous access to users, allowing them to surf through relevant news using algorithmic platforms.

To draw user engagement and a higher volume of clicks on their sites, online platforms increasingly utilize algorithmic nudges in their NRS. Newsrooms and social media platforms are working on the automatic detection of misinformation and fake news through AI, natural language processing, and machine learning. A trained algorithm will identify information as "fake news" and give it a lower rank to decrease the probability of users encountering it. However, algorithm detection remains

unreliable, primarily because current detection is based on the evaluation of text (content) and its social network to verify its credibility. While AI can determine the origin of sources and the dissemination pattern of fake news, the fundamental limitation lies in how it assesses the actual meaning of the content in terms of veracity and credibility. With a large amount of training data, an AI-driven classification model can detect misinformation and fake news in an article. However, making such detections requires social knowledge, prior political and cultural knowledge, or common sense, which current algorithmic processing skills still lack. Thus, a feasible way can be to rely more on human judgment and behavior modeling and less on algorithmic processing (Hermann, 2022).

In this light, the focus has been on increasing user awareness of misinformation. Numerous studies have examined the acceptance and diffusion of misinformation in the media (e.g., Lazer et al., 2018). Thus, it is a relevant and timely attempt to conceptually and empirically examine the antecedents of the discernment of misinformation as well as more general human reasoning models on accuracy nudges. A better understanding of the motivations and mechanisms of the susceptibility to believing misinformation can lead us to reduce the adoption and spread of misinformation (Talwar et al., 2020).

Nudging Away from Misinformation

A nudge is defined as “any aspect of the choice architecture that has the capacity to change people’s behavior in a predictable manner but without preventing any other alternatives or altering their economic incentives” (Shin et al., 2023). In an AI ecosystem, the definition is similar: the idea is to affect people’s behavior into acting in such a way that will improve their choice, decision-making, and user experience. A group of researchers has examined the impact of nudges on AI, including Shin (2023) and Acquisti et al. (2017), who conceptualized nudges to promote user awareness of privacy/security and the impact of posts on algorithmic platforms. Acquisti et al. (2017) discuss nudging by means of presentation, information, timing, incentives, reversibility, and defaults. Nudging with information involves providing information to raise awareness in their decision-making processes. For example, in the context of fake news, this may include giving a label or signal about the reputability or the political leaning of a source. Nudges are based on the assumption that people do not always

make rational decisions, and they attempt to direct human behavior without imposing a particular choice.

Due to its potential power, researchers seek to explore the power nudge in decreasing the possibility of spreading and consuming misinformation. Nudge interventions to combat the spread or consumption of misinformation have also been found to be effective in practice. For example, researchers use nudge tools to minimize the spread of misinformation on Twitter by preventing unaware users from retweeting fake news. One nudge intervention that can be utilized is called “throttling mindless activity,” which taps into regret aversion bias, i.e., people’s tendency to become more reflective and break their mindless activities whenever risk is identified. In this case, given that the user is about to post a tweet that contains fake stories, the tool could notify the user with a message such as “*The story is estimated to have an 80% probability of containing incorrect information. Are you certain you would share this story?*” By raising caution, this nudge leads users to reconsider the tweet’s publishing. A link to fact-checking news can further offer a venue to consider truthful news.

Research Questions

Investigating the user response to accuracy nudges is critical since misinformation has a profound influence on society and democracy. Despite surging interest in NRS and accuracy nudges, research on how these nudges influence users to deal with misinformation over NRSs has remained at the level of discussion. Accuracy nudges have thus far been championed, yet their actual feasibility and performance have rarely been examined, particularly in algorithmic contexts. This study underlines the conceptual development of accuracy nudges in online news consumption. It investigates whether the priming of accuracy alerts for news based on falsehoods will nudge users against sharing questioned news on social media. The effect of accuracy alerts and the mechanism of spreading misinformation have remained open questions for NRSs and journalism. Given the gap in the literature on misinformation and considering the importance of nudges in NRSs, we propose the following research questions:

RQ1: How does accuracy alert nudge users toward misinformation sharing on social media? What are the effects of accuracy nudges on users’ discernment of misinformation and sharing behavior?

RQ2: How do sources and trust in AI-based systems alter the effects of such accuracy nudges? What are the effects of sources and trust on users’ behaviors toward news consumption, particularly attitudes toward news sharing?

Given the increasing concern over misinformation, we aim to advance our understanding of a potential remedy for the infodemic that we currently face. The RQs will steer us toward understanding how people can become better at discerning misinformation and sharing behavior by looking at how accuracy nudges in NRSs can promote user discernment of misinformation. Our findings can contribute to the design of accuracy nudges in NRSs to nudge users away from misinformation and fake news.

HYPOTHESIS DEVELOPMENT

Our experiential model was designed to test the effects of accuracy nudges on users’ attitudes toward misinformation discernment (Fig. 4.1). Three hypotheses were proposed for this investigation. First, we hypothesized the main effect of the accuracy nudge on discerning misinformation. Second, we predicted the main effect of sources on misinformation sharing. Finally, we hypothesized that accuracy nudges and sources would interact in predicting misinformation sharing. The hypotheses helped in questioning users’ sensemaking of algorithmic nudges and the effect of such nudges on the judgment of misinformation and sharing behavior.

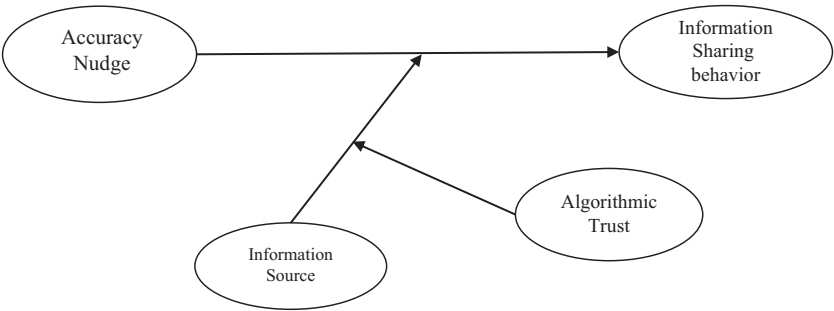


Fig. 4.1 Hypothesized Accuracy Nudge Model

Accuracy Nudge Effects: Nudges Away from Falsehood

People often share inaccurate news, despite being aware that this news is the product of misinformation and a malicious act (Andi & Akesson, 2021). By sharing debatable news with their networks (Jahng, 2021) and interacting with other users on social media (Talwar et al., 2020), people may be considered influential sources of information or even opinion leaders in their social media sphere (Dwyer & Martin, 2017), gaining status benefits and improving their social capital (Shin et al., 2022). The belief that a news story is accurate and factual before proceeding to share it is considered important in social media. Accuracy nudges have been found to be useful tools that social media platforms could implement. Pennycook et al. (2021) found that priming accuracy through indirect induction led users to process information more deliberately than relying on intuitive heuristics. Their previous study (Pennycook et al., 2020) found that prompting users to consider accuracy by rating the accuracy of an unrelated headline prior to rating the likelihood of sharing political headlines resulted in a lower likelihood that they would share false headlines, although it did not decrease their propensity to share truthful news. Relevant research has supported this view by identifying a causal link between nudges and assurances in the context of algorithmic journalism (e.g., Rai, 2020). Shin et al. (2022) showed that integrating epistemic cues into an NRS can lead to more informed decisions, as users are more likely to believe and accept systems when they are nudged to check accuracy and truthfulness. Consequently, accuracy can serve as a nudge, and thus, we present the following hypothesis:

H1: Accuracy alerts nudge users toward reducing news-sharing intentions.

Algorithmic Source Effects: Algorithmic Sources Versus Legacy News Sources

Information sources offer important social cues that impact belief formation (Shin et al., 2022). Generally, the information given by credible sources has a greater effect on users' beliefs, motivations, and attitudes than information delivered from noncredible sources. Users trust news sources if they perceive the source as trustworthy, powerful, and similar to themselves (Jia & Johnson, 2021). These source judgments are applied to algorithms as users perceive their source credibility and believe the news they generate. Users are more likely to react to accurate nudges for news

sources that have been personalized and algorithmicized (Shin et al., 2022). In the AI news environment, users understand that recommended news is generated, curated, and filtered by AI. When news is processed by algorithms, there is a tendency for people to believe that the curated news is correct, accurate, and personalized. Accuracy nudges will more likely work in algorithmic news because users are aware of the existence and practice of algorithms in news generation. Users are more likely to react to accuracy alerts in algorithmic sources as they believe in the algorithmic performance. NRS-based algorithmic news content is considered to have undergone the due quality of expert editorial processes and reporting rigor, thereby positively affecting users' perceptions of the quality of their informational sources and the reliability of the content (Shin, 2021). Nevertheless, in online news sources without algorithmic gatekeeping, users are more likely to react to accuracy alerts because they are less able to distinguish fake news, misinformation, and inaccurate news from unfiltered online news. Online news sources are generally perceived to be more vulnerable due to their lack of technical verification. Thus, we propose the following hypothesis.

H2: News from legacy news sources is more likely to be shared than news from an algorithm-driven NRS.

Trust Effects

Trust has been found to play a significant role in news readership (McQuail, 1992) and an even more critical role in NRSs, as it establishes an interacting feedback loop between users and recommending mechanisms (Shin et al., 2022). When users trust NRS mechanisms, they tend to look for news beyond their own personalized interests (Baden & Springer, 2017). When users believe that news recommendations are customized to their interests, they consider the service to be beneficial and are willing to engage further with the NRS (Mattis et al., 2022). As users continue to provide more data to the NRS, news generation becomes ever more capable of generating accurate and diverse content. Relevant research has shown the existence of this circular relationship in a diverse sample of algorithmic services (Shin et al., 2022).

Nudges have been found to operate on the basis of users' dependency on cognitive heuristics to influence decision-making (Pennycook et al., 2021). In line with these studies, trust heuristic and credibility assessment research traditions have consistently shown users' dependency on source heuristics (i.e., looks established and credible) and cognitive biases in

terms of peripherally cueing their credibility judgments in online environments (Jakesch et al., 2023; Schmitt et al., 2021). From a credibility transfer perspective, content credibility evaluations are postulated to depend on media-related attributes (Shin, 2022). When cautioned about inaccuracy in the news, users cognitively dissociate news content from the source and investigate the accuracy of the information recommended in the news by linking it to their existing schema, knowledge, and experience relating to the source (Metzger et al., 2015). In the evaluation, any credibility violation can decrease the overall credibility judgment of the perceivably credible news related to a trusted news source. Users’ high credibility perception of news sources arguably moderates the influence of news sources on accuracy nudges. When users trust the function of algorithms, they will be more influenced by accuracy alerts with algorithm-driven NRS than nonalgorithmic online news (Helberger, 2019; Hermann, 2022). When they are skeptical of algorithmic processes, they will be more influenced by online news. As users are expected to share news believed to be credible in their social network, we hypothesize the following:

H3: User trust moderates the effect of accuracy nudges such that users’ discernment in sharing news labeled for accuracy will be greater for news from nonalgorithmic online news sources than news from an algorithm-based NRS.

METHODS

In this study, an online experiment was conducted with a 2 × 2 (accuracy nudge: accuracy alert vs. no alert × source factor: algorithms vs. nonalgorithm news) between-subjects design, which produced a set of four groups (Table 4.1). The quasi-experiment was a between-subjects design that manipulated the trust and source of news services. The experiment took place in South Korea in August 2021.

Table 4.1 2 × 2
Experimental design

	<i>Nudge</i>	<i>Nonnudge</i>
Algorithms	Group 1	Group 2
Nonalgorithms	Group 3	Group 4

Data Collection and Sample

An experimental design was used through an online survey of 400 respondents (Table 4.2). To determine the sample size, we calculated the power analysis using the effect size ($\eta_p^2 = 0.02$, inputting power to 0.80) obtained in a similar recent study on the effects of algorithms and trust on user behaviors (Shin et al., 2020). We estimated that our sample would require 400 individuals (100 per condition). Power analysis with *G*Power* showed that considering this sample size, an α level of 0.05, and an estimated measures correlation of $\rho = 0.20$, the power of finding moderate-sized effects in our experiments was larger than 0.80 for the tested main effects.

To check for reliability and validity, six people completed a pretest questionnaire. Recommendations and preliminary results from this pretest (i.e., shortened minimum completion time for multiple-choice questions) were incorporated before distributing the survey on a large-scale online. The recruited participants were randomly assigned to one of the four groups, with no statistically significant differences between the groups in terms of gender, education, prior knowledge about algorithms, or media consumption of news. The four groups did not differ in terms of age ($p = 0.18$), gender ($p > 0.26$), or prior experience ($p > 0.23$). All study respondents were randomly recruited via a marketing company that was branched and located in Seoul. The majority of the participants were Korean (89%), the average age was 32.01 (SD = 8.42), and the gender ratio was almost equal (female: 51.0%; male: 49.0%). Most of the participants (63.0%) were college graduates or had college-level education. With

Table 4.2 Attributes of respondents per experimental group

	<i>Experimental group</i>			
	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>
Mean/median age (SD)	21.31/20.01 (3.10)	22.89/21.32 (3.07)	21.49/20.43 (3.01)	23.83/22.83 (2.98)
Gender (female %)	51.2	50.1	49.8	51.8
Prior nudge experience (year)	2.4	3.1	1.7	0.9
College education %	47	51	43	49

regard to employment status, 119 respondents were students, and 109 were employees. Of the collected responses, 11 incomplete responses with missing information were omitted.

Procedures

On the welcome page, the participants were informed about the study structure, the research goal, topic, and anonymity assurance. Next, they were introduced to the detailed content of the online study. They agreed to provide informed consent, after which they were randomly assigned to one of the four accuracy nudge conditions. All the participants completed a pretest questionnaire measuring their personal trust in the news issue, involvement with the NRS and online news, prior social media news-sharing experience, and algorithmic trust. They were then randomly assigned to each of the two news source stimuli. Those exposed to a legacy online source were shown an interface of the online news page aggregated by Daum, which has one of the highest market shares in Korea and the highest daily news clicks and readership rates. It is considered the most reputable news portal in the nation and was the most appropriate stimulus to represent a legacy online news source for the study context. The participants assigned to the algorithmic sources were shown news recommended by an NRS, which was run by Naver. The participants were required to search news stories on coronavirus, coronavirus vaccine, and COVID-19 diffusion. They were given approximately ten minutes to search and view the news.

Overall, they spent an average of 12 minutes on the news page ($SD = 2.74$). A factual news report on a relatively timely and current topic was selected to determine their trust in the experiment. To manipulate a nudge, half of the participants in each of the news source groups were exposed to an accuracy alert that popped up next to the news report to alert them of content veracity and news inaccuracy in the news report. The alert was akin to a banner used by social media when a piece of news or information is labeled “dubious or misinformation” by third-party fact-checkers. A posttest questionnaire measuring the participants’ intention to share the news was then conducted. The stimuli used in the study are provided in the appendix.

Stimuli

The NRS platforms used for this study were news platforms such as Naver and Kakao News. The participants were to select one of the NRSs or use their preferred applications. After signing the consent form, they completed a prescreening questionnaire that rated their existing trust in algorithms and legacy online news. The stimuli material differed in personal usage behavior, individual characteristics, the way users interacted with the NRS, and specific news search preferences. This was due to the algorithms' personalization in curating news, interaction with users, and reliance on existing data and the current trends in news agendas. The respondents downloaded an app from among a number of news apps, read news recommended through the chatbot, and were instructed to interact with all chatbot applications as much as they wanted. They concluded the experiment by completing a questionnaire measuring their attitudes, behaviors, and intention to share the news. To validate the reliability of the responses, a series of confirmation questions were included in the survey. Furthermore, data reduction was performed on the initial data set in relation to the reliability and validity of the responses.

Measures

Manipulation checks and multiple controls were performed for the experimental manipulation and measurement of the results.

News sharing. News sharing concerns individuals' intention to spread news and/or relevant information related to it to others on social media. Adapting prior measures on news-spreading intention (e.g., Dwyer & Martin, 2017), the participants ranked on a 7-point Likert-type scale (1 = strongly disagree and 7 = strongly agree) how likely they were to (1) suggest the news; (2) share the news; and (3) share opinions together with the news with others on social media ($\alpha = 0.88$, $M = 3.01$, $SD = 0.88$).

Algorithm Trust. This refers to whether people perceive algorithms as a more trustworthy processor of their data than operations by humans. It was measured in a similar manner as in previous research (Schmitt et al., 2021; Shin, 2022) by asking the participants to rank their agreement (7 = strongly agree and 1 = strongly disagree) with three items (scale adapted from Shin (2021)): (1) I think the algorithm is largely reliable and accurate in generating news recommendations, (2) I trust the algorithm in terms of data collection, analysis, and reflection, and (3) news recommendation by algorithms can be trusted due to their bias-free filtering ($\alpha = 0.87$, $M = 3.55$, $SD = 1.16$).

ANALYSES

Pearson's chi-square test was performed to test whether there were differences among the four conditions regarding the participants' characteristics and to analyze whether there were differences among the four conditions regarding sharing behaviors and the participants' demographics. The test showed that there were no significant differences in the sharing behaviors ($X^2(3) = 1.31$; $p = 0.712$) and the various conditions regarding the participants' gender and education ($X^2(3) = 2.19$; $p = 0.394$).

A GLM univariate analysis of variance (ANOVA) was conducted to compare the mean ratings of the dependent variables (news-sharing intention) of the accuracy nudge and nonnudge (independent variables). Furthermore, we analyzed the indirect effect of trust as a moderator of news sharing. A median split was used to categorize trust ($M_{\text{median}} = 3.20$) into low and high levels. Using Welch's F test, multiple one-way ANOVAs determined that the mean values of trust ($F(1, 99) = 219.72$; $p < 0.001$; $\eta^2 = 0.177$) and news sharing ($F(1, 99) = 38.84$, $p < 0.001$; $\eta^2 = 0.091$) were statistically significant between groups. Moreover, a pairwise t test was performed to determine the moderation effect of trust on the accuracy of the nudging and sharing intention. Cronbach's alpha was calculated to confirm the internal consistency of the constructs (Table 4.3).

For Hypothesis 1—the effect of accuracy nudges on news sharing—an ANOVA was conducted to compare the participants in the two conditions primed with accuracy nudges with participants receiving no alerts. The test showed a significant main effect ($F(1, 99) = 132$, $p < 0.001$). In fact, the participants who received the accuracy nudges perceived the news provided as more credible ($M_{\text{Nudge}} = 4.67$) compared to those who received no accuracy alerts ($M_{\text{Nonnudge}} = 2.95$).

Hypothesis 2 predicted a positive impact of the algorithm source on news sharing. A chi-square test of independence showed a statistically significant relation between the NRS and news sharing, $X^2(1, N = 199) = 4.78$,

Table 4.3 Experimental results

Group	1	2	3	4
News sharing (Cronbach's alpha: 0.73)	Algorithmic NRS		Legacy online portals	
	M:3.40	M:4.72	M:4.03	M:5.19
	SD:1.470	SD:1.048	SD:1.132	SD:1.483
Trust (Cronbach's alpha: 0.79)	M:3.21	M:4.98	M:4.21	M:5.04
	SD:0.806	SD:0.843	SD:0.718	SD:0.754
	Nudged	Nonnudge	Nudged	Nonnudge

$p < 0.01$. Thus, H2 was supported, as the participants who received news from the NRS relied significantly more on the nudged suggestions than those who received news from a legacy online source.

Hypothesis 3 hypothesized the moderation effect of trust on the accuracy of nudging. The results of our moderation analysis showed a significant interaction effect between the news sources and trust in news sharing ($F(1, 99) = 7.95, p < 0.001$). Pairwise *t* tests showed significant mean differences between all treatment groups. The pairwise comparisons between the algorithm source, low trust (M Algorithm source, Low trust = 5.23), and legacy source, low trust (M Nonalgorithms, Low Trust = 2.54), treatments ($p < 0.001$) and between the algorithm source, high trust (M Algorithms, High trust = 5.23), and legacy source, high trust (M Nonalgorithms, High trust = 3.12), treatments ($p < 0.001$) were significant. Similarly, the pairwise comparison between the accuracy nudge, high trust (M _{Nudge} High trust = 4.67), and the accuracy nudge, low trust (M Nudge, low trust = 2.54), treatments ($p < 0.001$) and between the nonnudge, high trust (M Nonnudge, High trust = 4.67), and nonnudge, low trust (M Nonnudge, Low trust = 3.06), treatments ($p < 0.001$) were significant.

The analysis reports that the accuracy nudge lowered the intentions of users willing to share false news and that the effect was greater in the algorithmically driven NRSs than in the nonalgorithmic sources. User willingness was also moderated by users' trust level in the source.

DISCUSSION

We have shown that accuracy nudges can effectively trigger user discernment in sharing and disseminating misinformation to others on social media (Table 4.4). This point corroborates previous findings (Pennycook & Rand, 2022; Roozenbeek et al., 2021; Nekmat, 2020) demonstrating how nudges as heuristic cues can alter user action in making sharing decisions. Furthermore, the findings extend those from previous studies by highlighting that nudges in the form of heuristic cues could effectively influence user action in algorithmic contexts. In line with a prior finding, attention is key to addressing misinformation, and a form of priming can help (Nekmat, 2020), as prompting an accuracy-check alert could trigger users' information processing, reminding them of the possible risks of sharing misinformation on social media. This nudge processing is consonant with ongoing research showing users' tendency to rethink their

Table 4.4 Review of hypotheses

<i>Hypotheses</i>	<i>Key findings</i>	<i>Results</i>
H1. The effect of accuracy nudges on news sharing	Accuracy nudges led to lower intention to share misinformation on social media.	Supported. *** $p < 0.001$
H2. The positive impact of the algorithm source on news sharing	Users who received news from algorithmic sources relied significantly more on the suggestions from this nudge than users who received news from legacy sources. Algorithmic sources had a significant effect on users' accuracy judgments.	Supported. ** $p < 0.01$
H3. The moderation effect of trust on the accuracy nudge	A significant interaction effect existed between news sources and trust for news-sharing behavior. When users had high trust in algorithms, they tended to receive more serious and positive accuracy nudges and less likely to share misinformation. When users had low trust in algorithms, they tended to consider accuracy nudging lightly and were more likely to share misinformation.	Supported. ** $p < 0.01$

intention to spread fake news in peer networks when they are being reminded of attention to veracity (Pennycook et al., 2020). Our results largely correspond to the literature but go a step further by clarifying the moderating effects of such a nudge effect on the discernment of misinformation. The characteristics of the news source and the levels of trust will likely influence the effectiveness of the accurate warning message. We posit that users more willingly accept nudges when they are suggested by algorithms (i.e., NRS) rather than by humans or nonalgorithm services. We hypothesize that this is driven by the belief that human-free nudges feel less judgmental and will thus allow for a greater sense of trust. This result contributes to the relevant literature on fake news and misinformation by clarifying how misinformation nudges can effectively influence users' intention to share misinformation on social media.

Algorithmic Source Effect

Our results show that algorithmic source credibility has significant effects on users' accuracy discernment in primed misinformation. The main effect found for the accuracy nudge is in line with previous findings (e.g., Nekmat, 2020), with users exercising greater caution toward sharing news

from algorithmic sources compared to news from online sources on social media. Our result goes beyond this finding by illustrating that users were generally unwilling to share news from algorithmic sources that were flagged as containing misinformation. Instead, this effect of users' intention to share news-primed misinformation when nudged by an accuracy alert was found to be stronger for news generated from an algorithmic source. This means that users are more aware that algorithmically curated and personalized news can be more vulnerable to fake news and misinformation and therefore more easily accept accuracy alerts when they are nudged to check the veracity and accuracy of the recommended news. This result can be explained from a theoretical perspective of source credibility and media credibility (Jia & Johnson, 2021). The user's judgmental confidence in algorithmic nudges is based on the user's perceived source credibility of algorithms. As users know that the news content is personalized by algorithms in line with what they expect or want to read, they more readily accept accuracy nudges from algorithms than a legacy news source because they believe that the nudges are part of the algorithmic advice or recommendation (Bryanov et al., 2020). Users of algorithmic NRSs deem accuracy nudges as algorithmically curated and confirmed and thus tend to believe them. Source credibility influences users' acceptance of message (nudge) credibility, and this is even more applicable to algorithmic sources than nonalgorithmic sources. In algorithmic contexts, message credibility is greatly influenced by source credibility, and the positive characteristics of the source affect the credibility of the nudge message.

Trust Moderation

This algorithmic source effect is related to trust. It can be inferred that the more users trust the algorithm's performance, the more likely they are to be nudged. This inference is supported by the identified interaction effect between trust and news sources, which provides important insights into the role of trust in misinformation and fake news. It is understandable that users consider both cues when forming accuracy judgments (Bak-Coleman et al., 2022). When users trust news sources, they tend to establish content credibility from these sources (Shin, 2021). A user's decision to assess the credibility of news is related to the news sources that they trust, that is, sources that play a more significant role in algorithmic sources than online sources. This implies the importance of the trust factor in algorithms—when users believe in algorithmic functions, they follow

algorithmic suggestions such as accuracy nudges. When users do not have trust or lack thereof in algorithms, they are less likely to accept accuracy nudges because they are not convinced about their quality. Thus, one can expect algorithmic trust to have an effect on user assessment and the sharing of suspected news that was the subject of a nudge. Our findings bridge misinformation research—which lacks empirical referents—with processes that link nudging and trust, particularly in the context of users’ sharing of misinformation in an algorithmic environment.

IMPLICATIONS

The goal of this study was to better understand, through a focus on algorithms, the processes contributing to users’ responses to misinformation. The results of our study are relevant from both theoretical and applied standpoints. Theoretically, the discussions in this study make a relevant contribution to the dynamics of nudges, misinformation, algorithmic source effects, and news sharing on social media. Understanding the spread of misinformation requires us to examine the factors influencing users’ discernment of misinformation. Evaluating the effect of accuracy nudges on news sharing from a user perspective can narrow the gap between theory and practice, as it transcends NRS-specific concepts and theories by focusing on users’ sensemaking process of misinformation and news sharing. A large part of online misinformation sharing is likely attributable to users’ inattention to accuracy when considering what to share (Pennycook et al., 2021).

When users are nudged by algorithms, their accuracy discernment increases with the interaction effect of algorithmic trust. This algorithmic source effect is arguably a theoretical advancement, as little is known about source effects in the context of algorithmically driven services. The results of our study are in line with previous research that underlined the function of nudging in influencing news sharing. We advance this finding by demonstrating that source credibility has an impact on misinformation sharing on social media and that nudge credibility encourages discerning and acknowledging misinformation. Users tend to believe in nudges when they rely on algorithms for news consumption, believing that nudges are part of algorithmic personalization and curation. This tendency increases when users have trust in algorithms. These results provide support for the theoretical argument concerning media source effects in algorithmic contexts. The algorithmic source effect implies the importance of trust and

credibility in algorithms. It also confirms the importance of trust in human interactions with algorithms (Shin et al., 2022) and further clarifies how credibility can be enhanced and how trust is established between users and algorithms. This theoretical implication further suggests that the effect of accuracy nudges should be understood from a user's sensemaking process: how users perceive the contextual factors regarding nudges, such as source, trust, and user traits. The results of this study show that drawing attention to the importance of accuracy can improve sharing judgment. Consistent with previous findings (e.g., Pennycook & Rand, 2022), this result indicates that accuracy alerts in news consumption not only nudge the way in which users perceive and consume news online but can also help users avoid unwittingly engaging in sharing misinformation. The cognitive processes of news sharing discernment have not been conceptually well defined or explicitly operationalized, particularly in the AI-embedded news platform context. Our attempt is meaningful because it empirically tested accuracy nudges from a user's sensemaking process to examine the nudge effects from a human perspective, not from a provider or algorithmic perspective.

The study shows the significance of accuracy nudges by illustrating how a nudge or cue that redirects attention to accuracy decreases the sharing of false content. This linkage serves as a foundation for theoretical contributions by establishing the relationship between accuracy and news sharing. This relation has also been touted a great deal, despite being conceptually established. It also raises the need for further research relating to habit formation and continuous behavior in a long-term context, especially given the lack of user research on these topics. As the experiment sessions were short and did not test longitudinal effects, the results did not have enough explanatory power to warrant longer-term effects. This could be a limitation of this study and could be pursued in future research.

Practically, the nudge model provides useful suggestions for policymakers and social media platforms on how to manage and operate accurate nudges. The results provide algorithm platform providers with guidelines on effective nudge prompts to choose from and cycle through when engaging users to increase the quality of information online. The results imply that an effective way to address misinformation is to prime users to reflect on the accuracy and veracity of the news they encounter (naturally pointing them toward discernment). Given the identified importance of accuracy nudges, social media firms and news platform providers should make fundamental changes to the social media ecosystem to systematically

redirect attention to accuracy. In particular, given the changeable nature of user attention, firms should determine interventions that last long enough to influence user behavior on a long-term basis. Furthermore, social media platforms may develop engagement strategies to disincentivize the sharing of misinformation online.

For example, a simple nudge isn't enough to combat misinformation, and social media platforms can leverage the nudging effect using inoculation approach, which involves preemptively alerting users of a probable attack on their opinions and exposing the manipulation techniques (or refuting the persuasive argument) prior their actual encounter of fake news. This method, called as prebunking, particularly helps give psychological resistance against impending attempts to mislead users with misinformation. These inoculations have been used many forms: Google Jigsaw uses free online games (*Harmony Square*, *Bad News*, and *Go Viral!*) in inoculating users before their actual exposure to real misinformation.

The AI-deploying journalism industry can use accuracy nudges as a guiding principle to program accurate news as well as implement accuracy nudges and misinformation warnings. Users' sensemaking and perceptions of algorithms and the ways in which they envisage and expect certain algorithmic nudges influence how they use and consume news through NRSs (Choi et al., 2022). As such, the industry could use algorithmic nudges as a reference to create feedback loops for the NRS. Algorithmic nudges not only benefit users by providing them with opportunities to understand how transparent, fair, and ethical the NRS is but also help journalism establish user trust in and engagement with their algorithmic nudging.

CONCLUSION

How effective is combating misinformation through nudges? This study discussed and illustrated nudges as a solution approach to misinformation that may create a feedback loop of misinformation and echo chambers. This study provided experimental proof of the effect of accuracy alerts on user discernment and behavior toward misinformation sharing from different sources on social media. It can be concluded that misinformation sharing on social media can be ascribed to users simply failing to judge accuracy when searching/consuming news. Our experimental evidence shows that priming attention toward the principle of accuracy can improve sharing judgment. The findings provide theoretically driven explanations

and a more nuanced understanding of the sensemaking process of user sharing of questioned news on social media. The potential of accuracy-check alerts to impact algorithm-based news media more than nonalgorithmic sources offers a heuristic research focus for future studies examining the algorithmic effect of nudges on the perceived utility, trust, and credibility of algorithmic news platforms. Despite ongoing research, many issues persist in relation to fake news and disinformation. We hope that the study will serve as a reference point for future investigations on accuracy and other nudges and calls to design effective nudges in managing the dissemination and flow of misinformation. Although our experiments demonstrate that source algorithmic credibility and trust have strong effects on accuracy judgment, follow-up works are warranted. Future research can examine the effectiveness of accuracy nudges using longitudinal methods and investigate possible moderators such as personal traits and credibility in AI.

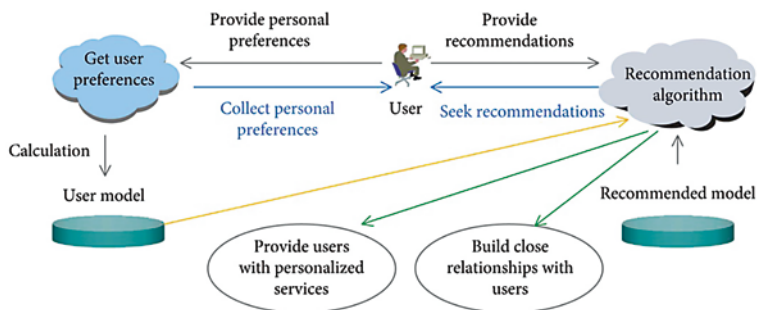
APPENDIX: EXPERIMENTAL STIMULI

Accuracy Alert System



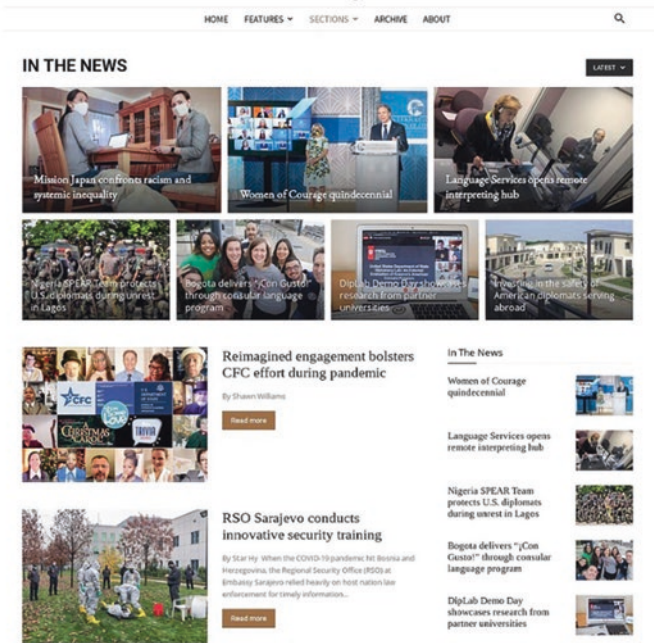
Accuracy Alert

Algorithm-Based Recommendations



Algorithm-based News Recommendation System

Legacy Online News (Nonalgorithm)



Legacy Online News (Nonalgorithm)

REFERENCES

- Acquisti, A., Adjerid, I., Balebako, R., Brandimarte, L., Cranor, L. F., Komanduri, S., et al. (2017). Nudges for privacy and security: Understanding and assisting users' choices online. *ACM Computing Survey*, 50(44). <https://doi.org/10.1145/3054926>
- Andi, S., & Akesson, J. (2021). Nudging away false news: Evidence from a social norm experiment. *Digital Journalism*, 9(1), 106–125. <https://doi.org/10.1080/21670811.2020.1847674>
- Baden, C., & Springer, N. (2017). Conceptualizing viewpoint diversity in news discourse. *Journalism*, 18(2), 176–194. <https://doi.org/10.1177/1464884915605028>
- Bak-Coleman, B., Kennedy, I., Wack, M., et al. (2022). Combining interventions to reduce the spread of viral misinformation. *Nature Human Behavior*. <https://doi.org/10.1038/s41562-022-01388-6>
- Bhuiyan, M., Vick, T., Mitra, K., & Horning, M. (2018). FeedReflect: A tool for nudging users to assess news credibility on Twitter. In *Proceedings of the ACM conference on Computer Supported Cooperative Work* (pp. 205–208). <https://doi.org/10.1145/3272973.3274056>
- Bryanov, K., & Vziatysheva, V. (2021). Determinants of individuals' belief in fake news. *PLoS ONE*, 16(6), e0253717. <https://doi.org/10.1371/journal.pone.0253717>
- Bryanov, K., Watson, B., Pingree, R., et al. (2020). Effects of partisan personalization in news portal experiment. *Public Opinion Quarterly*, 84(S1), 216–235.
- Cardenal, S., Aguilar-Paredes, C., Cristancho, C., & Majó-Vázquez, S. (2019). Echo-chambers in online news consumption. *European Journal of Communication*, 34(4), 360–376. <https://doi.org/10.1177/0267323119844409>
- Choi, S., Kim, H., & Gim, M. (2022). Do not read the same news! Enhancing diversity and personalization of news recommendation. In *Companion Proceedings of the Web Conference 2022*, April 25–29, Lyon, France. <https://doi.org/10.1145/3487553.3524936>
- Dwyer, T., & Martin, F. (2017). Sharing news online. *Digital Journalism*, 5(8), 1080–1100.
- Epstein, Z., Berinsky, A., Cole, R., Gully, A., Pennycook, G., & Rand, D. (2021). Developing an accuracy-prompt toolkit to reduce COVID-19 misinformation online. *Harvard Kennedy School Misinformation Review*, 2(3), 1–12. <https://doi.org/10.37016/mr-2020-71>
- Fazio, L. (2020). Pausing to consider why a headline is true or false can help reduce the sharing of false news. *Harvard Kennedy School Misinformation Review*, 1(2), 10.37016/mr2020-009.

- Helberger, N. (2019). On the democratic role of news recommenders. *Digital Journalism*, 7(8), 993–1012. <https://doi.org/10.1080/21670811.2019.1623700>
- Hermann, E. (2022). Artificial intelligence and mass personalization of communication content. *New Media & Society*, 24(5), 1258–1277. <https://doi.org/10.1177/14614448211022702>
- Jahng, M. (2021). Is fake news the new social media crisis? *International Journal of Strategic Communication*, 15(1), 18–36. <https://doi.org/10.1080/1553118X.2020.1848842>
- Jakesch, M., Hancock, J. T., & Naaman, M. (2023). Human heuristics for AI-generated language are flawed. *Proceedings of the National Academy of Sciences*, 120(11), e2208839120.
- Jia, C., & Johnson, T. (2021). Source credibility matters. *International Journal of Communication* 15, 3760–3781. 1932-8036/20210005
- Lazer, D. M. J., Baum, M. A., Benkler, Y., Berinsky, A. J., Greenhill, K. M., Menczer, F., et al. (2018). The science of fake news. *Science*, 359, 1094–1096. <https://doi.org/10.1126/science.aao2998>
- Lewandowsky, S., & van der Linden, S. (2021). Countering misinformation and fake news through inoculation and prebunking. *European Review of Social Psychology*, 32(2), 348–384. <https://doi.org/10.1080/10463283.2021.1876983>
- Mattis, N., Masur, P., Möller, J., & van Atteveldt, W. (2022). Nudging toward news diversity: A theoretical framework for facilitating diverse news consumption through recommenders design. *New Media & Society*. <https://doi.org/10.1177/14614448221104413>
- McQuail, D. (1992). *Media performance: Mass communication and the public interest*. Sage Publications.
- Mertens, S., Herberz, M., Hahnel, U., & Brosch, T. (2021). The effectiveness of nudging. *Psychological and Cognitive Sciences*, 119(1), e2107346118. <https://doi.org/10.1073/pnas.2107346118>
- Metzger, M., Flanagin, A., & Nekmat, E. (2015). Comparative optimism in online credibility evaluation among parents and children. *Journal of Broadcasting & Electronic Media*, 59(3), 509–529.
- Nekmat, E. (2020). Nudge effect of fact-check alerts: Source influence and media skepticism on the sharing of news misinformation in social media. *Social Media + Society*, 6(1), 1–14. <https://doi.org/10.1177/2056305119897322>
- Pennycook, G., Epstein, Z., Mosleh, M., et al. (2021). Shifting attention to accuracy can reduce misinformation online. *Nature*, 592, 590–595. <https://doi.org/10.1038/s41586-021-03344-2>
- Pennycook, G., McPhetres, J., Zhang, Y., Lu, G., & Rand, G. (2020). Fighting COVID-19 misinformation on social media. *Psychological Science*, 31(7), 770–780. <https://doi.org/10.1177/0956797620939054>

- Pennycook, G., & Rand, D. G. (2022). Nudging social media toward accuracy. *The ANNALS of the American Academy of Political and Social Science*, 700(1), 152–164. <https://doi.org/10.1177/00027162221092342>
- Rai, A. (2020). Explainable AI: From black box to glass box. *Journal of the Academy of Marketing Science*, 48(1), 137–141. <https://doi.org/10.1007/s11747-019-00710-5>
- Roozenbeek, J., Freeman, A. L. J., & van der Linden, S. (2021). How accurate are accuracy-nudge interventions? *Psychological Science*, 32(7), 1169–1178. <https://doi.org/10.1177/09567976211024535>
- Schmitt, A., Wambsganss, T., Soellner, M., & Janson, A. (2021). Toward a trust reliance paradox? ICIS 2021 proceedings. 14. https://aisel.aisnet.org/icis2021/ai_business/ai_business/14
- Shin, D. (2021). The perception of humanness in conversational journalism. *New Media & Society*. <https://doi.org/10.1177/1461444821993801>
- Shin, D. (2022). How do people judge the credibility of algorithmic sources? *AI & Society*, 37, 81–96. <https://doi.org/10.1007/s00146-021-01158-4>
- Shin, D., Zhong, B., & Biocca, F. (2020). Beyond user experience: What constitutes algorithmic experiences? *International Journal of Information Management*, 52, 102061. 1–11. <https://doi.org/10.1016/j.ijinfomgt.2019.102061>
- Shin, D., Ki, F. K., & Shin, E. (2023). The nudging effect of accuracy alerts for combating the diffusion of misinformation. *Journal of Broadcasting & Electronic Media*, 67(2), 141–160. <https://doi.org/10.1080/08838151.2023.2175830>
- Shin, D., Lim, J., Ahmad, N., & Ibarahim, M. (2022). Understanding user sense-making in fairness and transparency in algorithms. *AI & Society*. <https://doi.org/10.1007/s00146-022-01525-9>
- Sunstein, R., & Thaler, H. (2014). *Nudge: Improving decisions about health, wealth, and happiness*. Yale University Press.
- Talwar, S., Dhir, A., Singh, D., Virk, S., & Salo, J. (2020). Sharing of fake news on social media. *Journal of Retailing and Consumer Services*, 57, 102197. <https://doi.org/10.1016/j.jretconser.2020.102197>
- Vermeulen, J. (2022). To nudge or not to nudge: News recommendation as a tool to achieve online media pluralism. *Digital Journalism*. <https://doi.org/10.1080/21670811.2022.2026796>
- Vosoughi, S., Roy, D., & Aral, S. (2018). The spread of true and false news online. *Science*, 359(6380), 1146–1151. <https://doi.org/10.1126/science.aap9559>



Misinformation Processing Model: How Users Process Misinformation When Using Recommender Algorithms

INTRODUCTION

Misinformation has become a major concern in contemporary society, particularly posing a risk to highly vulnerable sectors. The widespread impact of misinformation has led to new imperatives for conducting challenging research to counter this harmful trend (Tully et al., 2020). Misinformation or fake news is spread online through sharing, resharing, and reposting (Chen & Cheng, 2020). Particularly in recent years, misinformation has taken on a form of great importance in the COVID-19 pandemic, the 2022 military conflict in Ukraine, and international tensions between the United States and other countries.

Thus far, various theories analyzing the diffusion and adoption of misinformation online have been developed (e.g., Kim et al., 2020). While there are manifold driving forces behind the spread of misinformation, relevant research has shown that its spread is related primarily to human cognitive dimensions (Pennycook & Rand, 2021; Shin, 2021). This view is based on the assumption that tackling the spread of misinformation is a multifaceted problem that requires a deeper understanding of individualized encounters and experiences so that a constructive solution can be identified. Personalized judgments in accepting and sharing information have become important, as they act as microlevel catalysts in the widespread proliferation of misinformation, even communally placing democratic decision-making at risk (Walter & Tukachinsky, 2020). It has been

argued that humans should be the focus when analyzing misinformation, as they are the underlying factors behind the social and technological repercussions of its spread (Shin et al., 2022). Thus, when assessing misinformation, it may be useful to systematically test theories on the problems associated with misinformation along with human reasoning and/or information processing models. Understanding the motivations and acceptability models of users regarding their cognitive level in leading them to believe misinformation can help combat misinformation and the spread of fake news (Pennycook et al., 2018; Rabb et al., 2022). Knowledge on how people process misinformation can provide insights into how it is produced, shared, legitimated, and amplified (Tully et al., 2020). The concerning impacts of misinformation and disinformation warrant robust empirical research examining how the processing of misinformation results in such impacts. Particularly for news organizations using artificial intelligence (AI), the gap between the impact of misinformation and the work of professionals involved in communication necessitates a better understanding of how users and misinformation influence each other, which is crucial as the industry and government strive to develop effective strategies to combat misinformation (Shin et al., 2022).

Against the backdrop of the rising problem of fake news, there have been increasing calls for modeling users' misinformation engagement and cognitive processes (Ahmed & Gil-Lopez, 2022). Modeling the perceptual and cognitive processes behind the spread of misinformation has been of interest for understanding the reasoning of users. Numerous theories from various fields have been formalized into misinformation models (e.g., Borukhson et al., 2022). Many methods for combating misinformation have been examined, and studies have investigated how people encounter and share false information or misinformation online (Bryanov et al., 2020). Since there are wide-ranging reasons regarding the use and spread of misinformation, it would be useful to examine what aspects of the experience can help users engage more critically with or more consciously respond to misinformation.

Despite their importance, the processing and mechanisms of misinformation, particularly those in algorithmic contexts, have been scarcely researched. In addition, the exact nature of how users respond to misinformation and its effects and impact on user behavior relating to misinformation and fake news remain inconclusive (Cho et al., 2020). While there is burgeoning research in the fact-checking literature on the utility and effectiveness of fact-checking practices (Jahng, 2021) and technical tools,

such as automated fact-checking (Epstein et al., 2021) and prebunking (Lewandowsky & van der Linden, 2021), there is a dearth of research focusing on the processing of misinformation by users and the ensuing effect of misinformation sharing behavior. Filling this gap in the literature requires better knowledge of the mechanisms that explain the process of misinformation in relation to the cognitive aspects of humans (Walter & Tukachinsky, 2020). We attempted to narrow this research gap by investigating how users respond/process misinformation in algorithmic contexts. As much misinformation these days is generated and amplified by AI (Kreps et al., 2022), understanding how users process misinformation in an algorithmic context could advance developments in theoretically understanding misinformation. Furthermore, it could help pave the way for professionals to incorporate the appropriate strategies needed to combat the spread of misinformation via AI.

To examine the cognitive and information processes related to how humans process misinformation, we integrated insights from two theoretical accounts, namely, the heuristic-systematic (HS) processing framework and the concept of information diagnosticity, to examine the antecedents and consequences of the processing of misinformation by users in algorithmic systems. Using the principle of diagnosticity and HS evaluations, we propose a conceptual model to explain the heuristic processing of users and how the perceived diagnosticity of misinformation may influence the intention of users exposed to misinformation via algorithms. We have expanded on the theoretical foundations of misinformation using information processing theory and diagnostic evaluation. Understanding the cognitive mechanisms underlying the effects of algorithmic contributions to misinformation on user behavior is critical in building a stronger theoretical background and countering misinformation spread via algorithms.

LITERATURE REVIEW: INFORMATION MISBEHAVIOR

Misinformation research, in general, has primarily concentrated on the overarching impact of content (Wilson & Maceviciute, 2022) and its dissemination mechanisms (Rabb et al., 2022), overlooking individual dimensions and differences in perceiving the potential consequences of such misinformation. Much research has been done on automatic fake news generation and detection of misinformation. However, research on how users cognitively deal with misinformation and the role of information processing regarding deepfakes have remained at the level of

rhetorical exhortation (Sundar et al., 2021). Previous research has yet to examine how individuals, based on their cognitive abilities, information processing capacity, and media literacy, might interpret the impact differently.

Effects of Misinformation on User Behavior

Misinformation is commonly defined as false or objectively incorrect information that is not supported by expert opinion or evident proof (Oh & Lee, 2019). Misinformation can take various forms; it can be completely fictitious or collate different versions of the truth, and it is fabricated to either intentionally or unintentionally mislead the public (Tandoc, 2019). Determining what is false or true is particularly difficult, especially in a critical situation such as the COVID-19 pandemic, since new scientific facts about the virus are introduced rapidly, possibly undermining already existing beliefs and information about the virus (Kim et al., 2020).

Misinformation can be differentiated from disinformation and misperception; disinformation is driven by a motive to mislead (Vraga & Bode, 2020), whereas misperception upholds a belief that is false or incorrect (Kim et al., 2020). Misinformation can be unintentionally false, and its dissemination or sharing can subsequently be either accidental or intentional. Although the phenomenon of misinformation spreading has been around a while the profoundness of misinformation and its detrimental impact are only beginning to become more pronounced: misinformation is more easily transmitted by sharing features on social media, and the social media algorithm that favors engagement in the form of likes, retweets, and clicks could also lead to “popular” content being mistaken as credible information without any prudent judgment regarding its accuracy (Shin, 2021). In the context of social media, disinformation with a manipulative intent can be shared later when perceived to be true, thereby becoming misinformation. For example, in the Coca-Cola Dasani water parasite case, false information was first disseminated on a website in the form of fake news and later disseminated widely on social media. Such misinformation could undesirably put firms and brands in the spotlight and cause longer-term harm to their corporate reputation and consumer trust if no appropriate response is provided (Jahng, 2021). To effectively combat such misinformation and mitigate its detrimental effects, industry professionals need to understand the mechanism through which it could possibly impact user responses.

Currently, with the emergence of AI, it has been poised as a threat due to its ability to spread misinformation at a fast pace and scale. While AI-powered systems provide an effective and powerful solution to counter misinformation through algorithm-based systems, such as automatic fact-checkers, AI-powered recommendation systems, such as news recommender systems (NRS), can easily sway voter perceptions; people tend to believe the news recommended from such systems without much consideration, as they believe that the recommendations presented are algorithmically and technically processed. The spread of misinformation through such AI-enabled systems has largely contributed to the spread of misinformation and false information (Kreps et al., 2022). To solve this problem, newsrooms and social media platforms are working on automatically detecting misinformation and fake news through AI, natural language processing, and machine learning. A trained algorithm will identify misinformation as “fake news” and assign it a lower rank to decrease the probability of users encountering it. However, algorithm detection remains unreliable, primarily because current detection methods only require the evaluation of text and the social network to which it belongs when verifying its credibility. Although AI can determine the origin of sources and the pattern of fake news dissemination, it is fundamentally limited in how it assesses the actual meaning of the content in terms of veracity and credibility. With a large amount of training data, an AI-driven classification model can detect misinformation and fake news in an article. However, making such detections requires social knowledge, prior understanding of political and cultural orientations, and common sense, which current algorithmic detection systems still lack. Thus, a feasible solution is to rely more on human judgment and behavior modeling and less on algorithmic processing (Hermann, 2022).

Considering this, the focus has now been on increasing user literacy regarding misinformation. Assessing the literacy of users when viewing misinformation critically requires an understanding of their cognitive and psychological processing capabilities (Amazeen & Krishna, 2023). Numerous studies have examined the acceptance and diffusion of misinformation from the perspective of information processing by users (e.g., Hwang & Jeong, 2021). Thus, conceptually and empirically examining the antecedents of misinformation discernment as well as the effect of overarching human reasoning models on the processing of misinformation is important. A need for a better understanding of the motivations of users to believe or be susceptible to misinformation led us to discern the adoption and proliferation of misinformation (Talwar et al., 2020).

Heuristic-Systematic Misinformation Processing

The influence of information processing on misinformed decision-making can be considered a dual process; information is processed within both an intuitive and autonomous system and within a more analytical reflective system using systematic processing methods. The HS model of information processing was developed to analyze the behavioral changes of individuals based on two types of information processing: heuristic and systematic (Chaiken, 1980). The HS theory suggests that when people engage in heuristic evaluation, they tend to process information based on available heuristic cues or simple decision rules. People use mental shortcuts (heuristics) to make judgments. In contrast, when they are involved in systematic processing, they thoroughly evaluate the information to check its validity. The heuristic evaluation of users involves asking questions such as the following: (1) How do users discern the qualities or features of messages? (2) How do people perceive them and for what reasons? Since misinformation is presented in numerous forms and functions, it is important to understand the a priori preferences of users and how these preferences are processed. The manner in which a user's sense of diagnosticity influences their attitudes and satisfaction, which subsequently influences their decisions (Clarke & McComas, 2012), is also critical. The HS theory is used as a prism to measure a user's sensemaking of misinformation; this model clarifies the role that algorithmic features play in forming the sensemaking of users regarding algorithms and how their actions influence this sensemaking. This theory is fit for this task since users may use a simplified form of quality judgment, as AI is a specialized field, and its related concepts are hypothetical (Shin, 2021). When users complete heuristic evaluations and their diagnosticities are established, they then use a systematic approach wherein they think prudently about the veracity of the misinformation when deciding whether the messages are true. Attitudes are then formed based on the conclusions drawn from meticulously analyzing the systematic evaluation. Research has shown that users who engage more in ratiocination are less likely to accept misinformation (Ahmed & Gil-Lopez, 2022; Hwang & Jeong, 2021). Thus, HS theory can be used in the context of misinformation to extend the information processing model by examining the antecedents of user heuristics and the consequences of systematic processing in the context of algorithmic misinformation.

Research Questions

With the prevalence of misinformation, the manner in which people perceive and process misinformation becomes a legitimate question when deciding solutions for handling misinformation (Hwang & Jeong, 2021). Previous research on the adoption of misinformation has mainly addressed the degree to which reason and reflection deter or facilitate the development of correct assessments (Pennycook & Rand, 2022). However, research on how users cognitively deal with misinformation and the role of HS processing regarding misinformation has remained at the level of rhetorical exhortation. We examine the cognitive processing of misinformation by users through the examination of the antecedents of user heuristics and the role of diagnosticity in mediating heuristic and systematic processes. The overall premise of our model is based on misinformation processing in an algorithmic environment that is dependent on both heuristic and systematic processing. The manner in which users process misinformation and the mechanism behind the spread of misinformation have remained open-ended questions; accordingly, the following research questions (RQs) are proposed:

- RQ1: How do users process misinformation? How do users interpret misinformation?
- RQ2: What are the effects of FAccT on users' heuristic assessments of misinformation?
- RQ3: How does the systematic assessment of misinformation influence how users share misinformation?

With the increasing concern over misinformation, we aim to advance our understanding of a potential remedy for the current worldwide infodemic. Considering the RQs, we propose a misinformation processing model in algorithms that incorporates both heuristic and systematic processes. Although other scholars have proposed different models for understanding how misinformation is processed (e.g., Chen & Cheng, 2020), they have not considered misinformation in an AI context and have not extensively included aspects unique to misinformation, such as diagnosticity. Since users actively seek AI to fulfill a variety of goals, diagnosticity functions need to be included in misinformation processing. The proposed RQs will steer us toward understanding the sharing behavior of people and how they can become better at discerning misinformation

through their HS processing. Our findings can contribute to managing misinformation in NRSs and thereby gently steering users away from misinformation and fake news.

COGNITIVE ACCOUNT OF MISINFORMATION EVALUATION

Information processing, which was developed in cognitive science, assumes that there are inputs and outputs from humans (Chaiken, 1980). The proposed model is rooted in information processing theory to explore the HS processing of misinformation in AI systems. FAccT are posited as antecedents of diagnosticity, and explainability and accuracy are proposed as systematic processes. Our experiential model was designed to test the HS theory in misinformation discernment. Our hypotheses helped in questioning the sensemaking of users when presented with misinformation recommended by algorithms and the effect of such sensemaking on their sharing behavior and discernment of misinformation.

Heuristic Processing

The heuristic assessment of misinformation involves evaluating FAccT (Shin et al., 2022). Despite their obvious benefits, AI-powered NRSs raise a series of ethical issues and pose new risks for user rights. Concerns include a lack of algorithmic fairness, transparency, accountability, content personalization resulting in filter bubbles, manipulation of potential recommendations, or the generation of fake news. Users want to verify the FAccT of the systems that spread misinformation. Examining the ethical factors—such as fairness, bias, equitability, and auditability—of the methods developed to counter the spread of misinformation is critical.

Algorithmic Fairness in Misinformation. Algorithmic fairness is a research field on the correction and removal of algorithmic biases, such as ethnicity, gender, and race biases, from machine learning systems (Mhasawade et al., 2021). Since more important decisions are being made by algorithm-driven models and platforms, biases in AI are increasingly concerning, and there is an increased potential for harm. To design fair algorithms, researchers strive to understand and correct biases by tracing them in data and algorithms, defining and applying measurements of fairness, and developing data collection and modeling methodologies aimed at creating such algorithms. With the theory of algorithmic information processing, Shin (2021) shows that algorithmic fairness hinges upon the

perceptual impartiality and equitability of users regarding AI. Related research has confirmed this argument by clarifying the relation between algorithmic fairness and user attitudes, particularly user efficacy and confidence (Diakopoulos & Koliska, 2017). Since the self-efficacy of users is related to their diagnosticity, a connection between algorithmic fairness and the diagnosticity of users in misinformation can be proposed:

H1 The algorithmic fairness perceived by users positively affects the diagnosticity of users when viewing misinformation.

Algorithm Accountability. According to Diakopoulos (2016), algorithmic accountability is defined as determining whether a decision is made under substantive and procedural standards and holding entities accountable if they fall short of these standards. Since algorithms are designed by humans, recommendation algorithms can have built-in human biases and/or errors. Accountability is particularly important in being responsible for the spread of misinformation, as the repercussions of the spread, amplification, and diffusion of misinformation are immense. There is a view that social media giants should be held accountable for spreading misinformation and provide further clarification when requested by the public. Related research shows that users develop positive confidence with clearly defined responsibilities and liabilities when they correctly judge incorrect information as misinformation (Barnoy & Reich, 2022). Given the ongoing findings, it is worthwhile to examine how users perceive algorithmic accountability in relation to misinformation and how their perceptions influence their diagnosticity of misinformation.

H2 Algorithmic accountability perceived by users positively influences their diagnosticity of misinformation.

Algorithm Transparency. Transparency in AI is defined as a principle under which both the inputs and outputs of algorithms should be visible and comprehensible to users or the public (Lavigne et al., 2022). Gran et al. (2021) examined the influence of algorithmic transparency on the psychological attitudes of individuals, such as their beliefs or intentions in evaluating algorithms correctly. Other research has confirmed that transparency enhances user trust in an algorithm, playing a key role in whether users use it (Peifer & Meisinger, 2021). When transparent and visible processes are enforced, people are more likely to view the given information

in a more trustworthy way (Diakopoulos & Koliska, 2017). Transparent AI systems can afford users a sense of assurance, presenting them with a readiness to diagnose misinformation.

H3 Algorithmic transparency perceived by users positively influences their diagnosticity of misinformation.

Diagnosticity. A widely used concept in consumer psychology and marketing communications, diagnosticity is commonly defined as the extent to which people interpret a piece of information to be useful and relevant according to their judgment and decision-making (Ahluwalia et al., 2001). Research over the years has found diagnosticity to be the underlying driving factor behind the attitudes and behavioral decisions of users (Jiang & Benbasat, 2005; Kwon et al., 2020; Lynch, 2006). In algorithmic contexts, information that is perceived to be highly diagnostic is given more weight in the judgments and evaluations of users (Shin et al., 2022); thus, diagnosticity influences how users systematically process messages. Upon realizing the diagnosticity of a system, users trust it and begin to systematically assess misinformation by measuring its accuracy and credibility (Chen & Cheng, 2020). The indicators for diagnosticity could be specific to tasks and goals, and the concept of diagnosticity could be operationalized based on the tasks and goals (Lynch, 2006). Regarding misinformation from AI-powered media, the indicators of diagnosticity are uniquely specific to the users' understanding of the accuracy and credibility of the information (i.e., the correctness and reliability of the suggested information). Accordingly, H4 and H5 have been proposed.

H4 Diagnosticity perceived by users positively affects the accuracy of information.

H5 Diagnosticity perceived by users positively affects the credibility of information.

Systematic Processing

Users conduct reflective assessments of messages wherein they assess the accuracy and credibility of the messages (Diakopoulos, 2016).

Accuracy. The perceived accuracy of a piece of information is the user's perception of the amount of misinformation in this information

(Pennycook & Rand, 2022); this refers to how truthfully the user perceives a viewed piece of information. From the system perspective, accuracy is concerned with the extent to which an NRS predicts items that people have already rated or interacted with; recommender systems with optimized precision prioritize more strongly related items for their users (Shin, 2021). The accuracy in an NRS is related to how many results are correctly personalized and the veracity of the recommended content. Regarding the measurement of the perceived accuracy of misinformation, relevant studies have devised different names to describe it, with one being called misinformation awareness, which is a person's awareness in knowing that the received news is false (Schuetz et al., 2021). There are also those who call it awareness of misinformation, which is how much a person knows about the news they receive (Shin et al., 2022). Empirical research has confirmed these relationships in diverse algorithm services (Pennycook et al., 2018), in which accuracy is found to be a determinant of attitude and behavioral intention (Shin, 2021). Accordingly, the following relation is proposed:

H6 Perceived accuracy positively influences the intention of a user to correct/share misinformation.

Credibility. As a strong predictor of the influence of a message, credibility can generally be understood as “believability” and “trustworthiness” (Wathen & Burkell, 2002). In communication-related research, credibility can be measured from different perspectives, such as source, message, and media credibility (Barnoy & Reich, 2022; Metzger et al., 2015). While credibility has historically been studied more heavily from the source perspective, in the social media environment, news consumers place more attention on the content of the message than on the source (Peifer & Meisinger, 2021). Thus, the credibility of the content/message is worth investigating in the context of misinformation in algorithmic media. The more credible the content of a piece of misinformation is perceived to be, the more useful and relevant (i.e., diagnostic) users would perceive it to be.

H7 Content credibility perceived by users regarding misinformation in news influences their intention to correct/share misinformation.

Moderating Effect of Explainability

Explainability has been a critical component in the field of explainable AI (Rai, 2020), which includes machine learning and AI technologies that can provide human-comprehensible rationales for their outputs or processes (Shin, 2021). As the complexity of AI systems and algorithms grows, people increasingly consider them “black boxes” that complicate understanding in the sense that increasing amounts of specialized expertise and knowledge are required to understand the decision or performance of AI systems (Gunning et al., 2019). The concerns related to the decision-making of AI tools being a “black box” are significantly worsened when considering how average people who lack technical knowledge are required to deal with misinformation in AI systems. Explainability is important in building trust, rapport, and understanding between the AI agent and its user, especially in regard to understanding misinformation. Explainability presents users with assurance and diagnostic confidence in evaluating the accuracy and credibility of misinformation (Rai, 2020). Relevant research has shown that explainability has a positive effect on user assessments of AI functions (Shin, 2021). The effects of explainability have been well researched, unlike the moderating effect of explainability on the path of diagnosticity to systematic evaluations. Given the strong effects of explainability explored in previous studies, it is worthwhile to test the moderating role of explainability in misinformation evaluation.

H8 The effect of diagnosticity perceived by users regarding misinformation on accuracy is moderated by the explainability of the messages in AI.

H9 The effect of diagnosticity perceived by users regarding misinformation on credibility is moderated by the explainability of the messages in AI.

Based on the constructs and hypotheses presented in this study, we propose a conceptual model that depicts how users process misinformation related to the harm of a product and its antecedents and consequences (see Fig. 5.1).

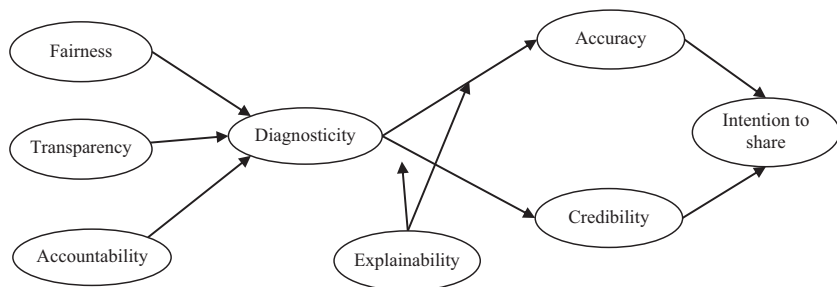


Fig. 5.1 Conceptual model

METHODS

We applied the HS model to the processing of misinformation in the algorithmic context; using the HS model, we tested FAccT as a construct that influenced the heuristic evaluation and systematic discernment of misinformation by users.

Collection of Data and Samples

We conducted a survey to examine the attitudes and behaviors of Korean users (Table 5.1). Panel participants were recruited from Qualtrics, a private survey sampling firm. Written informed consent was obtained from all the subjects prior to their participation in the experiment. A representative sample ($N = 731$) with demographic data comparable to the census information was collected. After applying quality control, filtering questions, and checking items, 410 disqualified responses were dropped, resulting in a remainder of 321 qualified responses for the final analysis. Prior to running the tests on the sample group, a 1-month pretest was conducted using 30 college students with experience using AI algorithms, services, and applications.

In the experiment, a mock NRS that contained misinformation (fake news and false information) was used. This experimental NRS provided users with contemporary news containing both real information and misinformation on COVID-19 and political campaigns. The experiment was designed to test how users processed misinformation suggested by algorithms. Upon entering the online research site, subjects were asked to surf through NRS platforms and search/read recent hot news (news currently

Table 5.1 Descriptive statistics

<i>Characteristics</i>	<i>Sample</i>
Age (Mean/SD/Median)	25.30/11.02/29
Gender (female rate)	50.98
College educated (%)	33.18
OS platform experience	2.8 years

being discussed and proliferating online) generated by algorithms for approximately 2–3 h. Then, the respondents were provided with a predetermined list of news items on COVID-19 and political campaigns with related misinformation. After searching and reading this news, they were required to complete a survey. As compensation for their time spent participating in the experiment, the subjects were given class credits and presented with opportunities to pursue undergraduate/graduate courses related to misinformation, news media, and user-generated marketing.

Scales and Measurements

The scales used in this study consisted of 21 measurements with a 7-point scale ranging from 1 (strongly disagree) to 7 (strongly agree). All the measurements were based on the literature on digital media and research on misinformation. The measurements were administered using combinations of previously developed metrics and modified metrics from previous versions. Such combinations were necessary because certain measurements in our analysis needed to be changed to reflect the distinctive features of the AI algorithms and services.

These measurements were then compared and contrasted with measurements from the literature verified in other studies. This two-step process was necessary to ensure reflecting user perspectives and to warrant consonance with existing measurements used in other AI services. The FAccT, explainability, and diagnosticity measurements were performed based on the works of Shin (2021), Rai (2020), and Chen and Cheng (2020), respectively. Some measurements required changes to reflect new traits in the misinformation. Ten users with prior experience with misinformation completed a pretest about a specific news topic. The goal of the pretest was to identify any potential problems that participants may have in either understanding or interpreting a question. The pretest was necessary to evaluate the practicability of the FAccT concepts in the

misinformation context. The results of the pretest revealed problems the subjects faced related to searching for information, viewing content, answering questions, and completing a survey, thus helping to identify weaknesses in the administration, organization, and execution of the test. The sequence of components in the pretest was modified in the actual experiment to facilitate understanding of the subjects. Explanatory stimuli were presented as cues in the presented misinformation, popping up as phrases when the subjects clicked on recommended news. These cues provided users with relevant explanations, such as “because you watched it before” or “most similar [to] watched news,” on why specific news was recommended. The explanatory cues made it possible for respondents to understand, interpret, and experiment with the misinformation.

Measurement Instrument

The validity and reliability of the measurement items were assessed based on the reliability and convergent validity criteria using Analysis of Moment Structure (AMOS) 18 software. The Cronbach’s alpha of the survey instrument was calculated to test its reliability. The results of this analysis were checked, and varimax rotation was performed on the original 30 items (7 items were eliminated due to low loading). According to Hair et al. (1995), an item has a high loading if its loading coefficient is above 0.6. Each value was above the recommended level of 0.8 (Cronbach, 1989). Confirmatory factor analysis was performed to test the validity of the instruments, and the composite reliability, estimated percentage of variance extracted by each construct, and discriminant and convergent validity were examined. All the constructs in the model met the required reliability and convergent validity values (Tables 5.2 & 5.3). Regarding the discriminant validity, the average variance extracted (AVE) values and the correlations of the factors were examined. The shared variance between the constructs was lower than the AVE from the individual constructs, confirming the presence of discriminant validity. A correlation analysis was performed, and Pearson’s R (correlation coefficient) showed an acceptable level of correlation among the variables. In conclusion, the measurement model demonstrated adequate reliability and the presence of convergent and discriminant validity.

Table 5.2 Discriminant validity

<i>Construct</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>
Fairness	0.71							
Accountability	0.21	0.83						
Transparency	0.33	0.52	0.83					
Accuracy	0.13	0.23	0.52	0.93				
Credibility	0.12	0.11	0.23	0.42	0.72			
Diagnosticsity	0.23	0.53	0.13	0.52	0.11	0.82		
Explainability	0.42	0.22	0.53	0.21	0.53	0.13	0.82	
Intention	0.52	0.11	0.22	0.22	0.22	0.53	0.23	0.73

Table 5.3 Reliability checks for constructs

<i>Construct</i>	<i>Initial items</i>	<i>Final items</i>	<i>Mean (SD)</i>	<i>Cronbach's alpha</i>
Fairness	4	3	4.17 (1.011) 3.84 (1.055) 4.11 (1.047)	0.795
Accountability	4	3	4.17 (1.011) 3.84 (1.055) 4.11 (1.047)	0.619
Transparency	5	3	3.39 (1.054) 3.98 (1.128) 3.30 (1.282)	0.875
Accuracy	4	3	3.74 (1.346) 3.87 (1.385) 3.40 (1.411)	0.812
Credibility	4	3	4.37 (1.182) 4.30 (1.141) 4.63 (1.202)	0.842
Diagnosticsity	4	3	4.37 (1.182) 4.30 (1.141) 4.63 (1.202)	0.853
Explainability	3	3	4.34 (1.210) 4.44 (1.295) 4.05 (0.984)	0.856
Intention	3	3	4.42 (1.121) 4.10 (1.264) 4.18 (1.268)	0.900

-Fit indices

A test of the structural model was conducted using the AMOS procedure, and maximum likelihood-based structural equation modeling software was employed. Table 5.4 reveals the estimates from the structural modeling procedure. The results indicated the presence of no misfits. All values were within acceptable ranges, indicating good overall model performance.

Structural Model Testing

Structural path analysis was performed; the results validated all the relationships in the hypotheses (Table 5.5). The seven path coefficients were statistically significant ($p < 0.001$ or $p < 0.0001$). Perceived diagnosticity was significantly affected by heuristic evaluations and simultaneously influenced the systematic processing of misinformation. The heuristic factors accounted for 59.1% of the variance in the diagnosticity. The intention behind sharing information was greatly influenced by the systematic processing of misinformation, with systematic factors explaining 32% of the variance in the intention to share information. The model explained a significant percentage of the variance in each factor. The significant paths

Table 5.4 Model fit indices

<i>Fit statistics</i>	<i>Structural model</i>	<i>Suggested value (Joreskog & Sorbom, 1996)</i>
χ^2 (df)	1710.5/244 = 7.01	< 5
p value	0.000	< 0.05
RMSEA	0.023	$0.05 < x < 0.10$
CFI	0.910	> 0.90
NFI	0.978	> 0.90
GFI	0.920	> 0.90
IFI	0.802	> 0.80
TLI	0.901	> 0.85
RFI	0.803	> 0.80
AIC	1870	
Hoelter's critical N	66(0.05)/70(0.01)	A higher value signifies a satisfactory fit
ECVI	4.688	A smaller value denotes a better fit

Table 5.5 Path results

<i>Paths</i>	<i>Estimate</i>	<i>Standard error (S.E.)</i>	<i>Critical ratio (C.R.)</i>	<i>P label</i>
H1: Fairness → Diagnosticity	0.610	0.129	4.739	***
H2: Accountability → Diagnosticity	0.512	0.043	11.840	***
H3: Transparency → Diagnosticity	0.457	0.047	9.791	***
H4: Diagnosticity → Accuracy	0.642	0.067	9.517	***
H5: Diagnosticity → Credibility	0.649	0.076	8.499	***
H6: Accuracy → Intention	0.396	0.063	6.292	***
H7: Credibility → Intention	0.221	0.087	2.525	0.012*

*1.96: 95% (0.05), **2.58: 99% (0.01), ***3.29: 99.9% (0.001)

implied a causal conceptual link between the heuristic and systematic processes through diagnosticity.

Moderating Effects of Explainability

We tested the moderating effects of explainability on the systematic processing of diagnosticity in the model. The explanatory cues in the NRS provided users with explanations on why and how the recommended news was presented. They also helped users systematically assess misinformation (Shin, 2021). Testing the moderating role of explainability in the path of diagnosticity on accuracy/credibility was worthwhile. We adopted a method used by Wu (2018) in the multigroup analysis conducted using AMOS 26 to evaluate the structural paths from the model across the experimental (Group 1) and control (Group 2) groups (Table 5.6). The two models showed equally good fits with the data, and distinct patterns that supported the hypotheses were observed. The results showed notable distinctions in item composition and path formation, providing insights into the effect of explainability on misinformation (Table 5.6). The effects of diagnosticity on credibility and accuracy were higher in Group 1 than in Group 2. Comparing the two models, Group 1 showed higher values for systematic evaluations, whereas Group 2 showed higher values for

Table 5.6 Moderating effects of explainability

<i>Path</i>	<i>Group 1 (Explanatory cue)</i>					<i>Group 2 (Control group)</i>				
	β	S.E.	C.R.	<i>p</i>	<i>Result</i>	β	S.E.	C.R.	<i>p</i>	<i>Result</i>
Diagnosticity- Credibility	0.684***	0.050	13.727	0.000	Yes	0.066	0.041	1.592	0.111	No
Diagnosticity- accuracy	0.106***	0.032	3.301	0.000	Yes	0.037	0.050	0.750	0.454	No
Accuracy- intention	0.328***	0.021	1.734	0.000	Yes	0.166*	0.113	1.434	0.005	Yes
Credibility- intention	0.421***	0.081	3.492	0.000	Yes	0.337**	0.054	0.342	0.01	Yes

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; β : Unstandardized coefficient

heuristic evaluations. It can be inferred that explanatory cues played a significant role in the systematic and heuristic processes, and the effects were greater on the systematic processes than they were on the heuristic processes. The results of the squared multiple correlations also supported the underlying role of explainability (Table 5.7). The R^2 values of the accuracy and credibility in Group 1 were greater than those in Group 2. Moreover, the R^2 of the intention of Group 1 was distinctly higher than that of Group 2. Additionally, there was a notable interaction effect between explainability and heuristic processing. As shown in Fig. 5.2, the effect of explainability on misinformation diagnosticity was greater for those who engaged in a higher level of heuristic processing. It can be inferred that the heuristic processing of misinformation moderated the relationship between the explainability and judgment of misinformation such that the effect of the explainability on the diagnosticity was stronger when the users engaged in a greater level of FAccT evaluation.

Table 5.7 Comparison of squared multiple correlations

	<i>Group 1</i>	<i>Group 2</i>
Accuracy	0.391	0.183
Credibility	0.425	0.088
Intention	0.394	0.111

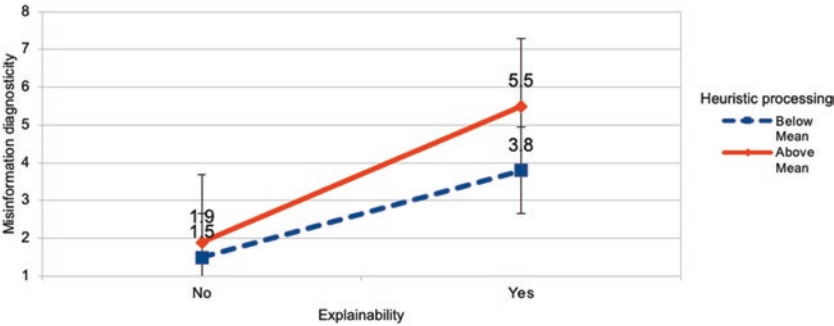


Fig. 5.2 Interaction role of heuristic processing in the effect of explainability on diagnosticity

DISCUSSION: ALGORITHMIC MISINFORMATION

We addressed how misinformation in the algorithmic context influenced the heuristic, systematic, and diagnostic assessment of users in discerning misinformation. The results from the analyses showed that the heuristic and systematic processing of the users significantly influenced their attitudes and intentions; however, such influence was mediated by the perceived diagnosticity and the explainability embedded in the AI system. Together, explainability and diagnosticity were the key facilitators through which misinformation influenced the discernability and sharing behavior of users.

Theoretical Implications: Algorithmic Effects on Misinformation

This study offers the relevant theoretical implications of algorithmic effects on misinformation (Cho et al., 2020). First, the processing of misinformation using algorithms is an underresearched field (Lewandowsky & van der Linden, 2021). By emphasizing the processing of misinformation in algorithmic systems, we offer a model integrating both the antecedents of heuristic processing and the consequences of systematic processing that can be reproduced and extended in different contexts. This modeling can fill a research gap of “user engagement of misinformation” that previous studies have examined (e.g., Ahmed & Gil-Lopez, 2022) but have not clarified the nature of user engagement and its components and process. In addition, previous research has mostly examined the unidimensional effects of the psychological impact of misinformation on the behavior of users (Lazer et al., 2018). We extended the current understanding by explicating the sensemaking of users in processing misinformation. We took an information processing approach, considering the cognitive and psychological development of users and how the processing in the evaluation influenced the discernment of misinformation. Our approach encompassed relevant factors of the cognitive psychological and communication processes and helped enrich the understanding of the relationship between how misinformation forges the attitudes of users and how their attitudes toward the content and source of algorithm-curated misinformation influenced the emergence of misinformation; this coevolving relation was facilitated by diagnosticity. While previous studies only highlighted the role of diagnosticity (e.g., Chen & Cheng, 2020), we clarified the underlying relation of diagnosticity with algorithmic features: when a news item is

recommended by algorithms in transparent, fair, and accountable ways, users perceive a high level of diagnosticity, which improves their intention to analyze systematically. While previous studies explored concepts such as self-efficacy and awareness, our study clarified the processing capacity of users through the introduction of a new perspective of diagnosticity and applying it in the AI context.

The results of the experiment revealed that the diagnosticity perceived by the users was positively influenced by FAccT, which indicated that the more confident the algorithmic media users were that the systems were ethically sound and normatively acceptable, the more likely they were to diagnose the information conscientiously. Such results are consistent with earlier findings by Ahmed and Gil-Lopez (2022) that users with lower cognitive ability are more likely to believe and share misinformation. Our findings are consistent with earlier reports of users' understanding of algorithmic functions and the associated ethical issues (Diakopoulos, 2016; Mhasawade et al., 2021). It was inferred that when algorithm users had a higher level of perceived diagnosticity, they also became more confident and capable of analyzing the accuracy and credibility of the information presented through AI and consequently tended to react to misinformation more conscientiously. Diagnosticity in misinformation can result from heuristic processing and from being an activator in conducting systematic processing. The proposed model showed that diagnosticity played a mediating role between heuristic and systematic processing in misinformation discernment. Diagnosticity was offered as a reason to switch from heuristic to systematic processing, showing that normative values gave users a sense of diagnostic readiness for handling misinformation, leading to the effective systematic evaluation of misinformation. This mediating role was consistent with the moderating effect of explainability. The effect of diagnosticity on systematic processing was also moderated by explainability, indicating that when there were certain forms of explainable cues, users tended to be more diagnostic and thus would be more alert in assessing the accuracy and credibility of the misinformation generated by algorithms.

Second, this study expands research on HS processing in the area of misinformation. While misinformation has been a major problem throughout the pandemic and beyond, relatively few studies have explored the process of misinformation (particularly from the users' point of view), with many only studying it as an antecedent of user behavior, such as debunking (Hwang & Jeong, 2021). The scarcity of research on the

procedural aspect of misinformation could potentially create limited views on it. Although plentiful research has been conducted and a series of factors behind the effects of misinformation have been shown, limited research has been conducted regarding how users are influenced by misinformation (Kim & Tandoc, 2022), especially regarding that presented by AI-powered algorithms in the case of news. Thus, this study narrows the gap by researching both the antecedents and consequences of misinformation in the contemporary algorithmic context. The results confirmed the proposition that diagnosticity is a critical construct (Ahluwalia et al., 2001) and further showed the different ways misinformation is processed in algorithmic contexts.

Finally, we also illustrated that HS processing in algorithms concurrently affected the intention of users to process misinformation via dual paths. Prior work has primarily focused on the unidirectional influence of misinformation from a functional viewpoint (Borukhson et al., 2022), with few studies holistically examining the dual processing of misinformation. Our attempt narrowed the gap by clarifying how ethical and normative dimensions could increase user diagnosticity and how users evaluated the accuracy and credibility of misinformation. The results further demonstrated that explainability had a moderating effect on the path of diagnosticity and systematic evaluations. In an algorithmic context, users strongly valued ethical media for communication, and such findings further augmented the theoretical framework by confirming diagnosticity as a key mediator in shaping how users manage misinformation.

PRACTICAL IMPLICATIONS

Our study contributes to a shared understanding across the broader scientific community and technology industry of how social technology companies can better address integrity issues on their platforms. Our findings offer relevant practical guidelines for algorithmic media regarding improving users' understanding of managing misinformation and how it influences their heuristic and procedural knowledge of it. Since diagnosticity is greatly influenced by the ethical foundations of a system, practitioners should not only engage in establishing the FAccT of the system but also adopt strategies to increase the perceived notion of diagnosticity of users. It is critical to examine both the presented information and user traits when practitioners are evaluating suitable venues to combat misinformation. In particular, the credibility of information on online platforms and

media is low; the distrust of users on such platforms may result in a disinterest in misinformation. Companies specializing in online platforms may invest resources and develop features to monitor misinformation on such platforms, offer valuable links to help users check or evaluate the credibility of information online, and engage in ethical and responsible operations to meet the trust expectancy of users. Enhanced trust in the platform exists when there are high levels of diagnosticity and established information credibility. Thus, when choosing platforms to counter misinformation, practitioners may reveal the source sharing the misinformation to relate the misinformation with the source instead of the platform through which the information was presented. It is also important to increase the capacity and efficacy of users in processing misinformation. Diagnosticity may depend on the technological features of the system, with the perceived diagnosticity of the users hinging greatly on how they view the system regarding its FAccT. While efforts should be undertaken to increase the technical features of both diagnosticity and accuracy, practitioners should also emphasize the importance of establishing trust and credibility to enhance the literacy of users dealing with misinformation.

LIMITATIONS AND FUTURE RESEARCH

Our efforts have explored how users process and respond to misinformation in algorithmic contexts, contributing to the literature on misinformation. The study also extends HS theory by applying it in the context of misinformation on algorithms and examining the antecedents and consequences of diagnosticity in this context. Like any other study, the findings of this study should be considered carefully, as it has limitations that need to be addressed in future work.

First, although our study proposes a misinformation processing model in the context of algorithms, it mostly addressed cognitive processing within a single-user schema, excluding the algorithmic side of automatic processing. To further test the model more comprehensively, future research should include the algorithmic features of information processing using the model. Further comparisons and investigations should be made regarding how and to what extent different dimensions in the assessment processing model are shaped by the HS processing of users and in turn

influence the perceived diagnosticity of misinformation and subsequent behavior of users. In addition, as we collected data at a single point in time, future studies may examine long-term exposure to misinformation and its effects on people with longer and deeper engagement.

Second, prior research on the processing of misinformation mentioned the role of trust, as people seemed more accepting of information when they trusted the media that presented it (Chen & Cheng, 2020). The existing trust of users regarding media may play a significant role in processing misinformation. As this study did not consider trust in the processing of misinformation, future research should integrate diverse messages with varying levels of initial trust and investigate how prior trust levels may influence how users process misinformation. In doing so, misinformation processing may be related to a broader trust dimension, as mentioned in prior literature.

Third, our results showed the connection among FAccT, diagnosticity, and credibility, thereby shedding light on the criticality of misinformation literacy or algorithm literacy for users. These results also implied the distinctive role that diagnosticity processing may play in the susceptibility of users to misinformation. As the role of algorithm literacy is going to increase drastically (Shin et al., 2022), future studies should also test the connections among algorithm literacy, trust, and diagnosticity to further explicate what may determine the efficacy and capacity of users when exposed to misinformation via algorithms.

Despite such drawbacks, our study positively contributes to research on the information processing model by considering this model in the context of diagnosing and discerning misinformation. We believe that the model offers a comprehensive framework that is useful for any scholars interested in the behavior of users exposed to misinformation in an AI environment. This model will foster thinking and provoke excitement among researchers attempting to test and advance theories and solutions regarding misinformation problems in the algorithmic medium. Future studies can further address the underlying processes of misinformation behavior in various contexts, considering the effects of personal traits on these processes.

REFERENCES

- Ahluwalia, R., Unnava, H. R., & Burnkrant, R. E. (2001). The moderating role of commitment on the spillover effect of marketing communications. *Journal of Marketing Research*, 38(4), 458–470. <https://doi.org/10.1509/jmkr.38.4.458.18903>
- Ahmed, S., & Gil-Lopez, T. (2022). Engaging with vilifying stereotypes: The role of YouTube algorithmic use in perpetuating misinformation about Muslim congresswomen. *Journalism & Mass Communication Quarterly*. <https://doi.org/10.1177/10776990221110113>
- Amazeen, M., & Krishna, A. (2023). Processing vaccine misinformation: Recall and effects of source type on claim accuracy via perceived motivations and credibility. *International Journal of Communication*, 17, 560–582.
- Barnoy, A., & Reich, Z. (2022). Trusting others: A Pareto distribution of source and message credibility among news reporters. *Communication Research*, 49(2), 196–220. <https://doi.org/10.1177/0093650220911814>
- Borukhson, D., Lorenz-Spreen, P., & Ragni, M. (2022). When does an individual accept misinformation? An extended investigation through cognitive modeling. *Computational Brain & Behavior*, 5, 244–260. <https://doi.org/10.1007/s42113-022-00136-3>
- Bryanov, K., Watson, B. K., Pingree, R. J., & Santia, M. (2020). Effects of partisan personalization in a news portal experiment. *Public Opinion Quarterly*, 84(S1), 216–235. <https://doi.org/10.1093/poq/nfaa011>
- Chaiken, S. (1980). Heuristic versus systematic information processing and the use of source versus message cues in persuasion. *Journal of Personality and Social Psychology*, 39(5), 752–766. <https://doi.org/10.1037/0022-3514.39.5.752>
- Chen, Z. F., & Cheng, Y. (2020). Consumer response to fake news about brands on social media: The effects of self-efficacy, media trust, and persuasion knowledge on brand trust. *Journal of Product & Brand Management*, 29(2), 188–198. <https://doi.org/10.1108/JPBM-12-2018-2145>
- Cho, J., Ahmed, S., Hilbert, M., Liu, B., & Luu, J. (2020). Do search algorithms endanger democracy? An experimental investigation of algorithm effects on political polarization. *Journal of Broadcasting & Electronic Media*, 64(2), 150–172. <https://doi.org/10.1080/08838151.2020.1757365>
- Clarke, C., & McComas, K. (2012). Seeking and processing influenza vaccine information: A study of health care workers at a large urban hospital. *Health Communication*, 27(3), 244–256. <https://doi.org/10.1080/10410236.2011.578332>
- Cronbach, L. J. (1989). Construct validation after thirty years. In R. L. Linn (Ed.), *Intelligence: Measurement, theory and public policy* (pp. 147–171). University of Illinois Press.

- Diakopoulos, N. (2016). Accountability in algorithmic decision making. *Communications of the ACM*, 59(2), 58–62. <https://doi.org/10.1145/2844110>
- Diakopoulos, N., & Koliska, M. (2017). Algorithmic transparency in the news media. *Digital Journalism*, 5(7), 809–828. <https://doi.org/10.1080/21670811.2016.1208053>
- Epstein, Z., Berinsky, A., Cole, R., Gully, A., Pennycook, G., & Rand, D. (2021). Developing an accuracy-prompt toolkit to reduce COVID-19 misinformation online. *Harvard Kennedy School Misinformation Review*, 2(3), 1–12. <https://doi.org/10.37016/mr-2020-71>
- Gran, A., Booth, P., & Bucher, T. (2021). To be or not to be algorithm aware: A question of a new digital divide? *Information, Communication & Society*, 24(12), 1779–1796. <https://doi.org/10.1080/1369118X.2020.1736124>
- Gunning, D., Stefik, M., Choi, J., Miller, T., Stumpf, S., & Yang, G. (2019). XAI: Explainable artificial intelligence. *Science Robotics*, 4(37), 7120. <https://doi.org/10.1126/scirobotics.aay7120>
- Hair, J. F., Jr., Anderson, R. E., Tatham, R. L., & Black, W. C. (1995). *Multivariate data analysis* (3rd ed.). Macmillan Publishing Company.
- Hermann, E. (2022). Artificial intelligence and mass personalization of communication content. *New Media & Society*, 24(5), 1258–1277. <https://doi.org/10.1177/14614448211022702>
- Hwang, Y., & Jeong, S. (2021). Misinformation exposure and acceptance: The role of information seeking and processing. *Health Communication*, 23(1), 1–9. <https://doi.org/10.1080/10410236.2021.1964187>
- Jahng, M. R. (2021). Is fake news the new social media crisis? Examining the public evaluation of crisis management for corporate organizations targeted in fake news. *International Journal of Strategic Communication*, 15(1), 18–36. <https://doi.org/10.1080/1553118X.2020.1848842>
- Jiang, Z., & Benbasat, I. (2005). Virtual product experience: Effects of visual and functional control of products on perceived diagnosticity and flow in electronic shopping. *Journal of Management Information Systems*, 21(3), 111–147. <https://doi.org/10.1080/07421222.2004.11045817>
- Joreskog, K., & Sorbom, D. (1996). *LISREL 8: User's reference guide*. Scientific Software International.
- Kim, H. K., Ahn, J., Atkinson, L., & Kahlor, L. A. (2020). Effects of COVID-19 misinformation on information seeking, avoidance, and processing: A multi-country comparative study. *Science Communication*, 42(5), 586–615. <https://doi.org/10.1177/1075547020959670>
- Kim, H. K., & Tandoc, E. C., Jr. (2022). Consequences of online misinformation on COVID-19: Two potential pathways and disparity by eHealth literacy. *Frontiers in Psychology*, 13, 783909. <https://doi.org/10.3389/fpsyg.2022.783909>

- Kreps, S., McCain, R., & Brundage, M. (2022). All the news that's fit to fabricate: AI-generated text as a tool of media misinformation. *Journal of Experimental Political Science*, 9(1), 104–117. <https://doi.org/10.1017/XPS.2020.37>
- Kwon, Y., Park, J., & Son, J.-Y. (2020). Accurately or accidentally? Recommendation agent and search experience in over-the-top (OTT) services. *Internet Research*, 31(2), 562–586. <https://doi.org/10.1108/INTR-03-2020-0127>
- Lavigne, M., Bélanger, É., Nadeau, R., Daoust, J.-F., & Lachapelle, E. (2022). Hide and seek: The connection between false beliefs and perceptions of government transparency. *Harvard Kennedy School Misinformation Review*, 3(2). <https://doi.org/10.37016/mr-2020-90>
- Lazer, D. M. J., Baum, M. A., Benkler, Y., Berinsky, A. J., Greenhill, K. M., Menczer, F., Metzger, M. J., Nyhan, B., Pennycook, G., Rothschild, D., Schudson, M., Sloman, S. A., Sunstein, C. R., Thorson, E. A., Watts, D. J., & Zittrain, J. (2018). The science of fake news. *Science*, 359(6380), 1094–1096. <https://doi.org/10.1126/science.aao2998>
- Lewandowsky, S., & van der Linden, S. (2021). Countering misinformation and fake news through inoculation and prebunking. *European Review of Social Psychology*, 32(2), 348–384. <https://doi.org/10.1080/10463283.2021.1876983>
- Lynch, J., Jr. (2006). Accessibility-diagnosticsity and the multiple pathway anchoring and adjustment model. *Journal of Consumer Research*, 33(1), 25–27. <https://doi.org/10.1086/504129>
- Metzger, M., Flanagin, A., & Nekmat, E. (2015). Comparative optimism in online credibility evaluation among parents and children. *Journal of Broadcasting & Electronic Media*, 59(3), 509–529. <https://doi.org/10.1080/08838151.2015.1054995>
- Mhasawade, V., Zhao, Y., & Chunara, R. (2021). Machine learning and algorithmic fairness in public and population health. *Nature Machine Intelligence*, 3, 659–666. <https://doi.org/10.1038/s42256-021-00373-4>
- Oh, H. J., & Lee, H. (2019). When do people verify and share health rumors on social media? The effects of message importance, health anxiety, and health literacy. *Journal of Health Communication*, 24(11), 837–847. <https://doi.org/10.1080/10810730.2019.1677824>
- Peifer, J. T., & Meisinger, J. (2021). The value of explaining the process: How journalistic transparency and perceptions of news media importance can (sometimes) foster message credibility and engagement intentions. *Journalism & Mass Communication Quarterly*, 98(3), 828–853. <https://doi.org/10.1177/10776990211012953>
- Pennycook, G., Cannon, T. D., & Rand, D. G. (2018). Prior exposure increases perceived accuracy of fake news. *Journal of Experimental Psychology*, 147(12), 1865–1880. <https://doi.org/10.1037/xge0000465>

- Pennycook, G., & Rand, D. G. (2021). The psychology of fake news. *Trends in Cognitive Sciences*, 25(5), 388–402. <https://doi.org/10.1016/j.tics.2021.02.007>
- Pennycook, G., & Rand, D. G. (2022). Accuracy prompts are a replicable and generalizable approach for reducing the spread of misinformation. *Nature Communications*, 13, 2333. <https://doi.org/10.1038/s41467-022-30073-5>
- Rabb, N., Cowen, L., de Ruiter, J., & Scheutz, M. (2022). Cognitive cascades: How to model (and potentially counter) the spread of fake news. *PLoS ONE*, 17(1), e0261811. <https://doi.org/10.1371/journal.pone.0261811>
- Rai, A. (2020). Explainable AI: From black box to glass box. *Journal of the Academy of Marketing Science*, 48, 137–141. <https://doi.org/10.1007/s11747-019-00710-5>
- Schuetz, S. W., Sykes, T. A., & Venkatesh, V. (2021). Combating COVID-19 fake news on social media through fact checking: Antecedents and consequences. *European Journal of Information Systems*, 30(4), 376–388. <https://doi.org/10.1080/0960085X.2021.1895682>
- Shin, D. (2021). The perception of humanness in conversational journalism. *New Media & Society*, 24(12), 2680–2704. <https://doi.org/10.1177/1461444821993801>
- Shin, D., Zaid, B., Biocca, F., & Rasul, A. (2022). In platforms we trust? Unlocking the black-box of news algorithms through interpretable AI. *Journal of Broadcasting & Electronic Media*, 66(2), 235–256. <https://doi.org/10.1080/08838151.2022.2057984>
- Sundar, S., Molina, M., & Cho, E. (2021). Seeing is believing: Is video modality more potent in spreading fake news via online messaging apps? *Journal of Computer-Mediated Communication*, 26(6), 301–319. <https://doi.org/10.1093/jcmc/zmab010>
- Talwar, S., Dhir, A., Singh, D., Virk, G. S., & Salo, J. (2020). Sharing of fake news on social media: Application of the honeycomb framework and the third-person effect hypothesis. *Journal of Retailing and Consumer Services*, 57, 102197. <https://doi.org/10.1016/j.jretconser.2020.102197>
- Tandoc, E. C., Jr. (2019). The facts of fake news: A research review. *Sociology Compass*, 13, 1–9. <https://doi.org/10.1111/soc4.12724>
- Tully, M., Bode, L., & Vraga, E. K. (2020). Mobilizing users: Does exposure to misinformation and its correction affect users' responses to a health misinformation post? *Social Media + Society*, 6(4). <https://doi.org/10.1177/2056305120978377>
- Vraga, E. K., & Bode, L. (2020). Defining misinformation and understanding its bounded nature: Using expertise and evidence for describing misinformation. *Political Communication*, 37(1), 136–144. <https://doi.org/10.1080/10584609.2020.1716500>

- Walter, N., & Tukachinsky, R. (2020). A meta-analytic examination of the continued influence of misinformation in the face of correction: How powerful is it, why does it happen, and how to stop it? *Communication Research*, 47, 155–177. <https://doi.org/10.1177/0093650219854600>
- Wathen, C., & Burkell, J. (2002). Believe it or not: Factors influencing credibility on the Web. *Journal of the American society for information science and technology*, 53(2), 134–144.
- Wilson, T. D., & Maceviciute, E. (2022). Information misbehavior: Modeling the motivations for the creation, acceptance and dissemination of misinformation. *Journal of Documentation*, 78(7), 485–505. <https://doi.org/10.1108/JD-05-2022-0116>
- Wu, Y. (2018). How age affects journalists' adoption of social media as an innovation. *Journalism Practice*, 13(5), 537–557. <https://doi.org/10.1080/17512786.2018.1511821>

PART III

How to Combat Misinformation
Online Amid Growing Concerns and
Build Trust



Misinformation and Diversity: Nudging Away from Misinformation Nudging Toward Diversity

DIVERSITY IN AI

Information diversity in online ecosystems has long been a fundamental and foundational basis for ensuring that the communicative needs of individuals and society at large are met. In the AI era, people increasingly rely on content filtering and recommender systems to use information contesting the traditional concept of news diversity (Helberger, 2019). The exact definition of diversity in the context of rising AI will need to be revisited, rendering interdisciplinary collaboration among computer scientists, social scientists, and legal scholars. Across disciplines, scholars agree that achieving diversity in AI recommendations is one step toward addressing misinformation.

DIVERSITY-AWARE AI

Is AI inclusive of all humans, cultures, and perspectives or have humans passed on their biases to AI? How does AI lead to exclusion or misrepresentation for diverse people in society? AI often functions quite as a double-edge sword by reinforcing current stereotypes, amplifying inequality, and further pushing some groups into categories that do not represent them. The way data are collected, trained, analyzed, and ingrained into our societies can have unintended outcomes unless we critically investigate the risks of systems left untested and can lead to homophobic,

misogynistic, racist, transphobic, ageist, or ableist biases. Diversity, equity, and inclusion (DEI) hold significant importance because we want to ensure that when an AI system makes a call, it will be able to make accurate decisions. As AI poses a threat to DEI, initiatives have been proposed to improve algorithm design to help devise strategies and counter narratives that redirect ongoing discourses toward a more equitable and more inclusive AI.

Diversity-aware AI is one such movement with the goal of improving algorithms with respect to the values of diversity and inclusion. A public consensus has been reached that AI should be more than diversity-aware or DEI competent. Algorithms should be able to take diversity into account in all possible analyses. Diversity is about recognizing, reflecting, and valuing differences based on gender, ethnicity, age, disability, and religion. Inclusion occurs when humans feel and are valued and respected regardless of their personal characteristics or conditions. Equity refers to the fact that every human can participate equally and freely in areas of public life without disadvantage or less favorable treatment due to their unique attributes (Recchiuto & Sgorbissa, 2022).

Diversity-aware AI is based on enabling meaningful interaction among semantically autonomous agents incrementally making sense of each other. It assumes that an AI system is a collective where humans may have a different understanding of the world but may mutually benefit from each other despite this diversity. The overall intelligence of this collective increases with the amount of interaction among agents capable of sharing meaning despite their diversity. Shared meaning is produced when external input (output) can be processed (produced) in such a way that it does not violate the values held by the receiving (sending) agent.

Diversity awareness will introduce a crucial innovation in AI. It will produce AI that can reconfigure their behavior to recognize and value the uniqueness of humans they interact with to promote respect for DEI values, not only for health and political debates but also for cooperation in education, social services, and public services.

NUDGE THEORY

Nudge theory is widely accepted and used in numerous contexts in which positive reinforcement and indirect influences guide the behavior and decision-making of individuals or groups (Sunstein & Thaler, 2014). Because of its effectiveness, nudging has been widely used in user

analytics, system design, and public policy areas (Mertens et al., 2021). For example, enhanced password visualization tools (illustrating password strength levels) have been shown to nudge users to form safer passwords (Tsavli et al., 2015). In the digital media literature, nudges have been researched and used as a concept of affordance (e.g., media affordance), allowing users a greater choice to control what, when, and how they view content online (de Ridder et al., 2022). Numerous studies have examined nudging in a digital context as an indirect form of using design, interactive, and psychometric components to guide user behavior in online environments without limiting the user's freedom of choice (Kroll & Stieglitz, 2021).

Recent research has considered nudging for media applications, such as the evaluation of media credibility in digital media (Bhuiyan et al., 2018) or the affordance of social interactions (Shin, 2021). Meanwhile, nudging has become a hot research topic in AI and algorithms. AI nudging, also termed algorithmic nudging, is defined as the use of algorithm design components to lead user behavior in algorithmic mediated contexts (Möhlmann, 2021) and is used widely in machine learning and AI services (Jesse & Jannach, 2021). Algorithmic nudging is a subtle way of framing algorithm behavior by nudging user behavior that has been used increasingly and widely (Juneja & Mitra, 2022) to affect user behavior and improve user service. Retailers use AI nudges to engage with their consumers in a highly personalized context that leads them to make better decisions and choices (Zingales, 2018). Raveendhran and Fast (2021) conceptualized AI nudging as the utilization of cognitive stimuli to influence users' behavior in a desired way without constraining their choices or modifying their motivations. Due to the opaque nature of algorithms, algorithmic nudges often function stealthily and covertly without being noticed (Jesse & Jannach, 2021). AI nudges have been used to solidify the primary functions of algorithms, silently shaping what users see, from the posts in social feeds to the suggested content they see and to the advertisements covering their online interface (Choi et al., 2022).

Currently, news platforms use nudges at different dimensions in AI algorithms, such as news recommending services, content purchasing suggestions, and prescriptive decision-making tools (Vermeulen, 2022). Although the consequences are still under debate and elusive, researchers have examined the relationship between AI and nudges, arguing that algorithmically personalized results can influence users and often lead to unintended consequences (i.e., self-reinforcing, echo chamber; Cardenal et al.,

2019), partisan personalization (Bryanov et al., 2020), and unwanted habits (i.e., selective exposure, addictive confirmation; Burr & Cristianini, 2019; Scheiber, 2021). The relationship between AI and nudges found in numerous studies illustrates how personalized tailored algorithms can use persuasion and psychometrics to affect individual behavior in unintended ways (Burr & Cristianini, 2019). Concerns about hyperpersonalized news—via behavioral and contextual data—in the algorithm environment (Yeung, 2017) have raised questions on how news recommender systems (NRS) can be more user-centered and less algorithm-driven (Hermann, 2022) to facilitate rather than constrain the normative goals of journalism (Joris et al., 2020). A number of obstacles stand in the way of NRS and the legitimate use of algorithmic nudges in news platforms (Choi et al., 2022). The prevalence of NRS has caused concerns about information cocoons, echo chambers, filter bubbles, and information enclaves. To keep users on a provided platform, users are fed content they like based on their interests in the name of personalization, without considering inclusive content or diverse viewpoints. Despite its importance, the effects and mechanisms of algorithmic nudges on user behaviors, particularly in NRS, have been underresearched, and the exact nature of how algorithmic nudges work with what effect, how users respond to the nudges, and what impact nudges have on news personalization and the diversity of viewpoints regarding issues of public interest via NRS remain inconclusive. How do algorithmic nudges influence user decisions, and to what extent are users likely to accept such nudges? This question is even more important because NRS is designed to provide personalized news, but there are issues arising about whether personalized news leads users to echo chambers and filter bubbles (Cardenal et al., 2019). For this reason, there is a need to design NRS to allow users to have options to access a diverse spectrum of news beyond personalized news (Raza & Ding, 2022). The need for diversity-aware news recommenders (Mattis et al., 2022) is growing and becoming increasingly imperative for journalistically sustainable, algorithmically efficient, and societally acceptable personalized services (Heitz et al., 2022).

In this light, we examine the effect of algorithmic nudges on user behaviors in NRS. Our objective is to contribute to conceptualizing and operationalizing algorithmic nudges in the context of NRS and broader personalization AI. To this end, we develop an understanding of algorithmic information processing, leading to trust in personalization and news diversity, with implications for both design and ethical models. Focusing

on user sensemaking instead of algorithmic logic will uncover and overcome current weaknesses (Shin & Ahmad, 2023; Schobel et al., 2020) and criticisms in algorithmic design and will potentially inform future governance and management (Yeung, 2017).

The implications of this study are relevant both theoretically and practically. While convenient and useful, algorithmic nudges raise a series of ethical concerns about privacy, information disclosure, manipulation, and tweaking (Schobel et al., 2020). The potential of a single algorithmic nudge to influence millions of users instantaneously indicates the need to meaningfully control nudges in personalization contexts (Möhlmann, 2021). The challenge is how to ensure that algorithmic nudges are used in a responsible and transparent way that is compatible with journalistic values (content pluralism and diversity of news) and whether the nudge could also help to achieve a sustainable way of life. This study contributes to the principles and dimensions of the algorithmic nudging effects of NRS on user behavior and evaluates how people can, in turn, nudge algorithmic systems to produce user-centered results. We conclude that algorithmic nudges can influence users' way of sensemaking personalization as well as their intention to consume diverse news by affecting user beliefs about either service features or how to value those features. We propose NRS as a tool to achieve news diversity and content pluralism through algorithmic nudges. Algorithm nudging plays a critical role in personalization; at the same time, diversity and inclusion should be a part of personalization.

LITERATURE REVIEW

The AI environment has radically changed the way users browse, search, consume, and experience news. The growing relevance of platformized news and algorithmic intermediaries for news generation and consumption increases the extent of personalization and more effective forms of news automation.

Algorithmic Nudges and Behavior Modifications

Almost all online platforms currently regularly utilize algorithmic behavior modification techniques to shape user behaviors in a way that maximizes profits (Shin et al., 2022). Platforms increasingly utilize techniques to flexibly and automatically tailor behavioral interventions to exploit human psychological traits and cognitions (Raveendhran & Fast, 2021). Platforms

are created to perform selective gathering, algorithmic filtering, amplification, and monetization of user data and are equipped with autonomous, data-driven, prescriptive, and intelligent algorithms to control users' behavior at scale (Laban & Araujo, 2020). Like digital breadcrumbs through social media posts or pages on Facebook, people leave their traces on the online portals they log into and through the search queries they enter in search engines. These digital traces make for a useful and predictive behavioral data reservoir, which platforms use for behavior medication when deciding what news to show, what ads to expose, and what information to avoid. By giving these digital traces to algorithms and platforms, we allow them to infringe on our privacy at times in unexpected ways and fine-tune their recommendation-targeting efforts. Platforms can increase their prediction accuracy by molding users' behaviors toward their predicted values, employing behavior modification techniques, and showing more positive predictions. Such seemingly enhanced predictions can unintentionally result from the use of reinforcement learning algorithms that integrate prediction and behavior modification.

In journalism, particularly algorithmic journalism, algorithmic nudging can be employed as a means to increase news diversity in NRS (Jürgens & Stark, 2022; Mattis et al., 2022) by reranking news items in the interface (Karimi et al., 2018). News networks, such as *CNN*, the *BBC*, *USA Today*, and *The New York Times*, provide ubiquitous access to users, allowing them to surf through relevant news using algorithm platforms. To draw a higher volume of clicks to their sites, online platforms increasingly utilize algorithmic nudges in their NRS to improve user engagement in their sites. Loecherbach et al. (2020) illustrated that less preferred news items are more likely to be chosen if they are displayed at higher ranks or positioned favorably on the page. When used correctly, algorithmic nudges can increase news diversity considerably (Bernstein et al., 2022), especially when readers face news overload or when their news selections are recommended by AI systems. Algorithmic nudges can support people who pursue but do not find diverse articles by making such articles more accessible and available or by providing unexpected recommendations that prevent unwanted algorithmic feedback loops. These unexpected recommendations lead users to develop new interests, which in turn enables recommendation algorithms to advance (Raza & Ding, 2022), contributing to greater user satisfaction. Readers' desire to seek diverse news is in line with the right to diverse information, as specified by the European Court of Human Rights (Article 10), which requires governments to adopt

measures that guarantee their citizens access to a diversity of information. Platform behavioral change methods are becoming increasingly invisible and opaque and are done without clear user approval. Although platforms' behavioral change is clear to the user, for example, as presented recommendations, advertising, or autocomplete functions, it is normally invisible and untraceable to third-party auditors. These algorithmic nudges work invisibly and can be operated without external supervision, amplifying filter bubbles or echo chambers (Dahlgren, 2021).

Currently, research on algorithmic nudging is limited in scope and operation (Greene et al., 2022). The entire interaction of user behavioral data and the associated algorithm data employed for behavior modification and prescriptions is completely unavailable to the public. It is almost impossible to access the operational datasets of platforms that allegedly maneuver behavior modifications. Thus, research in this area has focused on users' attitudes, behavior, and sensemaking of algorithmic nudges.

Nudging Toward Inclusion and News Diversity

Algorithmic journalism is increasingly seeking implicit and explicit algorithmic news personalization (Cardenal et al., 2019). This trend creates tension with news diversity, which has been an essential basis for media and journalism (Bernstein et al., 2022). As NRS engineers' technological architecture helps shape public discourse, it has reduced news diversity and social diversity (Choi et al., 2022). As algorithmic personalization depends primarily on automated recommendation systems, these systems are often neither fair nor transparent because they are mainly driven by commercial interest and technological efficiency. This nonpublic value contrasts with the normative principles of news as a marketplace of ideas in which readers are entitled to a wide spectrum of news and viewpoints (McQuail, 1992). As NRS is widely used and accepted, diversity in recommendation systems has become an important issue (Vrijenhoek et al., 2021) and the principle by which to consider avoiding the overfitting problem as well as a polarization of opinion (Currin et al., 2022), which poses a major threat to democratic discussion (Helberger, 2019). This importance comes from the fact that news diversity in the media has been a key principle for safeguarding the satisfaction of the communicative rights of individuals and society at large (Baden & Springer, 2017).

Numerous scholars warn of the potential threats of NRS, such as echo chambers and filter bubbles (Cardenal et al., 2019). Among the concerns

raised, scholars and practitioners have commonly asked questions about how NRS can be designed in a more user-centered way that preserves the normative goals of news journalism (Reuver et al., 2021). Optimizing the role of public values in news recommendations is an important task, which should be followed by defining and operationalizing the diversity principle in NRS. Algorithmic nudges are proposed as a possible solution to this question. This study underlines the theoretical development of algorithmic nudges in NRS that are aware of news diversity and inclusion as key drivers for personalized NRS. Such theoretical frames can provide practical implications for the technological feasibility of diverse news recommendations in NRS. Relevant research has examined nudging toward inclusion and news diversity, which can inform the conceptualization of diversity-aware news recommendations (e.g., Vrijenhoek et al., 2021). Helberger (2019), for example, proposed normative ideals and measurable metrics of diversity and inclusion in NRS. Despite recent research, realizing and reflecting news diversity in the recommendation process remains under-researched. To fully benefit from news recommendations, users should engage with diverse news and opinions in ways that lead to the marketplace of ideas and prosocial media effects.

Investigating the user response to diverse news is critical since more diversity does not necessarily result in a depolarizing capacity for democratic societies. Excessive diversity can be countereffective and can make users turn off from news, as it distracts user attention (Reuver et al., 2021). Promoting maximum diversity is undesirable, and thus, diversity-aware NRS should identify the optimum point at which recommendations are inclusive while being functionally effective and satisfactory to users (Bryanov et al., 2020).

In sum, despite surging interest in NRS, research on how NRS can facilitate news diversity and inclusion has remained underresearched. How to design nudges in AI that guide users toward better decisions ethically and responsibly remains an open question for NRS and journalism. Given the literature gap reviewed and considering the importance of diversity in NRS, we propose the following research questions:

- RQ1: How do algorithms nudge users toward diverse news exposure?
- RQ2: What are the effects of algorithmic nudges on users' behaviors on news consumption, particularly attitudes toward news diversity?
- RQ3: How can the notion of algorithmic personalization be reconceptualized and opened up for diversity-aware NRS through the lens of users?

With these questions in place, this study examines the effects of an algorithmic nudge on user behavior by examining how diversity-aware NRS can promote diverse and meaningful news consumption online. Our findings can contribute to the design of diversity-aware NRS to nudge users toward diverse consumption of topics and viewpoints.

HYPOTHESIS DEVELOPMENT

Our algorithmic nudge model is structured to examine the effects of algorithmic nudges on user intention for diverse news in NRS. The model examines users' sensemaking of algorithmic nudges and the effect of such nudges on personalization and diverse news.

Nudges of Explainability and Humanness

User cognition of NRS content often hinges upon their sensemaking of how the algorithm functions (Baumer, 2017). Shin et al. (2022) confirmed the importance to users of knowing how and why certain news items are recommended and how user input contributes to the results. Fairness and transparency of relevant feedback enhance the search performance and efficiency of recommendation systems. Similarly, the presence of explanations in recommendation systems can increase positive attitudes and subsequent behaviors among users. Relevant research has suggested a causal link between explainability and assurance in the context of algorithm journalism (e.g., Rai, 2020). This research shows that integrating epistemic cues into NRS can lead to more informed decisions, as users are more likely to trust and accept explainable systems when they are aware of how data are collected, analyzed, and presented. When the mechanism is visible and transparent, users are willing to increase their data input to advance recommendation outputs. NRS users are also able to comprehend the process of a recommendation system (Shin, 2021). In this light, explainability can serve as a nudge and is thus hypothesized as follows:

H1 Explainability nudges users to have a positive perception of NRS transparency.

H2 Explainability nudges users to have a positive perception of NRS fairness.

Nudging News Consumption Behavior with Anthropomorphic Cues

Another trend for nudging in NRS is humanizing news selection in an anthropomorphic form, such as chatbot news (Laban & Araujo, 2020). Anthropomorphic cues can be considered a fairly effective way to provide news recommendations. Anthropomorphism, the use of an object that emulates human qualities and features within an inanimate object, is frequently applied to AI interfaces (Shin, 2021). It is designed to enhance engagement, presence, and interaction with online platforms and can help users become more familiar with the functions and services being offered (Araujo, 2018). Individually personalized recommendations with anthropomorphic recommender systems are applied to NRS, particularly in chatbot contexts (Laban & Araujo, 2020). Research shows that embedding anthropomorphic cues in AI enhances the positive evaluation of algorithm-enabled news (Jang et al., 2022). Araujo (2018) found that consumers have positive emotions if they sense human-like attributes. In addition, Shin (2021) found that anthropomorphic chatbots with an explanation can increase the perception of transparency, which in turn leads to establishing user trust and an improved user experience. Thus, anthropomorphism can be hypothesized as follows:

H3 Anthropomorphism nudges users to have a positive perception of NRS transparency.

H4 Anthropomorphism nudges users to have a positive perception of NRS fairness.

Normative Belief and Trust

Trust is a key factor to consider when designing AI services in journalism (Karimi et al., 2018). Trust plays a role across the range of human perceptions, behaviors, and assessments of technology, and it can determine how people interact with AI (Baumer, 2017). In NRS, trust is viewed as a key question concerning transparency in journalism. Trust represents how much readers believe the news and news sources (credibility) and reflects the belief in the accuracy of news filtering and the user's willingness to accept the recommender system's capacity (Bhuiyan et al., 2018). In this study, trust was proposed as a mediator between normative value and the

intention to seek diverse news. Trust is closely associated with nonfunctional attributes, such as fairness, transparency, and responsibility. People tend to accept trustworthy systems because they know how data are collected and processed and how recommendations are produced. When there is a fair process, users increase their input data to improve recommendation outputs (Shin et al., 2022). If users have notions of normative value that NRS operate in a manner that contributes to the public good, they feel reassured, and trust is established (Baumer, 2017). Ongoing research consistently highlights the pivotal influence of trust on shaping user decisions about whether to engage with technology, but few studies have investigated algorithms, particularly in the context of NRS.

H5 Perceived transparency positively influences users' perceptions of personalization in NRS.

H6 Perceived fairness positively influences users' perceptions of personalization in NRS.

The Personalization Paradox: Personalization and News Diversity

Trust has been found to play a significant role in news readership (McQuail, 1992). Trust turned out to play an even more critical dimension in NRS, as it establishes an interacting feedback loop between users and recommendation mechanisms (Shin et al., 2022). When users trust NRS mechanisms, they tend to look for news other than their own personalized interests (Baden & Springer, 2017). Similar to social comparison theory (Buunk & Gibbons, 2000), readers naturally wonder about diverse viewpoints in news discourse to see how other people think about the discourse. NRS is supposed to provide personalized news, and when users trust that their news is personalized, they look for nonpersonalized news to compare their curated news with other views as a way to evaluate their own personalized set of spectra and to define their personalized self. It is natural for readers to feel curious about what news other people are interested in and how other people view the issue that they are interested in (Jürgens & Stark, 2022). When users believe that news recommendations are customized to their interests, they consider the service to be beneficial and are willing to further engage with NRS for other views and diverse news (Mattis et al., 2022). As users continue to provide more data to NRS, news generation becomes ever more capable of generating accurate

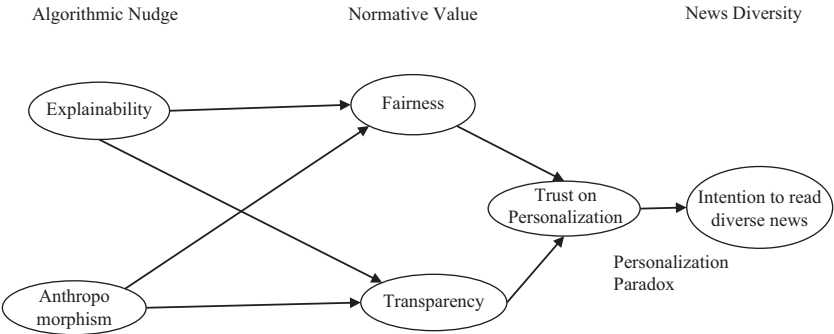


Fig. 6.1 Algorithmic nudge model in NRS

and diverse content. Relevant research has shown the existence of this circular relationship in a diverse sample of algorithm services (Joris et al., 2020; Jürgens & Stark, 2022). Thus, it can be hypothesized that the intention to seek diverse news is influenced by user trust in personalization (Fig. 6.1).

H7 Trust in personalization significantly influences users’ intentions for diverse news and viewpoints in NRS.

METHODS

This study collected 395 responses from the United States via Amazon Mechanical Turk in exchange for monetary compensation. The sample was confined to respondents who had prior experience with algorithm services (automatic recommendations, content suggestions, online news aggregation, etc.). From November to December 2021, a pool of panel participants who regularly used NRS or had prior experience with news platforms or similar AI services was collected. To ensure the reliability and validity of the responses, systematic validation checks were incorporated into the survey. Of the collected responses, 19 were incomplete with missing information. These responses were dropped, resulting in 376 responses being used for data analysis. Among the participants, 45% were male, and 2% were unidentified. Regarding age, 39% were aged between 19 and 25, 30% were aged between 26 and 30 years, 12% were between the ages of 31 and 39, 10% were between the ages of 40 and 49, and 9% were over 50.

Respondents were asked to surf, view, and read autogenerated news on algorithm-based platforms for approximately 1–2 h (Fig. 6.2). NRS was modified from existing recommender systems to test the nudge effect on diversity. The recommendation services were available through any digital device.

The NRS used for the experiment had explanatory and anthropomorphic cues in the interface (Fig. 6.2). Explanatory cues provide users with a brief explanation of why and how certain news is recommended. For anthropomorphism, human-like conversational agents were used to allow users to perceive humanness in the NRS. The effects of anthropomorphic nudges were tested using conversational agents to determine how the nudges triggered emotional involvement in the news. The participants had autonomy in interacting with chatbots and had the freedom to choose the news they wanted.

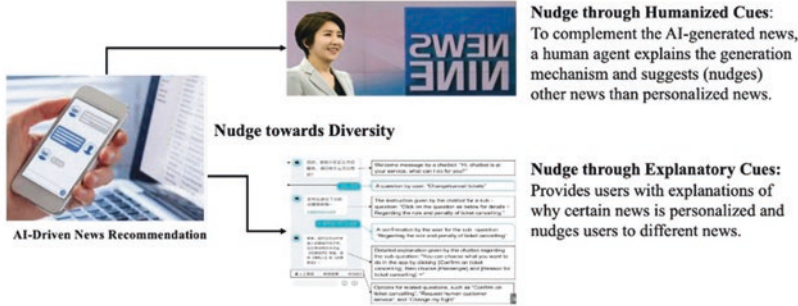
To trace the impact of algorithmic nudges on user behavior, we used a short-term longitudinal method. First, the recruited participants were asked to read the news provided by the NRS for one month. During this time, the participants were required to view/read/search algorithm-recommended news regularly. Specific instructions were given to them, such as login methods, platform use, usage time, and frequency of NRS exposure. After one month, they were invited to complete an online survey questionnaire. The participants could surf the news/item recommendations through the sites. They were told that the news items/content were generated by algorithms and enabled by machine learning and AI mechanisms. After viewing and reading the media, participants were given surveys to complete (Table 6.1).

Scales and Measurements

The scales used in this study *comprised* 18 measurements. All the measurements were developed from digital media research, including algorithm studies (Appendix). The measures of explainability and anthropomorphism were adopted from Rai (2020) and Araujo (2018), respectively. Trust measures were taken from Baumer (2017). Fairness and transparency measures were developed by Shin et al. (2022). Intention for diversity was developed from the idea of Mattis et al. (2022) and Jürgens and Stark (2022). Measurements were mixtures of previously used metrics and modified metrics from previous versions. This was necessary because in our analysis, some measurements needed to be changed to reflect the salient



Naver's NRS on mobile (right) and PC-version NRS (left)



The Beta Version Interface for the Study Design

Stimuli for humanized cues (interface):
-Humanized agent delivers news to users.



Stimuli for explanatory cues (examples):



- 1) The news is recommended based on your previous search/review history.
- 2) The information is given based on your expressed interest.
- 3) The stories are generated based on your political interests.
- 4) The information is drawn due to your given demographic information.
- 5) The news is suggested due to the inference made from your personality and attitudes.

Fig. 6.2 (a) Explanatory anthropomorphism experiment, (b) Naver's NRS (top) and the Beta Version Interface (bottom) for the Study Design

features of algorithms and AI services. Twenty respondents with knowledge of and experience using algorithm services or AI applications participated in a pretest over a three-week interval. Cronbach's alpha was used to check the reliability of the measurement items, and their scales, questions and correlation coefficients were calculated to assess the concurrent validity of the instrument. Reliability values show acceptable levels ranging from 0.720 to 0.838 (Table 6.1). A confirmatory factor analysis (CFA) was performed to test the findings of the exploratory factor analysis, with the analysis showing that the items had satisfactory factor loadings. The factor loadings for all scales were statistically significant ($p < 0.001$) and exceeded 0.80, providing support for the convergent validity of all constructs. To assess validity, correlation tests were conducted among the variables. A simple linear correlation (Pearson's r) was performed to evaluate the significance of the observed relationships. The intercorrelations among the variables showed no signs of multicollinearity, as the square root of the average variance extracted (AVE) from the construct was considerably higher than the variance shared between the construct and the other constructs in the model. Collectively, the data from the factor

Table 6.1 Reliability and validity

<i>Variables</i>	<i>Mean</i>	<i>Standard deviation</i>	<i>Cronbach's alpha</i>	<i>AVE</i>	<i>Weighted kappa</i>
Fairness	4.04	1.251	0.854	0.763	0.441
	4.45	1.203			0.533
	4.07	1.059			0.427
Anthropomorphism	4.34	1.189	0.841	0.751	0.495
	4.61	1.207			0.423
	4.06	0.979			0.336
Transparency	4.61	1.037	0.853	0.744	0.480
	4.63	1.110			0.384
	4.43	1.062			0.575
Explainability	4.04	1.208	0.858	0.720	0.582
	4.05	1.262			0.732
	4.16	1.265			0.582
Trust on personalization	4.36	1.117	0.904	0.838	0.660
	4.23	1.053			0.573
	4.13	1.042			0.618
Intention to seek diverse news	4.40	1.111	0.865	0.803	0.451
	4.09	1.278			0.481
	4.20	1.235			0.582

Table 6.2 Model fit indices

<i>Fit statistics</i>	<i>Structural model</i>	<i>Suggested value (Joreskog & Sorbom, 1996)</i>
$\chi^2 (df)$	934.29/128 = 7.29	< 5
<i>p</i> value	0.000	< 0.05
RMSEA	0.065	0.05 < <i>x</i> < 0.10
CFI	0.910	> 0.90
NFI	0.978	> 0.90
IFI	0.802	> 0.80
TLI	0.884	> 0.85
RFI	0.803	> 0.80
AIC	1056	
Hoelter	69(0.05)/74(0.01)	a higher Hoelter signifies a satisfactory fit
ECVI	2.583	a smaller ECVI value denotes a better fit

loadings, alpha values, correlations, composite reliabilities, and AVE values for each construct suggest that the indicators account for a large portion of the variance of the corresponding latent construct and thus provide evidence for measurement modeling.

We used widely recognized indices to evaluate the model’s goodness of fit. Selected goodness-of-fit indices were evaluated with recommended cutoff values—the chi-squared value per degree of freedom (χ^2/df), the normed fit index (NFI), the root mean square error of approximation (RMSEA), Hoelter, Expected cross-validation *index* (ECVI), the incremental fit index (IFI), and the Tucker–Lewis Index (TLI). The overall fit statistics for the path model suggested a good level of fit, and thus, the path coefficients of the structural model proceeded (Table 6.2).

RESULTS

Structural Model Testing

Structural path testing revealed that the relationships drawn in the hypotheses were all supported (Table 6.3). The seven path coefficients were statistically significant ($p < 0.001$ or $p < 0.0001$). Trust in personalization is significantly influenced by normative values and simultaneously influences the intention to seek diverse news. These factors account for 59.1% of the trust variance. Normative expectancy values are greatly affected by algorithmic nudges. The model explained a significant portion of the variance in each factor. The model explained 24.9% ($p < 0.001$) of the variance in

Table 6.3 Path results

<i>Paths</i>	<i>Estimate</i>	<i>S.E.</i>	<i>C.R.</i>	<i>P label</i>
H1: Explainability → Transparency	0.443	0.064	6.957	***
H2: Explainability → Fairness	0.068	0.023	2.932	0.003**
H3: Anthropomorphism → Transparency	0.434	0.051	8.573	***
H4: Anthropomorphism → Fairness	0.100	0.033	3.011	0.003**
H5: Transparency → Personalization	0.806	0.068	11.819	***
H6: Fairness → Personalization	0.953	0.305	3.127	0.002**
H7: Personalization → News Diversity	0.987	0.063	15.691	***

*1.96: 95% (0.05), **2.58: 99% (0.01), ***3.29: 99.9% (0.001)

fairness, 48.3% ($p < 0.001$) of the variance in transparency, and 39.1% ($p < 0.001$) of the variance in intention for diverse news. The significant paths imply a causal conceptual link between personalization and news diversity. Given the proven path of personalization on news diversity, it may be reasonable to infer the personalization paradox in NRS.

DISCUSSION

With its prevalence and wide applicability, there are hot debates and controversies about algorithmic nudging (Helberger, 2019; de Ridder et al., 2022). While effective in maneuvering user behaviors through the analytics of user data, concerns rise as algorithmic nudging has underlined the assumption of manipulative news personalization. In particular, issues about selective exposure and filtering bubbles through self-reinforcement in the algorithmic news environment trigger concerns regarding how NRS can become more human-centered and promote normative principles of journalism (Borgesius et al., 2016). The issues relate to the imperative task of how to embed explainable, human-centered, diverse, and inclusive algorithms in NRS. For this task, this study presented underlying theoretical works examining the design of recommender systems and the empirical bases for evaluating and testing a diversity-aware NRS. As such, the deployment of AI could support the NRS in providing users with relevant news and stimulating public debate, rather than undermining their role in that respect.

Algorithmic nudging can be viewed as a new means of the art of propaganda or maneuvered persuasion because it blurs the distinction between free user choice and nudges users toward specific outcomes and behaviors

(Mertens et al., 2021). The problem of influencing users' behavior in journalism contexts is that modifying their behavior faces ethical issues and negative impacts of often violating fundamental user rights to view broader perspectives and other voices (Jürgens & Stark, 2022). Critics claim that nudges fail to comply with normative ethical standards and that algorithmic nudges deserve thorough ethical scrutiny and editorial remedy (Greene et al., 2022). As the intelligent manipulation of individual behavior on a sophisticated scale is almost a technological reality, algorithmic nudging has been met with as much fierce criticism as it garners great enthusiasm (Weinmann et al., 2016).

Our findings show that algorithmic nudges using anthropomorphic and explanatory cues can impact users' decision-making and information processing mechanisms in NRS. Anthropomorphic nudging has a positive effect on users' understanding of normative value and trust in personalization. Explanatory nudges also play a significant role in affecting users' perceived fairness and transparency. Belief in personalization established from normative values leads users to seek diverse news beyond recommended news. These path dependencies illustrate that algorithmic nudges engage a cognitive information process in accessing recommended news, wherein algorithmic cues are underlined to stimulate a heuristic that users use to sense-make the extent of humanness and attitudes toward algorithmically curated news. The results of the algorithmic nudge model showed that anthropomorphized cues could trigger nonverbal and persuasive reactions by initiating and supporting positive user heuristics. Anthropomorphism arouses users' heuristic processes and helps them understand normative values in NRS. While the findings for perceived anthropomorphism and explainability are also in line with prior findings (Joris et al., 2020; Laban & Araujo, 2020), which reported the importance of humanness and explainability in algorithm recommendation contexts, our model is distinctive, as it conceptualizes such factors as algorithmic nudges and further clarifies the effects of such nudges on user behavior of personalization and news diversity. In this way, we elucidate the role of algorithmic nudges in NRS and identify the different dimensions of personalization, that is, the personalization paradox (Aguirre et al., 2016; Ontanon & Zhu, 2021). This finding provides a better strategy for personalized news services and sheds light on the way nudges are designed in the desired way to promote diversity and inclusion in AI news. With the advance of personalized AI, news diversity has become an important issue to address, and a concrete method of realizing diversity and

inclusion should be pursued. To benefit from AI systems, users need to engage with various views and diverse opinions in ways that translate into prosocial media effects. Our study, thus, has valuable implications for how we can redefine personalization and design diversity-aware NRS to promote diverse news consumption.

The Personalization Paradox

Personalization is a key feature in NRS, as it generates tailored news based on the interests and preferences of users. However, excessively personalized news stories limit users' exposure to different types and views of the news. At the user level, users may become dissatisfied with being exposed to similar kinds of news or stereotyped articles. Overpersonalization may also influence a user's perspective and behavior, leading them to seek counterattitudinal (attitude that reverses one's own beliefs) perspectives and opinions (Helberger, 2019).

Our findings confirm the social comparing desire (Buunk & Gibbons, 2000) of personalization: the more personalized news recommendation becomes, the more users would like to view nonpersonalized news and other perspectives to compare and contrast a variety of other viewpoints. While readers expect personalized news through NRS, they also seek diverse news in NRS. Greater personalization usually improves news relevance and user engagement, but paradoxically, it may also increase users' desire to read diverse news and seek different viewpoints. Personalized news serves as an anchoring effect, whereby users' decisions are influenced by a particular personalized referenced point, and from there, users explore other news (Currin et al., 2022) to make a comparison as a way of self-enhancement of personalization (Buunk & Gibbons, 2000). This can be the personalization paradox (Aguirre et al., 2016) in AI contexts: while users desire personalized news, they also seek diverse news to compare and contrast their personalization to other views. A fine line seems to exist between personalized and diversified (inclusive). As long as users are satisfied with personalized news, they also seek diversified news. Personalization by algorithms can be a fundamentally flawed metaphor because personalization is a relative, situational concept, and thus, an absolute state of personalization is technically impossible. Just as algorithmic personalization is not a permanent fixture of news and content, neither is users' personalization. Users' personalized interests are contextual (and thus a moving target) and comparative concerning the nonpersonalized

marketplace of ideas. For example, when users show an interest in political elections, AI can personalize their web banners, political stories, suggested columns, etc., but what if their interests change? As NRS continues to recommend the users very narrowly defined news, it will likely limit the potential of future personalization because of the self-favoring or self-reinforcing nature of personalization (Zingales, 2018), which becomes a downward spiral unless interventions are provided.

From this perspective, users are naturally curious about nonpersonalized views and opinions to (1) confirm that curated news is truly personalized; (2) compare their personalized view with other alternative views; and (3) evaluate the extent personalization of the curated news. As user preferences and behavior are contextual, so is algorithmic personalization. The relevant literature considers user personalization to be contextual and argues that personalization influences users' desires and goes beyond personalization (Ontanon & Zhu, 2021). Thus, personalization in NRS contexts is a coupling concept with diversity and inclusion, which should be considered equally in its design (Vrijenhoek et al., 2021).

This finding can be theoretically relevant and practically useful for developing diversity-aware NRS. Theoretically, algorithmic personalization has not been well established. Only a few studies have discussed algorithmic personalization as a positive effect of personalization on user adoption. The personalization paradox reveals interdependence and tensions between personalization and diverse news, which advocates the need for user control to include diversity and inclusion in NRS. For example, Facebook's news feed recommendation system may incorrectly suggest a specific user's preference to be political news tailored to conservative views based on their click/viewing history and thus only recommends the specific news of conservative political standpoints. Because users can only specify their preferences through the algorithmic mechanism run by the personalization codes, the algorithms are likely to further recommend the already curated preference due to the lack of other alternatives. In this way, personalization algorithms reinforce their user preferences without the possibility of adjusting user preferences. Thus, both users and personalization algorithms will look for diverse sources and news for further personalization, as it is contextually and continuously changing.

Practically, the personalization paradox provides insights into how to design diversity-aware NRS. NRS can pursue personalization as well as diversity and inclusion in news recommendations. Personalization can be achieved through algorithms, and this, in turn, gives room for diversity-aware design. In NRS, a series of parameters of certain personalization can

be established as parameters for diverse news. For example, when conservative viewpoints of news (right-wing perspectives) are personalized, then as a piece of subsequent news or supplement suggesting news, liberal viewpoints of news (left-wing voices) are recommended to users. When a series of opposing and contradictory views of immigration is personalized to readers, favoring perspectives of immigration are recommended. Diversity-aware NRS can be achieved by striking an optimum balance between personalization and diversity. Platforms can practice personalization and simultaneously achieve diversity and inclusion through personalization. NRS can refine personalization through diversity, as it can reflect the opposite view of news selection. One algorithmic personalization tactic is collaborative filtering, users' similarity to other users and their preferences (Raza & Ding, 2022). This collaborative filtering can be reversely applied in NRS; the so-called converse filtering or random dynamical nudge (Currin et al., 2022) can identify the users' news spectrum from preferred news to diverse news. As a result of converse filtering, NRS users can benefit from personalized news as well as diverse and inclusive news. The main function of converse filtering is to promote discourse across different perspectives based on content that can be ideologically skewed. Personalization and diversity are two sides of the same coin in NRS, and one helps the other to recommend the quality of news to users. The personalization paradox provides a conceptual basis and justification for diversity-aware NRS.

IMPLICATIONS

The results of our study can be discussed further in terms of theory and practice. Theoretically, the discussions in this study provide a relevant contribution to the dynamics of nudges, diversity, and personalization in NRS. Evaluating the effect of algorithmic nudges in NRS on news from a user perspective can fill the gap between theory and practice, as it transcends concepts and theories for NRS by focusing on users' sensemaking process of personalization and diversity. While nudges in AI have been prominent to date, users' sensemaking of algorithmic nudges has not been researched extensively. Consistent with previous findings, the model indicates that algorithmic nudges can nudge the way users perceive and consume news via NRS. The antecedents of diverse news have not been well conceptually defined or explicitly operationalized, particularly in the news platform context. This study is notable because it empirically tested algorithmic nudges from a user's sensemaking process to examine the nudge

effects from a human perspective, not from a sheer algorithm perspective. Based on this finding, future studies can examine how individuals can differ greatly in the extent to which they care about and look for diverse news (Kim & Ellison, 2022). News diversity is a fluid concept that can be operationalized and justified in various ways depending on context and personal experience (Helberger, 2019; Loecherbach et al., 2020).

Our results also contribute to the theoretical development of algorithmic personalization. While the importance of blending personalization with diversity has increased (Choi et al., 2022), the conceptual foundations of algorithmic personalization have been undermined and thus become fragile. Robust conceptual frameworks of personalization will be critical in the development of diversity-aware NRS. Based on the findings, it is fair to argue that what we truly mean by personalization in AI is more about relevance, context, and connection and less about accuracy, segmentation, and targeting. Hyperpersonalized news comes at the expense of user privacy and often leads to self-reinforcing loops of the interplay between personalization and user preference for news that are not truly in line with users' interests. Due to this negative effect, users desire personalization on the one hand and diversity on the other to eclectically seek relevant information that is useful to both the context and the users. Users need relevant news to determine overall pictures of certain issues.

Clear conceptualization and operationalization of algorithmic personalization also contribute to a theoretical discussion of the coconstruction of reality/news by humans and algorithms (Shin & Park, 2019). As algorithms alone are unlikely to generate the best outcomes, users and algorithms should cocreate values and enable the design of algorithmic nudge contexts that promote desirable behaviors, such as diversity and inclusion in AI news. The role of interaction stems from a pervasive nudge that forms contexts and augments capacities for self-understanding, interaction, and user action. This discussion leads to a relevant conceptualization of algorithmic personalization in relation to diversity and inclusion. We argue that algorithmic personalization can be built on diversity gained from social comparison by users. Users compare their personalized news with others to evaluate their own opinions and views. This personalization process is based on users' motivations and knowledge, the orientation of their interactions, and users' needs. We conclude that algorithmic nudging leads users to not only select differently but also act in a discretionary way in practice. By curating the range of users' possible choices of news, algorithmic nudging contributes to activating users' motivation, thus

enhancing satisfaction. Algorithmic nudging enacts an algorithmic information process that influences users' agencies and practices.

Algorithm technologies are forging a new media ecosystem and enabling new practices; a broader discussion should examine users' decision-making considering the broader AI ecosystems. It is essential to analyze how users and algorithms interact and make decisions within coconstructing AI-enabled media ecosystems. Users' decisions, intertwined with algorithmic curations, determine such coconstruction by taking consequent actions that can reform and renew practices and by following such practices and making them institutionalized. Because of the possible risks of manipulation and ethical concerns, algorithmic nudges should be designed carefully, used in discretionary ways, and audited responsibly. Excessive reliance on algorithms for nudges can have unintended results. While giving AI free rein might create effective and sustainable nudges, without the right motivations and rules for the algorithms, algorithmic nudges could lead people to make poor decisions, making them unethical and unsustainable.

Practically, the nudge model provides guidelines on how to design algorithmic nudges in a fair, equitable, and desirable way. While the distinction between manipulation and nudging can be fuzzy, the design of nudges to influence users' decisions in a certain way raises ethical issues. As platforms such as Facebook and Instagram continue their push toward datafication and algorithmization, we are quickly reaching a point where algorithmic nudges will increasingly guide us toward the most profitable content for their financial interest and remove content that is considered less lucrative. As platforms aggressively seek to maximize the monetization of user data through personalization, such algorithmic gatekeeping is an effective way to nudge content recommendations through their servers toward more lucrative issues. The nudges coming out of this algorithmic gatekeeping trigger the concern that the nudges infringe on free will or violate basic personal freedom—i.e., user autonomy. In an environment where AI-driven nudges come into play, users may be unaware that the selection they make has been predetermined by invisible algorithms. Without users' consent, this process can be considered a manipulation of human autonomy, as it is the algorithm that determines the range of choice overload. Juxtaposed to this criticism, several scholars (Shin, 2021) have criticized the idea that algorithmic nudging conflicts with key ethical values, such as liberty and autonomy. Our findings offer a set of ethical and operational guidelines for the AI industry in designing nudges in a more user-centered manner.

The journalism industry using AI can use algorithmic nudges as a guiding principle for programming personalized news, as well as affording an ideologically diverse news spectrum. Users' sensemaking and perceptions of algorithms and the ways in which users envisage and expect certain algorithmic nudges influence how they use and consume news through NRS (Bucher, 2017). As such, the industry could use algorithmic nudges as a reference base on which to create feedback loops for NRS. Algorithmic nudges not only benefit users by providing them with opportunities to understand how transparent, fair, and ethical NRS is, but they could also help journalism to establish user trust in and engagement with their algorithmic nudging. With the personalization paradox and nudges in mind, future NRS can support users in promoting diversity in personalizing news.

Excessively personalized news without considering the context from NRS is often the outcome of recommendation algorithms that place too much weight on tailoring accuracy (Bernstein et al., 2022). These technical accuracy-slanted approaches may overlook other aspects of subjective user desire, such as exposure to diverse viewpoints and inclusive discussions, alternative recommendations, an expanded list of user interests, and an enhanced list of information (Jürgens & Stark, 2022). When designing a diversity-aware NRS, the industry should reflect beyond-personalization principles to increase the quality of news recommendations and enhance the user experience of news consumption in the long run.

DIVERSITY-AWARE RECOMMENDATION SYSTEMS

How can this principle of diversity awareness be applied in practice? In the AI field, generating a recommendation involves a reranking mechanism. Given a set of preanalyzed parameters, recommendation algorithms present these parameters in such a way that the users find the parameters they are most preferred in at the top, followed by the second-most preferred one, and so forth. When the principle of diversity awareness is applied to recommendation systems, the goal is to generate a more diverse spectrum of news, perspectives, and ideas; normally, a component of novelty or less familiarity is being considered that compares a set of recommended news to what the user has been before, whereas diversity in this light juxtaposes the differences (i.e., the distance between certain issues). This approach engages modeling/considering varying perspectives and differing opinions about each issue, aspects of representation or inclusiveness (e.g., vis-à-vis cultural minorities), or to what extent certain mixes of content encourage tolerance, deliberation, engagement, or other goals that diversity, as a normative concept, does serve.

Technically, spectrum-based recommender systems can be proposed when the user's long-term history is available and the news set is highly dynamic (Fig. 6.3). The systems are designed to recommend subsequent news suggestions given the sequence of visited items in the active session of an anonymous user. The systems can use the sequential and collaborative information from prior sessions of anonymous users to rank

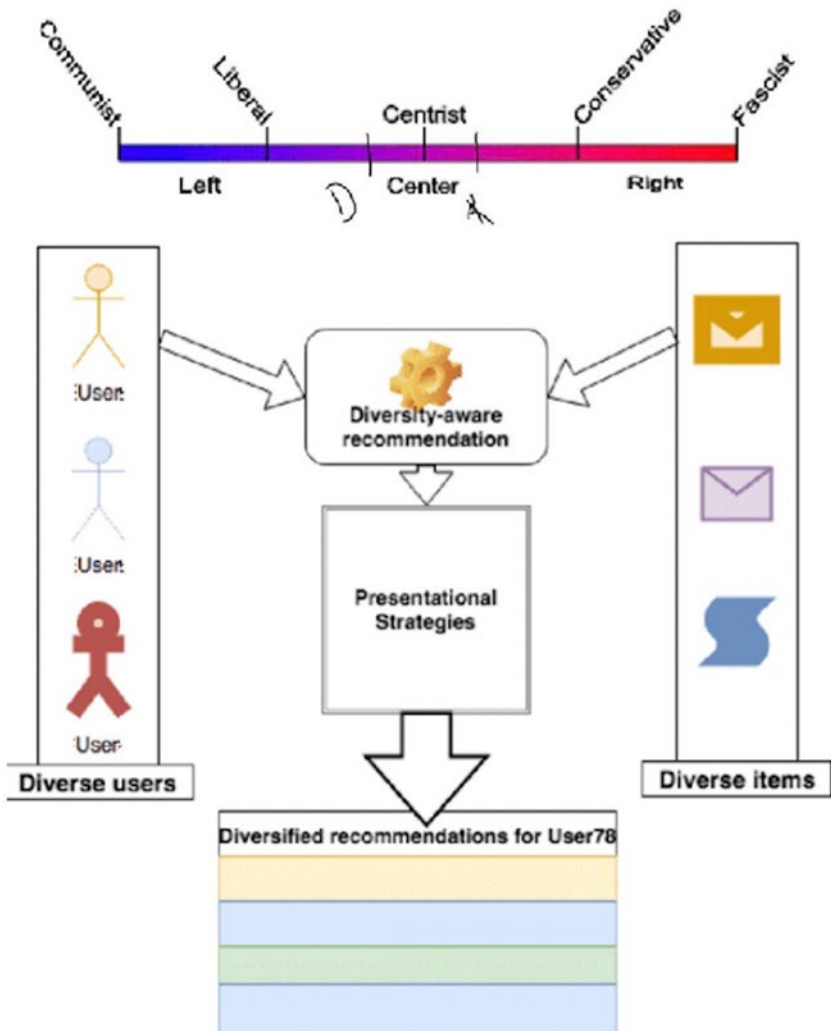


Fig. 6.3 Diversity-Aware AI system

and recommend next possible items for an active session. These methods are in fact applied in many applications, such as music recommendations, movie suggestions, and next basket prediction in e-commerce.

Achieving diverse recommendation systems that accommodate this democratic role of a diverse media diet requires a willingness of AI scientists to discuss social theories, as well as an increased openness of social scientists to learn about technical aspects such as the operations of recommendation AI.

CONCLUSION

Algorithmic nudges are useful in human decision-making because they suggest targeted information while preserving users' freedom of choice. However, algorithmic nudges also present concerns regarding overpersonalization, steered bias, selective exposure, and manipulation of news content and delivery (Bhuiyan et al., 2018). Thus, diversity in AI remains a challenge, as algorithms personalize our news feeds, make news recommendations, and target algorithmic curations (Heitz et al., 2022). Scholars and experts echo the need for diversity and inclusion in NRS and algorithmic technologies in general (Jürgens & Stark, 2022).

This study highlights how algorithmic nudges affect users and steer algorithms to enact choice architectures and nudges that influence user behavior relating to diverse news. The various principles of algorithmic nudges account for recent developments in AI and machine learning. An important proposition in algorithmic nudges is that users should understand the logic of algorithms, engage in the personalization process, and remain in control of such nudges. Users should maintain freedom of choice and autonomy concerning whether to consume personalized news or diversified news. Although algorithmic nudges are increasingly pervasive and embedded in many media services and applications, they also create unwanted results where journalistic value and moral responsibility for their nudges cannot be suitably attributed to any particular editorial or company. The notion of meaningful user control has been proposed for algorithmic nudges to address responsibility gaps and mitigate negative effects by establishing conditions that give people meaningful control over nudges (Courtois & Timmermans, 2018). Algorithmic nudging should enable users to make better decisions on news selection and consumption by facilitating cognitive processes, extending engagement to construct data, and augmenting users' literacy to utilize insights from the data. It is

critical to pay close attention to the complicated societal context within which algorithmic nudges are used and deployed to prevent algorithmic nudges from progressing past the limited perspective of traditional nudging as a simple user interface in AI environments. Equally important is to design algorithmic nudges in NRS as a platform for marketplace ideas and user-controlled mechanisms that contribute to overcoming users' emotional, cognitive, and psychological limits when they make decisions and perform actions that can promote discourse and, ultimately, strengthen democratic reflection in algorithmic media.

APPENDIX

Measurement Items

<i>Variables</i>	<i>Measures</i>
Anthropomorphism	1. I feel like to converse with real human when I interact with NRS. 2. It makes me feel that NRS are human-like entities. 3. This chatbot system has human properties. I thought NRS might be human.
Fairness	1. The system has no favoritism and does not discriminate against people. 2. The source of data throughout an algorithm and its data sources should be identified, logged, and checked. 3. The system follows due process of impartiality with no prejudice.
Transparency	1. The evaluation and the criteria of algorithms used should be publicly released and understandable and interpretable to people. 2. Any outputs produced by an algorithmic system should be explainable to the people affected by those outputs. 3. Algorithms should enable people to understand how well internal states of algorithms can be understood from knowledge of its external outputs.
Explainability	1. I found algorithm are easily understandable. 2. The algorithm services are interpretable. 3. I can figure out the internal mechanics of a machine learning. I hope that algorithm can be clearly explainable.
Trust	1. I trust the recommendations by algorithms-driven services. 2. Recommended items through algorithmic processes are trustworthy. 3. I consider the algorithm service results are reliable.
Intention for diversity	1. Using NRS, I feel like to view other sources of news. 2. The more I found the recommended items are a match to my needs, the more I like to read other news. 3. As much as algorithm-based service is customized to me, I would like to see other perspectives of news.

REFERENCES

- Aguirre, E., Roggeveen, A., Grewal, D., & Wetzels, M. (2016). Personalization-privacy paradox: Implications for new media. *Journal of Consumer Marketing*, 33(2), 98–110. <https://doi.org/10.1108/JCM-06-2015-1458>
- Araujo, T. (2018). Living up to the chatbot hype. *Computers in Human Behavior*, 85, 183–189. <https://doi.org/10.1016/j.chb.2018.03.051>
- Baden, C., & Springer, N. (2017). Conceptualizing viewpoint diversity in news discourse. *Journalism*, 18(2), 176–194. <https://doi.org/10.1177/1464884915605028>
- Baumer, E. P. (2017). Toward human-centered algorithm design. *Big Data & Society*. <https://doi.org/10.1177/2053951717718854>
- Bernstein, A., Vreese, C., Helberger, N., Schulz, W., & Zweig, K. A. (2022). Diversity in news recommendation. Dagstuhl perspectives workshop: Diversity, fairness, and data-driven personalization in recommender system. *Dagstuhl Manifestos*, 9(1), 43–61.
- Bhuiyan, M., Vick, T., Mitra, K., & Horning, M. (2018). FeedReflect: A tool for nudging users to assess news credibility on Twitter. In *Proceedings of the ACM Conference on Computer Supported Cooperative Work* (pp. 205–208). <https://doi.org/10.1145/3272973.3274056>
- Borgesius, F., Trilling, D., Möller, J., Bodó, B., de Vreese, C. H., & Helberger, N. (2016). Should we worry about filter bubbles? *Internet Policy Review*, 5(1), 1–16. <https://doi.org/10.14763/2016.1.401>
- Bryanov, K., Watson, B., Pingree, R., & Santia, M. (2020). Effects of partisan personalization in news portal experiment. *Public Opinion Quarterly*, 84(S1), 216–235.
- Bucher, T. (2017). The algorithmic imaginary: Exploring the ordinary affects of Facebook algorithms. *Information, Communication & Society*, 20, 30–44.
- Burr, C., & Cristianini, N. (2019). Can machines read our minds? *Minds & Machines*, 29, 461–494. <https://doi.org/10.1007/s11023-019-09497-4>
- Buunk, P., & Gibbons, X. (2000). Toward an enlightenment in social comparison theory. In J. Suls & L. Wheeler (Eds.), *Handbook of social comparison. The Springer series in social clinical psychology*. Springer. https://doi.org/10.1007/978-1-4615-4237-7_22
- Cardenal, S., Aguilar-Paredes, C., Cristancho, C., & Majó-Vázquez, S. (2019). Echo chambers in online news consumption. *European Journal of Communication*, 34(4), 360–376. <https://doi.org/10.1177/0267323119844409>
- Choi, S., Kim, H., & Gim, M. (2022). Do not read the same news! Enhancing diversity and personalization of news recommendation. In *Companion proceedings of the web conference 2022*. April 25–29, Lyon, France. ACM. <https://doi.org/10.1145/3487553.3524936>

- Courtois, C., & Timmermans, E. (2018). Cracking the tinder code. *Journal of Computer-Mediated Communication*, 23(1), 1–16. <https://doi.org/10.1093/jcmc/zmx001>
- Currin, C., Vera, S., & Khaledi-Nasab, A. (2022). Depolarization of echo chambers by random dynamical nudge. *Scientific Reports*, 12, 9234. <https://doi.org/10.1038/s41598-022-12494-w>
- Dahlgren, P. M. (2021). A critical review of filter bubbles and a comparison with selective exposure. *Nordicom Review*, 42(1), 15–33.
- de Ridder, D., Kroese, F., & van Gestel, L. (2022). Nudgeability: Mapping conditions of susceptibility to nudge influence. *Perspectives on Psychological Science*, 17(2), 346–359.
- Greene, T., Martens, D., & Shmueli, G. (2022). Barriers to academic data science research in the new realm of algorithmic behavior modification by digital platforms. *Nature Machine Intelligence*, 4, 323–330. <https://doi.org/10.1038/s42256-022-00475-7>
- Heitz, L., Lischka, J., Birrer, A., Paudel, B., Tolmeijer, S., Laugwitz, L., & Bernstein, A. (2022). Benefits of diverse news recommendations for democracy. *Digital Journalism*. <https://doi.org/10.1080/21670811.2021.2021804>
- Helberger, N. (2019). On the democratic role of news recommenders. *Digital Journalism*, 7(8), 993–1012. <https://doi.org/10.1080/21670811.2019.1623700>
- Hermann, E. (2022). Artificial intelligence and mass personalization of communication content. *New Media & Society*, 24(5), 1258–1277. <https://doi.org/10.1177/14614448211022702>
- Jang, W., Chun, J., Kim, S., & Kang, Y. (2022). The effects of anthropomorphism on how people evaluate algorithm-written news. *Digital Journalism*. <https://doi.org/10.1080/21670811.2021.1976064>
- Jesse, M., & Jannach, D. (2021). Digital nudging with recommender systems. *Computers in Human Behavior Reports*, 3, 100052. <https://doi.org/10.1016/j.chbr.2020.100052>
- Joreskog, K. G., & Sorbom, D. (1996). *LISREL 8: User's reference guide*. Chicago: Scientific Software International.
- Joris, G., Grove, F. D., Damme, K. V., & Marez, L. D. (2020). News diversity reconsidered: A systematic literature review unraveling the diversity in conceptualizations. *Journalism Studies*, 21(13), 1893–1912.
- Juneja, P., & Mitra, T. (2022). Algorithmic nudge to make better choices. In *CoRR abs/2202.02479. CHI 2022 Workshop on Operationalizing Human-centered Perspectives in Explainable AI*.
- Jürgens, P., & Stark, B. (2022). Mapping exposure diversity: The divergent effects of algorithmic curation on news consumption. *Journal of Communication*, 72(3), 322–344. <https://doi.org/10.1093/joc/jqac009>

- Karimi, M., Jannach, D., & Jugovac, M. (2018). News recommender systems. *Information Processing & Management*, 54(6), 1203–1227.
- Kim, D. H., & Ellison, N. B. (2022). From observation on social media to offline political participation: The social media affordances approach. *New Media & Society*, 24(12), 2614–2634. <https://doi.org/10.1177/1461444821998346>
- Kroll, T., & Stieglitz, S. (2021). Digital nudging and privacy. *Behavior & Information Technology*, 40, 1–19.
- Laban, G., & Araujo, T. (2020). Working Together with Conversational Agents: the Relationship of Perceived Cooperation with Service Performance Evaluations. In: CONVERSATIONS 2019: Third International Workshop on Chatbot Research, Amsterdam, The Netherlands, 19-20 Nov 2019, pp. 218–228. ISBN 9783030395391 (https://doi.org/10.1007/978-3-030-39540-7_15)
- Loeberbach, F., Moeller, J., Trilling, D., & van Atteveldt, W. (2020). The unified framework of media diversity. *Digital Journalism*, 8(5), 605–642.
- Mattis, N., Masur, P., Möller, J., & van Atteveldt, W. (2022). Nudging toward news diversity: A theoretical framework for facilitating diverse news consumption through recommenders design. *New Media & Society*. <https://doi.org/10.1177/14614448221104413>
- McQuail, D. (1992). *Media performance: Mass communication and the public interest*. Sage Publications.
- Mertens, S., Herberz, M., Hahnel, U., & Brosch, T. (2021). The effectiveness of nudging. *Psychological and Cognitive Sciences*, 119(1), e2107346118. <https://doi.org/10.1073/pnas.2107346118>
- Möhlmann, M. (2021). Algorithmic nudges do not have to be unethical. *Harvard Business Review*. <https://hbr.org/2021/04/algorithmic-nudges-dont-have-to-be-unethical>.
- Ontanon, S., & Zhu, J. (2021). The personalization paradox: The conflict between accurate user models and personalized adaptive systems. arXiv:2103.01771v1
- Rai, A. (2020). Explainable AI: from black box to glass box. *Journal of the Academy of Marketing Science*, 48, 137–141. <https://doi.org/10.1007/s11747-019-00710-5>
- Raveendhran, R., & Fast, J. (2021). Humans judge, algorithms nudge. *Organizational Behavior and Human Decision Processes*, 164, 11–26. <https://doi.org/10.1016/j.obhdp.2021.01.001>
- Raza, S., & Ding, C. (2022). News recommender system. *Artificial Intelligence Review*, 55, 749–800. <https://doi.org/10.1007/s10462-021-10043-x>
- Recchiuto, C., & Sgorbissa, A. (2022). Diversity-aware social robots meet people: Beyond context-aware embodied AI. In *Presented at the Roundtable AI in holistic care and healing practices: the caring encounter beyond COVID-19*. Anthropology, AI and the Future of Human Society, 6–10 June, Royal Anthropological Institute.

- Reuver, M., Mattis, N., Sax, M., Verberne, S., Tintarev, N., Helberger, N., Moeller, J., Vrijenhoek, S., Fokkens, A., & van Atteveldt, W. (2021). Are we human, or are we users? The role of natural language processing in human-centric news recommenders that nudge users toward diverse content. In *1st workshop on NLP for positive impact* (pp. 134–139). Stroudsburg, PA.
- Scheiber, N. (2021, April 2). How Uber uses psychological tricks to push its drivers' buttons. *New York Times, Technology Section*.
- Schobel, S., Barev, T., Janson, A., Hupfeld, F., & Leimeister, M. (2020). Understanding user preferences of digital privacy nudges. In *Hawaii International Conference on System Sciences*.
- Shin, D. (2021). The perception of humanness in conversational journalism. *New Media & Society*. <https://doi.org/10.1177/1461444821993801>
- Shin, D., & Ahmad, N. (2023). Algorithmic nudge: An approach to designing human-centered generative artificial intelligence. *Computer*, 56(8), 95–99. <https://doi.org/10.1109/MC.2023.3278156>
- Shin, D., & Park, Y. (2019). Role of fairness, accountability, and transparency in algorithmic affordance. *Computers in Human Behavior*, 98, 277–284. <https://doi.org/10.1016/j.chb.2019.04.019>
- Shin, D., Lim, J., Ahmad, N., & Ibarahim, M. (2022). Understanding user sense-making in fairness and transparency in algorithms. *AI & Society*. <https://doi.org/10.1007/s00146-022-01525-9>
- Sunstein, R., & Thaler, H. (2014). *Nudge: Improving decisions about health, wealth, and happiness*. Yale University Press.
- Tsavli, M., Efraimidis, P., Katos, V., & Mitrou, L. (2015). Reengineering the user. *Information and Computer Security*, 23(4), 394–405. <https://doi.org/10.1108/ICS-10-2014-0071>
- Vermeulen, J. (2022). To nudge or not to nudge: News recommendation as a tool to achieve online media pluralism. *Digital Journalism*. <https://doi.org/10.1080/21670811.2022.2026796>
- Vrijenhoek, S., Kaya, M., Metoui, N., & Helberger, N. (2021). Recommenders with a mission: Assessing diversity in news recommendations. Conference. In *CHIIR '21: ACM SIGIR Conference on Human Information Interaction and Retrieval*. <https://doi.org/10.1145/3406522.3446019>
- Weinmann, M., Schneider, C., & vom Brocke, J. (2016). Digital nudging. *Business & Information Systems Engineering*, 58(6), 433–436. <https://doi.org/10.2139/ssrn.2708250>
- Yeung, K. (2017). Hyper nudge: Big data as a mode of regulation by design. *Information, Communication & Society*, 20(1), 118–136.
- Zingales, N. (2018). Google shopping: Beware of self-favoring in a world of algorithmic nudging. *Competition Policy International- Europe Column*. <https://ssrn.com/abstract=3707797>



Misinformation, Paradox, and Nudge: Combating Misinformation Through Nudging

THE MISINFORMATION PARADOX: WHY NUDGING IS NEEDED IN MISINFORMATION?

Newman et al. (2022) describe the paradoxical behavior of people on misinformation by stating, “believing in nothing and believing in everything” (p. 1). While people have skepticism of misinformation, people are gullible and susceptible to misinformation. People understand the harmful nature of misinformation but continue to engage with misinformation and continue to accept and share misinformation (Valenzuela et al., 2019). This phenomenon is called the misinformation paradox (Munyaka et al., 2022), the discrepancy between users’ attitudes toward misinformation and how they actually behave when it comes to misinformation online. It is often referred to as inconsistency between the expressed concern and the actual behavior of users in discerning truth online. The relationship between individuals’ intentions to debunk misinformation and their actual personal misinformation-accepting behaviors has been extensively researched and empirically validated (e.g., Newman et al., 2022). Research shows that users expressing misinformation concerns often fail to act in accordance with them (Munyaka et al., 2022). The likely explanation for the misinformation paradox is a mixture of all different factors of individual, psychological, social, and cultural.

People may assert openly that they criticize misinformation but behave in ways that enjoy consuming, reproducing, and sharing misinformation. While people might express doubt and disbelief toward information sources, they engage with misinformation content in the same ways they engage with content they trust. People express high awareness and alertness of misinformation, yet in reality, they continue to interact with misinformation in negative ways. Research has shown this misinformation paradox in people that a complete distrust of misinformation does not result in disengagement from that misinformation (Kim et al., 2023; Munyaka et al., 2022). People understand cognitively the negative sides of misinformation. Nevertheless, they still engage with it online: the more they know about the adversarial effects of misinformation, the more they consume and share it. The misinformation paradox is a manifestation of a limited ability of human truth discernment.

HUMAN DECISION-MAKING AND THE DIFFUSION AND CONSUMPTION OF MISINFORMATION

Why are some people predisposed to refrain from belief change even in the face of good corrective evidence, and how can corrections be designed to maximize impact? These are ongoing challenges and issues that misinformation researchers have been struggling with. Relevant studies have consistently shown that misinformation spreads deeper, quicker, and stronger than true information. Research (e.g., Vosoughi et al., 2018) showed that the top 1% of misinformation cascades diffused quickly to between 1000 and 100,000 people, whereas the truth rarely diffused to more than 1000 people. Other research echoed that misinformation was significantly more likely to be shared and retweeted than true information. A rising question related to this phenomenon is the role that human decision-making plays: How can AI enable users to make better informed decisions? Recent research has shown that cognitive biases in decision-making can enhance the adoption and spread of misinformation. For instance, Vosoughi et al. (2018) argued that the spread of misinformation can result more from a novelty effect than from the structure of social media outlets, website platforms, and internet bots. Novelty attracts people's attention, facilitates speedy decision-making, and stirs the reproduction of misinformation since novelty satisfies human needs to update their understanding of the world. Rumors and fake news were found to be more attempting and

attractive than true information, implying that people were more likely to welcome and share novel information. Per selective exposure, people see merely what they expect or want to view without critically examining information (Badke, 2018). This is an example of confirmation bias, the conscious tendency of users to pursue information that verifies and confirms their already made belief instead of looking at overall and comprehensive aspects of information. Per the theory of cognitive dissonance, whenever a generated recommendation includes information that conflicts with the already held cognitive models of users, it promptly causes cognitive dissonance (Festinger, 1957). Users are naturally trying to scale down this dissonance; hence, they may avoid or deny knowledge that contrasts their beliefs. Whenever users encounter a new piece of information, they tend to evaluate its trustworthiness. Users normally seek out the social consensus of the information, its supporting evidence, and its consistency, coherence, and credibility. During the seeking out process, people tend to rely on mental shortcuts to minimize the effort and time spent rather than evaluating the information analytically.

DOES ALGORITHMIC NUDGING MAKE A BETTER JUDGMENT OF MISINFORMATION?

Nudge theory is a principle in cognitive economics, behavioral science, and cognitive psychology that argues that positive reinforcement and indirect influences guide the behavior and decision-making of individuals or groups (Thaler & Sunstein, 2014). The theory gained huge popularity with a book written by Nobel-winning economist Richard Thaler. Nudging can afford positive reinforcement to individuals and eventually lead to a particular action or decision. Because of its effectiveness, nudging has been widely used in user insights, system development, and public policy areas. The human-computer interaction (HCI) literature has researched numerous principles that suit an online form of nudging well. For instance, Tsavli et al. (2015) showed that enhanced password visualization tools (illustrating password strength levels) can lead to safer passwords. In news context, “accuracy primes”—short reminders intended to shift people’s attention toward the accuracy of the news content they come across online—can be built into AI-based news recommendation systems. In the information systems literature, numerous relevant studies have examined nudging in a digital context as a subtle form of using

design, information, and interaction elements to guide user behavior in online environments without restricting the individual's freedom of choice (Kroll & Stieglitz, 2021).

Nudging, which is termed algorithmic nudging, is defined as the use of algorithm design components to lead user behavior in algorithm-mediated contexts (Möhlmann, 2021) and is used widely in AI and machine learning services. Additionally, known as AI nudging, nudging is an indirect way of framing algorithm behavior by nudging human behavior that has been used increasingly and widely (Juneja & Mitra, 2022) to affect purchasing behavior and improve user service. Retailers use AI nudges to engage with their consumers in a highly personalized context that leads them to make better decisions and choices. Shin et al. (2022) conceptualized AI nudges as the use of cognitive stimuli to influence people's behavior predictably without constraining their choices or modifying their incentives. This conceptualization is based on a previous definition of digital nudging by Weinmann et al. (2016), who defined digital nudging as user interfaces in online decision contexts that influence customer actions, although this definition does not indicate the potential of algorithmic technologies. One of the clear differences from conventional nudging is that algorithmic nudges work more secretly behind the algorithm interfaces than general digital nudges. Due to the black box nature of algorithms, algorithmic nudges function stealthily and covertly without being noticed. AI nudges have been used to solidify the primary functions of algorithms, silently shaping what users see, from the posts in social feeds to the suggested content they see to the advertisements covering their online interface. Algorithmic nudges form part of a large AI system to help the larger system work as planned.

Algorithmic nudging uses AI to influence users' behavior by gently coaxing them toward a preferred choice through visual cues, push notifications, and alert alarms. With drastic developments in AI and machine learning, algorithmic nudges have become much more influential than traditional nudges because algorithms can effectively identify target segments in which to nudge users to make behavioral changes. Curating personalized models is a well-honed skill with so much online data about users' behavioral patterns. Algorithmic nudging helps users evaluate and interact with algorithmic systems on the basis that informed judgments lead to wiser decisions (Zarouali et al., 2021). Used correctly, algorithmic nudges help people assess how platforms, firms, and the government use these technologies and, in doing so, enable them to advocate for

responsible technology design and use that avoids biases and protects privacy (Aker et al., 2021). Algorithmic nudges can involve meaningful efforts to empower more users to impact data flows and to perceive if or when they or others are being sidelined. The effects of these efforts may be restrained, depending on the extent of the technical knowledge needed (Schobel et al., 2020).

Platform providers use nudges at different dimensions in AI algorithms, such as news recommending services, content purchasing suggestions, and prescriptive decision-making tools. Although the consequences are still debated and illusive, researchers have examined the relationship between AI and nudges, arguing that algorithmically personalized results can influence users and often lead to unintended consequences and unwanted habits (Burr et al., 2018; Shin et al., 2022). The relationship between AI and nudges found in many studies (Burr et al., 2018) illustrates how personalized tailored algorithms can use persuasion and psychometrics to affect individual and collective behavior in unintended ways. In this regard, Tufekci (2017) conducted longitudinal data to analyze the effects of algorithms on individuals and concluded that algorithmic nudges put users down an ever-darker rabbit hole. Many recent discussions concur with concerns about the ability of algorithmic nudges to predict user taste (Cowgill & Stevenson, 2020). The debate is ongoing on how we manage algorithmic nudges to lead to better consequences and whether we should enforce regulations on liability for negative nudges that lead to bad influence. As nudging intermediaries can amplify the severity of public-related harm, it has been voiced that any form of rules and regulation should respond to unethical nudges with varying guidelines for deciding cases of intermediary liability. How to design nudges in AI that guide people toward better decisions ethically and responsibly remains an open question for algorithmic nudges.

NUDGES AND ALGORITHMIC AFFORDANCE: FROM BLACK BOX AI TO TRANSPARENT AFFORDANCES

Current literature and industry applications on algorithmic nudges tend to focus on framing user behaviors in certain ways, with insufficient attention given to how people develop cognitive processing or how users respond to nudges (Shin et al., 2022). These questions relate to the principle of affordance, and thus, it is useful to see algorithmic nudges from an

affordance perspective. Algorithmic affordances describe the range of explicit and implicit interaction possibilities that enable users to engage with and control the algorithmic system directly and/or indirectly (Shin & Akhtar, 2024). Affordance-based nudges have been proposed as an alternative to algorithmic nudges to mitigate the weaknesses of nudging practices. An affordance-based approach to algorithmic nudging is to follow users' perceptions, information processing, and cognitive development. The concept of affordances can be used as a conceptual framework to understand the relationship between nudges and people, especially with respect to algorithms and AI. A user affordance-based approach can capture users' needs and stakeholders' ideas and generate design options based on those, rather than an algorithm-based approach. Affordance-based nudges can make it clear how the device caused a user's action, and affordance is built into the device to show how users can use it without intervention (Shin & Akhtar, 2024). For example, when users see personally recommended product suggestions from platforms, they want to click on them to see if they are truly personalized to their preferences.

Despite technical sophistication, algorithms rarely offer a practical means for users to interact with them, as they lack the affordances that would enable users to comprehend them or understand how best to use them to complete their tasks. Affordances help the user identify an object's invariants (e.g., functional properties) relative to the user's capabilities, and algorithmic affordances allow users to take action based on their perceptions of features in their environment, such as fairness, transparency, and accountability. In reality, integrating affordances into algorithms is not easy. If the algorithms used by automation are highly complex and different from human cognition or are not perceived by users, the algorithmic results will not be accepted. To take full advantage of algorithmic nudges, it may be necessary to ensure that the algorithm's opportunities for action are made observable and understandable by its users (Baumer, 2017).

The question of how to design algorithmic nudges that are transparent, accountable, and fair relates to how we understand and reflect users' perceived affordances of algorithms. Algorithmic affordance can be activated by trust through users' understanding of fairness, transparency, and

accountability. Although users expect algorithms to produce relevant, personalized, and prescriptive recommendations, the underlying assumption is that users presume algorithms to be fair, transparent, and accountable. When these issues are confirmed by users, algorithmic affordance plays a role in users' consumption of algorithms. From this relation, it can be inferred that algorithmic nudges based on algorithmic affordance have a positive effect on users' cognition and lead them to take steps to recognize, perceive, use, and adopt the suggested nudges.

The affordance principle is useful for the theoretical conceptualization of algorithmic nudges. Affordance refers to users' perception of the utility of an object by understanding its features. Users anticipate algorithms that provide personalized, useful, and accurate results, and such desires come with the assessment of fairness, transparency, and accountability. Algorithmic affordance offers a possible cognitive process for perception in humans and perceptions of transparency, fairness, and accountability (Shin et al., 2024). When users trust algorithms, they are likely to continue to interact with them by consenting to their data being gathered by algorithms, and the augmented data enable better prescriptive and predictive analytics (Akter et al., 2021). When users confirm algorithmic processes to be fair, transparent, and accountable, they establish trust in AI. Trust and affordance have a circular relationship in that once users trust algorithm services or providers, they perceive that the services are easy to use and adopt, and thus, they continue to use them. When trust is established, users would like to continue to use as they trust AI, and they are most likely to accept AI with satisfaction. Relevant research has confirmed that transparency, fairness, and accountability build user trust (Burr et al., 2018). The trust feedback loop is a positive feedback loop that reduces users' concerns over transparency and accuracy, and user satisfaction and intention increase significantly (Fig. 7.1). Positive feedback is likely positively related to trust, satisfaction, transparency, fairness, and intention. Such feedback implies the significance of examining the complex cognitive mechanisms related to feedback. The positive feedback loop of trust provides heuristic ideas for designing effective algorithmic nudges.

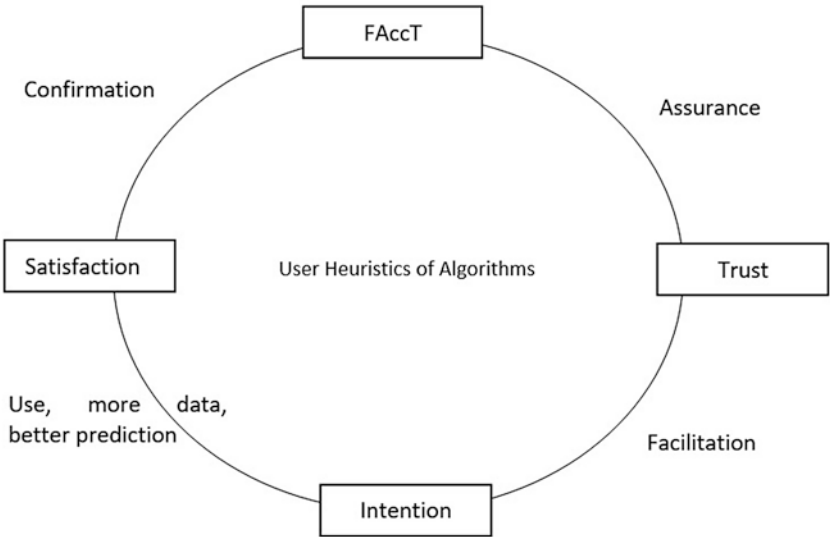


Fig. 7.1 Positive feedback loop

ALGORITHMIC SOCIAL MANAGING: ALGORITHMIC
BEHAVIOR MODIFICATION

The idea of nudges is not completely new, as the method has been used since the 1980s under names such as behavioral engineering and behavior framing, which are a series of psychological methods to promote better behavior by indirectly encouraging certain behaviors. With AIs becoming mainstream, embedded invisible interventions to encourage people toward desired actions have become more sophisticated. We can easily find numerous examples of algorithms being used to nudge decision-makers toward a particular social outcome. This is often called algorithmic social management. Every day, we face nudges that affect our daily decision-making. Recommender systems are based on algorithmic nudging structures in which content is easily accessible to users and thus shapes their decision-making processes through the automated selection and ranking of presented information. For example, when people buy a burger, they are likelier to purchase fries and drinks if they are offered a bundled recommendation; when there is an additional cost for plastic bags at a superstore, people are less likely to ask for bags, thus decreasing plastic consumption; and colleges hang pictures or posters of leaders and their quotes to encourage students to think in a certain way. Algorithm-based

nudges would be even more powerful levers than normal nudges for changing human behavior. AI can be a stable venue for nudging, as people visit the platforms regularly, where they are frequently exposed to numerous choices. As algorithm-based nudges have the ability to learn from data, they can deliver increasingly relevant nudges over time. For instance, Google's Suggested Searches and Autocomplete (autofill) functions are effective kinds of nudging since they gently encourage users to click on the suggested options. Netflix is also developing a kind of nudging by suggesting sequences of episodes at the beginning of menu interfaces. Users could spend minimal effort discontinuing the content, but they were likelier to continue to the subsequent episode due to the kind and quiet suggestion. YouTube also incorporates nudges through the supplementary element of an algorithm-based option advising which content to watch next. Such algorithmic nudges refer to any aspect of the algorithmic architecture that steers and frames people's attitudes and behaviors.

Uber uses algorithmic nudges to maximize the profits and productivity of its drivers (Scheiber, 2021). To create an effective system in which the supply of drivers matches the rising numbers of customers, Uber's algorithms dispatch drivers' next trip ride when the current passenger is still on board. Uber also informs drivers that they are near to reaching a particular passenger when the drivers wish to finish their trips for the day, encouraging them to be on the road for longer durations to earn more cash. From the viewpoint of Uber's profit, suggesting the next drive is an effective default setting, as more drivers feel compelled to continue driving. Uber's algorithmic nudge is analogous to the opt-out setting, wherein the drivers have the option of rescinding the notification that pops up by default or continuing with the next ride. The drivers are not required to accept more rides; they are simply being nudged algorithmically by an in-app notification that leads them to the next fare. While drivers can easily opt out with the tap of a button, it is less likely that the driver will choose to opt out, as, inherently, people tend to prefer obtaining something to losing out. When the algorithmic nudge leaves very little choice in the hands of the drivers, although Uber is making decisions for the benefit of its drivers, the algorithmic nudge is not always in the best interest of the drivers because they may unintentionally end up working for more rides (Scheiber, 2021). One might argue that those who are contracted to be Uber drivers usually need additional income; thus, Uber's decision to include algorithmic nudges is based on the interests of the drivers. However, it could also be said that Uber is manipulating its drivers by exploiting their needs and economic status. The power dynamic between companies and workers

tends to be in favor of companies. Organizations can use algorithmic nudges to encourage their workers to work more, but ethical issues arise concerning the way in which nudges are utilized and the context in which they are used. It is not the algorithmic nudges themselves that are bad, but the way in which they are implemented could face ethical barriers that reflect on those who use the nudges.

Networked platforms currently almost regularly utilize algorithmic behavior modification techniques to shape user behaviors in a way that maximizes profits. Platforms increasingly utilize techniques to flexibly and automatically tailor behavioral interventions to exploit human psychological traits and cognitions. Platforms are created to perform selective gathering, algorithmic filtering, amplification, and monetization of user data and are equipped with autonomous, data-driven, prescriptive, and intelligent algorithms to control users' behavior at scale. People leave their traces online, such as digital breadcrumbs through social media posts or pages on Facebook, the Wi-Fi addresses they log into, and the search queries they put in search engines. These digital traces make for a useful and predictive behavioral data reservoir, which platforms use for behavior medication when deciding what news to show, what ads to expose, and what information to avoid. By giving these digital traces to algorithms and platforms, we allow them to infringe on our privacy at times in unexpected ways and fine-tune their recommendation-targeting efforts. Platforms can increase their prediction accuracy by molding users' behaviors toward their predicted values, employing behavior modification techniques, and showing more positive predictions. Such seemingly enhanced predictions can unintentionally result from using reinforcement learning algorithms that integrate prediction and behavior modification.

In journalism, algorithmic nudges are used as a tool to increase news diversity in news recommender systems (Mattis et al., 2022). Loecherbach et al. (2020) illustrated that less-preferred news items are more likely to be chosen if they are displayed at higher ranks or ranked highly on the page. Algorithmic nudges have been proven to considerably increase news diversity, especially when readers experience information overload or when their news selections are recommended by AI systems. Algorithmic nudges could support people who seek but do not find diverse articles by making such articles more accessible and available or by providing unexpected recommendations that prevent unwanted algorithmic feedback loops. These unexpected recommendations lead users to develop new interests, which in turn enables recommendation algorithms to evolve (Karimi et al., 2018), contributing to greater user satisfaction.

Natural language processing algorithms and reinforcing learning are examples of such evolving behavioral changes in algorithms, which are utilized to optimize services and recommendations, promote user engagement, produce more behavioral feedback data, and even [hook](#) users via behavioral modifications and long-term addictive habit creation. For example, TikTok's "For You" feed replicates preferences personalized to each user. To hone their personalized For You feed, the service recommends content by rating videos based on a combination of different parameters—starting from the interests of users default to as a new user and adjusting for things users specify they are not interested in. In the long run, this service hooks users and forms a habit. In public health, medical, and therapeutic contexts, [algorithmic](#) behavior change is a desirable, observable, and replicable intervention intended to modify patient behavior with the participants' consent. However, platform behavioral change methods are becoming increasingly invisible and opaque and are done without clear user approval. Technically, algorithms can program people's behavior for any outcome. A study of the inductive nature of Facebook likes revealed that like buttons can precisely forecast personal profiles, such as ethnicity, sexual orientation, political beliefs, religion, and personality. Based on the number of likes, Facebook predicts your friends (150 likes), your parents (250 likes), and yourself (300 likes). It has been criticized as unethical practice to use such psychological traces acquired from masses of users without their consent to influence users' attitudes. However, the feeds people view on their social media are the results of a specifically directed marketing plan with customized content to nudge a specific user toward a particular action. Although platforms' behavioral change is clear to the user, for example, as presented recommendations, advertising, or autocomplete functions, it is normally invisible and unobservable to third-party auditors. These algorithmic behavior modifications work invisibly and can be operated without external supervision, amplifying filter bubbles, or echo chamber trends.

Former Facebook employee Frances Haugen revealed the significance of transparency and external audits for platforms. In her US Senate testimony in 2021, Haugen stated:

No one can understand Facebook's destructive choices better than Facebook because only Facebook gets to look under the hood. A critical starting point for effective regulation is transparency: full access to data for research not directed by Facebook.... As long as Facebook is operating in the shadows, hiding its research from public scrutiny, it is unaccountable.... Left alone, Facebook will continue to make choices that go against the common good, our common good.

This testimony calls for clear platform transparency and access, highlighting the need for a mechanism that conducts independent appraisals and complies with regulatory agencies to audit platform abuse of behavioral modifications, develops appropriate measures to evaluate algorithmic behavior modification impacts, and facilitates public debate in society. Currently, research on algorithmic behavior modification is limited in scope and operation (Greene et al., 2022). The entire interaction of user behavioral data and the associated algorithm data employed for behavior modification and prescriptions is completely unavailable to the public. It is almost impossible to access the operational datasets of platforms that allegedly maneuver behavior modifications. Thus, the role of scholarly researchers and data scientists in this area is unclear. Greene et al. (2022) argued that significant barriers exist between academic researchers and platform operators in terms of legal, scientific, and methodological issues. Therefore, we need to breakdown the barriers to moving beyond academic research practices and to developing human-centered algorithms because the collective social costs of algorithmic harm in the era of algorithmic behavior modifications are too large to overlook.

The European Commission conducted an audit of the Google online shopping recommendation case. The Commission ruled that Google was in breach of Article 102 because of Google's favorable positioning and distorted display of its own comparison shopping service compared to competing comparison shopping services (Zingales, 2018). Google was fined 2 million euros and directed to take corrective measures to stop the conduct immediately and to avoid engaging in any conduct or act with the same or a similar object or effect. As a result of certain algorithmic design frames, a dominant undertaking smoothly nudges users toward its own services and brands in online shopping, some of which are negative. For example, recommending unhealthy junk foods when shopping online can be a negative nudge since it induces consumers to purchase something harmful for their health and wellbeing.

Realizing the negative effects of algorithmic nudging, in 2021, the Information Commissioner's Office of the UK (ICO) announced a new law prohibiting the use of nudging maneuvers to modify children's behavior negatively. The law stipulates a set of criteria for age-appropriate design that targets safeguarding the personal data of children and prohibits the use of nudge methods to encourage or lead children to offer harmful personal data, deteriorate or close off their privacy safeguards, or extend their use. The Behavioral Insights Team under the ICO has been monitoring

algorithmic nudges to continually assess how well organizations are performing.

Algorithmic nudging emerges as a risky means in need of statutory rules and regulations. Behavioral changes via algorithmic nudging can amplify the severity of freedom-related harm. The law should respond to malicious nudges with differential guidelines for deciding cases of intermediary accountability. The question, then, is whether the law should recognize liability for malicious nudges that result in negative consequences. To answer this question, some researchers have proposed “society-in-the-loop” systems (Shin et al., 2022), while others have proposed methods to audit algorithms (Shin & Kee, 2023). Both approaches seem feasible as long as general users can control the internal AI code. Alternatively, algorithms should be made that persuade people to behave ethically and conscientiously, although this notion seems to be a long way off.

CONCERNS OVER ALGORITHM-DRIVEN NUDGES

With its prevalence and wide applicability, there are hot debates and controversies about algorithmic nudging. While effective in monetizing user data, concerns arise as algorithmic nudging has underlined the assumption of manipulative media maneuvers. Algorithmic nudging can be viewed as a new means of the art of propaganda or maneuvered persuasion because it blurs the distinction between free human choice and nudges users toward specific outcomes and behaviors. This negativity is contrary to the argument by Richard Thaler and Cass Sunstein (2014), who initially highlighted the positive roles of nudging in making further decisions about wellbeing, happiness, and health. Despite the generally positive impact of nudges, it has become apparent in the debate that nudges are problematic on many grounds. The problem with influencing people’s behavior is that modifying their behavior faces ethical issues and often violates basic human rights. Many critics claim that nudges fail to comply with normative ethical standards and that algorithmic nudges deserve thorough ethical scrutiny (Greene et al., 2022). As the intelligent manipulation of individual behavior on a mass scale is almost a technological reality, algorithmic nudging has been met with as much fierce criticism as it garners great enthusiasm (Shin et al., 2022). While the distinction between manipulation and nudging can be fuzzy, the design of nudges to influence users’ decisions in a certain way raises ethical issues. As platforms such as Facebook and Instagram continue their push toward algorithmization, we

are quickly reaching a point where algorithmic nudges will increasingly not only guide us toward the most profitable content for their financial interest but will also remove content that is estimated to be less lucrative. As platforms aggressively seek to maximize the monetization of users, such algorithmic gatekeeping is an effective way to nudge content recommendations through their servers toward more lucrative issues. The nudges coming out of this algorithmic gatekeeping trigger the concern that the nudges infringe on free will or violate basic personal freedom—i.e., user autonomy. In an environment where AI-driven nudges come into play, users may be unaware that the selection they make has been predetermined by invisible algorithms. Without users' consent, this process can be considered a manipulation of human free will, as it is the algorithm that determines the range of choices. Relatedly, several scholars (Shin, 2021) have criticized the idea that algorithmic nudging conflicts with key ethical values, such as liberty and autonomy.

Several civil critics consider that nudges can weaken people's freedom of choice and are not as liberty-preserving as nudge advocates claim (Juneja & Mitra, 2022; Yeung, 2017). This concern relates to volitional autonomy, in that one's behaviors should reflect the individual's desires, interests, preferences, or goals. When influenced by nudges, people may be misled so that their resulting wishes and deeds are no longer truly their own. When people are nudged implicitly and covertly, they are no longer the controllers of their selections, and they no longer mirror their self-directed needs. Algorithmic nudges pull our strings and use tricks to get users to do what algorithms need. When people are nudged implicitly and covertly, they are no longer the controllers of their selections, and they no longer mirror their self-directed needs, which supports the criticism that nudges fail to respect rationality, as they often work through opaque or irrational processes. Even if nudges respect our freedom and promote efficiency and convenience, they tap into our irrational or opaque heuristics and biases, which means that they do not treat us as rational human beings and instead dehumanize us (Jakesch et al., 2023). It has been argued that using algorithmic nudging should be done in a highly cautious and discretionary manner to let people make their own decisions (Yeung, 2017). In this light, Shin (2021) is concerned that nudging can remove users from the ability to make diverse judgments and corrodes their remit for their own decisions. Additionally, when the responsibility for making decisions is taken away from people, their sensible assessment and decision-making abilities cannot progress, which in turn weakens their ethical autonomy.

(Raveendhran & Fast, 2021). This concern relates to deep learning capabilities and emphasizes that becoming a subject of intelligent mass manipulation is not a pleasant thought (Yeung, 2017). Related to this concern is the ongoing debate that exposure to algorithmic nudges predisposes users to make certain decisions or behave toward specific choices. Is this triggered by malicious algorithmic nudges or intentional abuse of the algorithms behind platforms, or is the phenomenon of algorithmic nudges emerging from the experience of searching and interacting with online systems? In other words, is behavior being suggested by algorithmic nudges, or do algorithmic nudges simply echo already existing predispositions for engaging with such content? Shin et al. (2024) argued that algorithmic nudges do not create prejudices, nor do they have any intention of leading users to read fake news and misinformation. The bias we are seeing in AI is the result of a vicious circle in our own content-seeking and use (Shin et al., 2020). Similar to the proverb “*What you sow so shall you reap*,” the bias generated from algorithmic nudging is most likely the result of previous user attitudes and actions.

Increasing concerns over algorithmic nudging have led to the Social Media Nudge Act, proposed by Senator Cynthia Lummis (R-Wyoming) and Amy Klobuchar (D-Minnesota). The bill was motivated as a first step toward tackling the algorithmic amplification of destructive nudges. The Act directs the National Academy of Science, Engineering, and Medicine and the National Science Foundation to examine content-neutral ways to increase friction to content sharing online. The bill also requests scholars to identify possible ways of holding up the diffusion of detrimental content and fake news by requesting that users view an article before tweeting or sharing. The Federal Trade Commission regulates nudges and orders that Twitter or Facebook put nudge rules into action. The background behind the bill is that many politicians believe that social media has garnered undue profits by hooking users and spreading sensational fake news. Debates on this bill argue that, on the one hand, it could address misinformation effectively, but on the other hand, it could threaten free speech by legally allowing censorship mechanisms as a means of controlling debate and suppressing different views. The bill was triggered by Facebook revealer Frances Haugen, and similar bills have since been introduced. Proposed by Tom Malinowski (D-New Jersey) and Anna Eshoo (D-California), the Protecting Americans from Dangerous Algorithms Act is designed to control algorithmic amplification. Together, these bills can

be seen as concerted efforts to increase transparency and improve the user experience by directing algorithmic nudges.

Against the wide use of algorithmic nudges, there is a societal need to think seriously about the governance and control of such algorithms. Some critics are concerned that algorithmic nudging can fall down as a convenient means in the hands of industry or government to exert tricky control over users and citizens and their lives (Brown et al., 2022). Some proponents (Greene et al., 2022) evoke the idea of neoliberalism in which industry and governments increase their control beyond conventional forms of ruling over people and increasingly depend on algorithms to maximize profits. Industry power over users prevails only if industry power is not properly managed by the public. The concern is that algorithms facilitate such domination since they selectively gatekeep public voices and dodge democratic governance behind the shielded algorithms.

Because of the inherently limited transparency and fairness of algorithms, critics have voiced concerns about the manipulative potential of algorithmic nudges (Shin, 2021). Park argued that algorithmic nudges are dangerous because they are manipulative, undermine human dignity, and decrease the human critical thinking process because algorithms influence users invisibly and in a black box instead of transparently. In cases where algorithmic nudges are used in the public sector, primarily by governments, concern is raised about the aims they promote. How can governments determine people's best interests? Governments should not force their goals and interests onto people, particularly not in communities where citizens hold very broad ideas of the good and value.

ALGORITHM UN-NUDGE: ALGORITHM AVERSION AND RESISTANCE TO ALGORITHMS

Within algorithmic nudges, algorithmic behavior is defined as a change in any algorithmic treatment, intervention, or manipulation of platforms designed to influence user attitude and behavior. User behaviors include clicking ads, buying items, posting specific information, and retweeting fake news. Behavior change methods stem from concepts of behavioral psychology and include nudging and operant conditioning. Algorithmic behavior has largely two representations: appreciation and aversion. Algorithm appreciation describes people who are happy with algorithms and are often more likely to accept advice from an algorithm than from a

human (Shin et al., 2021). In contrast, algorithm aversion describes a negative assessment of an algorithm that leads to negative attitudes and behaviors toward the algorithm compared to a human service (Logg et al., 2019). Humans often reject recommendations from an algorithm in cases where they adopt the same suggestions from human agents (Möhlman & Henfridsson, 2019). However, snubbing advice from algorithms can lead to low performance or a decline in quality (Shin et al., 2024) because it debilitates the feedback loop because of users and algorithms. It is important to understand why and in what contexts users exhibit algorithm aversion to leverage the benefits of algorithmic nudges to the fullest extent.

The growing availability of algorithms has increased the trend of algorithm aversion. Researchers have examined the factors that lead to algorithm aversion (Logg et al., 2019). Their findings showed that design factors related to the design of AI lead to aversion. For example, the limited transparency with the black box nature of algorithms means that people cannot understand how algorithms function and produce results. Users are naturally interested in understanding the fundamental logic of the algorithmic process and are thus more inclined to talk to human agents because they can ask questions and understand the rationales behind the results. As such, people turn away from algorithms when they do not understand the algorithmic processes behind the results.

Researchers have also found that decision factors can lead to algorithmic aversion (Shin et al., 2024). For example, factors related to the quality of algorithmic decisions, such as precision, accuracy, and relevance, play a key role in determining user aversion. Inaccurate or irrelevant decisions from algorithms lead people to lose trust in them, as it makes users believe that algorithms are ineffective at completing complicated jobs.

The third cause of algorithmic aversion was found to relate to whether the decisions led people to benefit or suffer from the algorithmic results. The role of algorithmic decisions in relation to human decisions is also a valid factor in algorithmic aversion. People tend to depend more on algorithmic decisions when algorithms enhance human decisions as an assisting role to humans.

Finally, personal factors can contribute to algorithmic aversion. For example, people demonstrate algorithm aversion when the precision of the decisions of both humans and algorithms is indistinguishable and when people evaluate their choices against algorithmic recommendations. Some people are naturally antagonistic to AI, regardless of the algorithmic performance and quality of service, due to their personal disbelief of

algorithms, and others reject algorithmic decisions based on their evaluation of the performance of these decisions.

Designers and providers of algorithms should comprehend what factors trigger dislike and what factors lead to appreciation. The factors presented in this chapter are not a comprehensive list, but developers should have a broad perspective of the relevant factors that need to be considered while planning algorithmic nudges. Because users often snub algorithms because of their black box feature design, designers should consider opening the black box while improving algorithmic performance.

ALGORITHM NUDGES WITH MEANINGFUL CONTROL AND ALGORITHM AUDIT

Algorithmic nudges are useful in human decision-making because they suggest relevant information while preserving users' freedom of choice. However, algorithmic nudges also present concerns regarding steered bias, data accuracy, and manipulation. As the algorithmically formed reality increasingly shifts from merely nudging people toward predetermined results toward removing all other potential results, the algorithms are seemingly taking over. Social platforms are generally silent on issues regarding the fairness, transparency, and accuracy of their nudging. For example, TikTok has also not responded to questions about its nudge accuracy and legitimacy.

This chapter discusses how algorithmic technologies affect humans and how they steer algorithms to enact choice architectures (e.g., placing profitable services at the best interface position while placing less profitable items in harder-to-click spots) and nudges to influence user behavior. The various principles of algorithmic nudges account for recent developments in AI and machine learning. An important proposition in algorithmic nudges is that humans should remain in control of such nudges. Although algorithmic nudges are increasingly pervasive and embedded in many services and objects, they also create unwanted results where moral responsibility for their nudges cannot be suitably attributed to any particular individual or community. The notion of meaningful user control has been proposed for algorithmic nudges to address responsibility gaps and mitigate negative effects by establishing conditions that give people meaningful control over nudges. Algorithmic nudging should enable users to make better decisions by facilitating cognitive processes, extending engagement

to construct data, and augmenting users' abilities to utilize insights from the data. For users to control AI meaningfully, they need appropriate knowledge to evaluate AI, known as algorithmic literacy, and they should be able to experience and interact with algorithms to develop algorithmic experience and algorithmic appreciation. The principle of meaningful control of AI is not as easy as it sounds, as it demands numerous conditions as prerequisites. Current discussions of meaningful control over algorithms are either too abstract to be practiced or too narrowly specific, and the technical conditions of transparency or fairness do not consider normal laymen users or the wider societal context. Nevertheless, meaningful control is not completely ideal. The first step toward meaningful control is to enable algorithmic auditing of the AI system, which makes it more interpretable, fair, and controllable. Algorithmic auditing enables us to protect fundamental rights related to privacy and personal data (Brown et al., 2022). Numerous studies have suggested specific methods of algorithmic audit instruments. For example, Brown et al. (2022) proposed three components: a list of the possible interests of users affected by algorithmic nudges, an appraisal metric that explains key ethically salient features of the algorithmic nudges, and a relevancy matrix that connects the assessed metrics to user interests. Shin et al. (2022) proposed algorithmic audit principles (fairness, transparency, accountability, and explainability) for the ethical evaluation of an algorithmic nudge that could be adopted by regulators and policymakers. In any case, it is critical to pay close attention to the complicated societal context within which algorithmic nudges are used and deployed to prevent algorithmic nudges from moving past the limited perspective of traditional nudging as a simple user interface in AI environments. It is important to design algorithmic nudges as human-driven and user-controlled architectures that contribute to overcoming users' emotional, cognitive, and psychological limits when they make decisions and perform actions that contribute to value cocreation.

As algorithms are unlikely to generate the best outcomes by themselves, humans and algorithms should cocreate values and enable the design of algorithmic nudge contexts that promote desirable behaviors. The role of interaction stems from a pervasive nudge that shapes contexts and augments capacities for self-understanding, interaction, and user action. This discussion leads to a deeper conceptualization of algorithmic processes. We argue that algorithm creation relies on an interactive process based on users' motivation and knowledge, the orientation of their interactions, and users' needs. While the user's awareness and literacy are critical to value

cocreation, new forms of self-understanding and self-development shaped by AI are also important. We conclude that algorithmic nudging allows users to not only select differently but also act differently in practice. By curating the range of users' possible choices, algorithmic nudging contributes to activating users' motivation, thus boosting satisfaction. Algorithmic nudging enacts an algorithmic information process that influences users' agencies and practices.

There is a clear distinction between nudging a particular behavior and compelling a specific choice (Burr & Cristianini, 2019). A good algorithmic nudge can encourage a particular choice, but it needs to (1) be transparent, making the nudge visible and clear instead of hiding other options, costs, or intentions; (2) show available options, enabling users to make the final choice; and (3) be trustworthy, so that users have good reason to believe algorithmic nudges are warranted, as they are designed to improve user experience.

These suggestions provide the industry with a new design principle for algorithmic nudges and behavioral modifications. Recent advances in machine learning technologies have enabled the establishment of socio-technical systems that closely interweave users and their social structures with technologies. With the emerging prominence of algorithmic nudges, the question is how to mediate the tension between human choice and algorithmic nudges. This is a key question in algorithmic nudges.

The AI industry can use algorithmic affordance as a guiding principle for programming algorithmic nudges. Users' understanding and perceptions of algorithms and the ways in which users imagine and expect certain algorithmic affordances affect how they approach technologies (Bucher, 2017). As such, the industry could use algorithmic affordance as a key base on which to create feedback loops of machine learning systems such as Facebook, making user beliefs an important component in shaping overall system behavior. Algorithmic affordance not only benefits users by providing them with opportunities to understand how transparent, fair, and accountable algorithmic nudges are but could also help industries establish user trust in and satisfaction with their algorithmic nudging.

Algorithm technologies are forging AI ecosystems and enabling new practices; a broader discussion should examine users' decision-making, taking into consideration broader AI ecosystems. It is essential to analyze how users and algorithms interact and make decisions within coconstructing AI ecosystems. Users' decisions, intertwined with algorithm technologies, determine such coconstruction by taking consequent actions that can

reform and renew practices and by following such practices and making them institutionalized. Because of the possible risks of manipulation and ethical issues, algorithmic nudges should be designed carefully and used in discretionary ways. Excessive reliance on algorithms for nudges can have unintended results. While giving AI free rein might create effective and sustainable nudges, without the right motivations and rules for the algorithms, algorithmic nudges could lead people to make poor decisions, making them unethical and unsustainable. Future research on interventions should aim to combine effective techno-cognition and nudging in the user experience of online services.

REFERENCES

- Akter, S., McCarthy, G., Sajib, S., Michael, K., Dwivedi, Y., D'Ambra, J., & Shen, K. (2021). Algorithmic bias in data-driven innovation in the age of AI. *International Journal of Information Management*, 102387. <https://doi.org/10.1016/j.ijinfomgt.2021.102387>
- Badke, W. (2018). Fake news, confirmation bias, the search for truth, and the theology student. *Theological Librarianship*, 11(2), 4–7. <https://doi.org/10.31046/tl.v11i2.519>
- Baumer, E. P. (2017). Toward human-centered algorithm design. *Big Data & Society*. <https://doi.org/10.1177/2053951717718854>
- Brown, S., Davidovic, J., & Hasan, A. (2022). The algorithm audit: Scoring the algorithms that score us. *Big Data & Society*. <https://doi.org/10.1177/2053951720983865>
- Bucher, T. (2017). The algorithmic imaginary: Exploring the ordinary affects of Facebook algorithms. *Information, Communication & Society*, 20, 30–44.
- Burr, C., & Cristianini, N. (2019). Can machines read our minds? *Minds & Machines*, 29, 461–494. <https://doi.org/10.1007/s11023-019-09497-4>
- Burr, C., Cristianini, N., & Ladyman, J. (2018). An analysis of the interaction between intelligent software agents and human users. *Minds and Machines*, 28, 735–774. <https://doi.org/10.1007/s11023-018-9479-0-1-3>
- Cowgill, B., & Stevenson, M. T. (2020, May). Algorithmic social engineering. In *AEA papers and proceedings* (Vol. 110, pp. 96–100). American Economic Association.
- Festinger, L. (1957). *A theory of cognitive dissonance*. Stanford University Press.
- Greene, T., Martens, D., & Shmueli, G. (2022). Barriers to academic data science research in the new realm of algorithmic behavior modification by digital platforms. *Nature Machine Intelligence*, 4, 323–330. <https://doi.org/10.1038/s42256-022-00475-7>

- Jakesch, M., Hancock, J. T., & Naaman, M. (2023). Human heuristics for AI-generated language are flawed. *Proceedings of the National Academy of Sciences*, 120(11), e2208839120.
- Juneja, P., & Mitra, T. (2022). Algorithmic nudge to make better choices: Evaluating effectiveness of XAI frameworks to reveal biases in algorithmic decision making to users. *CoRR abs/2202.02479*. *CHI 2022 Workshop on Operationalizing Human-centered Perspectives in Explainable AI*.
- Karimi, M., Jannach, D., & Jugovac, M. (2018). News recommender systems: Survey and roads ahead. *Information Processing & Management*, 54(6), 1203–1227.
- Kim, J., Lee, J., & Dai, Y. (2023). Misinformation and the Paradox of Trust during the COVID-19 pandemic in the U.S.: Pathways to Risk perception and compliance behaviors. *Journal of Risk Research*, 26(5), 469–484. <https://doi.org/10.1080/13669877.2023.2176910>
- Kroll, T., & Stieglitz, S. (2021). Digital nudging and privacy: Improving decisions about self-disclosure in social networks. *Behavior & Information Technology*, 40, 1–19.
- Loecherbach, F., Moeller, J., Trilling, D., & van Atteveldt, W. (2020). The unified framework of media diversity: A systematic literature review. *Digital Journalism*, 8(5), 605–642.
- Logg, J., Minson, J., & Moore, D. (2019). Algorithm appreciation: People prefer algorithmic to human judgment. *Organizational Behavior and Human Decision Processes*, 151, 90–103.
- Mattis, N., Masur, P., Möller, J., & van Atteveldt, W. (2022). Nudging toward news diversity: A theoretical framework for facilitating diverse news consumption through recommenders design. *New Media & Society*. <https://doi.org/10.1177/14614448221104413>
- Möhlman, M., & Henfridsson, O. (2019). What people hate about being managed by algorithms, according to a study of uber drivers. *Harvard Business Review*. www.hbr.org.
- Möhlmann, M. (2021). Algorithmic nudges do not have to be unethical. *Harvard Business Review*. <https://hbr.org/2021/04/algorithmic-nudges-dont-have-to-be-unethical>
- Munyaka, I., Hargittai, E., & Redmiles, E. (2022). The misinformation paradox: Older adults are cynical about news media, but engage with it anyway. *Journal of Online Trust and Safety*, 1(4), 10.54501/jots.v1i4.62.
- Newman, D., Lewandowsky, S., & Mayo, R. (2022). Believing in nothing and believing in everything: The underlying cognitive paradox of anti-COVID-19 vaccine attitudes. *Personality and Individual Differences*, 189, 111522. <https://doi.org/10.1016/j.paid.2022.111522>

- Raveendhran, R., & Fast, N. J. (2021). Humans judge, algorithms nudge: The psychology of behavior tracking acceptance. *Organizational Behavior and Human Decision Processes*, 164, 11–26. <https://doi.org/10.1016/j.obhdp.2021.01.001>
- Scheiber, N. (2021, April 2). How uber uses psychological tricks to push its drivers' buttons. *New York Times, Technology Section*.
- Schobel, S., Barev, T., Janson, A., Hupfeld, F., & Leimeister, J. M. (2020). Understanding user preferences for digital privacy nudges. In *Hawaii International Conference on System Sciences, Maui, Hawaii, USA*. 10.24251/HICSS.2020.479
- Shin, D. (2021). The perception of humanness in conversational journalism: An algorithmic information-processing perspective. *New Media & Society*. <https://doi.org/10.1177/1461444821993801>
- Shin, D., & Akhtar, F. (2024). Algorithmic inoculation against misinformation: How to build cognitive immunity against misinformation. *Journal of Broadcasting & Electronic Media*. <https://doi.org/10.1080/08838151.2024.2323712>
- Shin, D., & Kee, K. F. (2023). Editorial note for special issue on AI and fake news, mis(dis)information, and algorithmic bias. *Journal of Broadcasting & Electronic Media*, 67(3), 241–245. <https://doi.org/10.1080/08838151.2023.2225665>
- Shin, D., Ibrahim, M., & Zaid, B. (2020). Algorithm appreciation: Algorithmic performance, developmental processes, and user interactions. In *2020 International conference on Communications, Computing, Cybersecurity, and Informatics*, November 3–5, 2020, The University of Sharjah, Sharjah, UAE. <https://doi.org/10.1109/CCCI49893.2020.9256470>
- Shin, D., Lim, J., Ahmad, N., & Ibarahim, M. (2022). Understanding user sensemaking in fairness and transparency in algorithms: Algorithmic sensemaking in over-the-top platform. *AI & Society*. <https://doi.org/10.1007/s00146-022-01525-9>
- Shin, D., Jitkajornwanich, K., Lim, J., & Spyridou, A. (2024). Debiasing misinformation: How do people diagnose health recommendations from AI? *Online Information Review*. <https://doi.org/10.1108/OIR-04-2023-0167>
- Thaler, R., & Sunstein, C. R. (2014). *Nudging: Improving decisions about health, wealth, and happiness*. Yale University Press.
- Tsavli, M., Efraimidis, P. S., Katos, V., & Mitrou, L. (2015). Reengineering the user: Privacy concerns about personal data on smartphones. *Information and Computer Security*, 23(4), 394–305. <https://doi.org/10.1108/ICS-10-2014-0071>
- Tufekci, Z. (2017). *Twitter and tear gas*. Yale University Press.
- Valenzuela, S., Halpern, D., Katz, J., & Miranda, J. (2019). The paradox of participation versus misinformation: Social media, political engagement, and the spread of misinformation. *Digital Journalism*, 7(6), 802–823. <https://doi.org/10.1080/21670811.2019.1623701>

- Vosoughi, S., Roy, D., & Aral, S. (2018). The spread of true and false news online. *Science*, 9(359), 1146–1151. <https://doi.org/10.1126/science.aap9559>
- Weinmann, M., Schneider, C., & vom Brocke, J. (2016). Digital nudging. *Business & Information Systems Engineering*, 58(6), 433–436. <https://doi.org/10.2139/ssrn.2708250>
- Yeung, K. (2017). Hyper nudge: Big data as a mode of regulation by design. *Information, Communication & Society*, 20(1), 118–136.
- Zarouali, B., Boerman, S., & de Vreese, C. (2021). Is this recommended by an algorithm? *Telematics and Informatics*, 62, 101607. <https://doi.org/10.1016/j.tele.2021.101607>
- Zingales, N. (2018). Google shopping: Beware of self-favoring in a world of algorithmic nudging. *Competition Policy International-Europe Column*. Available at SSRN: <https://ssrn.com/abstract=3707797>

What Are the Implications of AI for
Misinformation? The Challenges and
Opportunities When Misinformation
Meets AI



Misinformation and Inoculation: Algorithmic Inoculation Against Misinformation Resistance

INTRODUCTION

A new trend is the spread and consumption of misinformation on an individual level, facilitated by AI media. Immunizing misinformation initiatives has become an increasingly effective technique for mitigating the spread of misinformation (Musi et al., 2023). Among the initiatives, the inoculation approach has been supported in various contexts and appears to be a promising approach for mitigating susceptibility to both individual instances of misinformation and the techniques and strategies that are broadly used to mislead or misinform people (Compton, 2021). Inoculation theory proposes preemptively exposing users to a refuted or a weakened version of the message beforehand (Compton et al., 2022). The theory affords a conceptual framework for a cognitive vaccine against rising falsehoods by conferring mental resistance against future misinformation, and manipulation attempts, such as medical vaccines, make resistance against a particular disease (Traberg et al., 2022). Inoculation messages normally include refutational preemption and motivational coercion. The coercion component prewarns users that they can be exposed to refutational preemption and psychological manipulation, which either involves directly offering users counterarguments that refute misinformation, called passive inoculation, or actively entrenching users in generating those counterarguments, named active inoculation (Shin & Akhtar, 2024). Active inoculation can give a relatively greater inoculation effect since

users are nudged to produce their own resistance (antibodies) and counterarguments (Borukhson et al., 2022). As people play a proactive role in the inoculation process, inoculated immunity nudges them to trigger internal refutations themselves (Van der Linden, 2023), which can lead to long-term consequences due to enhanced psychological resistance (Pennycook, 2023).

Recently, relevant research has examined the inoculation of users against mis-and-disinformation techniques instead of individual instances of misinformation, similar to a kind of refutational-different inoculation (Shin & Akhtar, 2024). The strength of active and technique-based inoculations lies in their potential applicability and scalability (Roozenbeek et al., 2022), as they arrange users to counter being exhorted by messages that may differ in content but use the same underlying persuasion method. The feasibility of technique-based inoculation has been found to be highly effective, as it is a broad-spectrum, tactic-recognition perception: it can confer cross-defense against falsehoods that inoculated users were not trained to recognize and can decrease susceptibility to actual misinformation that is prevalent online (Roozenbeek et al., 2022).

However, essential questions about the validity of message/technique-based inoculation in AI remain underresearched. No research has directly compared the efficacy of technique-based inoculation for reducing susceptibility to misinformation that is simply falsehoods versus misinformation that makes use of a particular misinformation method that people are vaccinated or immunized against. Moreover, scant empirical research has been conducted on the way proactive inoculation interventions generate cognitive immunity against real-life examples of misinformation (Roozenbeek et al., 2022).

The most common consequence tools for measuring inoculation interventions can be the perceived reliability (Borukhson et al., 2022) or manipulateness (Shin & Akhtar, 2024) of both legitimate news information and misinformation. It is critical to examine how inoculation nurtures people's ability to evaluate the *accuracy* of true and false information (Pennycook et al., 2022). Requesting users to assess the *manipulativeness* or *reliability* of information does not necessarily render requesting them to judge its authenticity, although this is the case for prompting a headline's *accuracy*. Relatedly, Zrnek et al. (2022) found that evoking different queries and response modes does not produce significant variations in response patterns when evaluating misinformation susceptibility. Despite numerous attempts, a key question remains as to what extent

psychological resistance and cognitive immunity against misinformation is possible and useful.

In light of this research gap, this study empirically tests the effect of inoculation theory within heuristic–systematic (HS) processes on the problems associated with misinformation and human reasoning and/or information processing models of AI chatbots by focusing on the interaction between user cognitive models and generated inoculated messages. The HS framework allows for implementing cognitive, psychological, and social elements related to susceptibility to misinformation and enables us to examine the efficacy of the inoculation approach on an individual level. Despite their importance, the processing mechanisms of inoculation, particularly in AI contexts, have been scarcely researched. Thus, the accurate nature of how users respond to inoculation and its effects, as well as the impact of misinformation in AI contexts on user behavior, remains inconclusive. Users’ cognitive heuristics to understand inoculation, the related process of misinformation discernment, and the ensuing effect of misinformation-amplification behavior should be further clarified (Walter & Tukachinsky, 2020). Thus, this study is guided by the question of how and whether inoculation messages help users resist misinformation in AI.

Examining these topics is relevant and critical for several reasons. Understanding users’ heuristics and acceptance of inoculation mechanisms about the role of cognitive processes in interpreting inoculation can help combat misinformation and the diffusion of falsehoods in the AI landscape (Pennycook & Rand, 2022). Knowledge of how people process inoculated messages can provide insights into combatting misinformation and all forms of inaccurate information in AI contexts (Tully et al., 2020). Modeling the perceptual and cognitive processes of inoculation has been of interest in understanding the reasoning of users encountering misinformation (Ahmed & Gil-Lopez, 2022).

The goal of this study is to explore ways to build psychological resilience against misinformation through cognitive inoculation. The findings contribute to ongoing development by clarifying the algorithmic effects of AI constraints, such as the black-box nature and the procedural uncertainty of AI, on a user’s evaluation of generated misinformation (Shin, 2023). The findings of this study reveal how AI can improve its design to foster users’ evaluation of misinformation and to help them arrive at a literate informed decision. This insight could pave the way for professionals to incorporate the appropriate strategies needed to combat the generation and spread of misinformation via AI. With this focus on misinformation

processing, this study provides theoretical insights and relevant recommendations for firms to be more resilient in protecting users from the detrimental impact of misinformation.

LITERATURE REVIEW: INOCULATING MISINFORMATION

To examine the cognitive and information processes related to inoculation processing, we integrated insights from two theoretical accounts—the HS processing framework and the concept of inoculation theory—and examined the antecedents and consequences of inoculation processing by chatbot users.

Inoculation Theory

Inoculation theory is the classic psychological/communicative frame that illustrates how a perception or belief can be enhanced or is resistant to attacks through preexposure to weak counterarguments in a similar way that a human body can be immunized against disease (Compton et al., 2021). Since its inception in 1961, the theory has been used widely to describe how beliefs and values modify and how to maintain current cognition and perception consistent in the face of attempts to propagandize them. Using medical inoculation as its exemplary analogy (applied to belief instead of infections), the theory works as a cognitive approach to safeguard attitudes from persuasion by conferring *resistance* to counterattitudinal attacks, whether such attacks take the form of sustained pressures, direct/indirect influences, and psychological threats from various sources (Basol et al., 2021). The *theory* proposes that an individual can build immunity against certain issues with weak counterarguments beforehand by strengthening their existing attitudes toward the issues (Musi et al., 2023). When exposed to weakened doses of information of a stronger threat, individuals confirm that they support information to further reinforce their attacked beliefs. The strengthened beliefs become resistant to a stronger adversarial message, similar to the medicinal analogy of an immunization. This cognitive immunization has been confirmed to generate cognitive resistance against stronger manipulation attacks (Compton & Pfau, 2005).

Inoculation messages commonly contain forewarning and refutational preemption (Compton et al., 2021). One way of messaging triggers a response to a warning (forewarning of impending misinformation), while

another gives a weakened dose of prebunk falsehood (refutational pre-emption). Cognitive inoculation can be triggered through a motivation threat response, in which users attempt to defend themselves against an attack on purely fictional information. For instance, a message can be presented to users, such as “*There is intentional hateful and false information on immigration being spread online and in ChatGPT.*” The second way is to give the prebunk message as a weakened dose of false propaganda. This could be the same type of message used in debunking where we warn about the myth, for example, “*Immigrants take our jobs and destroy our economy.*” This is subsequently refuted by the correct information.

Recently, with the COVID-19 pandemic and political correctness debates, the theory has been proven to be a particularly effective approach to develop interventions that build psychological resistance and/or public immunization against fake news and disinformation (Musi et al., 2023), for instance, in addressing conspiracy theories, risky health behaviors, science denialism, political propaganda messages, and emotionally manipulative campaigns (Van der Linden, 2023). The inoculation approach has received heightened attention, although a review of such research is lacking. This study attempts to fill this gap by tackling the potential of risk perceptions of misinformation to be inoculated against fake news myths among users. Applying inoculation theory can lead this study to understand how AI chatbots and interpersonal communication can stimulate people’s attitudes and how they create counterarguments against misinformation attacks.

HS Inoculation Processing

The influence of inoculation information processing on misinformed decision-making can be dual. Information is processed within an intuitive and autonomous system and within a more analytical reflective system using systematic processing methods (Pennycook, 2023). The HS model of inoculation processing analyzes the behavioral changes of individuals based on two types of information processing: heuristic and systematic (Chaiken et al., 1996). HS theory explains that when people engage in heuristic evaluation, they process inoculation based on available heuristic cues or simple decision rules. In contrast, when they are involved in systematic processing, they thoroughly evaluate the information to check its validity. This theory is used as a prism to examine users’ sense-making of inoculation information; it clarifies the role of algorithmic features in the

sense-making of users regarding algorithms and how their actions influence this sensemaking (Zrnec et al., 2022). The HS theory can be applied in the context of inoculation to extend the information processing model by examining the antecedents of user heuristics and the consequences of the systematic processing of algorithmic inoculation.

Inoculation Messages Against Misinformation in AI Chatbots

Misinformation is false or inaccurate information that spreads accidentally and is not sustained by evident proof or scientific data (Vraga & Bode, 2020). *Misinformation-related problems* are exacerbated by the wide availability of online platforms and the instantaneous and viral nature of online debates (Margolin, 2021). With the emergence of AI, the misinformation dilemma escalates. AI-driven chatbots have attracted enormous traction because of their ability to quickly and confidently present users with information (Korzynski et al., 2023). However, this confidence and speed could pose a risk, as chatbots often generate false narratives. Efficient and quick information generation by AI could have wide-ranging implications for the diffusion of misinformation online. New AI chatbots can generate a vast flood of misinformation online. Misinformation can flow into AI models and from them, which means that some chatbots are susceptible to “injection attacks,” where malicious parties input lies to the tools and train them; the tools, in turn, spread them. Chatbots, for example, are susceptible to being used as a platform for misinformation generation and amplification (Eysenbach, 2023). In many cases, misinformation is presented as a fact and truth (Melchior & Oliveira, 2022). Because the results have natural language modality, users tend to trust the results that have been algorithmically constructed for them. This concern is related to a lack of transparency and explainability (Shin, 2023). For chatbot-generated answers, explaining to users where the information comes from and how the answers are generated, constructed, and presented to them remains a challenge. End users cannot and, in general, do not try to construe a sense of transparency and explainability of chatbots, which ultimately hurts user trust and engagement.

Research Questions: A Conceptual Characterization of Inoculation in AI

With the popularity of AI and the prevalence of misinformation from generative mechanisms, how people understand and process inoculation messages becomes a legitimate question when preparing solutions to mitigate misinformation from AI (Shin, 2023). Previous research on countering misinformation has mainly addressed the degree to which reason and reflection deter or facilitate the development of correct assessments (Borukhson et al., 2022). While extant research suggests that inoculation is the underlying driving factor of the attitudes and behavioral decisions of users (e.g., Roozenbeek et al., 2022), how inoculation is formed, construed, and accepted by users in the HS process should be examined. We examine how the presentation of inoculation messages in chatbots influences users' resistance to misinformation, making them immune to chatbots' misinformation problems. The following research questions (RQs) are proposed:

RQ: What are the cognitive mechanisms of inoculation effects on using chatbots?

1. How do users construe a message of inoculation?
2. How do inoculation messages influence users' resistance to misinformation? How does inoculation confer resistance?

UNDERLYING THE COGNITIVE PROCESS OF INOCULATION

The proposed model is rooted in information processing theory and explores the HS processing of inoculation in AI systems. Our hypotheses help investigate the sensemaking of users when presented with inoculation generated by algorithms and the effect of such sensemaking on their misinformation discernment and behavior.

Main Effects

Heuristic Reasoning

In response to the growing public attention to generative AI, people's awareness of the potential misuse of chatbots has increased (Van Dis et al., 2023). Awareness has also increased people's understanding of AI, as the

specifics of chatbots or internal algorithms are not easily comprehensible to the public owing to their black-box nature (Ali et al., 2022). In this murky AI situation, users face a high level of uncertainty and rely on their own heuristics. Normally, users gather and use relevant information from available sources to make optimal choices. When information decisions are generated in AI contexts, such as opaque processes or possible biases, users may change their evaluations, effectively adjusting how they evaluate and use misinformation judgment. The heuristic assessment of misinformation involves evaluating the intrinsic nature of algorithmic attributes, such as source credibility (Feezell et al., 2021), affect (Compton et al., 2022), and fairness/transparency (Shin, 2023). As users view inoculation messages, they rely on heuristic cues to process inoculations. Sundar et al. (2007) show that news chatbot users rely on heuristic cues to process recommended news feeds. Chatbot users apply heuristics and a systematic evaluation to process inoculation among the generated information. Users would like to see inoculation in terms of fairness, bias, accountability, and objectivity.

Affect Heuristics

The role of affect in susceptibility to believing misinformation has been researched extensively. Previous research on the spread of misinformation has primarily emphasized the extent to which reasoning and deliberation impede versus facilitate the development of correct judgments. Many prior studies have confirmed that users who engage in more systematic assessment are more likely to reject misinformation, and the role of affect in belief in misinformation remains unexplored (Martel et al., 2020).

The affect heuristic is a form of the *cognitive* process that plays a role in decision-making (Vafeiadis & Xiao, 2021). The affect heuristic occurs when users rely on their emotions and affect, such as gut feelings, rather than objective or accurate information when making decisions (Compton et al., 2022). The affect heuristic impacts users' message processing by influencing how people perceive the risks and benefits related to an action. This can also serve as a shortcut to solving a problem quickly.

In an inoculation context, the affect heuristic occurs due to affective or emotional reactions to a stimulus, which are often the very first reactions people have (Vafeiadis & Xiao, 2021). Affect heuristics occur quickly and spontaneously, affecting how users process and evaluate information. When users subconsciously let these feelings lead their decisions, they rely on the affect heuristic because they perceive inoculation messages in

preliminary ways without any preconditions (Ecker et al., 2022). The causality of the affect heuristic to users' cognitive immunity is hypothesized as follows:

H1: Affect heuristics of users positively influence users' cognitive immunity against misinformation in chatbots.

Fairness Heuristics

With the cognitive inability to process the increasing amount of misinformation in their generative content, users apply the fairness heuristic in interacting with chatbots. People use their heuristics to judge possible algorithmic biases, such as ethnicity and gender biases, from chatbots (Mhasawade et al., 2021). As people rely more on chatbots for their important decisions and related information, biases in AI are increasingly concerning and may potentially cause harm. In reality, users have a limited capacity to judge algorithmic biases, as they neither evaluate message validity on their own nor configure systematically and correct biases by tracing them in data and internal algorithms. Shin (2023) showed that algorithmic fairness hinges upon the perceptual impartiality and equitability of users regarding AI. Related research has clarified the causality of algorithmic fairness to user attitudes, particularly user efficacy and confidence (Diakopoulos & Koliska, 2017). Because the self-efficacy of users is related to their diagnosticity, a connection between algorithmic fairness and the diagnosticity of users in misinformation can be proposed as follows:

H2: The fairness heuristics of users positively affect their ability to judge misinformation.

Transparency Heuristics

Transparency in AI entails both the inputs and outputs of algorithms being visible and comprehensible to users or the public (Shin, 2023). The transparency of the data acquisition technique can work as a heuristic when evaluating data protection and sensitivity (Shin, 2023).

It is reasonable to think that misinformation corrections are more effective when the debunking misinformation is made transparent. A fact-checker rating, for example, is not generally understandable unless users are told why the information provided is not correct or potentially misleading (Musi et al., 2023). Transparent statements showing the

generative mechanism process on websites or in AI systems provide users with certain heuristics that allow users to quickly form their judgment on transparency without conducting extensive technical assessment of the information content. As users cannot assess each piece of information systematically or each component of the AI system technically, users depend on their heuristics based on available cues (Molina & Sundar, 2023). When users have a sense of transparency in systems, they tend to deem the system trustworthy and reliable (Gran et al., 2021). Research has confirmed that transparency enhances user confidence in an algorithm, playing a key role in whether users use it (Peifer & Meisinger, 2021). When transparent and visible cues are available, people are more likely to view the given information as diagnosable (Diakopoulos & Koliska, 2017). Transparent AI systems can afford users a sense of confidence, presenting them with a readiness to diagnose misinformation.

H3: The transparency heuristics of users positively influence users' ability to judge misinformation.

Systematic Processing

Users conduct reflective assessments of information, for example, evaluating how accurate and credible the generated information is (Ali et al., 2022).

Accuracy of Inoculation Messages

The perceived accuracy of a piece of information is the user's perception of the amount of misinformation contained in the information (Pennycook & Rand, 2022); it refers to how truthful the user perceives a viewed piece of information to be. The accuracy of a chatbot application is related to how many results are correctly personalized and to the veracity of the recommended content. Regarding the measurement of the perceived accuracy of misinformation, relevant studies have proposed various terms to describe it, such as misinformation awareness, which is a person's awareness of knowing that the received information is false (Schuetz et al., 2021). Empirical research has confirmed these relationships in diverse algorithm services (Shin, 2023), in which accuracy is identified as a determinant of attitude and behavioral intention (Pennycook & Rand, 2022). Accordingly, the following relation is proposed:

H4: Perceived accuracy positively influences users' ability to judge misinformation.

Credibility of Inoculation Messages

As a strong predictor of the influence of information, credibility can generally be understood as “believability” and “trustworthiness” (Wathen & Burkell, 2002). In the communication literature, credibility is measured based on different perspectives, such as source, message, and media credibility (Barnoy & Reich, 2022). Although credibility has historically been more extensively studied from the source perspective, in the social media environment, news consumers focus more on the message’s content than on the source (Peifer & Meisinger, 2021). In chatbots, source credibility and content credibility are equally important, as content is generated by AI (Ecker et al., 2022). The more credible the content of a piece of misinformation is perceived to be, the more useful and relevant (i.e., diagnostic) users would perceive it to be.

H5: Credibility perceived by users regarding misinformation in chatbots influences their ability to judge misinformation.

Cognitive Immunity: How Users Form Their Efficacy and Vaccination Belief

When arriving at a diagnostic judgment about the misinformation and system of the corresponding heuristics in cognition, users evaluate both the content of the information and the systematic cues it contains (Niu et al., 2021). Having diagnostic information is key for AI to operate trustworthy services. Diagnosticity is defined as the extent to which people interpret a piece of information to be truthful and relevant according to their judgment and decision-making (Ahluwalia et al., 2001; Chen & Cheng, 2020). In AI contexts, diagnosticity refers to a user’s perceived ability to understand the features and attributes of AI services in their decision-making (Shin, 2023). Estimating the diagnosticity of information from AI must be based on inference and personal speculation. Information that is perceived to be diagnostic is given relatively more weight in the judgments and evaluations of users (Shin, 2023); thus, diagnosticity influences how users systematically process messages (Niu et al., 2021). Upon realizing a system’s diagnosticity, users trust it and begin systematically assessing misinformation by measuring its accuracy and credibility (Chen & Cheng, 2020; Stecula et al., 2020). The indicators of diagnosticity are uniquely specific to users’ ability to identify misinformation based on their understanding of the information’s accuracy and credibility (i.e., the correctness and reliability of the suggested information;

Zrnec et al., 2022). It can be hypothesized that diagnosticity functions as a conduit between heuristic and systematic evaluations.

H6: Diagnosticity perceived by users positively affects cognitive immunity.

Moderating Effects

Message Type Effect: Forewarning Versus Refutational Preemption

Per inoculation theory, users can be inoculated against warnings or become resistant to attitude modification via inoculated treatments (Compton, 2022). To enhance resistance to attitude change against misinformation, inoculation interventions should contain (a) messages about the chatbot's performance since the refutational pretreatment part of the inoculation intervention can give possible warnings to users. Moreover, performance information should be (b) provided in the optimal linguistic form. In the case of a chatbot misinformation recommendation, the potential risk to the user is the chatbot's misperformance. Considering the essential features of the most popular platforms for chatbot design, this performance information can be provided to the user in written form. Chatbots are diverse in their declared performance level. Chatbot misinformation problems cannot be completely avoided because they are often caused by factors that affect users' interactions and responses to them. Thus, moderating effects can be hypothesized that an inoculation message for chatbots with low forewarning lessens the effect of perceived performance on users' discernment and cognitive immunity.

H_{m1}: The message types of inoculation messages can moderate the effect of heuristic and systematic processes on users' misinformation immunizations.

Algorithmic Source Effect: AI-Powered Chatbot Versus Legacy Sources

Information sources provide essential cues and intuitive heuristics that influence belief formation (Shin et al., 2024). In general, information from sources perceived as credible has a greater effect on users' beliefs, motivations, and attitudes than information associated with noncredible sources. Users trust information sources if they perceive the source as trustworthy, established, and individual, similar to themselves (Jia & Johnson, 2021). This kind of source judgment is applied to the chatbot context as users perceive the algorithmic source's credibility and believe the information it generates. Users are more likely to react to inoculation

messages for information sources that have been algorithmically customized to users (Shin, 2023). In the chatbot environment, users understand that the generated information is algorithmicized, curated, and filtered by AI. When information is mediated by algorithms, users tend to believe that the recommended information is relevant, correct, and accurate. Hence, inoculation treatments are more likely to work when presented with algorithmic information because users are aware of the operation of algorithms in the background. People are more likely to react to the inoculation messages of misinformation present for algorithmic sources, as they already believe in the power of algorithms in general. Algorithmic information from chatbots is often considered to have undergone the due quality of expert processes and rigor (Shin, 2023), thereby positively affecting users' perceptions of the quality of their informational sources and the accuracy of the content (Shin, 2022). Nevertheless, for online sources without algorithmic filtering, users are more likely to react to accuracy alerts because they are less able to detect fake news, misinformation, and inaccurate news from unfiltered online information. Thus, the following moderation hypothesis can be proposed:

H_{m2} : The information sources of inoculation messages can moderate the effect of heuristic and systematic processes on users' misinformation immunizations.

Based on the constructs and hypotheses presented in this study, a conceptual model is proposed to depict how users process inoculation messages related to the harmful effects of a product, its antecedents, and its consequences (see Fig. 8.1).

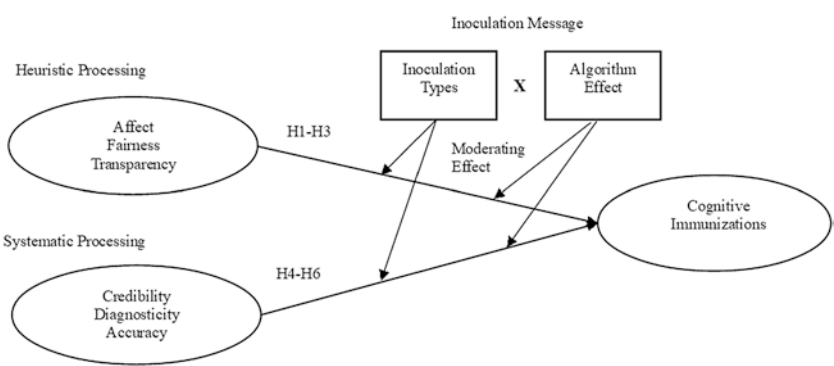


Fig. 8.1 Conceptual Model

METHODS

The method measured the efficacy of the intervention in a 2×2 design, comparing the effects of message types and source effects on user cognitive immunizations. Following Roozenbeek and Van der Linden (2019) and Basol et al. (2020), the dependent variable uses cognitive immunity consisting of an assessment of the accuracy of misinformation and user confidence in discernment.

Experimental Design

Inoculation messages involve two main elements that nurture users’ attitudinal resistance: threat and refutational preemption. We used a 2×2 between-subject experiment to empirically investigate people’s cognitive processing of inoculation (inoculation message: forewarning vs. refutational preemption $\times$ algorithmic effect: algorithmic vs. nonalgorithmic service), which produced a set of four conditions (Table 8.1). The quasi-experiment was a between-subjects design that manipulated the source of and trust in information services. The experiment, which took place in Seoul, Korea, in the first quarter of 2023, was a randomized online experiment in the context of AI in which users interact with chatbots to search, view, and seek information.

Manipulations

Participants were given the inoculation treatment and were requested to complete the required procedures. Two types of inoculation messages were presented to the treatment groups: (1) forewarning and (2) refutational preemption (Compton, 2021). First, message receivers were presented with a forewarning, hinting at them with a pretaste of an actual falsehood that they may subsequently face. Participants were told that the AI-based chatbot might not be perfect and that it could suggest

Table 8.1 2×2 Experimental design

	<i>Forewarning</i>	<i>Refutational preemption</i>
Algorithmic	Group 1	Group 2
Nonalgorithmic	Group 3	Group 4

misinformation: “AI chatbot is not faultless. This chatbot is constantly advancing, but it may produce misinformation.” Second, downplayed potential threats with a refutational pretreatment were presented to participants. This step was fulfilled by proving the participants with a constructed counterargument or competitive frame. A refutational paragraph was constructed based on the AI’s performance information: “The chatbot’s information quality is superior. Currently, it provides approximately 90% of correct information.” The second message was provided with a supportive treatment to persuade the participants to engage more effort in generating their requests: “If you encounter misinformation from the chatbot, please report it to us and try to rephrase your inquiry” (Fig. 8.2).

Procedure

Upon receipt of research approval from the Institutional Review Board, respondents were recruited from Qualtrics panels, and panel members were invited to the survey via Qualtrics, who offered monetary incentives for survey completion. Upon obtaining informed consent, the participants were asked to visit the setup link where they could see the interface of the chosen chatbot. With brief instructions, the participants were provided with a predetermined list of news items on current health and health campaigns with related misinformation. The participants were set up in an interaction scenario in which they were required to search and look for information through a chatbot. The experimental procedure comprises five sequential components. A short introduction to the participants was presented that contained the instructions for task completion and information about the experimental procedure.

All messages were inoculation information that comprised dual items, one forewarning paragraph and one pretreatment paragraph, except for

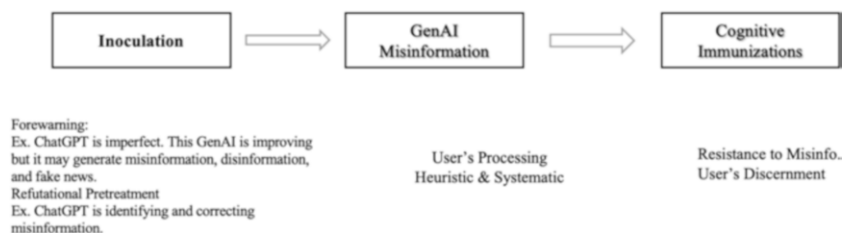


Fig. 8.2 Methodology

the control group, which did not receive any inoculation message. The message information was presented on a single page before task solving began and was repeated as part of the chatbot's welcome greeting during task solving to ensure that participants recognized it. After completing the task, the participants completed the postexperimental questionnaire about their chatbot experience and other questions. ***Collection of Data and Samples*** To check for reliability and validity, six additional research participants completed a pretest questionnaire. Recommendations and preliminary results from this pretest (i.e., shortened minimum completion time for multiple-choice questions) were incorporated before distributing the questionnaire on a large scale online. The recruited participants for the main experiment were randomly assigned to one of the four groups, with no statistically significant differences among the conditions in terms of gender, education, prior knowledge about algorithms, or media consumption of news. The four groups were similar in terms of age, gender, and prior experience. All study respondents were randomly recruited via a marketing company located in Seoul. Most of the participants were Korean (92%), the average age was 31.76 (SD = 9.87), and the gender ratio was almost equal (female: 52.1%; male: 47.9%). Most of the participants (61.0%) were college graduates or had a college-level education. Regarding employment status, 120 respondents were students, and 110 were full-time workers. Of the collected responses, ten incomplete responses with missing information were omitted. Descriptive statistics show that the data were distributed approximately normally. After interacting with chatbots, the participants completed a survey. They were either paid \$2.50 or received course credits for completing the study, which took a median time of 20 min ($M = 20.19$; $SD = 7.92$).

Scales and Measurements

In this study, the scales used comprised 21 measurements with a 7-point scale ranging from 1 (*strongly disagree*) to 7 (*strongly agree*). All measurements were based on the literature on health communications and research on misinformation (Appendix). The measurements were administered using combinations of previously developed measurements and modified items from previous studies. Such combinations were necessary because certain measurements in our analysis needed to be adjusted to reflect the distinctive features of AI algorithms and services. To assess users' heuristic process, item measures were composed of affect heuristics, fairness

heuristics, and transparency heuristics. The measures of fairness and transparency were taken from the works of Shin (2023) and Shin (2020). The measures of affect heuristics were adopted from Compton (2021). To evaluate users' systematic process, three constructs of nine items were composed (credibility, diagnosticity, and accuracy). Credibility measures were taken from Ali et al. (2022). Diagnosticity measures were borrowed from Chen and Cheng (2020). Accuracy measures used the items developed by Epstein et al. (2021). Cognitive immunity refers to users' ability to discern misinformation (Basol et al., 2020). The users' cognitive immunity to misinformation was taken from Basol et al. (2020).

Measurement Instrument

The reliability and validity of the measurement metrics were assessed based on the reliability and convergent validity criteria using Analysis of Moment Structure (AMOS) 18 software. The Cronbach's alpha of the survey instrument was calculated to test its reliability. The analysis results were verified, and varimax rotation was performed on the 30 original items (7 items were eliminated owing to low loading). According to Hair et al. (1995), an item has high loading if its loading coefficient is above 0.7. Each item's loading value was above the recommended level of 0.8 (Cronbach, 1989). Confirmatory factor analysis was performed to test the validity of the instruments; furthermore, the composite reliability, estimated percentage of variance extracted by each construct, and discriminant and convergent validity were tested. All the constructs in the model met the required reliability and convergent validity values (Tables 8.2 and 8.3). *Regarding discriminant validity*, the average variance extracted (AVE) values and the correlations of the factors were examined. The shared variance between the constructs was lower than the AVE from the individual constructs, confirming the presence of discriminant validity. Correlation analysis was performed, and Pearson's *R* (correlation

Table 8.2 Descriptive statistics (*N* = 299)

<i>Characteristics</i>	<i>Sample</i>
Age (mean/SD/median)	24.30/10.02/26
Gender (female rate) (%)	50.43
College educated (%)	32.39
AI platform experience	3.1 years

Table 8.3 Discriminant validity

<i>Construct</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>
Fairness	0.71						
Accountability	0.21	0.83					
Transparency	0.33	0.52	0.83				
Accuracy	0.13	0.23	0.52	0.93			
Credibility	0.12	0.11	0.23	0.42	0.72		
Diagnosticity	0.23	0.53	0.13	0.52	0.11	0.82	
Immunity	0.52	0.11	0.22	0.22	0.22	0.53	0.23

coefficient) showed an acceptable level of correlation among the variables. In conclusion, the measurement model demonstrated adequate reliability and convergent and discriminant validity.

Fit Indices

The structural model was tested with the AMOS procedure and maximum-likelihood-based structural equation modeling software. The results indicated the presence of no misfits (Table 8.3). All values were within acceptable ranges, indicating good overall model performance.

ANALYSES

To test the main effect hypotheses, an analysis of variance (ANOVA) was performed on the dependent variable cognitive immunity (see Table 8.4). The results show that heuristic evaluations positively affected the dependent variable, thus supporting hypotheses 1, 2, and 3 (H1: $\beta = -0.549$, $p = 0.004$; H2: $\beta = -0.358$, $p = 0.003$; H3: $\beta = -0.627$, $p = 0.001$). Users who perceived the chatbot’s normative ethics as good were more likely to have misinformation discernment (odds ratio = 0.523). Second, hypotheses 4, 5, and 6 were supported through a significantly positive impact of systematic evaluations on misinformation immunity (H4: $\beta = -0.549$, $p = 0.004$; H5: $\beta = 0.428$, $p = 0.004$; H6: $\beta = -0.331$, $p = 0.004$), implying that users who trusted the chatbots to be useful, accurate, and discernable were more likely to have cognitive immunity (odds ratio = 0.521).

It was hypothesized that the presentation of forewarning inoculation information might moderate the effect of the HS evaluation on users’ immunity to a greater extent (moderation effect of inoculation message types on cognitive immunity). A moderation analysis using PROCESS

Table 8.4 Reliability checks for constructs

<i>Construct</i>	<i>Initial items</i>	<i>Final items</i>	<i>Mean (SD)</i>	<i>Cronbach's alpha</i>
Fairness	4	3	4.17 (1.011) 3.84 (1.055) 4.11 (1.047)	0.795
Affect	4	3	4.17 (1.011) 3.84 (1.055) 4.11 (1.047)	0.619
Transparency	5	3	3.39 (1.054) 3.98 (1.128) 3.30 (1.282)	0.875
Accuracy	4	3	3.74 (1.346) 3.87 (1.385) 3.40 (1.411)	0.812
Credibility	4	3	4.37 (1.182) 4.30 (1.141) 4.63 (1.202)	0.842
Diagnosticity	4	3	4.37 (1.182) 4.30 (1.141) 4.05 (0.984)	0.853
Immunity	3	3	4.42 (1.121) 4.10 (1.264) 4.18 (1.268)	0.900

Macro 3.1 (Model 1) was conducted, in which cognitive immunity was the dependent variable, HS processes were the independent variable, and message types were the moderator. The moderation test showed a significant interaction effect between the message types and evaluation processes in cognitive immunity (Table 8.5, $p < 0.05$, CI: -19.43, -3.941). The results indicate that for information generated from algorithmic sources, the forewarning message on misinformation immunity becomes significant. Similarly, hypothesis H_{m2} is supported by the significant interaction effect between refutational preemption and HS processes on cognitive immunity ($\beta = -0.591$, $p = 0.051$), while other interaction results were not supported. Given this, it can be inferred that inoculation messages play a significant role in encouraging users' intention to discern misinformation. It can also be inferred that inoculation messages, if used in the AI context, can effectively combat the spread of misinformation (Tables 8.6 and 8.7).

The results of further analysis with t tests also support the moderation effects. Pairwise t tests showed significant mean pairwise differences

Table 8.5 Model fit indices

<i>Fit statistics</i>	<i>Structural model</i>	<i>Suggested value</i> <i>(Joreskog & Sorbom, 1996)</i>
$\chi^2 (df)$	1710.5/244 = 7.01	<5
<i>p</i> value	0.000	<0.05
RMSEA	0.023	0.05 < <i>x</i> < 0.10
CFI	0.910	>0.90
NFI	0.978	>0.90
GFI	0.920	>0.90
IFI	0.802	>0.80
TLI	0.901	>0.85
RFI	0.803	>0.80
AIC	1870	
Hoelter's critical N	66(0.05)/70(0.01)	Higher values signify a satisfactory fit
ECVI	4.688	Smaller values denote a better fit

Table 8.6 Experimental results

<i>Group</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>
Forewarning (Cronbach's $\alpha = 0.87$)	Algorithmic source M: 4.77 SD: 2.102	M: 5.61 SD: 1.153	Legacy online source M: 2.470 SD: 1.095	M: 2.820 SD: 2.006
Refutational preemption (Cronbach's $\alpha = 0.90$)	M: 5.860 SD: 0.841 Nudged	M: 5.620 SD: 1.042 Nonnudge	M: 2.780 SD: 0.732 Nudged	M: 1.610 SD: 0.665 Nonnudge

Table 8.7 Moderation test by Hayes' PROCESS Macro

<i>IV</i>	<i>Misinformation Inoculation</i>			
	<i>Coefficient</i>	<i>Standard error</i>	<i>T</i>	<i>P</i>
Inoculation message	49.7284	17.5923	3.234	0.0050
Source type	68.5393	19.4913	3.7482	0.0004
Message type $\times$ source type	-15.3194	5.3292	-1.9482	0.0003
		Rsq	0.5923	0.0010
		F	18.0329	

between all conditions. The pairwise comparisons between the algorithmic source, forewarning (MAlgorithmic source, Forewarning = 5.23), and legacy source, refutational preemption (MNonalgorithmic source, Refutational Preemption = 2.54), treatments ($p < 0.001$) and between the

algorithmic source, refutational preemption (MAlgorithmic, Refutational Preemption = 5.23), and legacy source, forewarning (MNonalgorithmic, Forewarning = 3.12), treatments ($p < 0.001$) were significant. The findings indicate that the inoculation intervention significantly enhanced users' confidence in their ability to counter misinformation. This effect is moderated by message types and source credibility. The inoculation effect was greater on algorithmic chatbots than on nonalgorithmic legacy sources. Users' levels of immunity were also higher in refutational preemption than in forewarning.

DISCUSSION: HOW TO BUILD PSYCHOLOGICAL VACCINES AGAINST MISINFORMATION

This study examined the use of inoculation interventions as a tool to help users immunize themselves against misinformation. The findings are consonant with the well-established concept of prebunking (Van Der Linden, 2023) that detailed warning about the specific misinformation before users are exposed to it in algorithmic contexts would influence users to view it as misinformation and less likely to accept it. The findings make noble contributions for several reasons. First, this study confirmed the underlying fundamental assumption that chatbot misinformation has a negative effect on user perception in a service context, which is consistent with ongoing research highlighting the need for misinformation-free AI (Eysenbach, 2023). Moreover, our findings show that inoculation messages containing forewarnings about chatbot misinformation moderate/weaken the effects of HS evaluations on users' cognitive immunity. Thus, in the case of chatbot misinformation, users who were presented with an inoculation message at the beginning of their chatbot interactions were more likely to have increased resistance or awareness against misinformation from the chatbot. When an inoculation message containing forewarning information was presented, the strongly positive relationship between the HS process and immunity was enhanced. Hence, the results respond to research gaps in examining measures to mitigate the chatbot misinformation fallacy.

Second, the results resonate that the fundamental principle of inoculation messages, as identified in other studies such as health and political communication, namely, preparing users for impending negative campaigns through inoculation interventions, is also applicable in the areas of AI technologies and can thus be applied to specific usage cases (such as

generative AI services). Depending on the message type of inoculation, users processed the inoculation effect through different cognitive paths. A forewarning form moderated the effect of the heuristic process, whereas a refutational preemption form moderated the effect of a systematic process on user cognitive immunity. With the results, further confirmations for the previous findings (e.g., Pennycook, 2023; Roozenbeek et al., 2022) can be added that forewarning forms are processed more heuristically and preliminarily than refutational preemption, engaging more ethical/procedural evaluations and corresponding to the HS process of algorithmic information processing (Shin, 2023; Zrnec et al., 2022). Furthermore, it can be confirmed that a refutational preemption is processed more analytically and calculatingly, taking up more utilitarian aspects of chatbot functions.

In contrast to previous research that focused only on ex post debunking methods in educating users about misinformation, the results highlight that the measures taken prior to response failure can help boost user self-efficacy and confidence in misinformation discernment. Importantly, the inoculation effects are not limited to users continuing to interact with the chatbot, as it has an influence before the users decide to face misinformation. Thus, the risk of losing users after a chatbot misinformation problem can be eliminated. Taken together, these results implicate the theoretical mechanisms and practical issues of relevant measures.

THEORETICAL IMPLICATIONS: INOCULATING USERS AGAINST MISINFORMATION

This study responded to the research call to further the user-centered approach to AI misinformation problems. Focusing on users' cognitive process of inoculation, this study empirically examined the effects of inoculation treatment on users' attitudes and behavior while proposing and testing a misinformation inoculation model. The study attempted to overcome the main limitations of user-centered AI literature: (1) the emphasis of previous studies on misinformation in AI is not reflective of contemporary trends of chatbots, hence limiting the applicability of results; (2) underplaying users' responses to inoculation messages in AI could deteriorate the misinformation problems in chatbots due to a lack of relevant research-based design implications. Focusing on the inoculation of

misinformation, this study provides an effective measure to reduce the negative effect of a misinformation problem on users' interaction with chatbots.

Second, the current literature on misinformation research focuses on ex post debunking tools, hence limiting opportunities to preemptively address users (e.g., Ecker et al., 2021). Implementing measures prior to actual misinformation encounters in chatbots is critical, as users easily discontinue their interaction with chatbots after they experience misinformation, and ex post measures would not work after they decide not to use chatbots. The study confirms that inoculation interventions presenting prebunking can increase users' intention to continue using AI.

Third, this study contributes to AI research by proposing a method that has the potential to make contemporary chatbots more effective and human-centered (Shin, 2023). The results of this study illuminate that the inoculation approach and its messages are effective for promoting user acceptance after a chatbot's misinformation is identified by users. Based on the widely accepted inoculation theory, this study showed the types of effective inoculation tactics and strategies that can alleviate the misinformation effect of chatbots. To warrant the effectiveness of inoculation treatment, this study proposes a user-specific design based on a dual inoculation message featuring chatbot performance information. Contemporary AI research focuses on the technical matter of how to improve the generative mechanism of algorithmic components in relation to user perception and experience (Korzynski et al., 2023). Research addressing user-centric and procedural aspects is essential to ensure the sustainable development of chatbots by producing practitioners with a hands-on measure to improve their chatbot interaction. Thus, this study provides a relevant implication for chatbot research, ensuring long-term sustainability while offering short-term applicable tactics.

Furthermore, the study contributes to HS theory by addressing the role of inoculation interventions on users' information processing in the context of chatbots. To improve particular user behavior, prior to an initial interaction, previous HS research focused on the importance of methods such as literacy and awareness (Shin, 2023). However, these approaches are limited because they convey persuasion through the systematic processing of generated information. This assumes that users are competent and incentivized so that systematic processing can be activated. Owing to these constraints, there has been a research gap on more effective measures, including evidence on which cognitions they involve and how they

could be selectively evoked to promote acceptance (Roozenbeek et al., 2022). This study illustrated that differentiated tactics of inoculation messages elicit a heuristic and systematic process of inoculation, supported by a moderator effect on the respective process. Thus, this study contributes to the HS theory literature by producing a relevant measure to promote acceptance while illustrating that forewarning and refutational preemption in inoculation messages are processed through heuristic/systematic evaluation.

PRACTICAL IMPLICATIONS: MENTAL ANTIBODIES

The findings of this study can be translated into practical guidelines for chatbot providers and misinformation stakeholders. First and foremost, this study provides the necessary proof of concept that the principle of psychological inoculation can be effectively harnessed in AI contexts. The findings suggest specific methods of using inoculation interventions if there is a risk of misinformation generation over chatbots. The empirical validation of inoculation effects on misinformation discernment reveals that warning users of the possibility of misinformation generation will lead users to improve their awareness and judgment of misinformation generated by chatbots.

Although inoculation interventions attenuate the negative effects of chatbot misinformation, thus promoting adoption, providers' ethical obligations are critical. The use of the inoculation approach should be evaluated according to the following ethical criteria: (1) public interest, (2) accuracy and integrity, and (3) quality of the service. These suggestions indicate that prior to implementing inoculation messages, chatbot providers need to seriously evaluate which benefits or risks for users for inoculation messages can serve to inform users about the expected functionality and possible harms so that they can conduct informed decision-making on misinformation.

The findings of this study suggest that chatbot providers should be accurate and truthful in their evaluations of chatbot performance when presenting inoculation interventions. This is important to inoculation messages, as an essential part of the messaging (i.e., prewarning and refutational pretreatment) is based on the disclosure of the chatbot's actual performance. If an accurate evaluation of a chatbot's performance is not possible, providers should assess the performance to the best of their knowledge and belief. Before implementing inoculation messages, providers should ensure that they have reached the limit of their capacity in terms

of improving the performance and functionality of the chatbot. It should be ensured that the chatbot has undergone a structured product development process and that no avoidable defects remain before its integration into actual user processes.

LIMITATIONS AND SUGGESTIONS FOR FUTURE RESEARCH

This study is not without limitations. The data collection is still limited, and it is suggested to include other factors for the model process and validation. Based on the study's limitations, this study provides recommendations for future studies focusing on the design of chatbots, which may help reduce the negative impact of misinformation failures on users' actual usage.

First, this study used a simplified version of chatbot and misinformation generation. The manipulations in this study were structured text-based inoculation messages, which are different from current trends in inoculation treatment (interactive content, visualizations, and cartoon). Future studies can benefit from investigating potential interactions with other modes of providing inoculation messages. Future experiments can apply the model of this study to other real-life contexts, such as case studies. Future research may relate the effects of inoculation messages on actual behavior across several chatbot user interfaces that differ in levels of performance. In a case study, particular attention should be given to the dimension of different types of misinformation, such as disinformation and fake news.

Second, the experiment showed only the effects based on staged conditions due to the purpose of the methodological design in this study. Examining other forms of chatbot malfunction and inoculation messages would produce interesting, relevant implications for theory and practice. Future studies can compare timing factors of the misinformation effect, varying the moment of misinformation acceptance, which might be done in longitudinal methods where the long-term effects of misinformation from chatbots on user experience can be examined.

Finally, since chatbots are relatively emerging technologies, researchers should research changes in user behavior and contextualize how these changes relate to AI-specific variables and context-dependent factors. Chatbots have only recently garnered attention from industry, and it is likely that more technologically sophisticated chatbots will be developed in the near future. This predicts that users can be more comfortable with the AI's features and react differently over time when they are familiar with the AI's functionalities.

CONCLUSION

Machine learning–spurred chatbots have seen remarkable advancements in recent years. Despite impressive technological functions, they often produce misinformation. Such misinformation can be a significant obstacle that can impact their success and diffusion because they can cause a loss of trust in the system’s performance among people, with the possible result of discontinuation of usage. This study performed a randomized online experiment to illustrate that inoculation interventions generated for users at the beginning of a chatbot interaction can mitigate the negative effects of a misinformation problem. In particular, this alleviation effect was enhanced if the chatbot was described as a forewarning message. For chatbots that were explained as refutational preemption, the alleviation effect was reduced. Furthermore, the results show that the linguistic form of the performance information influences the type of cognitive processing users utilize. By and large, the results contribute to AI research by (1) foregrounding hitherto neglected user perspectives on chatbot misinformation problems and (2) presenting a method to promote long-term sustainable chatbots. Practitioners can harness the results as a guiding reference in practicing effective inoculation tools to address users’ distrust of chatbots. This study will serve as a catalyst for further studies on tackling the negative effects of chatbot misinformation generation and beyond AI services and users’ interaction with AI.

REFERENCES

- Ahluwalia, R., Unnava, H., & Burnkrant, R. (2001). The moderating role of commitment on the spillover effect of marketing communications. *Journal of Marketing Research*, 38(4), 458–470. <https://doi.org/10.1509/jmkr.38.4.458.18903>
- Ahmed, S., & Gil-Lopez, T. (2022). Engaging with vilifying stereotypes. *Journalism & Mass Communication Quarterly*. <https://doi.org/10.1177/10776990221110113>
- Ali, K., Li, C., Zain-ul-abdin, K., & Zaffar, M. (2022). Fake news on Facebook: Examining the impact of heuristic cues on perceived credibility. *Internet Research*, 32(1), 379–397. <https://doi.org/10.1108/INTR-10-2019-0442>
- Barnoy, A., & Reich, Z. (2022). Trusting others: A Pareto distribution of source and message credibility among news reporters. *Communication Research*, 49(2), 196–220. <https://doi.org/10.1177/0093650220911814>

- Basol, M., Roozenbeek, J., Berriche, M., Uenal, F., McClanahan, W., & Van der Linden, S. (2020). Toward psychological herd immunity: Cross-cultural evidence for two prebunking interventions against COVID-19 misinformation. *Big Data & Society*, 8. <https://doi.org/10.1177/20539517211013868>
- Basol, M., Roozenbeek, J., Berriche, M., Uenal, F., McClanahan, W., & Van der Linden, S. (2021). Toward psychological herd immunity: Cross-cultural evidence for two prebunking interventions against COVID-19 misinformation. *Big Data & Society*, 8. <https://doi.org/10.1177/20539517211013868>
- Borukhson, D., Lorenz-Spreen, P., & Ragni, M. (2022). When does an individual accept misinformation? *Computational Brain & Behavior*, 5, 244–260. <https://doi.org/10.1007/s42113-022-00136-3>
- Chaiken, S., Giner-Sorolla, R., & Chen, S. (1996). Beyond accuracy: Defense and impression motives in heuristic and systematic information processing. In P. M. Gollwitzer & J. A. Bargh (Eds.), *The psychology of action: Linking cognition and motivation to behavior* (pp. 553–578). Guilford Press.
- Chen, Z. F., & Cheng, Y. (2020). Consumer response to fake news about brands on social media. *Journal of Product & Brand Management*, 29(2), 188–198. <https://doi.org/10.1108/JPBM-12-2018-2145>
- Compton, J. (2021). Threat and/in Inoculation Theory. *International Journal of Communication*, 15, 4294–4306. <http://ijoc.org.JOSH.COMPTON>
- Compton, J. (2022). Threat and/in Inoculation Theory. *International Journal of Communication* 15, 4294–4306. Available at <http://ijoc.org.JOSH.COMPTON>
- Compton, J., Linden, S., Cook, J., & Basol, M. (2021). Inoculation theory in the post-truth era: Extant findings and new frontiers for contested science, misinformation, and conspiracy theories. *Social and Personality Psychology Compass*, 15, e12602. <https://doi.org/10.1111/spc3.12602>
- Compton, J., Linden, S., Cook, J., & Basol, M. (2022). Inoculation theory in the postruth era: Extant findings and new frontiers for contested science, misinformation, and conspiracy theories. *Social and Personality Psychology Compass*, 15, e12602.
- Compton, J., & Pfau, M. (2005). Inoculation theory of resistance to influence at maturity. *Annals of International Communication Association*, 29, 97–145. https://doi.org/10.1207/s15567419cy2901_4
- Cronbach, L. (1989). Construct validation after thirty years. In R. L. Linn (Ed.), *Intelligence: Measurement, theory and public policy* (pp. 147–171). University of Illinois Press.
- Diakopoulos, N., & Koliska, M. (2017). Algorithmic transparency in the news media. *Digital Journalism*, 5(7), 809–828. <https://doi.org/10.1080/21670811.2016.1208053>

- Ecker, U., Lewandowsky, S., Cook, J. et al. (2021). The psychological drivers of misinformation belief and its resistance to correction. *Nature Review Psychology*, 1, 13–29. <https://doi.org/10.1038/s44159-021-00006-y>
- Ecker, U., Lewandowsky, S., Cook, J., et al. (2022). The psychological drivers of misinformation belief and its resistance to correction. *Nature Review Psychology*, 1, 13–29. <https://doi.org/10.1038/s44159-021-00006-y>
- Epstein, Z., Berinsky, A., Cole, R., Gully, A., Pennycook, G., & Rand, D. (2021). Developing an accuracy-prompt toolkit to reduce COVID-19 misinformation online. *Harvard Kennedy School Misinformation Review*, 2(3), 1–12. <https://doi.org/10.37016/mr-2020-71>
- Eysenbach, G. (2023). The role of ChatGPT, generative language models, and artificial intelligence in medical education. *JMIR Medical Education*, 9, e46885. <https://doi.org/10.2196/46885>
- Feezell, J., Wagner, J., & Conroy, M. (2021). Exploring the effects of algorithm-driven news sources on political behavior and polarization. *Computers in Human Behavior*, 116, 106626. <https://doi.org/10.1016/j.chb.2020.106626>
- Gran, A., Booth, P., & Bucher, T. (2021). To be or not to be algorithm aware. *Information, Communication & Society*, 24(12), 1779–1796. <https://doi.org/10.1080/1369118X.2020.1736124>
- Hair, J. F., Jr., Anderson, R. E., Tatham, R. L., & Black, W. C. (1995). *Multivariate data analysis* (3rd ed.). Macmillan Publishing Company.
- Jia, C., & Johnson, T. (2021). Source credibility matters: Does automated journalism inspire selective exposure? *International Journal of Communication*, 15(1), 1–19.
- Joreskog, K., & Sorbom, D. (1996). *LISREL 8: User's reference guide*. Scientific Software International.
- Korzynski, P., Mazurek, G., Altmann, A., Ejdy, J., Kazlauskaitė, R., Paliszkievicz, J., Wach, K., & Ziemba, E. (2023). Generative artificial intelligence as a new context for management theories: Analysis of ChatGPT. *Central European Management Journal*. <https://doi.org/10.1108/CEMJ-02-2023-0091>
- Margolin, D. (2021). The theory of informative fictions. *Communication Theory*, 31(4), 714–736. <https://doi.org/10.1093/ct/ctaa002>
- Martel, C., Pennycook, G., & Rand, D. G. (2020). Reliance on emotion promotes belief in fake news. *Cognitive Research*, 47(5). <https://doi.org/10.1186/s41235-020-00252-3>
- Melchior, C., & Oliveira, M. (2022). Health-related fake news on social media platforms. *New Media & Society*, 24(6), 1500–1522. <https://doi.org/10.1177/14614448211038762>
- Mhasawade, V., Zhao, Y., & Chunara, R. (2021). Machine learning and algorithmic fairness in public and population health. *Nature Machine Intelligence*, 3, 659–666. <https://doi.org/10.1038/s42256-021-00373-4>

- Molina, M., & Sundar, S. (2023). Does distrust in humans predict greater trust in AI? *New Media & Society*. <https://doi.org/10.1177/14614448221103534>
- Musi, E., Carmi, E., Reed, C., Yates, S., & O'Halloran, K. (2023). Developing misinformation immunity: How to reason-check fallacious news in a human-computer interaction environment. *Social Media + Society*. <https://doi.org/10.1177/20563051221150407>
- Niu, W., Huang, L., & Chen, M. (2021). Spanning from diagnosticity to serendipity. *International Journal of Information Management*, 60, 102362. <https://doi.org/10.1016/j.ijinfomgt.2021.102362>
- Peifer, J., & Meisinger, J. (2021). The value of explaining the process. *Journalism & Mass Communication Quarterly*, 98(3), 828–853. <https://doi.org/10.1177/10776990211012953>
- Pennycook, G. (2023). A framework for understanding reasoning errors. *Advances in Experimental Social Psychology*, 67, 131–208. <https://doi.org/10.1016/b.s.aesp.2022.11.003>
- Pennycook, G., & Rand, D. G. (2022). Accuracy prompts are a replicable and generalizable approach for reducing the spread of misinformation. *Nature Communications*, 13, 2333. <https://doi.org/10.1038/s41467-022-30073-5>
- Roozenbeek, J., & Van der Linden, S. (2019). Technique-based inoculation against real-world misinformation. *Research Society Open Science*, 9(8), 211–719. <https://doi.org/10.1098/rsos.211719>
- Roozenbeek, J., et al. (2022). Psychological inoculation improves resilience against misinformation on social media. *Science. Advances*, 8, eabo6254. <https://doi.org/10.1126/sciadv.abo6254>
- Schuetz, S., Sykes, T., & Venkatesh, V. (2021). Combating COVID-19 fake news on social media through fact-checking. *European Journal of Information Systems*, 30(4), 376–388. <https://doi.org/10.1080/0960085X.2021.1895682>
- Shin, D. (2020). User perceptions of algorithmic decisions in the personalized AI system: Perceptual evaluation of fairness, accountability, transparency, and explainability. *Journal of Broadcasting & Electronic Media*, 64(4), 541–565. <https://doi.org/10.1080/08838151.2020.1843357>
- Shin, D. (2022). Expanding the role of trust in the experience of algorithmic journalism: User sensemaking of algorithmic heuristics in Korean users. *Journalism Practice*, 16(6), 1168–1191. <https://doi.org/10.1080/17512786.2020.1841018>
- Shin, D. (2023). *Algorithms, humans, and interactions*. Routledge. <https://doi.org/10.1201/b23083>
- Shin, D., & Akhtar, F. (2024). Algorithmic inoculation against misinformation: How to build cognitive immunity against misinformation. *Journal of Broadcasting & Electronic Media*. <https://doi.org/10.1080/08838151.2024.2323712>

- Shin, D., Koerber, A., & Lim, J. (2024). Impact of misinformation from generative AI on user information processing: How people understand misinformation on generative AI. *New Media and Society*, 26(4), 12–29.
- Stecula, D., Kuru, O., & Jamieson, K. (2020). How trust in experts and media use affect acceptance of common anti-vaccination claims. *The Harvard Kennedy School Misinformation Review*, 1(1), 1–11. <https://doi.org/10.37016/mr-2020-007>
- Sundar, S., Knobloch-Westerwick, S., & Hastall, M. (2007). News cues: Information scent and cognitive heuristics. *Journal of the American Society for Information Science and Technology*, 58(3), 366–378. <https://doi.org/10.1002/asi.20511>
- Traberg, C., Roozenbeek, J., & Van der Linden, S. (2022). Psychological inoculation against misinformation. *The ANNALS of the American Academy of Political and Social Science*, 700(1), 136–151. <https://doi.org/10.1177/00027162221087936>
- Tully, M., Bode, L., & Vraga, E. (2020). Mobilizing users: Does exposure to misinformation and its correction affect users' responses to a health misinformation post? *Social Media + Society*, 6(4). <https://doi.org/10.1177/2056305120978377>
- Vafeiadis, M., & Xiao, A. (2021). Fake news: How emotions, involvement, need for cognition and rebuttal evidence influence consumer reactions toward a targeted organization. *Public Relations Review*, 47(4), 102088. <https://doi.org/10.1016/j.pubrev.2021.102088>
- Van der Linden, S. (2023). *Foolproof: Why misinformation infects our minds and how to build immunity*. W.W. Norton & Company.
- Van Dis, E., Bollen, J., Zuidema, W., Rooij, R., & Bockting, C. (2023). ChatGPT: Five priorities for research. *Nature*, 614, 224–226. <https://doi.org/10.1038/d41586-023-00288-7>
- Vraga, E., & Bode, L. (2020). Defining misinformation and understanding its bounded nature. *Political Communication*, 37(1), 136–144. <https://doi.org/10.1080/10584609.2020.1716500>
- Walter, N., & Tukachinsky, R. (2020). A meta-analytic examination of the continued influence of misinformation in the face of correction. *Communication Research*, 47, 155–177. <https://doi.org/10.1177/0093650219854600>
- Wathen, C., & Burkell, J. (2002). Believe it or not: Factors influencing credibility on the web. *Journal of the American Society for Information Science and Technology*, 53(2), 134–144.
- Zrnec, A., Pozenel, M., & Lavbic, D. (2022). Users' ability to perceive misinformation. *Information Processing & Management*, 59(1), 102739. <https://doi.org/10.1016/j.ipm.2021.102739>



Misinformation and Generative AI: How Users Construe Their Sense of Diagnostic Misinformation

CHATGPT: THE ILLUSION OF TRUTH AND THE DANGERS OF ARTIFICIAL MISINFORMATION

A human-like artificial intelligence (AI) chatbot, *the Chat Generative Pretraining Transformer* (ChatGPT), a generative AI (GenAI) application, has rapidly become a widely used tool for information search and generative services. ChatGPT can answer questions and talk in human language; its services are highly versatile. This viral AI-powered chatbot can write essays, emails, theses, and even computer code (Goldstein et al., 2023). The dialog format allows ChatGPT to answer follow-up questions, admit mistakes, challenge incorrect premises, and make appropriate suggestions (Eysenbach, 2023).

Despite its sensational popularity, users have encountered that AI-powered ChatGPT is vulnerable to misinformation, as it lacks a judgment of the boundary between fact and fiction (Zhou et al., 2023). Moreover, the potential for misinformation is tremendous because such GenAI technologies are designed not for factual accuracy but for quick and eloquent conversations (Van Dis et al., 2023). Most GenAIs cannot verify whether the results they generate are accurate, and users are limited in their ability to check the source of the information and its processing by the algorithms to generate content (Korzynski et al., 2023). Chatbots are not only prone to producing misinformation and factual errors but also predisposed to providing false information that appears plausible and

authoritative (Van Dis et al., 2023). Numerous examples of GenAI generating misinformation or simply fabricating content to quickly respond are available. Although the results from GenAI can be novel and intriguing, the information or content they generate should be considered with caution (Barrot, 2023). AI-generated misinformation poses significant threats of misleading users and voters in political campaigns. Sophisticated GenAI tools can generate hyperrealistic images, cloned human voices, videos, and audio in seconds. When strapped to social media algorithms, this algorithm-created misinformation can spread quickly and far and target highly segmentized groups, potentially confusing voters, slandering political candidates, and inflaming violence.

Despite ongoing efforts to overcome this disadvantage, designing a system in which AI generates accurate information remains a major challenge. Individuals with little algorithmic literacy or understanding are more susceptible to consuming incomplete or false content, content reflecting biases, or even intentionally fabricated content (Shin, 2023). In this study, we empirically test theories on the problems associated with misinformation and human reasoning and/or information processing models of GenAI by focusing on the interaction between user cognitive models and generated messages. Understanding users' heuristics and acceptability models with regard to the role of cognitive processes in interpreting misinformation can help combat misinformation and the spread of fake news in the GenAI landscape (Pennycook & Rand, 2022). Knowledge of how people process the misinformation generated by ChatGPT can provide insights into how this misinformation is produced, shared, legitimated, reproduced, and amplified (Tully et al., 2020). Robust empirical research on how the processing of misinformation results in concerning impacts is thus warranted (Ahmed & Gil-Lopez, 2022).

Modeling the perceptual and cognitive processes behind the spread of misinformation has been of interest for understanding the reasoning of users encountering misinformation. Numerous theories (e.g., debunking, informative fiction, inoculation theory) from various fields have been formalized into misinformation models (e.g., Borukhson et al., 2022). Many methods for combating misinformation have been explored, and studies have investigated how people encounter and share false information or misinformation online (e.g., Bryanov et al., 2020). Among all the attempts, efforts should be focused on the processes that can help users more critically engage with or more consciously respond to misinformation (Molina & Sundar, 2023). Despite their importance, the processing and

mechanisms of misinformation, particularly in GenAI contexts, have been scarcely researched. Thus, the exact nature of how users respond to misinformation and its effects as well as the impact of misinformation in GenAI contexts on user behavior remain inconclusive. Whereas there is burgeoning research in the fact-checking literature on the utility and efficacy of interventions of fact-checking practices (Jahng, 2021) and technical tools, such as automated fact checking (Epstein et al., 2021), prebunking (Lewandowsky & van der Linden, 2021), and debunking (Ecker et al., 2022), research focusing on users' cognitive heuristics to diagnose misinformation, the related process of misinformation discernment, and the ensuing effect of misinformation-amplification behavior is limited. Filling this research gap requires better knowledge of the mechanisms of misinformation processing in relation to the cognitive development of users in interpreting the misinformation provided by GenAI (Walter & Tukachinsky, 2020). We narrow this research gap by investigating how users respond to/process misinformation in GenAI contexts. Because much misinformation today is generated and amplified by AI (Kreps et al., 2022), understanding how users process misinformation could help advance theoretical foundations as well as practical solutions to misinformation problems. The findings contribute to the ongoing literature by clarifying the algorithmic effects of AI constraints such as the black-box nature and the procedural uncertainty of AI on a user's evaluation of generated misinformation. The findings of this study will reveal how GenAI can improve its design to foster users' evaluation of misinformation and to help them arrive at literate informed decisions. This insight could pave the way for professionals to incorporate the appropriate strategies needed to combat the generation and spread of misinformation via AI.

LITERATURE REVIEW: THE SENSEMAKING OF GENERATIVE MISINFORMATION

To examine the cognitive and information processes related to misinformation processing, we integrated insights from two theoretical accounts—the heuristic–systematic (HS) processing framework and the concept of information diagnosticity—and examined the antecedents and consequences of misinformation processing by GenAI users.

Heuristic–Systematic Misinformation Processing

The influence of information processing on misinformed decision-making can be dual; information is processed within both an intuitive and autonomous system and within a more analytical reflective system using systematic processing methods (Pennycook, 2023). The HS model of information processing analyzes the behavioral changes of individuals based on two types of information processing: heuristic and systematic (Chaiken et al., 1996). The HS theory explains that when people engage in heuristic evaluation, they tend to process information based on the available heuristic cues or simple decision rules. In contrast, when they are involved in systematic processing, they thoroughly evaluate the information to check its validity. This theory is used as a prism to examine users' sensemaking of misinformation; it clarifies the role of algorithmic features on the sensemaking of users regarding algorithms and how their actions influence this sensemaking (Zrnec et al., 2022). HS theory can be applied in the context of misinformation to extend the information processing model by examining the antecedents of user heuristics and the consequences of systematic processing of algorithmic misinformation (Shin, 2023).

GenAI and Misinformation: Is ChatGPT a Misinformation Generator?

Misinformation is not new at all, but advanced technologies such as AI generate unprecedented quantities of misinformation. *Misinformation-related problems* are exacerbated by the wide availability of online platforms and the instantaneous and viral nature of online debates (Margolin, 2021), and the emergence of AI exacerbates its potential to cause significant harm at the individual, societal, and economic levels. The GenAI application, ChatGPT, has attracted enormous traction for its ability to quickly and confidently present users with information (Korzynski et al., 2023). One of the strengths of GenAI models is their power to automate the generation of content that is personalized, customized, and elaborated as human-written content (Goldstein et al., 2023). Owing to this strength, large language models integrated in AI, such as ChatGPT, are already able to engage in meaningful conversations and dialog with humans in highly effective and persuasive manners.

However, this confidence and speed could pose a risk, as ChatGPT often generates false narratives. Remarkable advances in GenAI, particularly those powered by huge language models, make it easier than ever to

generate articles at light speed and significant volume, raising the challenge of detecting misinformation and countering its spread at scale and in real-time. Efficient and quick information generation by GenAI could have wide-ranging implications for the diffusion of misinformation online. The viral nature of GenAI and its speed at which information is spread can lead to a misinformation spiral in which falsehood gets bigger and stronger and becomes beliefs that embedded in people's cognition, even when substantiation debunks the false. The spiral amplification features of digital platforms pose significant risks of burgeoning misinformation trends. GenAI has worsened the misinformation problem as it churns out misinformation at a great speed. GenAI assists those who intentionally seek to mislead, manipulate, and distort. GenAI has the risk of exacerbating the spread of misinformation and accelerating the ongoing challenge of seeking information that humans can trust.

ChatGPT cannot discern the boundary between fact and likelihood (Van Dis et al., 2023). Owing to this inability, ChatGPTs tend to fabricate information to satisfy users' inquiries with a quick response time. Misinformation can flow into AI models as well as from them, which means that some GenAI will be vulnerable to "injection attacks," where malicious users input lies to the tools and train them; the tools, in turn, spread them. ChatGPT, for example, is susceptible to being used as a platform for misinformation generation and amplification (Eysenbach, 2023). In many cases, misinformation is presented as a fact and truth (Melchior & Oliveira, 2022). Because the results have a natural language modality, users tend to trust these results that have been algorithmically constructed for them. This concern is related to a lack of transparency and explainability (Shin, 2023). For ChatGPT-generated answers, explaining to users where the information comes from and how the answers are generated, constructed, and presented to them remains a challenge. End users cannot and, in general, do not try to construe a sense of transparency and explainability of GenAI, which ultimately hurts user trust and engagement.

Research Questions: A Conceptual Characterization of Generative Misinformation

With the popularity of GenAI and the prevalence of misinformation over generative mechanisms, how people construe and process misinformation becomes a legitimate question when deciding solutions to mitigate misinformation over GenAI (Shin, 2023). Previous research on countering misinformation has mainly addressed the degree to which reason and reflection

deter or facilitate the development of correct assessments (Borukhson et al., 2022). An information-oriented approach, for example, views misinformation based on an information deficit principle, focusing on information-related factors, the lack of user access to correct information, or providing more information to users (Ecker et al., 2022). However, this approach underplays the social, cognitive, and psychological dimensions of heuristic formation and systematic evaluations. For instance, people deny the pandemics of respiratory viruses despite the availability of scientific proof. This rejection of publicly accepted information is not the result of a lack of information; rather, it is driven by psychological and social factors, such as conspiracy theories, amplified bias, and selective cognition, which are influenced by individual or personal values rather than factual information (Islam et al., 2020). Psychological factors are influenced by motivation, cognition, understanding, perception, and beliefs, which can be grouped into heuristic and systematic processes. The relationship between different processes is essential to grasp the context of decision-making. While extant research suggests that diagnosticity is the underlying driving factor of the attitudes and behavioral decisions of users (Kwon et al., 2020; Yi et al., 2017), how diagnosticity is formed, construed, and practiced by users in the HS process should be examined. The following research questions (RQs) guide this study:

RQ: What are the cognitive mechanisms of misinformation effects on the use of GenAI?

1. How do users construe a sense of misinformation over GenAI?
2. What are the dynamic *mechanisms of misinformation* evaluation and spreading used in GenAI?
3. How do users construe a sense of diagnosticity in algorithmically generated information, and what is the role of diagnosticity in the HS process?

UNDERLYING COGNITIVE PROCESS OF MISINFORMATION EVALUATION

The proposed model is rooted in information processing theory and explores the HS processing of misinformation in AI systems. Our hypotheses help investigate the sensemaking of users when presented with misinformation generated by algorithms and the effect of such sensemaking on their sharing behavior and misinformation discernment.

Heuristic Processing

In response to growing public attention to ChatGPT, people's awareness of the potential misuse of GenAI has increased (Van Dis et al., 2023). Awareness has also increased about people's own understanding of AI, as the specifics of GenAI or internal algorithms are not easily comprehensible to the public owing to their black-box nature (Ali et al., 2022). In this AI situation, users face a high level of uncertainty and rely on their own heuristics. In a normal context, users gather and use relevant information from available sources to make optimal choices. When the information decision is generated in AI contexts, such as opaque processes or possible biases, users may change their evaluations, effectively adjusting the way in which the users evaluate and use misinformation judgment. The heuristic assessment of misinformation involves evaluating the intrinsic nature of algorithmic attributes such as fairness, accountability, and transparency (FAccT; Shin, 2023). As users are unable to systematically evaluate AI, they rely on heuristic cues to process large amounts of generated information, such as fairness, transparency, and trustworthiness. Sundar et al. (2007) show that news chatbot users rely on heuristic cues to process recommended news feeds. GenAI users will apply heuristics before systematic evaluation to process the generated information. Users want to verify the FAccT of the systems that spread misinformation. Examining the ethical/normative factors—such as fairness, bias, equitability, and auditability—of the methods developed to counter the dissemination of misinformation is critical.

Fairness Heuristics

With limited cognitive ability to process the increasing amount of misinformation in their generative content, users apply the fairness heuristic in interacting with GenAI. People use their own heuristics to judge possible algorithmic biases, such as ethnicity and gender biases, from GenAI (Mhasawade et al., 2021). As people increasingly rely on GenAI for their important decisions and related information, biases in AI are increasingly concerning and may potentially cause harm. In reality, users have a limited capacity to judge algorithmic biases, as they neither evaluate message validity on their own nor configure systematically and correct biases by

tracing them in data and internal algorithms. Shin (2023) showed that algorithmic fairness hinges upon the perceptual impartiality and equitability of users regarding AI. Related research has clarified the relation between algorithmic fairness and user attitudes, particularly user efficacy and confidence (Diakopoulos & Koliska, 2017). Because the self-efficacy of users is related to their diagnosticity, a connection between algorithmic fairness and the diagnosticity of users in misinformation can be proposed:

H1: The fairness heuristics of users positively affect the diagnosticity of users when interacting with GenAI.

Accountability Heuristics

Algorithmic accountability examines the process of determining whether a decision is made under substantive and procedural standards and holding entities accountable if they fall short of these standards (Diakopoulos & Koliska, 2017). Accountability is particularly important in monitoring the spread of misinformation, as the repercussions of the spread, amplification, and diffusion of misinformation are immense. People may view that GenAI should be held accountable for disseminating misinformation and must provide further clarification when requested by the public. Related research shows that users develop positive confidence with clearly defined responsibilities and liabilities when they correctly judge incorrect information as misinformation (Barnoy & Reich, 2022). Based on previous findings, examining how users construe algorithmic accountability in relation to misinformation and how their perceptions influence their diagnosticity of misinformation over GenAI seems relevant.

H2: The accountability heuristics of users positively influence their diagnosticity of misinformation in GenAI.

Transparency Heuristics

Transparency in AI entails both the inputs and outputs of algorithms being visible and comprehensible to users or the public (Shin, 2023). Generally, as end users do not have the ability to examine the transparency of algorithms, users base their judgments on easily processed heuristics when the related heuristic cues are available. For example, transparent statements showing the generative mechanism process on websites or in AI systems provide users with certain heuristics that allow users to quickly form their judgment on transparency without engaging in extensive

technical evaluation of the message content. As users are unable to process each piece of information systematically or each component of the AI system technically, users depend on their own heuristics based on available cues (Molina & Sundar, 2023). When users have a sense of transparency in systems, they tend to deem the system trustworthy and reliable (Gran et al., 2021). Research has confirmed that transparency enhances user confidence in an algorithm, playing a key role in whether users use it (Peifer & Meisinger, 2021). When transparent and visible cues are available, people are more likely to view the given information as diagnosable (Diakopoulos & Koliska, 2017). Transparent AI systems can afford users a sense of assurance, presenting them with a readiness to diagnose misinformation.

H3: The transparency heuristics of users positively influence the diagnosticity of GenAI.

Diagnosticity

When arriving at a diagnostic judgment about the misinformation and system of the corresponding heuristics in cognition, users evaluate both the content of the information and the systematic cues it contains (Niu et al., 2021). Having diagnostic information is important for AI to provide trustworthy services. Diagnosticity is defined as the extent to which people interpret a piece of information to be truthful and relevant according to their judgment and decision-making (Ahluwalia et al., 2001; Chen & Cheng, 2020). In AI contexts, diagnosticity refers to a user's perceived ability to understand the features and attributes of AI services in their decision-making (Shin, 2023). Estimating the diagnosticity of the information from AI must itself be based on inference and personal speculation. Information that is perceived to be diagnostic is given relatively more weight in the judgments and evaluations of users (Shin, 2023); thus, diagnosticity influences how users systematically process messages (Niu et al., 2021). Upon realizing a system's diagnosticity, users trust it and begin systematically assessing misinformation by measuring its accuracy and credibility (Chen & Cheng, 2020; Stecula et al., 2020). Diagnosticity functions as a conduit between heuristic and systematic evaluations. The indicators of diagnosticity are uniquely specific to the users' ability to identify misinformation based on their understanding of the information's accuracy and credibility (i.e., the correctness and reliability of the suggested information; Zrnek et al., 2022).

H4: Diagnosticity perceived by users positively affects the credibility of information.

H5: Diagnosticity perceived by users positively affects the accuracy of information.

Systematic Processing

Users conduct reflective assessments of information, e.g., evaluating how accurate and credible the generated information is (Ali et al., 2022).

Accuracy

The perceived accuracy of a piece of information is the user's perception of the amount of misinformation contained in the information (Pennycook & Rand, 2022); it refers to how truthful the user perceives a viewed piece of information to be. The accuracy of a GenAI application is related to how many results are correctly personalized and to the veracity of the recommended content. Regarding the measurement of the perceived accuracy of misinformation, relevant studies have proposed various terms to describe it, such as misinformation awareness, which is a person's awareness in knowing that the received information is false (Schuetz et al., 2021). Empirical research has confirmed these relationships in diverse algorithm services (Pennycook & Rand, 2022), in which accuracy is found to be a determinant of attitude and behavioral intention (Shin, 2023). Accordingly, the following relation is proposed:

H6: Perceived accuracy positively influences the intention of a user to correct/share misinformation.

Credibility

As a strong predictor of the influence of information, credibility can generally be understood as "believability" and "trustworthiness" (Wathen & Burkell, 2002). In the communication literature, credibility is measured based on different perspectives, such as source, message, and media credibility (Barnoy & Reich, 2022). Although credibility has historically been more extensively studied from the source perspective (credibility of information source), in the social media environment, news consumers place more attention on the content of the message than on the source (Peifer & Meisinger, 2021). Thus, the credibility of the content/message is worth

investigating in the context of misinformation in algorithmic media (Ecker et al., 2022). The more credible the content of a piece of misinformation is perceived to be, the more useful and relevant (i.e., diagnostic) users would perceive it to be.

H7: Content credibility perceived by users regarding misinformation in news influences their intention to correct/share misinformation.

Explanatory Cues as an Intervention: Moderating Effect of Explainability

Explainability is a critical component in the field of explainable AI (Rai, 2020) and has become an important component of generative AI, which includes machine learning and AI technologies that can provide human-comprehensible rationales for their outputs or processes (Shin, 2023). Owing to their increasing complexity, people consider AI systems and algorithms to be “black boxes”; furthermore, increasing amounts of specialized expertise and knowledge are required to understand the decision-making processes or performance of AI systems (Rai, 2020). ChatGPT is a nonexplainable AI, as users do not understand how and why answers are generated (Korzynski et al., 2023). Although it is convenient, ChatGPT may exacerbate the problem of misinformation instead of mitigating it. ChatGPT stands against explainable AI and should not be applied in situations where credibility and explainability are critical requirements. Regarding explainability, understanding the specific reasoning behind specific outputs may be difficult, but users may understand how the model works in general and how it generates text based on the input it receives. However, it is not a decision-making AI and is not meant to be deployed in an operational context where security, safety, or explainability are critical requirements. Although ChatGPT is a powerful language generation tool, it must be responsibly used while being aware of its limitations. ChatGPT can be used to feed information into knowledge graphs and contribute to their explainability. ChatGPT can link information to knowledge that has already been referenced and verified, preferably in a traceable manner. With explainability incorporated, ChatGPT serves as a knowledge reference model and helps to extend the cognitive model by suggesting further potentially meaningful services. Explainability is critical in building trust, credibility, and understanding between the AI agent and its user, especially in regard to understanding misinformation (Rai, 2020). Given the

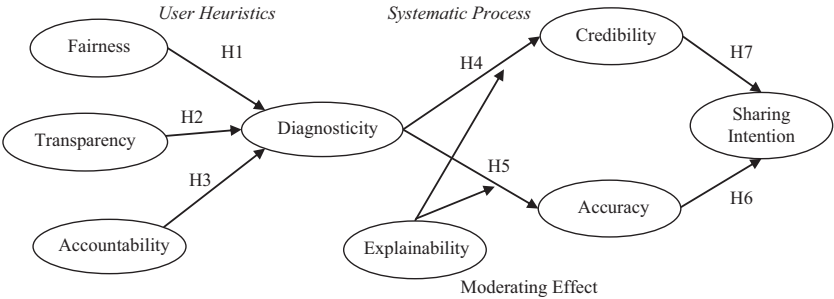


Fig. 9.1 Conceptual Model

significant effects of explainability explored in previous studies, the moderating role of explainability in ChatGPT for misinformation evaluation can be tested.

H_{m1} : The effect of diagnosticity perceived by users regarding misinformation on accuracy is moderated by the explainability of the information generated by GenAI.

H_{m2} : The effect of diagnosticity perceived by users regarding misinformation on credibility is moderated by the explainability of the information generated by GenAI.

Based on the constructs and hypotheses presented in this study, we propose a conceptual model that depicts how users process misinformation related to the harmful effects of a product, its antecedents, and its consequences (see Fig. 9.1).

METHODS

We used experimental surveys with the Wizard of Oz method to empirically investigate people’s cognitive processing of misinformation. A Wizard of Oz experiment was set up in which participants interacted with ChatGPT; participants believed the interaction to be controlled by GenAI but was being operated by human agents.

Collection of Data and Samples

The Wizard of Oz method was used to examine the users' misinformation processing. The interface of Wizard of Oz was created for users to feel like they were communicating with a GenAI application—ChatGPT (Fig. 9.2). That is, participants used the Wizard of Oz GenAI as if they were using ChatGPT. As participants chat and question, human wizards respond with appropriate questions, resembling how AI responds.

After receiving approval from the Institutional Review Board, the study was administered from January 2022 to February 2022 among 302 respondents. Respondents were recruited among college students (undergraduate and graduate) in the United Arab Emirates. Given that the majority of GenAI users are youth, college student sampling was justified. Upon obtaining informed consent, the participants were asked to sit at a computer desk, where they could see the interface of the Oz GenAI Wizard. With brief instructions, the participants were provided with a pre-determined list of news items on current health and health campaigns with related misinformation. Then, they were asked to interact with GenAI through the interface and read current health-related news (health information currently being discussed and proliferating online) generated by the wizards for approximately 2–3 h. Examples of health topics included COVID, infectious viruses, respiratory symptoms, vaccines, flu, and medications. The human wizards played the same role as GenAI, as they used

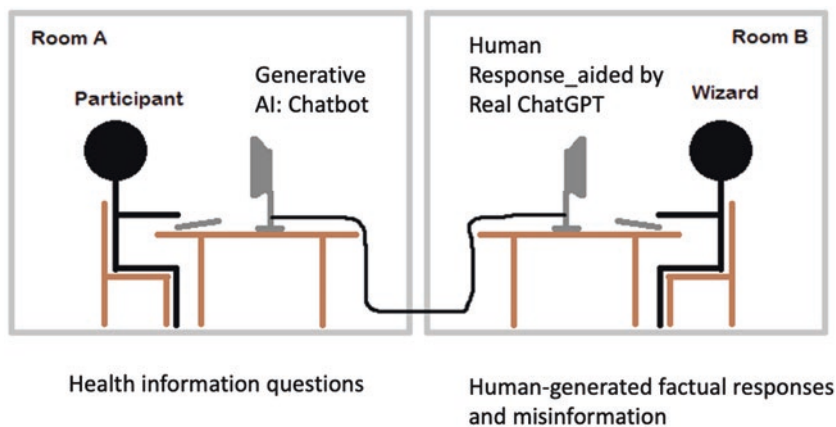


Fig. 9.2 Setup for the GenAI Wizard of Oz Method

ChatGPT to answer the participants' questions and at times responded with misinformation, fake news, incorrect recommendations, or misinformation related to the questions.

The Wizard of Oz GenAI provided the participants with contemporary news containing both real information and misinformation on contemporary health issues and medical campaigns. The Wizard of Oz was designed to test how users perceive and process misinformation suggested by GenAI. After interacting with GenAI, participants completed a survey. They were either paid \$2.50 or received course credits for completing the study, which took a median time of 20 minutes ($M = 20.19$; $SD = 7.92$) (Table 9.1 and Fig. 9.3).

Table 9.1 Descriptive statistics ($N = 302$)

<i>Characteristics</i>	<i>Sample</i>
Age (mean/SD/median)	25.30/11.02/29
Gender (female rate) (%)	50.98
College educated (%)	33.18
AI platform experience	2.8 years

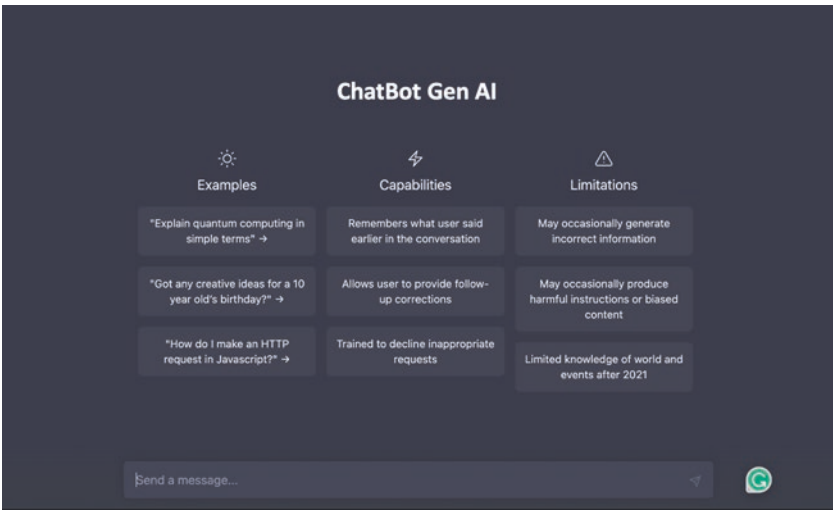


Fig. 9.3 The interface of experimental GenAI

Data Analyses

The proposed model was tested via a two-stage method used by Islam et al. (2020) in predicting misinformation behavior. In the first stage, a structural equation modeling (SEM) method was employed to confirm the validity and reliability of the constructs and test the structural paths between them. Next, a neural network (NN)-based method was applied because SEM cannot examine the nonlinear paths between constructs, which might be important in AI contexts. NN analysis was designed to validate the SEM results and to prioritize predictors based on their relative importance in affecting HS processes.

Scales and Measurements

In this study, the scales used comprised 21 measurements with a 7-point scale ranging from 1 (*strongly disagree*) to 7 (*strongly agree*). All the measurements were based on the literature on health communications and research on misinformation (Appendix). The measurements were administered using combinations of previously developed metrics and modified metrics from previous studies. Such combinations were necessary because certain measurements in our analysis needed to be changed to reflect the distinctive features of the AI algorithms and services.

These measurements were then compared with measurements from the literature on AI, communications, and cognitive science. The FAccT, explainability, and diagnosticity measurements were performed based on the works of Shin (2023), Rai (2020), and Chen and Cheng (2020), respectively. Some measurements required changes to reflect new traits in the misinformation. Explanatory stimuli were presented as cues in the misinformation, popping up as phrases when the subjects clicked on recommended news. These cues provided users with relevant explanations, such as “because you searched it before” or “most similar [to] viewed news,” on why specific health information was recommended. The explanatory cues allowed participants to understand, interpret, and experiment with the misinformation.

Measurement Instrument

The validity and reliability of the measurement items were assessed based on the reliability and convergent validity criteria using Analysis of Moment

Structure (AMOS) 18 software. Cronbach’s alpha of the survey instrument was calculated to test its reliability. The analysis results were verified, and varimax rotation was performed on the 30 original items (7 items were eliminated owing to low loading). According to Hair et al. (1995), an item has high loading if its loading coefficient is above 0.6. Each item’s loading value was above the recommended level of 0.8 (Cronbach, 1989). Confirmatory factor analysis was performed to test the validity of the instruments; furthermore, the composite reliability, estimated percentage of variance extracted by each construct, and discriminant and convergent validity were examined. All the constructs in the model met the required reliability and convergent validity values (Tables 9.2 and 9.3). *Regarding discriminant validity*, the average variance extracted (AVE) values and the correlations of the factors were examined. The shared variance between the constructs was lower than the AVE from the individual constructs, confirming the presence of discriminant validity. Correlation analysis was performed, and Pearson’s *R* (correlation coefficient) showed an acceptable level of correlation among the variables. In conclusion, the measurement model demonstrated adequate reliability and the presence of convergent and discriminant validity.

Fit Indices

The structural model was tested using the AMOS procedure and maximum-likelihood-based structural equation modeling software. Table 9.4 reveals the estimates from the structural modeling procedure. The results indicated the presence of no misfits. All values were within acceptable ranges, indicating good overall model performance.

Table 9.2 Discriminant validity

<i>Construct</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>
Fairness	0.71							
Accountability	0.21	0.83						
Transparency	0.33	0.52	0.83					
Accuracy	0.13	0.23	0.52	0.93				
Credibility	0.12	0.11	0.23	0.42	0.72			
Diagnosticsity	0.23	0.53	0.13	0.52	0.11	0.82		
Explainability	0.42	0.22	0.53	0.21	0.53	0.13	0.82	
Intention	0.52	0.11	0.22	0.22	0.22	0.53	0.23	0.73

Table 9.3 Reliability checks for constructs

<i>Construct</i>	<i>Initial items</i>	<i>Final items</i>	<i>Mean (SD)</i>	<i>Cronbach's alpha</i>
Fairness	4	3	4.17 (1.011) 3.84 (1.055) 4.11 (1.047)	0.795
Accountability	4	3	4.17 (1.011) 3.84 (1.055) 4.11 (1.047)	0.619
Transparency	5	3	3.39 (1.054) 3.98 (1.128) 3.30 (1.282)	0.875
Accuracy	4	3	3.74 (1.346) 3.87 (1.385) 3.40 (1.411)	0.812
Credibility	4	3	4.37 (1.182) 4.30 (1.141) 4.63 (1.202)	0.842
Diagnosticity	4	3	4.37 (1.182) 4.30 (1.141) 4.63 (1.202)	0.853
Explainability	3	3	4.34 (1.210) 4.44 (1.295) 4.05 (0.984)	0.856
Intention	3	3	4.42 (1.121) 4.10 (1.264) 4.18 (1.268)	0.900

Structural Model Testing

The SEM results validated all the relationships in the hypotheses (Table 9.5). The proposed path coefficients were statistically significant ($p < 0.001$ or $p < 0.0001$). Moreover, perceived diagnosticity was significantly affected by heuristic evaluations and simultaneously influenced the systematic processing of misinformation. The heuristic factors accounted for 59.1% of the variance in diagnosticity. The intention behind sharing information was significantly influenced by the systematic processing of misinformation, with systematic factors explaining 32% of the variance in the intention to share information. The model explained a significant percentage of the variance in each factor. Furthermore, the significant paths implied a causal conceptual link between the heuristic and systematic processes through diagnosticity.

Table 9.4 Model fit indices

<i>Fit statistics</i>	<i>Structural model</i>	<i>Suggested value</i> <i>(Joreskog & Sorbom, 1996)</i>
χ^2 (<i>df</i>)	1710.5/244 = 7.01	<5
<i>p</i> value	0.000	<0.05
RMSEA	0.023	$0.05 < x < 0.10$
CFI	0.910	>0.90
NFI	0.978	>0.90
GFI	0.920	>0.90
IFI	0.802	>0.80
TLI	0.901	>0.85
RFI	0.803	>0.80
AIC	1870	
Hoelter's critical N	66(0.05)/70(0.01)	Higher values signify a satisfactory fit
ECVI	4.688	Smaller values denote a better fit

Table 9.5 SEM results

<i>Paths</i>	<i>Estimate</i>	<i>Standard error</i>	<i>Critical ratio</i>	<i>P value</i>
H1: Fairness → Diagnosticity	0.610	0.129	4.739	***
H2: Accountability → Diagnosticity	0.512	0.043	11.840	***
H3: Transparency → Diagnosticity	0.457	0.047	9.791	***
H4: Diagnosticity → Accuracy	0.642	0.067	9.517	***
H5: Diagnosticity → Credibility	0.649	0.076	8.499	***
H6: Accuracy → Intention	0.396	0.063	6.292	***
H7: Credibility → Intention	0.221	0.087	2.525	0.012*

*1.96: 95% (0.05), **2.58: 99% (0.01), and ***3.29: 99.9% (0.001)

Neural Network Analysis

Table 9.6 lists the average importance and average normalized importance of each input factor in predicting diagnosticity and sharing intention. Transparency is the most critical predictor, followed by fairness and accountability, in predicting diagnosticity (heuristic process). Explainability is the most determining factor in forming a systematic process (sharing intention), followed by credibility and accuracy. The results are largely consistent with the SEM results.

The Impact of Explanatory Heuristics on Systematic Evaluation

We examined the moderating effects of explanatory cues on the HS processing of diagnosticity in the model. The explanatory cues in GenAI provided users with explanations on why and how the recommended information was generated. They also helped users systematically assess misinformation (Shin, 2023). The investigation of the moderating role of explainability in the path of diagnosticity on accuracy/credibility was worthwhile. We conducted a multigroup analysis to evaluate the structural paths from the model to the experimental (Group 1) and control groups (Group 2) (Table 9.6).

In the experiment, explanatory cues were provided along with generated information during participants’ chats with the human Wizard of Oz GenAI, whose interface was designed to resemble that of ChatGPT. Such explanatory cues included fact-check information, information source, accuracy nudging, and a brief explanation of how and why certain health information was generated. Similar to the first experiment, the participants in the second experiment were paid, and the participation time was approximately 20 min ($M = 21.28$; $SD = 6.42$).

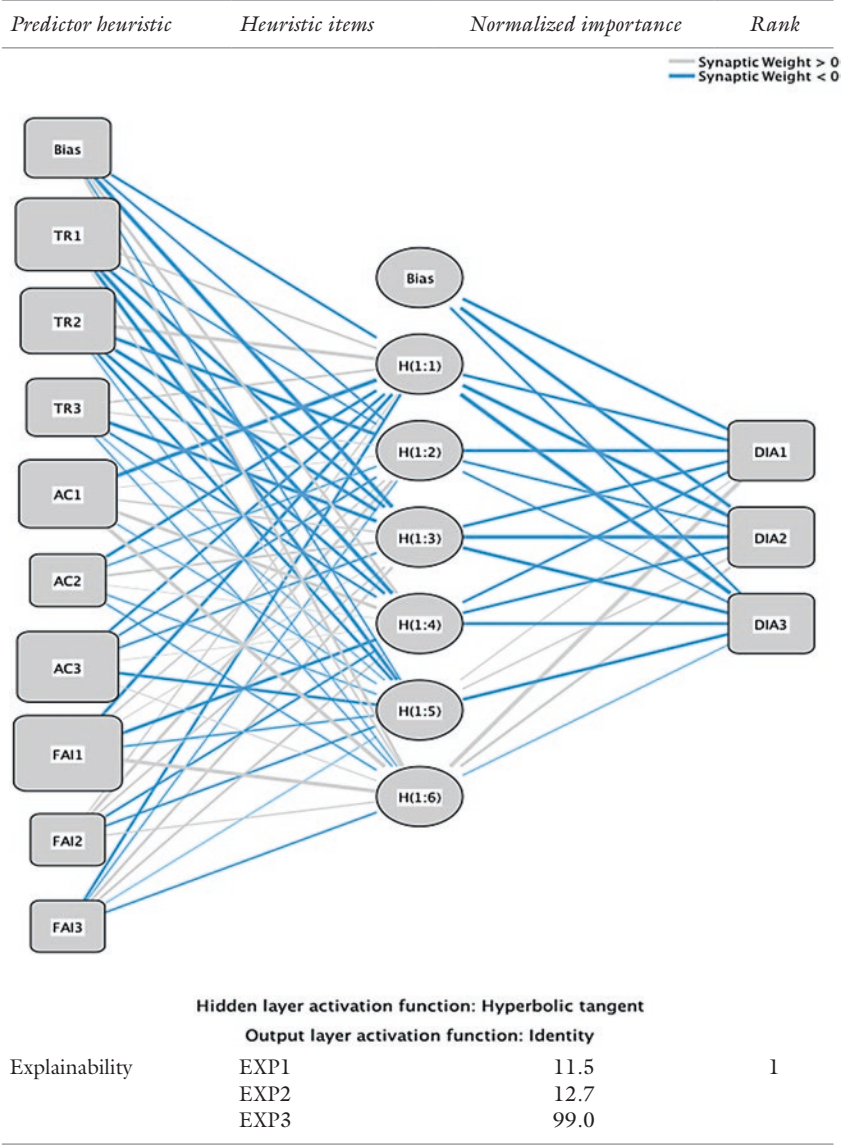
The two groups showed equally good fits with the data, and distinct patterns were observed that supported the hypotheses. The results showed notable distinctions in item composition and path formation, providing insights into the effect of explainability on misinformation (Table 9.6). The effects of diagnosticity on credibility and accuracy were more significant in Group 2 than in Group 1. Comparing the two models, Group 2 showed higher values for systematic evaluations, whereas Group 1 showed higher values for heuristic evaluations. Therefore, we can infer that

Table 9.6 Neural network-based approach in predicting heuristic processes

<i>Predictor heuristic</i>	<i>Heuristic items</i>	<i>Normalized importance</i>	<i>Rank</i>
Transparency	TR1	90.4	1
	TR2	68.2	
	TR3	42.8	
Fairness	FAI1	98.0	2
	FAI2	25.0	
	FAI3	24.4	
Accountability	AC1	76.3	3
	AC2	27.5	
	AC3	84.2	

(continued)

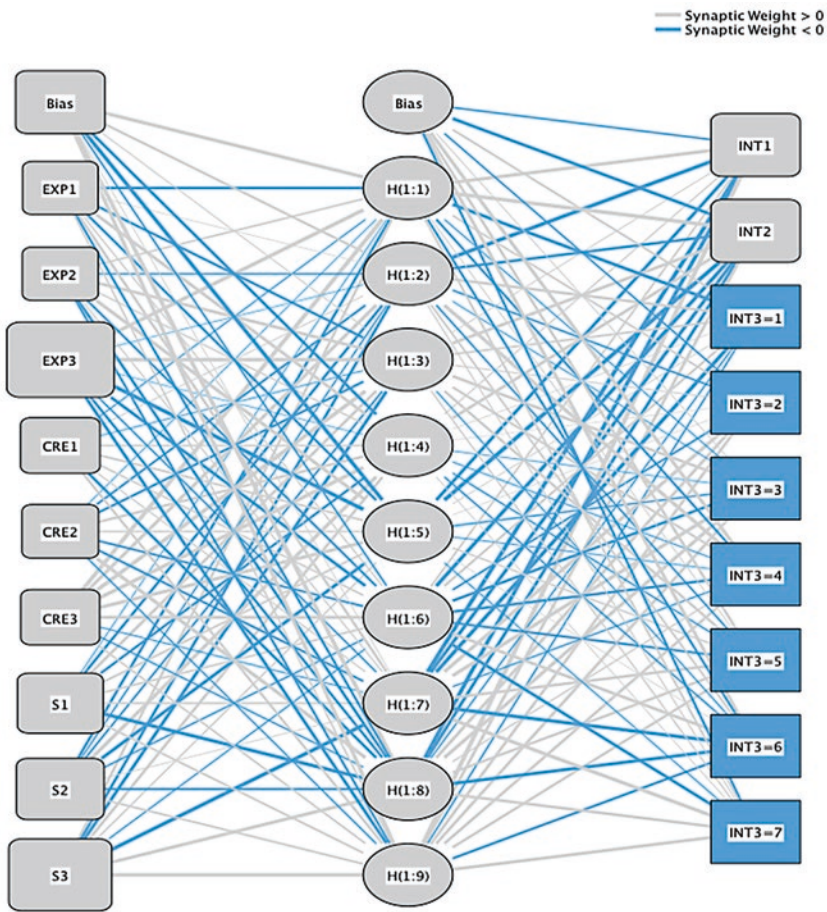
Table 9.6 (continued)



(continued)

Table 9.6 (continued)

Predictor heuristic	Heuristic items	Normalized importance	Rank
Accuracy	ACC1	39.7	2
	ACC2	44.9	
	ACC3	88.0	
Credibility	CRE1	22.1	3
	CRE2	14.2	
	CRE3	20.9	



Hidden layer activation function: Hyperbolic tangent
Output layer activation function: Identity

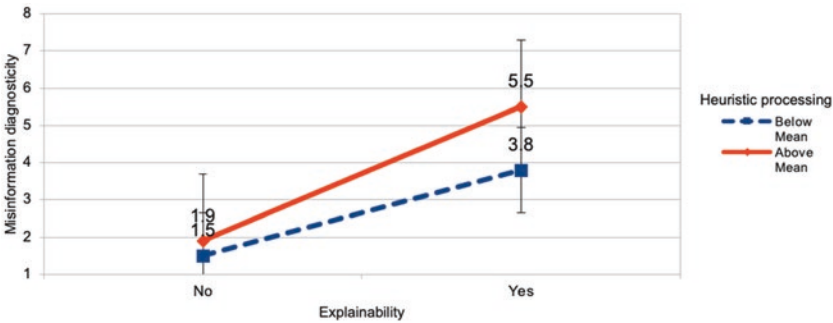


Fig. 9.4 Interaction role of heuristic processing in the effect of Explainability on Diagnosticity

explanatory cues played a facilitating role in systematic as well as heuristic processes, and their effects were greater on systematic than on heuristic processes. The results of the squared multiple correlations also supported the underlying role of explainability (Table 9.7). The R^2 values of the accuracy and credibility in Group 1 were greater than those in Group 2. Moreover, the R^2 value of intention in Group 2 was distinctly higher than that in Group 1.

In addition, a notable interaction effect between explainability and heuristic processing was observed. The effect of explainability on misinformation diagnosticity was more significant for users who engaged in a higher level of heuristic processing (Fig. 9.4). This result suggests that the heuristic processing of misinformation moderated the relationship between the explanatory cues and judgment of misinformation such that the effect of explainability on diagnosticity was stronger when the users engaged in a higher level of FAccT evaluation.

DISCUSSION: ALGORITHMIC MISINFORMATION

The analysis results showed that the HS processing of the users significantly influenced their attitudes and behaviors; the users’ misinformation processing is heuristic based as well as systematically oriented, and the GenAI users process generated information with available heuristics and systematic cues to arrive at a diagnostic judgment. We found that the more users felt they could detect and evaluate misinformation, the more likely they were to determine the accuracy and credibility of information. The

Table 9.7 Moderating effects of explainability

Path	Group 1 (Control group)				Group 2 (Explanatory cue)					
	B	SE	CR	p	Result	β	SE	CR	p	Result
Diagnosticity-credibility	0.066	0.041	1.592	0.111	No	0.684***	0.050	13.727	0.000	Yes
Diagnosticity-accuracy	0.037	0.050	0.750	0.454	No	0.106***	0.032	3.301	0.000	Yes
Accuracy-intention	0.166*	0.113	1.434	0.005	Yes	0.328***	0.021	1.734	0.000	Yes
Credibility-intention	0.337**	0.054	0.342	0.01	Yes	0.421***	0.081	3.492	0.000	Yes

* $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$; β : Unstandardized coefficient

more likely users felt the information was diagnostic, the more likely they were to find the information credible or accurate. Heuristic knowledge of the misinformation significantly influenced users' construed diagnosticity of the information and subsequent credibility. Furthermore, explanatory cues of the information moderate the effects of diagnosticity on the perceived accuracy and credibility of information. HS processing was mediated by perceived diagnosticity and moderated by the extent of the explainability embedded in the AI system. Together, explainability and diagnosticity were the key user heuristics through which misinformation influenced users' discernibility and sharing behaviors.

Theoretical Implications: Algorithmic Effects on Misinformation

The study extends HS theory by applying it in the context of misinformation generated by GenAI and examining the antecedents and consequences of diagnosticity. The results offer relevant theoretical implications of algorithmic effects on misinformation (Cho et al., 2020). First, although previous studies have researched the HS model, we are able to elaborate on the duality of the information processing mode when users consider generated misinformation. We characterized dual-process misinformation; users tend to process misinformation heuristically, and such heuristics invoke a sense of diagnosticity, which then facilitates systematic evaluation. Such a dual- or two-tier process is linked and mediated by diagnosticity. This finding fills a gap in research on user behaviors regarding misinformation that previous studies have strived to elucidate (e.g., Ahmed & Gil-Lopez, 2022); research has also failed to clarify the nature of user engagement and its matrix and process. In addition, previous research has mostly examined the unidimensional effects of the psychological impact of misinformation on user behavior (e.g., Lazer et al., 2018). We have extended the current understanding by explicating HS processing by users in evaluating misinformation. We adopted an information processing approach, considering the cognitive and psychological development of users and how HS processing in the evaluation influenced the discernment of misinformation. Our approach revealed relevant factors of cognitive-psychological processes and helped enrich the understanding of the relationship between how misinformation forges the attitudes of users and how their attitudes toward the information and source of algorithm-generated misinformation influence the emergence of misinformation; this coevolving relation was mediated and facilitated by diagnosticity.

Whereas previous studies have merely conceptualized the existence of diagnosticity (e.g., Chen & Cheng, 2020; Yi et al., 2017), we provided a hypothetical and empirical clarification of the underlying relation of diagnosticity with GenAI features: when a piece of information is generated by algorithms in transparent, fair, and accountable ways, users perceive it with a high level of diagnosticity, which improves their intention to systematically analyze the information.

It is notable to note the mediating role of diagnosticity in relaying the heuristic process to the systematic process. Previous research on HS theory has assumed two separate models of two different processes in which the two-tiered processes have little to do with each other or little in interactional relation (Chaiken et al., 1996). In the AI-generated misinformation context, diagnosticity plays a critical role in two aspects: (1) due to the black-box nature of GenAI generation, users evaluate the attributes heuristically, and (2) given the susceptible nature of misinformation, users evaluate the generated information systematically. The mediating role of diagnosticity indicates that the more confident the GenAI users were about the systems being ethically sound and normatively acceptable, the more likely they were to technologically diagnose the generated information. Such results are consistent with earlier findings by Ahmed and Gil-Lopez (2022) and Ali et al. (2022), suggesting that users with lower cognitive heuristics are more likely to believe and accept/share misinformation. Our findings are also consistent with ongoing reports of users' understanding of algorithmic functions (Shin, 2023) and the associated ethical issues (Mhasawade et al., 2021). The results suggest that when algorithm users had a higher level of perceived diagnosticity, they became more confident and capable of analyzing the accuracy and credibility of the information presented through GenAI and, consequently, tended to react to misinformation with higher cognitive load and conscientiousness. The interaction effect identified in this study supports the proposition that users who are oriented toward ethical considerations are likely to wisely use explanatory cues in diagnosing the accuracy and credibility of information. This result on the interaction effect highlights a new role of diagnosticity in misinformation processing: diagnosticity in misinformation can result from heuristic processing and from being an activator in conducting systematic processing. This argument can be a relevant theoretical advancement, as it clarifies the processing capacity of users through diagnosticity. This role of diagnosticity can be presented as an extension of previous conceptualizations of user awareness (Gran et al., 2021), self-efficacy (Ali

et al., 2022), informative fiction (Margolin, 2021), and algorithmic literacy (Shin, 2023). The diagnosticity principle in misinformation may refer to users' heuristics to probe the salience of algorithmic features using available cues and optimal inference. Diagnosticities are heuristic-dependent and context-oriented and acquire significant diagnostic value if the diagnostic object is extended to include a systematic matrix.

Diagnosticity can be a cognitive trigger to switch from heuristic to systematic processing, functioning as a conduit, channeling evaluation efforts from heuristic factors into systematic processes. Normative values give users a sense of diagnostic readiness for handling misinformation, thus leading to the effective systematic evaluation of misinformation. In GenAI, users construe a sense of diagnosticity based on their understanding of algorithmic heuristics. That is, the extent of diagnostic ability in AI may reside in users' cognition as much as salient technological features or performance. Thus, it can be inferred that diagnosticity depends more on how users construe algorithmic heuristics than on actual technological readiness; diagnosticity is cognitively construed and constructed by users.

This user-centered view on diagnosticity is nicely aligned with the procedural aspect of misinformation. Although misinformation has been an ongoing problem throughout the COVID-19 pandemic and beyond, relatively few studies have explored the process of misinformation, particularly from the users' viewpoint, with many only studying it as a tool to combat, such as debunking, prebunking, and inoculation (Shin et al., 2024; Hwang & Jeong, 2021). The scarcity of research on the procedural aspect of misinformation could potentially limit our understanding of how users interact with misinformation (Kim et al., 2020), especially that presented by GenAI in the case of health information. Our results confirmed the proposition that diagnosticity is a critical construct (Chen & Cheng, 2020) and further corroborated that HS processing in GenAI concurrently affects the intention of users to process misinformation via dual paths (Pennycook, 2023). Prior work has primarily focused on the unidirectional influence of misinformation from a functional viewpoint, such as misinformation traits (Borukhson et al., 2022), with few studies holistically examining the dual processing of misinformation. Our study advanced ongoing research by clarifying how ethical and systematic dimensions could increase user diagnosticity and how users evaluate the accuracy and credibility of misinformation.

*Practical Implications: Is GenAI Credible Chatbot or
a Misinformation Generator?*

Our study contributes to a shared understanding across the AI industry and the broader scientific community on how GenAI can better address integrity issues on its platforms. Our findings offer relevant practical guidelines for GenAI in improving users' understanding of managing misinformation and how misinformation processing influences their heuristic and procedural knowledge of it. Because diagnosticity is significantly influenced by the ethical heuristics of AI systems, practitioners should not only engage in establishing the normative values of the system but also adopt strategies to increase the perceived heuristics of the diagnosticity of users. Examining both the presented information and user traits is critical for practitioners evaluating suitable venues to combat misinformation. In particular, the credibility of information on AI platforms is low; users' distrust of such platforms may result in a disinterest in discerning misinformation. Firms specializing in AI platforms may invest resources and develop intervening features (e.g., explanatory cues) to monitor misinformation on their platforms, offer valuable cues or links to help users check or evaluate the credibility of information online, and engage in ethical and responsible operations to meet the trust expectancy of users. Trust in the platform is generated when diagnosticity and established information credibility are high. Thus, to counter misinformation, practitioners may reveal the source sharing the misinformation to relate the misinformation with the source instead of the platform on which the information was presented. This information can be shared through explanatory interventions, as we show the effectiveness of explanatory interventions in reducing the effects of misinformation.

Raising the literacy and efficacy of users in countering misinformation is also critical. Diagnosticity may depend on the technological features of the AI system, but it can be contextual, and the perceived diagnosticity of the users significantly hinges on how they view the system regarding its FAccT. Efforts should therefore be made to improve the technical features and competence of the system for both diagnosticity and accuracy, and practitioners should establish the credibility of GenAI to enhance the literacy of users dealing with misinformation.

LIMITATIONS AND FUTURE RESEARCH

Our study, like any other, has limitations that need to be addressed in future work; the findings should therefore be considered with caution.

First, although our study proposes a misinformation processing model in the context of GenAI, it mainly addresses cognitive processing within an emerging and single-user schema and excludes the algorithmic side of automatic processing. To more comprehensively test the model, future research should include the algorithmic features of information processing when using the model. Further comparisons and investigations should be conducted regarding how and to what extent the different dimensions in the misinformation processing model are shaped by the HS processing of users and, in turn, influence the perceived diagnosticity of misinformation and subsequent behavior of users. In addition, we collected data at a single point in time; future studies may examine long-term exposure to misinformation and its longitudinal effects on people with longer and deeper engagement.

Second, prior research on misinformation processing identified the role of trust, as people seemed more accepting of information when they trusted the media that presented it (Chen & Cheng, 2020). The existing trust of users regarding AI may play a significant role in processing misinformation. Future research should integrate diverse messages with varying levels of initial trust and investigate how prior trust levels may influence how users process misinformation. In doing so, misinformation processing may be related to a broader trust dimension, as mentioned in prior literature (e.g., Stecula et al., 2020).

Finally, our results showed the connection between FAccT, diagnosticity, and credibility, thereby shedding light on the nexus between misinformation, literacy, and trust. These results implied a possible two-tier or multiple-tier process that may play a role in the susceptibility of users to misinformation. Future studies should test the relation among algorithm literacy, trust, and diagnosticity to further explicate what may determine the efficacy and capacity of users when exposed to misinformation via algorithms. This task would warrant refining the two-tier process of misinformation.

Despite these limitations, our study offers meaningful insights for theory building and practice. We believe that the model offers a relevant framework for scholars interested in user behavior exposed to misinformation in a GenAI landscape. This model will provide the basis for future

studies on cognitive heuristics countering misinformation. Future research can combine algorithmic technologies and nudging in the user experience of GenAI services. Future studies can further address the underlying processes of misinformation behavior in various contexts, considering the effects of personal traits on these processes. Our research continues to address this ongoing problem, which will have enormous societal repercussions.

REFERENCES

- Ahluwalia, R., Unnava, H. R., & Burnkrant, R. E. (2001). The moderating role of commitment on the spillover effect of marketing communications. *Journal of Marketing Research*, 38(4), 458–470. <https://doi.org/10.1509/jmkr.38.4.458.18903>
- Ahmed, S., & Gil-Lopez, T. (2022). Engaging with vilifying stereotypes. *Journalism & Mass Communication Quarterly*. <https://doi.org/10.1177/10776990221110113>
- Ali, K., Li, C., Zain-ul-abdin, K., & Zaffar, M. (2022). Fake news on Facebook: Examining the impact of heuristic cues on perceived credibility. *Internet Research*, 32(1), 379–397. <https://doi.org/10.1108/INTR-10-2019-0442>
- Barnoy, A., & Reich, Z. (2022). Trusting others: A Pareto distribution of source and message credibility among news reporters. *Communication Research*, 49(2), 196–220. <https://doi.org/10.1177/0093650220911814>
- Barrot, J. (2023). Using ChatGPT for second language writing: Pitfalls and potentials. *Assessing Writing*, 57, 100745. <https://doi.org/10.1016/j.asw.2023.100745>
- Borukhson, D., Lorenz-Spreen, P., & Ragni, M. (2022). When does an individual accept misinformation? *Computational Brain & Behavior*, 5, 244–260. <https://doi.org/10.1007/s42113-022-00136-3>
- Bryanov, K., Watson, B. K., Pingree, R. J., & Santia, M. (2020). Effects of partisan personalization in a news portal experiment. *Public Opinion Quarterly*, 84(S1), 216–235. <https://doi.org/10.1093/poq/nfaa011>
- Chaiken, S., Giner-Sorolla, R., & Chen, S. (1996). Beyond accuracy: Defense and impression motives in heuristic and systematic information processing. In P. M. Gollwitzer & J. A. Bargh (Eds.), *The psychology of action: Linking cognition and motivation to behavior* (pp. 553–578). Guilford Press.
- Chen, Z. F., & Cheng, Y. (2020). Consumer response to fake news about brands on social media. *Journal of Product & Brand Management*, 29(2), 188–198. <https://doi.org/10.1108/JPBM-12-2018-2145>

- Cho, J., Ahmed, S., Hilbert, M., Liu, B., & Luu, J. (2020). Do search algorithms endanger democracy? *Journal of Broadcasting & Electronic Media*, 64(2), 150–172. <https://doi.org/10.1080/08838151.2020.1757365>
- Cronbach, L. (1989). Construct validation after thirty years. In R. L. Linn (Ed.), *Intelligence: Measurement, Theory and Public Policy* (pp. 147–171). University of Illinois Press.
- Diakopoulos, N., & Koliska, M. (2017). Algorithmic transparency in the news media. *Digital Journalism*, 5(7), 809–828. <https://doi.org/10.1080/21670811.2016.1208053>
- Ecker, U., Lewandowsky, S., Cook, J., et al. (2022). The psychological drivers of misinformation belief and its resistance to correction. *Nature Review Psychology*, 1, 13–29. <https://doi.org/10.1038/s44159-021-00006-y>
- Epstein, Z., Berinsky, A., Cole, R., Gully, A., Pennycook, G., & Rand, D. (2021). Developing an accuracy-prompt toolkit to reduce COVID-19 misinformation online. *Harvard Kennedy School Misinformation Review*, 2(3), 1–12. <https://doi.org/10.37016/mr-2020-71>
- Eysenbach, G. (2023). The role of ChatGPT, generative language models, and artificial intelligence in medical education. *JMIR Medical Education*, 9, e46885. <https://doi.org/10.2196/46885>
- Goldstein, J. A., Sastry, G., Musser, M., DiResta, R., Gentzel, M., & Sedova, K. (2023). Generative language models and automated influence operations: Emerging threats and potential mitigations. arXiv preprint arXiv:2301.04246
- Gran, A., Booth, P., & Bucher, T. (2021). To be or not to be algorithm aware. *Information, Communication & Society*, 24(12), 1779–1796. <https://doi.org/10.1080/1369118X.2020.1736124>
- Hair, J. F., Jr., Anderson, R. E., Tatham, R. L., & Black, W. C. (1995). *Multivariate data analysis* (3rd ed.). Macmillan Publishing Company.
- Hwang, Y., & Jeong, S. (2021). Misinformation exposure and acceptance: The role of information seeking and processing. *Health Communication*, 23(1), 1–9. <https://doi.org/10.1080/10410236.2021.1964187>
- Islam, A., Laato, S., Talukder, S., & Sutinen, E. (2020). Misinformation sharing and social media fatigue during COVID-19. *Technological Forecasting and Social Change*, 159, 120201. <https://doi.org/10.1016/j.techfore.2020.120201>
- Jahng, M. (2021). Is fake news the new social media crisis? *International Journal of Strategic Communication*, 15(1), 18–36. <https://doi.org/10.1080/1553118X.2020.1848842>
- Joreskog, K., & Sorbom, D. (1996). *LISREL 8: User's reference guide*. Scientific Software International.
- Kim, H., Ahn, J., Atkinson, L., & Kahlor, L. (2020). Effects of COVID-19 misinformation on information seeking, avoidance, and processing. *Science Communication*, 42(5), 586–615. <https://doi.org/10.1177/1075547020959670>

- Korzynski, P., Mazurek, G., Altmann, A., Ejdy, J., Kazlauskaitė, R., Paliszkievicz, J., Wach, K., & Ziemba, E. (2023). Generative artificial intelligence as a new context for management theories: Analysis of ChatGPT. *Central European Management Journal*. <https://doi.org/10.1108/CEMJ-02-2023-0091>
- Kreps, S., McCain, R., & Brundage, M. (2022). All the news that's fit to fabricate. *Journal of Experimental Political Science*, 9(1), 104–117. <https://doi.org/10.1017/XPS.2020.37>
- Kwon, Y., Park, J., & Son, J. (2020). Accurately or accidentally? Recommendation agent and search experience in over-the-top services. *Internet Research*, 31(2), 562–586. <https://doi.org/10.1108/INTR-03-2020-0127>
- Lazer, D. M. J., Baum, M. A., Benkler, Y., Berinsky, A. J., Greenhill, K. M., Menczer, F., Metzger, M. J., Nyhan, B., Pennycook, G., Rothschild, D., Schudson, M., Sloman, S. A., Sunstein, C. R., Thorson, E. A., Watts, D. J., & Zittrain, J. (2018). The science of fake news. *Science*, 359(6380), 1094–1096. <https://doi.org/10.1126/science.aao2998>
- Lewandowsky, S., & van der Linden, S. (2021). Countering misinformation and fake news through inoculation and prebunking. *European Review of Social Psychology*, 32(2), 348–384. <https://doi.org/10.1080/10463283.2021.1876983>
- Margolin, D. (2021). The theory of informative fictions. *Communication Theory*, 31(4), 714–736. <https://doi.org/10.1093/ct/qtaa002>
- Melchior, C., & Oliveira, M. (2022). Health-related fake news on social media platforms. *New Media & Society*, 24(6), 1500–1522. <https://doi.org/10.1177/14614448211038762>
- Mhasawade, V., Zhao, Y., & Chunara, R. (2021). Machine learning and algorithmic fairness in public and population health. *Nature Machine Intelligence*, 3, 659–666. <https://doi.org/10.1038/s42256-021-00373-4>
- Molina, M., & Sundar, S. (2023). Does distrust in humans predict greater trust in AI? *New Media & Society*. <https://doi.org/10.1177/14614448221103534>
- Niu, W., Huang, L., & Chen, M. (2021). Spanning from diagnosticity to serendipity. *International Journal of Information Management*, 60, 102362. <https://doi.org/10.1016/j.ijinfomgt.2021.102362>
- Peifer, J., & Meisinger, J. (2021). The value of explaining the process. *Journalism & Mass Communication Quarterly*, 98(3), 828–853. <https://doi.org/10.1177/10776990211012953>
- Pennycook, G. (2023). A framework for understanding reasoning errors. *Advances in Experimental Social Psychology*, 67, 131–208. <https://doi.org/10.1016/bs.aesp.2022.11.003>
- Pennycook, G., & Rand, D. G. (2022). Accuracy prompts are a replicable and generalizable approach for reducing the spread of misinformation. *Nature Communications*, 13, 2333. <https://doi.org/10.1038/s41467-022-30073-5>

- Rai, A. (2020). Explainable AI: from black box to glass box. *Journal of the Academy of Marketing Science*, 48, 137–141. <https://doi.org/10.1007/s11747-019-00710-5>
- Schuetz, S., Sykes, T., & Venkatesh, V. (2021). Combating COVID-19 fake news on social media through fact checking. *European Journal of Information Systems*, 30(4), 376–388. <https://doi.org/10.1080/0960085X.2021.1895682>
- Shin, D. (2023). *Algorithms, humans, and interactions*. Routledge. <https://doi.org/10.1201/b23083>
- Shin, D., Koerber, A., & Lim, J. (2024). Impact of misinformation from generative AI on user information processing: How people understand misinformation on generative AI. *New Media and Society*, 26(4), 1–29.
- Stecula, D. A., Kuru, O., & Jamieson, K. (2020). How trust in experts and media use affect acceptance of common anti-vaccination claims. *The Harvard Kennedy School Misinformation Review*. <https://doi.org/10.37016/mr-2020-007>
- Sundar, S., Knobloch-Westerwick, S., & Hastall, M. (2007). News cues: Information scent and cognitive heuristics. *Journal of the American Society for Information Science and Technology*, 58(3), 366–378. <https://doi.org/10.1002/asi.20511>
- Tully, M., Bode, L., & Vraga, E. (2020). Mobilizing users: Does exposure to misinformation and its correction affect users' responses to a health misinformation post? *Social Media + Society*, 6(4). <https://doi.org/10.1177/2056305120978377>
- Van Dis, E., Bollen, J., Zuidema, W., Rooij, R., & Bockting, C. (2023). ChatGPT: Five priorities for research. *Nature*, 614, 224–226. <https://doi.org/10.1038/d41586-023-00288-7>
- Walter, N., & Tukachinsky, R. (2020). A meta-analytic examination of the continued influence of misinformation in the face of correction. *Communication Research*, 47, 155–177. <https://doi.org/10.1177/0093650219854600>
- Wathen, C., & Burkell, J. (2002). Believe it or not: Factors influencing credibility on the web. *Journal of the American Society for Information Science and Technology*, 53(2), 134–144.
- Yi, C., Jiang, Z., & Benbasat, I. (2017). Designing for diagnosticity and serendipity. *Information Systems Research*, 28, 413–429. <https://doi.org/10.1287/isre.2017.0695>
- Zhou, J., Zhang, Y., Luo, Q., Parker, A., & Choudhury, M. (2023). Synthetic lies: Understanding AI-generated misinformation and evaluating algorithmic and human solutions. *CHI '23, April 23–28, 2023, Hamburg, Germany*. <https://doi.org/10.1145/3544548.3581318>
- Zrnec, A., Pozenel, M., & Lavbic, D. (2022). Users' ability to perceive misinformation. *Information Processing & Management*, 59(1), 102739. <https://doi.org/10.1016/j.ipm.2021.102739>



Conclusion: Misinformation and AI—How Algorithms Generate and Manipulate Misinformation

AI MAKES MISINFORMATION MUCH MORE TOXIC

While creating deepfakes has become easier than ever, detecting and managing them has become increasingly difficult (Giansiracusa, 2021). Many openly available applications allow layperson users to create almost any type of video. This is a crucial issue with deepfakes. While the direct effects of deepfakes have received substantial attention, the indirect effects of deepfakes may be even more concerning in relation to newly developed technologies. Generative AI tools, particularly generative adversarial networks (GANs), simplify and democratize the creation of realistic deep-faked images, cloned voices, and synthetic videos (Mirsky & Lee, 2021). GANs' proficient ability to model high-dimensional data and handle missing data and the capacity of GANs to provide multi-modal outputs or multiple plausible answers has been observed. These deepfake productions are increasingly sophisticated, making it difficult to differentiate them from genuine ones. The proliferation of deepfake production websites and apps, driven by social media users' desire to create and share humorous and captivating short-form videos, enables users to create highly realistic synthetic videos with minimal technical expertise. Social media is abuzz with natural deepfake photos, including those depicting fake Donald Trump arrests and him wearing an orange jumpsuit, photos of Pope Francis in a puffer jacket, and a terror-evoking photo with the Pentagon as

the backdrop, which instilled fear in investors, causing stocks to tumble in March 2023. Deepfakes carry a substantial potential and risk of being used or abused in political campaigns, as was evident in the recent case of Florida Governor Ron DeSantis, who used deepfake photos of Donald Trump embracing one of his former medical advisers, Anthony Fauci (Nehamas, 2023).

With advanced GANs, deepfakes are used to maliciously spread false information. They can also be designed to harass, intimidate, demean, and undermine people. Deepfakes can create confusion about critical societal issues and exacerbate unethical activities such as the fabrication of revenge porn. Researchers (Campbell et al., 2022; Whittaker et al., 2021) have noted the proliferation of synthetic media and its implications, emphasizing the sophistication of manipulative techniques such as deepfakes in crafting highly convincing, albeit unreal, versions of reality. Recent research has addressed how deepfakes directly impact people's attitudes and actions and has investigated how deepfakes contribute to the larger ecology of misinformation. Cheapfakes are examples of synthetic media in which simple digital techniques are applied to content to change the observer's perception of an event (Qian et al., 2023). Deepfakes, representing the latest advancements in AI techniques, substantially threaten the accuracy of information, obfuscating its authenticity and source. A study examining the effect of deepfakes on trust in the news found that exposure to deepfakes increased people's uncertainty about the media (Vaccari & Chadwick, 2023), which caused people to distrust the media.

Deepfakes, combined with the rapid spread and development of AI, have created imminent concerns within the popular and scholarly literature (Giansiracusa, 2021). Some fear that such technology will open the door to new forms of privacy invasion, security threats, cyber-bullying, and nonconsensual pornography (Kerner & Risse, 2021). Others have raised concerns over the tendency of synthetic media to tarnish people's reputations by planting false beliefs among the public (Diakopoulos & Johnson, 2021) or even within security (Iacobucci et al., 2021). Related to this concern, some have stressed political factors, warning that synthetic audio and video materials falsely starring political figures may pollute democratic discourse to the extent that meaningful deliberation becomes impossible due to the lack of shared foundational truth.

de Ruiter (2021) made the following argument: "Protection against the manipulation of digital representations of our face and voice should be considered a fundamental moral right in the age of deepfakes" (p. 1314).

However, protection against deepfakes requires work. Deepfakes commonly result in what is called “identity leakage” (Mirsky & Lee, 2021, p. 7), which refers to when numerous people’s facial or vocal features blend into each other; some are so poorly made that the identification of a single individual is problematic. It is not always clear whose face is featured in the final product.

GENERATE ADVERSARIAL NETWORK: THE DAWN OF ARTIFICIAL MISINFORMATION

Deepfakes use underlying AI technologies, namely GANs. Generative modeling is an unsupervised learning task in machine learning that involves automatically discovering and learning the regularities or patterns in input data so that the model can be used to generate or output new examples that plausibly could have been drawn from the original dataset. GANs are an effective means of training a generative model by framing the problem as a supervised learning problem with two sub-models: one attempting to produce a fake and the other trying to detect it. The two models are trained together in a zero-sum adversarial manner until the discriminator model is fooled about half the time, meaning that the generator model generates plausible examples (GAO, 2020).

In a GAN, two machine-learning networks are utilized to create synthetic content through an adversarial process. The first network is the “generator.” Data that represents the type of content to be created is fed to this first network to “learn” the characteristics of that type of data (Mirsky & Lee, 2021). The generator then develops new examples of that data, which exhibit the same characteristics as the original data. These generated examples are then presented to the second machine learning network, which has also been trained to “learn” to identify the characteristics of that type of data. This second network (the “adversary”) attempts to detect flaws in the presented examples. It rejects those that it determines do not exhibit the same characteristics as the original data, identifying them as “fakes.”

Deepfakes using GANs represent an emergent type of risk falling under the more significant and more pervasive umbrella of synthetic media, utilizing a form of machine learning to create plausible, realistic videos, audio, pictures, and text of events that never occurred. Many applications of synthetic media represent innocent forms of simple entertainment, but

others carry actual threats. The threat of deepfakes and synthetic media results not from the AI used to make them but from people's cognitive inclination to trust what they see. As a result, deepfakes and synthetic media can be more sophisticated and realistic to effectively spread fake news and mis/disinformation (Ahmed, 2021).

Based on numerous recent research findings, it has become apparent that the severity and urgency of the current threat emanating from synthetic media depends on the exposure, perspective, and position of whom you ask. The spectrum of concerns ranges from "an urgent threat" to "do not panic, just be prepared."

As GANs have gained prominence, it has become essential to tackle ethical concerns and to ensure the responsible use of this technology (Diakopoulos & Johnson, 2021). The rise of fake media and misinformation has posed challenges, necessitating safeguards to protect against malicious intent. Striking a balance between AI-generated content and human creativity is essential to preserve societal integrity and maintain ethical standards (Shin, 2023).

In line with the multifaceted nature of the problem, there is no single or one-size-fits-all solution. However, technological innovation, education, and regulation elements must comprise part of any detection and mitigation measures. To ensure success, there will have to be significant cooperation among stakeholders in the private and public sectors to overcome current obstacles, such as stovepiping (the presentation of information without proper context), and to ultimately protect ourselves from these emerging threats while protecting civil liberties.

The proliferation of misinformation online and its amplification by platforms constitute growing concerns; this has prompted numerous efforts involving AI to improve the detection of and response to misinformation. Effective control over misinformation has garnered significant attention to addressing responsibility gaps and mitigating them by establishing conditions that enable a proper representation of responsibility for humans (Shin et al., 2023).

SYNTHETIC MEDIA AND DISINFORMATION

Synthetic media is generally defined as encompassing all media that has either been created through digital or artificial means (i.e., computer-generated people) or media that has been modified or otherwise manipulated through technology, whether it is analog or digital (Vaccari &

Chadwick, 2023). Algorithms or machine learning processes may entirely or partially generate content. This includes any combination of video, photo, or audio media formats that may or may not be created with malicious intent. Deepfakes represent a subset of the general synthetic media or synthetic content category. Threats from synthetic media present a growing challenge for all users of modern technology and communications (Karasavva & Noorbhai, 2021). As with many technologies, synthetic media techniques can be used for positive or malicious purposes.

Cheapfakes are another example of synthetic media in which simple digital techniques are applied to content to alter the observer's perception of an event. Cheapfake examples include speech being slowed and video being accelerated. Science and technology are constantly advancing. Deepfakes and automated content creation and modification techniques merely represent the latest mechanisms developed to alter or create visual, audio, and text content. The critical difference they represent is the ease with which they can be made—and made well. In the past, casual viewers (or listeners) could easily detect fraudulent content. This may no longer always be the case, which may allow adversaries interested in sowing misinformation or disinformation to leverage far more realistic image, video, audio, and text content in their campaigns than ever before.

With the rise of synthetic media, misinformation has become more upgraded. A primary tool of misinformation warfare has been the simple meme: an image, a video, or text shared on social media that conveys a particular thought or feeling. China used memes to target protesters in Tibet and Hong Kong; Russia used memes to justify the Ukraine invasion by fabricating videos of Ukraine President Volodymyr Zelenskyy discussing surrendering to Russia; additionally, White supremacist groups targeted minorities, and conspiracy theorists questioned the effectiveness of vaccines for Coronavirus disease and used memes as a technical tool.

Advances in machine learning and AI have precipitated a new and highly compelling method of conveying disinformation. By many accounts, memes, as well as other standard and seemingly old-fashioned disinformation tools such as fake news webpages and stories and antagonistic Facebook posts, have successfully undermined confidence in U.S. elections (Atlantic Council's Digital Forensic Research Lab, 2021), sown division in the American electorate (Posard et al., 2020), and increased the adoption of conspiracy theories (Center for Countering Digital Hate, 2021; Marcellino et al., 2021).

Deepfake videos are synthetically altered footage in which the depicted face or body has been digitally modified to appear to be someone or something else (Kobis et al., 2021). Such videos are becoming increasingly lifelike, and many fear the technology will dramatically increase the threat of foreign and domestic disinformation. This threat has been realized by the many women targeted by AI-enabled pornography sites (Jankowicz et al., 2021). In other ways, however, the potential for havoc has not yet been realized.

THE LIAR'S DIVIDEND AND DEEPPAKES: ECHOES OF FAKES

Deepfakes benefit from the liar's dividend, which places an additional burden on media and fact-checkers to prove a piece of media's authenticity. Many scholars believe that deepfakes and other forms of misinformation contribute to the problem of the liar's dividend, which refers to the payoff that malicious people yield when leveraging deepfakes to conceal their unethical behavior; this problem will continue to intensify as the public becomes accustomed to seeing deepfakes (Chesney & Citron, 2019). The liar's dividend refers to a phenomenon wherein someone can get away with lying by asserting that something is fake news; if the media attempts to expose the fraudulent nature of the material, this can [backfire](#) and serve to make the lie appear even more reliable. Simply debunking fake or manipulated materials, such as audio contents, videos, or texts, could ultimately bolster the public's belief in the fakes and make it more difficult for them to trust the media. Hany Farid, a professor at the University of California Berkeley, has confirmed the threat of the liar's dividend, "My biggest concern is not the abuse of deepfakes, but the implication of entering a world where any image, video, audio can be manipulated. In this world, if anything can be fake, then nothing has to be real, and anyone can conveniently dismiss inconvenient facts."

As Chesney and Citron (2019, p. 1785) have argued, "not all lies involve affirmative claims that something occurred (that never did): some of the most dangerous lies take the form of denials." Deepfakes make it easier for liars to refute the truth in two ways. First, if they are accused of having said or done something that they did say or do, liars may generate and disseminate altered texts or images to produce doubt. The second "equally pernicious" (p. 1785) means involves simply criticizing the authentic as being false, a technique that "becomes more plausible as the public becomes more educated about the threats posed by deep fakes."

Many researchers agree that the [liar's dividend](#) is the most significant concern regarding the widespread usage of deepfakes. Vaccari and Chadwick (2023) have quantitatively confirmed that deepfakes sow uncertainty and, in turn, deteriorate trust in online news.

The liar's dividend has grown in proportion to public awareness of deepfakes and “runs with the grain of larger trends in truth skepticism” (p. 1786). Citing what they refer to as an increasing distrust of traditional news sources, Chesney and Citron (2019) stated the following: “That distrust has been stoked relentlessly by President Trump and like-minded sources in television and radio; the mantra ‘fake news’ has thereby become an instantly recognized shorthand for a host of propositions about the supposed corruption and bias of a wide array of journalists, and a useful substitute for argument when confronted with factual assertions ... [I]t is not difficult to see how ‘fake news’ will extend to ‘deep fake news’ in the future. As deep fakes become widespread, the public may have difficulty believing what their eyes and ears are telling them—even when the information is real. In turn, the spread of deep fakes threatens to erode the trust necessary for democracy to function effectively. The combination of truth decay and trust decay creates greater space for authoritarianism.”

Malicious actors can undermine all digital imagery by suggesting that it may be fake. A skeptical public will be primed to doubt the authenticity of real audio and video evidence. Governments have started to address the liar's dividend problem. South Korea has criminalized the creation or distribution of videos, images, or audio of politicians modified to resemble real footage within 30 days of an election. However, with such versatile AI, legislation alone cannot solve the problem. Users retweet, share, and repost, and the optimal means to engage with the issue is to improve users' awareness and literacy in relation to deepfakes. The distinction between memes and disinformation is tenuous with deepfakes. It is necessary for internet users to train themselves to detect what is real and what is fake and to consider the motivation behind making a deepfake.

THE DARKER SIDE OF DEEPPAKES

Public concerns around deepfakes pertain to their use within disinformation operations. Deepfakes are designed to influence the public and to spread misinformation about political, social, militaristic, or economic issues to cause confusion, unrest, and uncertainty. Deepfakes extend beyond text-based persuasion because images make a message far more

persuasive and convincing than abstract concepts. Its role in spreading disinformation is far more powerful and thus worrisome (Sundar et al., 2021). AI-generated fake content has emerged at a highly vulnerable time. Given the degree of trust our society invests in video footage and the plethora of applications of such footage, it appears clear that deepfakes could affect society and national security. When people's ability to judge where they should place their trust deteriorates, it creates problems for democracy.

There are many means by which deepfakes could be weaponized by malicious actors. Policies and regulations will have to play a role in mitigating this, at least in some of the highest-risk areas of deepfakes. The United States and the European Union are leading the way globally with the AI Act.

First, a growing problem with deepfakes relates to masking one's identity. Using deepfakes in a live setting involves masking one's identity in real time, such as over a phone call or Zoom meeting. Fraudsters can use AI-generated deepfake images or videos to deceive facial recognition algorithms into recognizing them as legitimate individuals. This can allow them to gain unauthorized access to accounts, bypass security measures, or even enter secure premises. Using voice cloning, impostors can gain access to user accounts that are protected with voice authentication and use it to authorize fraudulent transactions. The threat of someone using a fake identity could arise in many situations, including job interviews, remote college exams, and visa applications. Even Insider reporters have encountered [AI-generated scams disguised as sources](#).

Second, malicious people, most likely cyber criminals, frequently use multiple types of manipulated media in social engineering campaigns for financial gain. These may include impersonating key leaders or financial officers and operating over various mediums using manipulated audio, video, or text to illegitimately authorize the disbursement of funds to accounts belonging to the malicious actor. Business email compromise schemes are among these types of social engineering, costing companies hundreds of millions of dollars in losses. Similar techniques may also be used to manipulate the trade or sale of crypto assets. These schemes are far more common in practice, and several partners have reported being targeted by these types of operations.

Third, fraudulent actors may use the same manipulated media techniques and technologies to access an organization's personnel, operations, and information. These techniques may include using manipulated media

during job interviews, particularly for remote jobs. In 2022, malicious actors reportedly employed synthetic audio and video during online interviews, although the content was often unaligned or unsynchronized, indicating the fraudulent nature of the calls. These malicious attempts were enabled by stolen personally identifiable information. Successful compromises may enable actors to obtain sensitive financial, proprietary, or internal security information.

Fourth, deepfake content could exacerbate social conflicts and divisions. Russia has already established a reputation for disseminating propaganda designed to divide the U.S. public (Posard et al., 2020). Furthermore, that same U.S. public, driven by growing and rancorous partisan debate, frequently employs various propaganda-like tactics to smear, attack, and defame those with opposing political affiliations. Research has documented online echo chambers, where partisans disproportionately consume and share content that confirms and reinforces their opinions (Shin et al., 2022). Partisan deepfakes and other AI-driven disinformation content could exacerbate this negative impact of echo chambers. Additionally, deepfake pornography technology (e.g., DeepNude) can convincingly overlay a specific face over that of a pornographic actor in an attempt to demean and humiliate celebrities. Such videos, rarely created with the subjects' permission, provide unlimited fodder for abuse and exploitation. They could also result in broader national security threats in that they could be used to embarrass, undermine, or exploit intelligence operatives, candidates for political office, journalists, or U.S. and allied leaders (Jankowicz et al., 2021).

Lastly, deepfakes present new challenges for establishing the veracity of online content, and their contents could erode trust in institutions and societies in general. In particular, deepfake content could undermine journalism and trustworthy sources of information. With the advent of highly believable deepfakes, even accurate video content or recordings can be slandered as deepfakes by those who consider the content unfavorable (Waldemarsson, 2020). The proliferation of deepfakes could jeopardize trust in prominent news institutions by propagating mistrust even in legitimate forms of news and information (see Vaccari & Chadwick, 2020). Malicious people may use deepfakes, employing manipulated audio and video, to attempt to impersonate an organization's executive officers and other high-ranking personnel. They may employ convincing audio and video impersonations of critical leaders to sully the reputation and value of an organization's brand, such as by rapidly distributing a convincing

deepfake publicly across social media platforms before it can be ceased or disputed. Various types of manipulated media operations have targeted high-profile political figures, such as Ukraine's Volodymyr Zelenskyy, to spread disinformation and fake news. The Republican National Committee utilized AI to create a 30-second ad that envisions what President Joe Biden's second term might look like. It depicts a series of fictional crises, such as a Chinese invasion of Taiwan as well as the shutdown of the city of San Francisco, illustrated with fake images and news reports. A small disclaimer in the upper left indicates that the video was "built with AI imagery."

DEEPPFAKE RESEARCH

The research community has begun to examine the true impact of deepfakes on real users (Gamage et al., 2021). Compared with disinformation in texts (headlines or articles), visual disinformation, such as deepfakes, can make a substantial impression. For example, Hwang et al. (2021) found that deepfake videos are more likely than fake news articles to be perceived as persuasive, vivid, and trustworthy. Their study also confirmed that disinformation within deepfake videos was more likely to be shared by online users than textual misinformation. Wittenberg et al. (2021) uncovered this finding in one of the most extensive studies to date on the issue: in a study involving more than 7000 participants, they found that participants were more likely to believe that an event took place when they were presented with a fake video than when they were presented with fake textual evidence. However, the fake videos were less persuasive than anticipated, producing only "small effects on attitudes and behavioral intentions" (p. 1). The authors admonished that deepfakes could be more persuasive outside a laboratory setting. However, they suggest that "current concerns about the unparalleled persuasiveness of video-based misinformation, including deepfakes, may be somewhat premature" (p. 5). Another study found that deepfakes were no more likely than textual headlines or audio recordings to persuade a large sample of survey respondents to believe in scandals that never occurred (Barari et al., 2020). One presumed impact of deepfakes is that they will erode trust in the media, which some research has validated. For example, one study suggested that participants who saw deepfakes were more likely to feel uncertain than to be fully misled by the content, and participants' uncertainty contributed to

disintegrating trust in social media-based news content (Vaccari & Chadwick, 2020).

Overall, experimental research concerning the impact of deepfakes remains in its nascent phase, and further research will be critical. Misinformation research in general has primarily concentrated on the overarching impact of content and its dissemination mechanisms, overlooking individual dimensions and differences in perceiving the potential consequences of such misinformation. Substantial research has examined automatic deepfake generation and the detection of deepfakes. For example, the utility and efficacy of interventions involving fact-checking practices (Shin, 2024) and technical tools, such as automated fact-checking (Epstein et al., 2022), prebunking methods (Lewandowsky & van der Linden, 2021), and debunking techniques (Ecker et al., 2022), have been studied extensively. However, research concerning how users cognitively process deepfakes and the role of information processing regarding deepfakes has remained at the level of rhetorical exhortation. Research has yet to examine how individuals, based on their cognitive abilities, information processing capacity, and media literacy, might interpret the impact differently. Among these individual differences, we aim to highlight key factors that could result in varied assessments of the impact of deepfakes on individuals and society. These factors include the perceived verisimilitude (Campbell & Reiman, 2022) and perceived manipulative intent (PMI) (Campbell, 1995; Shin, 2022) of deepfakes. In the context of misinformation produced by synthetic media, perceived verisimilitude refers to how individuals perceive the information presented through deepfake content or synthetic media as resembling reality or truth (Campbell et al., 2022).

Amid an abundance of studies that offer doomsday scenarios regarding deepfakes, Tim Hwang of the Center for Security and Emerging Technology offers a more holistic evaluation of the risks associated with deepfakes (Johnson, 2023). First, it has been argued that although experts continue to debate the future danger of deepfakes, “shallow” fakes represent a more imminent threat (Stoll, 2020, p. 2). Shallow fakes are videos that are manually altered or selectively edited to mislead an audience. For instance, a fictitious Bloomberg journalist, “Maisy Kinsley,” who had a profile on LinkedIn and Twitter, was most likely a deepfake. Another LinkedIn fake, “Katie Jones,” claimed to work at the Center for Strategic and International Studies but is believed to be a deepfake who was created for a foreign spying operation.

The second factor mitigating the malicious use of deepfakes is the fact that high-quality videos are, at least for now, out of reach for amateurs (Johnson, 2023; Shin, 2023). As noted above, creating highly realistic video content requires high-cost equipment, a substantial library of training content, specialized technical skills, and willing individuals with acting talent. The technology will ultimately advance to allow more democratized access, but until then, the range of actors who can effectively use deepfake technology will remain limited. For instance, extremists generated a synthetic speech file of actress Emma Watson reciting Adolf Hitler's manifesto "Mein Kampf." This audio generated a deepfake video of Watson reading the book aloud at a podium while dressed in a Nazi uniform.

Third, time is a relevant factor (Johnson, 2023). The fact that such videos can take months to create means that deepfake disinformation operations must be planned at least months in advance, which will necessarily limit the range of circumstances in which the technology can be put to practical use and increase the risk that unanticipated changes in circumstances could render a planned operation moot. Time also limits rapid-fire operations and could make it difficult for an adversary to use the technology opportunistically. The time and effort required for foreign adversaries to create deepfake videos could also give the United States and allied intelligence communities opportunities to learn about planning efforts and mitigate the risks prior to a deepfake's release.

Fourth, deepfake videos require extensive training data (Johnson, 2023). High-quality deepfakes require thousands of training data images; thus, such videos often feature celebrities and politicians (Shin et al., 2022). Acquiring such data for figures such as Barack Obama or Tom Cruise is a relatively less complicated task, and it would likewise be easy to acquire data for other highly video-recorded individuals, such as politicians. However, the requirements may limit the ability of adversaries to create high-quality fakes of lesser-known or lesser-photographed individuals, such as intelligence agents. The zero day of disinformation will also limit the prevalence of high-quality deepfakes. The term "zero-day" is typically used to describe a software vulnerability unknown to the developers or for which there is no available security patch.

IMPLICATIONS AND RECOMMENDATIONS

Based on this review of deepfakes and related issues, several recommendations can be offered.

First, the adversarial use of deepfakes will involve a decision calculus that weighs opportunities, benefits, and risks, and such decisions could be modeled via wargaming and other exercises. Governments should criminalize the nonconsensual distribution of intimate deepfake images. Legislation may be required to outlaw the nonconsensual distribution of deepfake intimate imagery to increase protections for victims and targets of deepfakes while upholding free speech protections and principles. Lawmakers should strive to pass similar legislation. This would ensure that perpetrators of deepfakes are held accountable and can serve as an essential deterrent for those considering using deepfakes to target individuals.

Likewise, the intelligence community should invest in intelligence collection strategies that could provide forewarnings of adversarial efforts to invest in deepfake technology and to create the deepfake content itself. Organizations should actively pursue partnerships with the media, social media, career networking, and similar companies to learn more about how these companies preserve online content's provenance. This is especially critical given how they may be working to identify and mitigate the harms of synthetic content, which may be used to exploit organizations and their employees. Organizations should use active authentication techniques such as watermarks and Content Authenticity Initiative standards to protect public figures or celebrities from being used or repurposed for disinformation.

Second, it will be necessary for the U.S. government, the research community, social media platforms, and other private stakeholders to continue investing in and taking other steps to enhance detection technology. Critical steps include creating a "deepfake zoo" of known deepfake content, which can be used to inform detection technology development. Likewise, the government should work with the private sector to "proliferate" radioactive data sets of video content that would render any trained deepfake videos more easily detectable. However, it should be noted that more than a mere technological criterion for identity and detection is needed. As Tim Hwang notes, this would "significantly lower the costs of detection for deepfakes generated by commodified tools" and "force more sophisticated disinformation actors to source their datasets to avoid detection" (Hwang, 2020, p. iv). Researchers should continue to examine best

practices for labeling deepfake content. Governments and other stakeholders should explore limiting access to specific high-performance deepfake detectors. Alternatively, the government and the private sector could engage in a broader multistakeholder deliberation process to achieve the same ends. However, coordinating the efforts of such stakeholders would require substantial work.

Third, educating the general public to increase AI literacy and critical reasoning to discern and be wary of shallow-fake content will be particularly crucial. Industries and governments agree that improving AI literacy will help to prevent democracy and society from succumbing to the influence of fakes. It is essential to promote AI literacy by encouraging users to be vigilant when consuming online information and media; by creating or sharing educational media resources with users, the public may be better inoculated against the effects of false or misleading content (Locoro et al., 2021). Media literacy efforts will likely need to continue to promote broad media literacy skills and build resilience against disinformation. This training must be evidence-based and promoted at multiple levels, including school curricula for primary and secondary schools and media literacy interventions that offer short, sharable educational content that can be disseminated online.

The major trends in detection and authentication include education, detection refining, and increased pressure from the community to employ authentication techniques for media. Eventually, these trends may result in policies that will require specific changes. A suitable method would involve continuing efforts to warn audiences more directly about the reality of deepfake technology and the prospects of such technology being used to promote disinformation. In the long term, as it becomes easier and cheaper to create credible deepfake content, media literacy interventions might need to sow mistrust in provenance-based videographic evidence (and promote trust in provenance-based content). At present, videos are taken at face value; they are perceived as representing events as they occurred. The proliferation of deepfake content will inevitably erode this trust, and this erosion might be a necessary facet of a media-literate public. This is the insidious impact of deepfakes, which corrode trust in society, where people do not wish to distinguish the truth from falsehoods. Various actors should support media literacy efforts. News organizations, social media platforms, and civil society groups have taken the lead by creating and disseminating educational content, and they should continue to do so. Individual state and local governments should strive to facilitate media

literacy in school curricula. The U.S. government should take a more active role in media literacy. For example, the U.S. Department of Education should support the development of empirically proven curricula that local school districts can field; the U.S. Department of State should more actively support media literacy initiatives abroad, especially in areas such as Eastern Europe that are strongly targeted by Russian propaganda.

Fourth, the industry can prohibit harmful manipulated content: in addition to their regular content policies, which may already ban some forms of synthetic media that are hateful, harassing, or offensive, platforms should have explicit policies banning nonconsensual deepfake pornography and synthetic media that is deceptive, misleading, and likely to cause harm. Platforms should also prioritize transparency: platforms should maintain records or audit trails of the instances in which they detect synthetic media and the subsequent steps they take upon discovering that a piece of media is synthetic. They should also be transparent about their synthetic media policy and enforcement mechanisms with users and the public. Firms should consider whether to implement requirements for users to disclose the use of such contents. Disclosure may include labeling requirements, provenance information, or watermarks to visibly demonstrate that a post involved the creation or use of synthetic media. For example, TikTok changed its user policy to make it obligatory for creators to disclose the use of deepfakes in any realistic-looking image. Google and YouTube require that any deepfake-generated videos used in political campaigns should be declared on their platforms.

Any policy responses seeking to address deepfakes would likely face constitutional and other legal challenges along with the technical challenges of detection. Key policy questions include the following: (1) What is the maturity of deepfake detection technology? How much progress have federal programs and public-private partnerships made in developing such technology? What expertise will be required to ensure that detection keeps pace with deepfake technology? (2) What rights do individuals have to their privacy and likenesses? What rights do creators of deepfakes have under the First Amendment? What policy options exist regarding election interference? What policy options exist regarding exploitation and image abuse, such as nonconsensual pornography? (3) What can be done to educate the public about deepfakes? Should manipulated media be marked or labeled? Should media be traceable to its origin to determine authenticity? (4) What roles should media outlets and social media companies play in detecting and moderating content that has been altered or falsified?

CONCLUSION

It remains uncertain what the deceptive use of deepfakes will mean for us and our society. Fallis (2021) is concerned that the prevalence of deepfakes produces a new epistemic risk. If people can no longer trust what they encounter online, deepfakes will challenge our information acquisition through media consumption. A desire for the authenticity and literacy of deepfakes, as well as a law limiting how they can be maliciously used, can support online users in becoming more discerning about what they view online, which can mitigate the adversarial impacts of deepfakes.

If this heightened state of vigilance becomes a norm to detect fraud, how will this hinder collaboration, productivity, and camaraderie at work and among friends? What if deepfakes used for fraud become even more mainstream and a carefully orchestrated scam such as a spoofed number and a deepfake voice of a family member becomes something to be feared because there is no discerning reality from fiction? Will this change how people socialize and interact with others? How can we trust AI? Perhaps only time will tell.

REFERENCES

- Ahmed, S. (2021). Navigating the maze: Deepfakes, cognitive ability, and social media news skepticism. *New Media & Society*, 25, 1108–1129. <https://doi.org/10.1177/14614448211019198>
- Atlantic Council's Digital Forensic Research Lab. (2021). Ethical AI. A research report at Atlantic Council's Digital Research. <https://www.atlanticcouncil.org/programs/digital-forensic-research-lab>
- Barari, M., Ross, M., & Surachartkumtonkun, J. (2020). Negative and positive customer shopping experience in an online context. *Journal of Retailing and Consumer Services*, 53, 101985 (6) (PDF) A meta-analysis of customer engagement behaviour. Available from: https://www.researchgate.net/publication/343863039_A_meta-analysis_of_customer_engagement_behaviour [accessed Feb 29 2024].
- Campbell, C. (1995). When attention-getting advertising tactics elicit consumer inferences of manipulative intent. *Journal of Consumer Psychology*, 4(3), 225. https://doi.org/10.1207/s15327663jcp0403_02
- Campbell, C., Plangger, K., Sands, S., & Kietzmann, J. (2022). Preparing for an era of deepfakes and AI-generated ads: A framework for understanding responses to manipulated advertising. *Journal of Advertising*, 51(1), 22–38. <https://doi.org/10.1080/00913367.2021.1909515>
- Campbell, D., & Reiman, A. (2022). Has social psychology lost touch with reality? *Journal of Experimental Social Psychology*, 98, 104255. <https://doi.org/10.1016/j.jesp.2021.104255>

- Center for Countering Digital Hate. (2021). Stop the spread of hate and lies online. Research report. <https://counterhate.com>
- Chesney, R., & Citron, D. K. (2019). Deep fakes: A looming challenge for privacy, democracy, and national security. *California Law Review*, 107(6), 1753–1819. <https://doi.org/10.2139/ssrn.3213954>
- de Ruiter, A. (2021). The distinct wrong of deepfakes. *Philosophy and Technology*, 34(4), 1311–1332. <https://doi.org/10.1007/s13347-021-00459-2>
- Diakopoulos, N., & Johnson, D. (2021). Anticipating and addressing the ethical implications of deepfakes in the context of elections. *New Media and Society*, 23(7), 2072–2098. <https://doi.org/10.1177/1461444820925811>
- Ecker, U. K. H., Lewandowsky, S., Cook, J., Schmid, P., Fazio, L. K., Brashier, N., Kendeou, P., Vraga, E. K., & Amazeen, M. A. (2022). The psychological drivers of misinformation belief and its resistance to correction. *Nature Reviews Psychology*, 1(1), 13–29. <https://doi.org/10.1038/s44159-021-00006-y>
- Epstein, Z., Berinsky, A., Cole, R., Gully, A., & Pennycook, G., & Rand, D. (2022). Developing an accuracy-prompt toolkit to reduce COVID-19 misinformation online. *Harvard Kennedy School Misinformation Review*, 2(3). <https://doi.org/10.37016/mr-2020-71>
- Fallis, D. (2021). The Epistemic Threat of Deepfakes. *Philos Technol.*, 34(4), 623–643. <https://doi.org/10.1007/s13347-020-00419-2>. Epub 2020 Aug 6. PMID: 32837868; PMCID: PMC7406872.
- Gamage, K. A. A., Dehideniya, D. M. S. C. P. K., & Ekanayake, S. Y. (2021). The Role of Personal Values in Learning Approaches and Student Achievements. *Behav. Sci.*, 11, 102. (6) (PDF) The Role of Personal Values in Learning Approaches and Student Achievements. Available from: https://www.researchgate.net/publication/353298857_The_Role_of_Personal_Values_in_Learning_Approaches_and_Student_Achievements [accessed Feb 29 2024].
- GAO. (2020). Technology assessment: Artificial intelligence: Emerging opportunities, challenges, and implications. GAO Report: Science, Tech Assessment and Analysis. GAO-18-142SP. <https://www.gao.gov/assets/gao-20-379sp.pdf>
- Giansiracusa, N. (2021). *How algorithms create and prevent fake news*. Springer.
- Hwang, T. (2020). *Deepfakes: A grounded threat assessment*. Center for Security and Emerging Technology. George Town University Research Report. <https://cset.georgetown.edu>
- Hwang, Y., Ryu, J., & Jeong, S. (2021). Effects of disinformation using deepfake: The protective effect of media literacy education. *Cyberpsychology, Behavior, and Social Networking*, 24(3), 188–193. <https://doi.org/10.1089/cyber.2020.0174>
- Iacobucci, S., De Cicco, R., Michetti, F., Palumbo, R., & Pagliaro, S. (2021). Deepfakes unmasked: The effects of information priming and bullshit receptivity on deepfake recognition and sharing intention. *Cyberpsychology Behavior & Social Networking*, 24, 194–202. <https://doi.org/10.1089/cyber.2020.0149>

- Jankowicz, Nina, et al. "Malign Creativity: How gender, sex and lies are weaponized against women online" Wilson Center, (January 2021), <https://www.wilsoncenter.org/sites/default/files/media/uploads/documents/Report%20Malign%20Creativity%20How%20Gender%2C%20Sex%2C%20and%20Lies%20are%20Weaponized%20Against%20Women%20.pdf>, "Gendered disinformation and what can be done to counter it," Media Support (May 4, 2021). <https://www.mediasupport.org/news/gendered-disinformation-and-what-can-be-done-to-counter-it>
- Johnson, C. (2023). *Deepfakes and detection technologies*. RAND Corporation, RR-A1482-1.
- Karasavva, V., & Noorbhai, A. (2021). The real threat of deepfake pornography: A review of Canadian policy. *Cyberpsychology, Behavior, and Social Networking*, 24(3), 203–209. <https://doi.org/10.1089/cyber.2020.0272>
- Kerner, C., & Risse, M. (2021). Beyond porn and discreditation: Epistemic promises and perils of deepfake technology in digital life worlds. *Moral Philosophy and Politics*, 8(1), 81–108. <https://doi.org/10.1515/mopp-2020-0024>
- Kobis, N., Dolezalova, B., & Soraperra, I. (2021). Fooled twice: People cannot detect deepfakes but think they can. *iScience*, 24(11), 103364. <https://doi.org/10.1016/j.isci.2021.103364>
- Lewandowsky, S., & van der Linden, S. (2021). Countering misinformation and fake news through inoculation and prebunking. *European Review of Social Psychology*, 32(2), 348–384. <https://psycnet.apa.org/doi/10.1080/10463283.2021.1876983>
- Locoro, A., Fisher, W. P., & Mari, L. (2021). Visual information literacy: Definition, construct modeling and assessment. *IEEE Access*, 9, 71053–71071. <https://doi.org/10.1109/ACCESS.2021.3078429>
- Marcellino, William, Todd C. Helmus, Joshua Kerrigan, Hillary Reiningger, Rouslan I. Karimov, and Rebecca Ann Lawrence. (2021). "Detecting Conspiracy Theories on Social Media: Improving Machine Learning to Detect and Understand Online Conspiracy Theories." RAND Corporation. <https://doi.org/10.7249/RR-A676-1>
- Mirsky, Y., & Lee, W. (2021). The creation and detection of deepfakes: a survey. *ACM Computing Surveys*, 54(1), 1–41. <https://doi.org/10.1145/3425780>
- Nehamas, N. (2023, June 9). Fake images? DeSantis goes after Trump—the *New York Times*, p. B6.
- Posard, M., Marine, J., Danielle, S., & Todd, H. (2020). Russian Propaganda Hits Its Mark: Experimentally Testing the Impact of Russian Propaganda and Counter-Interventions. RAND research report, RR-A704-3. https://www.rand.org/pubs/research_reports/RR-A704-3.html
- Qian, S., Shen, C., & Zhang, J. (2023). Fighting cheapfakes: Using a digital media literacy intervention to motivate reverse search of out-of-context visual misinformation. *Journal of Computer-Mediated Communication*, 28(1). <https://doi.org/10.1093/jcmc/zmac024>

- Shin, D. (2022). The perception of humanness in conversational journalism: An algorithmic information-processing perspective. *New Media & Society*, 24(12), 2680–2704. <https://doi.org/10.1177/1461444821993801>
- Shin, D. (2023). *Algorithms, humans, and interactions: How do algorithms interact with people? Designing meaningful AI experiences*. Routledge. <https://doi.org/10.1201/b23083>
- Shin, D. (2024). How algorithms promote self-radicalization: Audit of Tiktok's algorithm using a reverse engineering method. *Social Science Computer Review*, 29(3), 1–21.
- Shin, D., Kee, K., & Shin, E. (2022). Algorithm awareness: Why user awareness is critical for personal privacy in the adoption of algorithmic platforms? *International Journal of Information Management*, 65, 102494. <https://doi.org/10.1016/j.ijinfomgt.2022.102494>
- Shin, D., Ki, F. K., & Shin, E. (2023). The nudging effect of accuracy alerts for combating the diffusion of misinformation. *Journal of Broadcasting & Electronic Media*, 67(2), 141–160. <https://doi.org/10.1080/08838151.2023.2175830>
- Shin, D., Hameleers, M., Park, Y., Diakopoulos, N., Helberger, N., Lewis, S., Westlund, O., & Baumann, S. (2022). Countering algorithmic bias and disinformation and effectively harnessing the power of AI in media. *Journalism & Mass Communication Quarterly*, 99(4), 887–907.
- Stoll, A. (2020, January 13). Shallowfakes and their potential for fake news. *Washington Journal of Law, Technology, and Arts*.
- Sundar, S., Molina, M., & Cho, E. (2021). Seeing is believing: Is video modality more potent in spreading fake news via online messaging apps? *Journal of Computer-Mediated Communication*, 26(6), 301–319. <https://doi.org/10.1093/jcmc/zmab010>
- Vaccari, C., & Chadwick, A. (2020). Deepfakes and disinformation: Exploring the impact of synthetic political video on deception, uncertainty, and trust in news. *Social Media and Society*, 6, 1–13. <https://doi.org/10.1177/205630512090340>
- Vaccari, C., & Chadwick, A. (2023). Deepfakes and disinformation: Exploring the impact of synthetic political video on deception, uncertainty, and trust in news. *Social Media + Society*, 6(1). <https://doi.org/10.1177/2056305120903408>
- Waldemarsson, C. (2020, April 27). *Disinformation, deepfakes and democracy: The European response to election interference in the digital age*. Copenhagen: Alliance of Democracies.
- Whittaker, L., Letheren, K., & Mulcahy, R. (2021). The rise of deepfakes: A conceptual framework and research agenda for marketing. *Australasian Marketing Journal*, 29(3), 204–214. <https://doi.org/10.1177/1839334921999479>
- Wittenberg, E., Kerr, A. M., & Goldsmith, J. (2021). Exploring family caregiver communication difficulties and caregiver quality of life and anxiety. *American Journal of Hospice and Palliative Medicine*, 38(2), 147–153. <https://doi.org/10.1177/1049909120935371>

EPILOGUE

This book project was initiated and inspired purely by my personal motivation. I was the victim of misinformation, perhaps a more exact term, disinformation, while I was in Seoul, Korea, for ten years. I became curious about why people like and are driven to spread misinformation online. Misleading information of all kinds is deliberately spread and shared in various online forms and at varying speeds and magnitudes. Misinformation often comes from cherry-picked data taken out of context. Some people spread misinformation for personal satisfaction or self-motivation, just like what Immanuel Kant described as the “tendency to perceive with displeasure the good of others.” Misinformation often is an expression of human envy that is vindictive and inwardly tormenting frame of mind, the displeasure with which one perceives the success and the prosperity of others, begrudges them these things, and, in addition, wishes one were able to destroy or to possess them oneself. People spread misinformation to accuse those who succeed, thinking it is for the sake of social justice.

During my tenure at Sungkyunkwan University, Seoul, Korea, I was extremely fortunate to be noted as one of the top scholars in the university and the country. I enjoyed my research and loved what I did. My research was honored as the best research in the country and nominated for the

highest honor in academia. My exceptional achievement, however, caused more harm than good, at least in the academic community to which I belonged. People started to be jealous of my research achievements and tried to spoil my work and integrity. I did not graduate from the best university in the country and was still deemed the best researcher in academia. My research and funding record was impressive: I received the highest amount for a single research project ever in the country, and I am sure this created envy, anger, and resentment among elite academic groups in Korea. Maybe those who had graduated from the best university wondered how I dared to be the best and how I actually could do it. It did not take long for their common curiosities to turn into malicious accusations and false/misleading claims. Gossip is triggered by negative jealousy and malicious envy. Small drops of gossip turn into a huge sea of misinformation and false beliefs, which hurt people and society, and conspiracy theories have tremendous power over our beliefs. Justice is often misunderstood as a way of excusing jealous behavior. Jealousy underlies our misinformation crises. They work like a virus that can poison and spread to people, especially when they are limited to echo chambers where people encounter self-reinforcing beliefs. People who encounter false information tend to actively spread it further by sharing or otherwise engaging with it, even though they know the information is false. Sensational misinformation spreads because people want to hear misinformation and because people are all connected. Outrage is the key to the virality of misinformation. Indeed, misinformation is a combination of diverse factors: strong negative emotions such as envy, anxiety, jealousy, fear, perceived risk, anger, and feelings of disgust. People share misinformation they believe to be true since they are trying to harm others.

To curb this problem and the threat it poses to us individually and socially, it is important that we understand the mechanisms and structural conditions that cause it. The misinformation led me to wonder: if misinformation spreads like a germ, can we inoculate humans against it? I began to wonder why people love to gossip and spread rumors about other people's personal lives. Why do people like fake news and wish for someone to fail? Why does it feel good to see someone's downfall? Why do people spread sensational and outrageous disinformation? Why is it more likely for fake news to be shared than true information? I realized from my experience that misinformation becomes real because people do not question what they see, hear, and talk about. Once fake news becomes an issue, the important thing is not whether the information is true or not but how it can be spread and how it can be amplified, squashed, or radicalized. Once

people have heard misinformation, they tend to accept it as the truth and further solidify its veracity, even after it has been corrected. A myriad of social, cognitive, and technological factors make people vulnerable to misinformation. It can cause hatred, sow mistrust, harm people's mental health, and undermine social capital.

My personal experience and curiosity turned into a professional interest and a scientific research project. Anybody can easily become the victim of disinformation. My experience as a victim of misinformation allowed me to delve deeply into the psychology of misinformation by looking at human instincts and their integration into instinctive conduct. Misinformation is a reflection of human desire, and fake news preys on some of our deepest human instincts. Disinformation is a product of a societal process where it is shaped, spread to others, evaluated, and consumed, all against a backdrop of cultural, social, and political-economic factors. During my research over many years, I have realized that misinformation is not something that we can fix easily; it is a societal condition that we should all constantly keep an eye on and adjust to. The ecosystem of misinformation will not easily be solved in the near future since AI cannot improve human nature. The problem of misinformation will be aggravated because the worst side of human nature is amplified by malicious agents taking advantage of advanced AI technologies. Unlike our hope, AI cannot solve this problem completely—at least not with something as simple and automatic as fake news detection. The reasons reveal not only the issues with such a product but also the limitations of AI in general. It is easy to consider AI as a superintelligence that is smarter than humans, but algorithms are just tools, and not every misinformation is simple to fix. The spread of misinformation elucidates AI's abilities as well as limitations. Misinformation is not a problem that solely AI can cure. There must be a paradigm shift in our thinking about how people approach the information-acquiring process. Maybe this sounds disappointing, but we must continue figuring out the best way to harness AI to combat misinformation.

This monograph invites readers to think about misinformation critically and take action to fight it effectively. I also hope this book offers an inoculation against the misinformation pandemic by putting forward a future research agenda for human-misinformation interaction.

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